

Selection Plots: Config Comparisons

May 5, 2026

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1 bdt_nom vs bdt_tightbdt50

Config diff (A $\rightarrow$ B)

```

--- config_bdt_nom.yaml
+++ config_bdt_tightbdt50.yaml
@@ -1,293 +0,0 @@
-#config.yaml
-#bdt nominal
-input:
-  photon_jet_file_root_dir: "/sphenix/user/shuhangli/ppg12/FunWithxgboost/"
-  photon_jet_file_branch_dir: "/bdt_split.root"
-  #photon_jet_file_branch_dir: "/bdt_tree.root"
-  data_file: "/sphenix/user/shuhangli/ppg12/anatreemaker/macro_maketree/data/ana521/condorout/part.*.root"
-  tree: "slimtree"
-  cluster_node_name: "CLUSTERINFO_CEMC"
-  bdt_model_name: "base_E"
-  # Optional: use different BDT models in different ET bins.
-  # bdt_et_bin_edges has N+1 values; bdt_et_bin_models has N values.
-  # Clusters outside all bins use bdt_model_name as fallback.
-  bdt_et_bin_edges: [8, 15, 35]
-  bdt_et_bin_models: ["base_v3E", "base_v3E"]
-  unfold_input: "results/MC_efficiency_photon5.root"
-
-output:
-  eff_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/MC_efficiency"
-  response_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/MC_response"
-  data_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/data_histo"
-  var_type: "bdt_nom"
-  final_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/Photon_final"
-
-analysis:
-  isbackground: 0
-  eta_bins: [-0.7, 0.7]
-  #eta_bins: [-0.7, 0.7]
-  vertex_cut: 60.0
-  #vertex_cut: 60.0
-  truth_iso_max: 4.0
-  reco_iso_min: -20.0
-  reco_iso_max_b: 0.490
-  reco_iso_max_s: 0.037
-  #reco_iso_max_b: 0
-  #reco_iso_max_s: 0.0293956
-  #reco_iso_max_b: 0.330117
-  #reco_iso_max_s: 0.025
-  #reco_iso_max_b: 1.08128
-  #reco_iso_max_s: 0.0299107
-  #reco_iso_max_b: -0.109478
-  #reco_iso_max_s: 0.0369506
-  #reco_iso_max_b: 3.0
-  #reco_iso_max_s: 0.0
-  reco_noniso_min_shift: 0.8
-  reco_noniso_max: 20.0
-

```

```

- # L1 trigger-efficiency correction on data clusters: weight *= 1/eps_L1(ET^reco)
- # Gumbel-CDF parametrization eps_L1(ET) = p0 * exp(-exp(-(ET - mu)/beta))
- # measured in plot_trigger_bit30.C with scaled[10]/live[30].
- apply_trigger_eff_correction: 1
- trigger_eff_p0: 1.000
- trigger_eff_mu: -2.04
- trigger_eff_beta: 3.33
- reco_min_ET: 5.0
- cone_size: 3
- eff_dR: 0.1
- trigger_used: 30
- pT_bins: [10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 32, 36]
- pT_bins_truth: [8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 32, 36, 45]
- use_topo_iso: 2 # 0: standard cone (size set by cone_size), 1: cluster_iso_topo_03, 2: cluster_iso_topo_04
- iso_threshold: 1
- iso_hcalonly: 0
- iso_emcalinnerr: 0.0
- n_nt_fail: 1
- weta_fail: 0
- wphi_fail: 0
- et1_fail: 0
- e11_to_e33_fail: 0
- e32_to_e35_fail: 0
- bdt_fail: 0
- weta_on: 1
- wphi_on: 0
- et1_on: 0
- e11_to_e33_on: 0
- e32_to_e35_on: 0
- et2_on: 0
- et3_on: 0
- et4_on: 0
- bdt_on: 1
- mc_iso_shift: 0.1
- mc_iso_scale: 1.2
- vertex_reweight_on: 1
- # Path to data vertex scan output file (h_vertexz). Used for on-the-fly vertex reweighting.
- # Leave empty ("") to use the pre-made vertex_reweight_file instead.
- vertex_scan_data_file: ""
- # Data-driven iterative truth-vertex reweight (standalone tool in
- # efficiencytool/truth_vertex_reweight/). When on, forces vertex_reweight_on
- # off and applies w(z_h) (single MC) or w(z_h)*w(z_mb) (double MC) per event.
- # Flip truth_vertex_reweight_on to 1 to switch the nominal 1.5 mrad
- # analysis from the f(z_r) reco-level reweight to the truth-level one.
- truth_vertex_reweight_on: 0
- truth_vertex_reweight_file: "/sphenix/user/shuhangli/ppg12/efficiencytool/truth_vertex_reweight/output"
- fit_option: 1
- cluster_escale: 1.0
- cluster_eres: 0.04
- mc_purity_correction: 0
- mc_purity_correction_file: ""
-
- # Data-driven tower mask (phi-symmetry z<-2 at tight-selection level).
- # Nominal production applies this mask symmetrically to data and MC to

```

```
- # correct for MC not emulating all Run-24 data dead towers. Switched on
- # as nominal on 2026-04-23. Other single-level masks (preselect, common,
- # OR) are available as systematic cross-checks via VARIANTS in
- # make_bdt_variations.py.
- tower_mask_on: 1
- tower_mask_file: "/sphenix/user/shuhangli/ppg12/efficiencytool/tower_masks_bdt_nom.root"
- tower_mask_name: "mask_phisymm_tight"
-
- tight:
-   reta77_min: 0.0
-   reta77_max: 1.0
-
-   rhad33_max: 1.0
-   rhad33_min: 0.00
-
-   w72_max: 1.0
-   w72_min: 0.0
-
-   re11_E_max: 1.0
-   re11_E_min: 0.0
-
-   CNN_max: 1.0
-   CNN_min: 0.0
-   #CNN_min: 0.8
-
-   #correlation
-   et1_max: 1.0
-   et1_min: 0.5
-   et1_min_b: 0.5
-   et1_min_s: 0.0
-   #correlation
-   e32_over_e35_max: 1.0
-   e32_over_e35_min: 0.8
-
-   #correlation
-   weta_cogx_max: 1.0
-   weta_cogx_min: 0.0
-   weta_cogx_max_b: 1.0
-   weta_cogx_max_s: 0.0
-
-   wphi_cogx_max: 1.0
-   wphi_cogx_min: 0.0
-   wphi_cogx_max_b: 1.0
-   wphi_cogx_max_s: 0.0
-
-   #correlation
-   prob_max: 1.0
-   prob_min: 0.0
-
-   #small correlation
-   e11_over_e33_max: 1.0
-   e11_over_e33_min: 0.0
-

```

```
- #small correlation
- et4_max: 1.0
- et4_min: 0.0
-
- et2_max: 1.0
- et2_min: 0.0
-
- et3_max: 1.0
- et3_min: 0.0
-
- #w32 small correlation
- w32_max: 1.0
- w32_min: 0.0
-
- #bdt
- bdt_max: 1.0
- bdt_min: 0.7
- bdt_min_slope: -0.003333333333333336
- bdt_min_intercept: 0.8333333333333334
-
-
- non_tight:
-   reta77_min: 0.0
-   reta77_max: 1.0
-
-   rhad33_max: 1.0
-   rhad33_min: 0.00
-
-   w72_max: 1.0
-   w72_min: 0.0
-
-   re11_E_max: 1.0
-   re11_E_min: 0.0
-
-   CNN_max: 1.0
-   CNN_min: 0.0
-   #correlation
-   et1_max: 1.0
-   et1_min: 0.6
-   #correlation
-   e32_over_e35_max: 1.0
-   e32_over_e35_min: 0.8
-
-   #correlation
-   weta_cogx_max: 1.0
-   weta_cogx_min: 0.0
-   weta_cogx_max_b: 1.0
-   weta_cogx_max_s: 0.0
-
-
-   wphi_cogx_max: 1.0
-   wphi_cogx_min: 0.0
-   wphi_cogx_max_b: 1.0
-   wphi_cogx_max_s: 0.0
```

```

-
- #correlation
- prob_max: 1.0
- prob_min: 0.0
-
- #small correlation
- e11_over_e33_max: 1.0
- e11_over_e33_min: 0.0
-
- #small correlation
- et4_max: 1.0
- et4_min: 0.0
-
- #w32 small correlation
- w32_max: 1.0
- w32_min: 0.0
-
- #bdt
- bdt_max: 0.5
- bdt_max_slope: 0.003333333333333336
- bdt_max_intercept: 0.6666666666666666
- bdt_min: 0.02
-
- bdt_min_intercept: 0.7333333333333333
- bdt_min_slope: -0.01333333333333333
-
- common:
- wr_cogx_bound: 0.0
- cluster_weta_cogx_bound: 2.0
-
- npb_cut_on: 1
- npb_score_cut: 0.5
-
- prob_max: 1.0
- prob_min: 0.0
-
- e11_over_e33_max: 0.98
-
- e11_over_e33_min: 0.0
-
- e32_over_e35_max: 1.0
- e32_over_e35_min: 0.8
-
- et1_max: 1.0
- et1_min: 0.6
-
-
-
-
-
-
- # Run range for data processing (-1 = no bound)
- run_min: 47289
- run_max: 54000

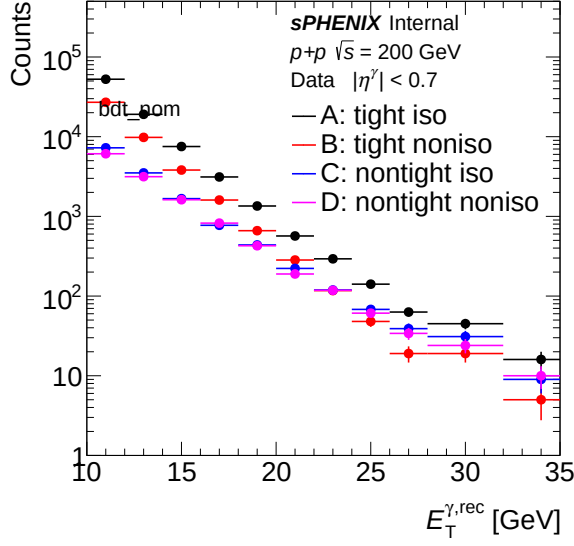
```

```

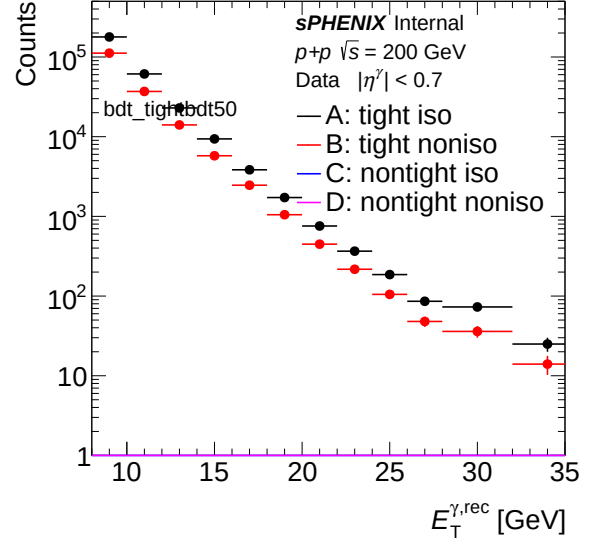
- # Optional run list file (one run number per line). If set, only listed runs are processed.
- run_list_file: ""
- # Total luminosity (pb^-1) for the selected run range.
- # Nominal switched to all-z (beam-delivered) 64.3718 pb^-1 on 2026-04-22;
- # paired with vertex_cut_truth=9999 to widen the MBD-eff truth denominator.
- lumi: 64.3718
-
- unfold:
-   resultit: 2
-   resultleak: 1
-   reweight: 1
-   # Truth-pT reweight for unfolding response prior. Re-derived 2026-04-27
-   # from R = (data unfolded with iter=1, no-truth-reweight variant) / mc_truth_prior
-   # (both unit-area normalized) on allz nominal with iso 0.490+0.037*ET and the
-   # L1 trigger-efficiency correction enabled. Fit range [8,35] GeV, rational(2,2)
-   # form (parameter-bounded). Iter=1 captures the prior-bias correction on the
-   # first round of iterative-prior reweighting. Loader: RecoEffCalculator_TTreeReader.C ~line 606.
-   truth_reweight:
-     formula: "([0] + [1]*x + [3]*x*x) / (1 + [2]*x + [4]*x*x)"
-     params: [4.05592, -0.984728, -0.478818, 0.0723232, 0.0522681]
-     xmin: 0
-     xmax: 100
-     clamp_pT_min: 10
-     clamp_pT_max: 30
-
-
- lumi_target: 64.3718
- vertex_cut_truth: 9999.0

```

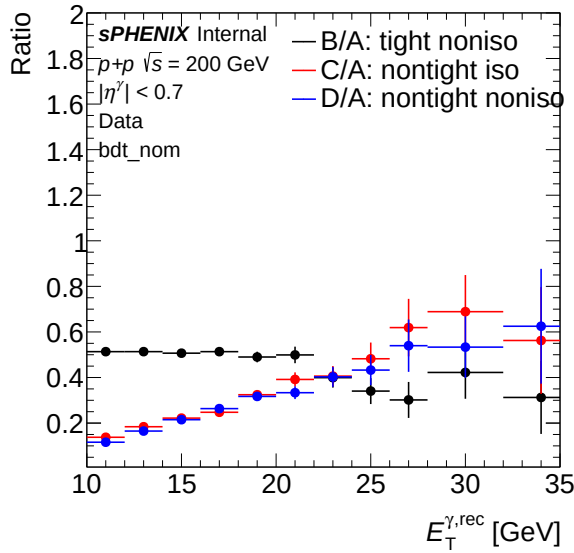
1.1 Sideband (Data)



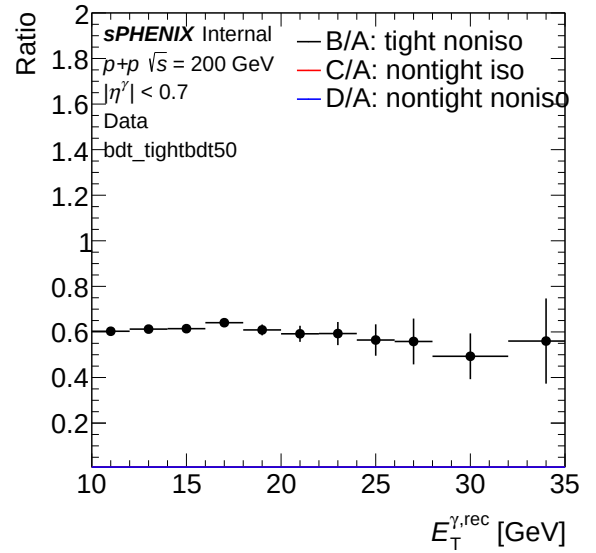
(a) Sideband ET spectrum (data) — bdt_nom



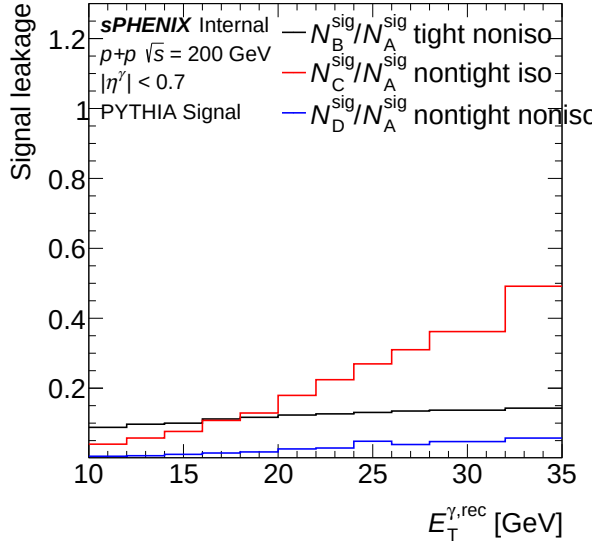
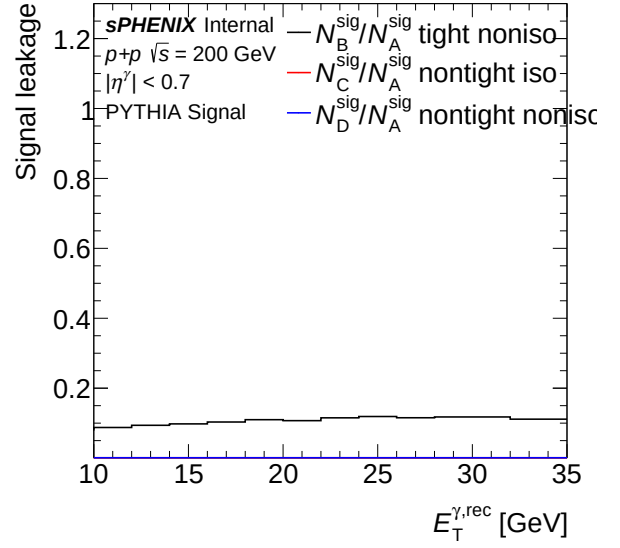
(b) Sideband ET spectrum (data) — bdt_tightbdt50



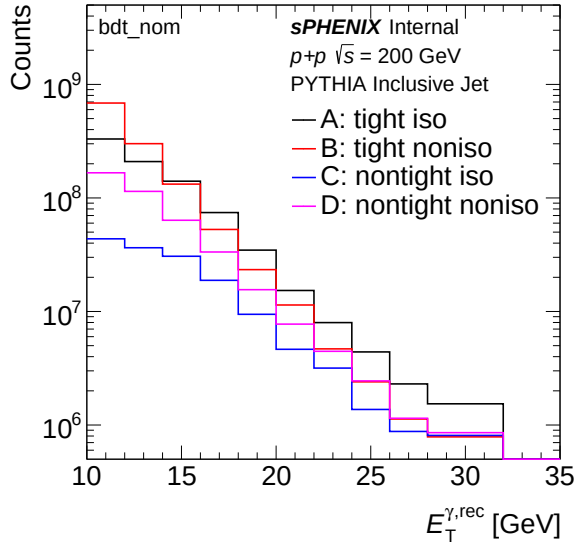
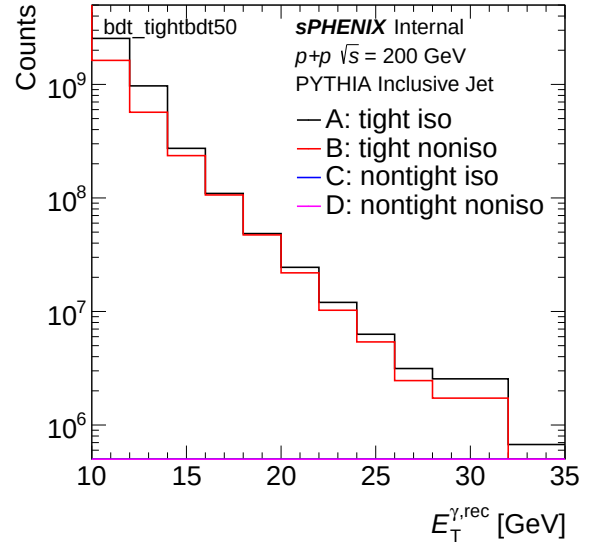
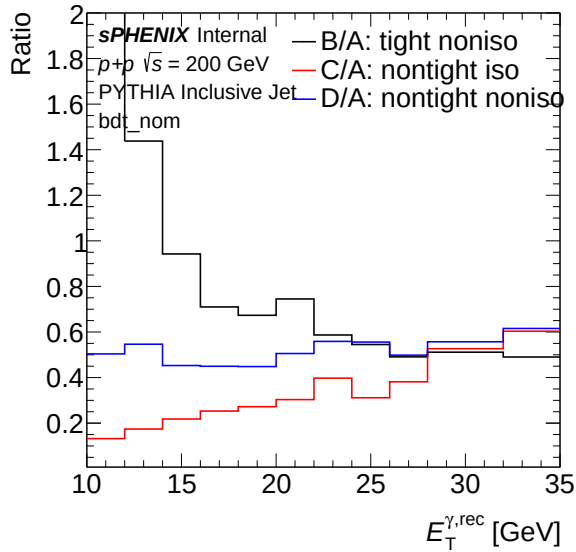
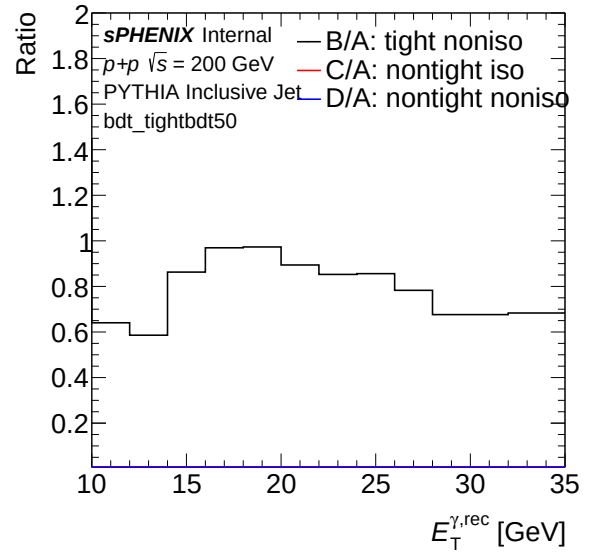
(a) Sideband ET ratio (data) — bdt_nom

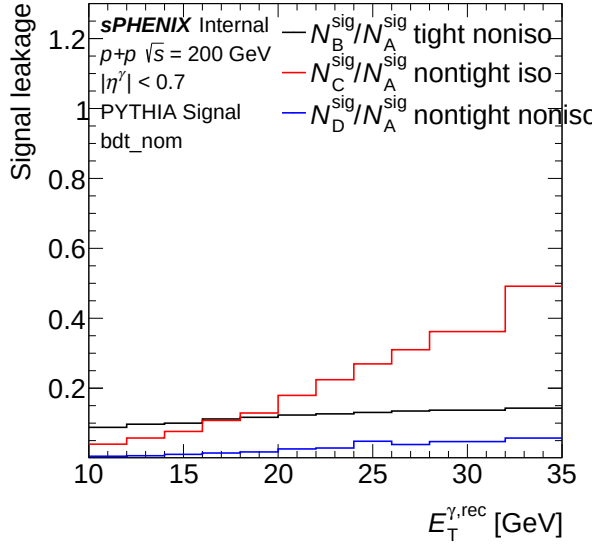


(b) Sideband ET ratio (data) — bdt_tightbdt50

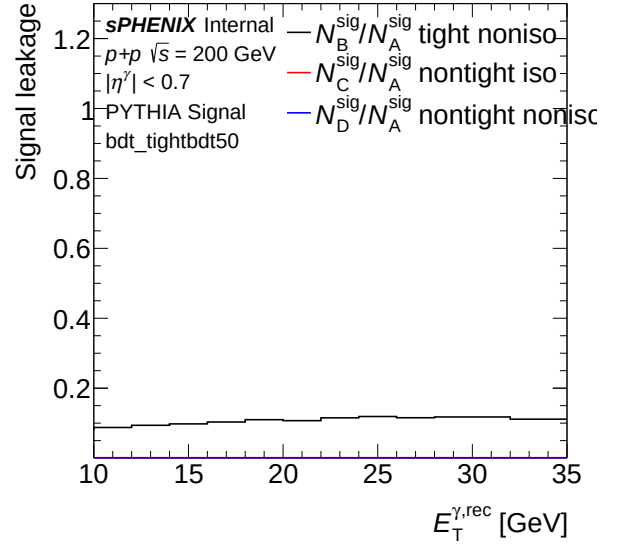
(a) Leakage fraction vs ET (data) — `bdt_nom`(b) Leakage fraction vs ET (data) — `bdt_tightbdt50`

1.2 Sideband (MC)

(a) Sideband ET spectrum (MC) — `bdt_nom`(b) Sideband ET spectrum (MC) — `bdt_tightbdt50`(a) Sideband ET ratio (MC) — `bdt_nom`(b) Sideband ET ratio (MC) — `bdt_tightbdt50`

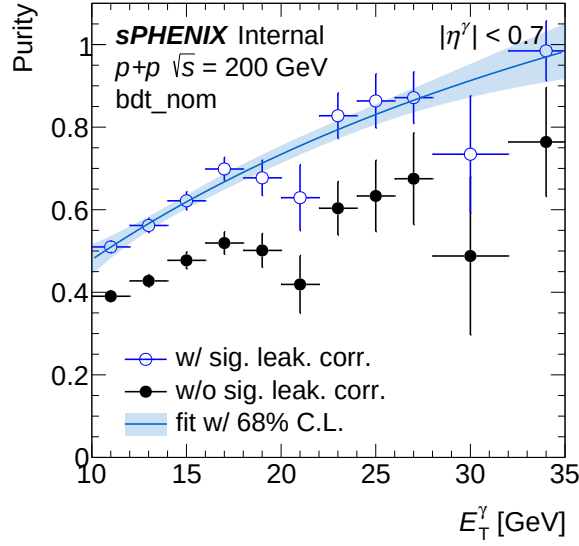


(a) Leakage fraction vs ET (MC) — bdt_nom

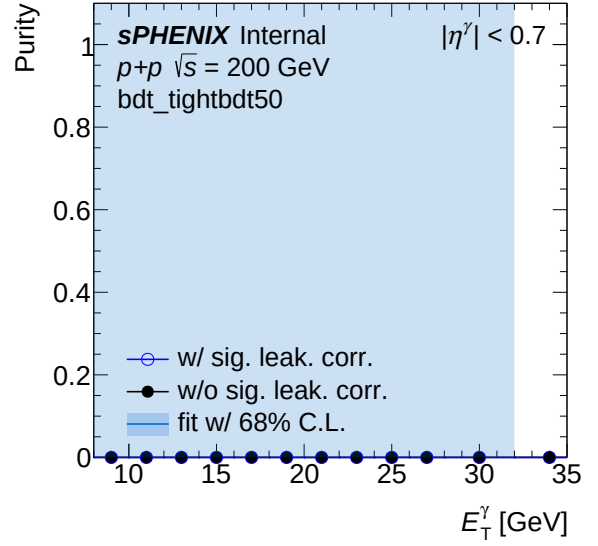


(b) Leakage fraction vs ET (MC) — bdt_tightbdt50

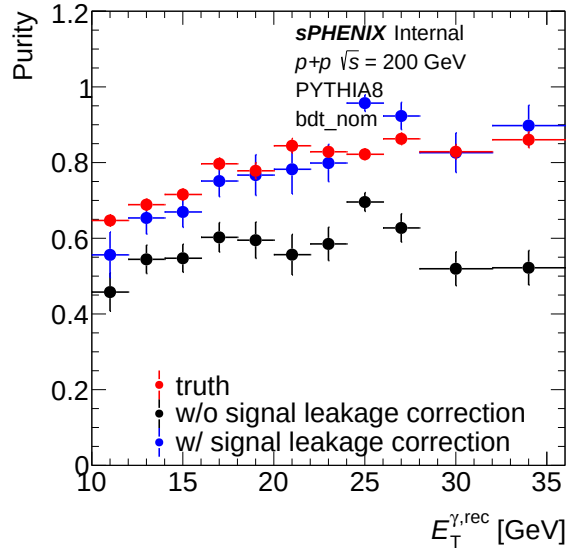
1.3 Purity



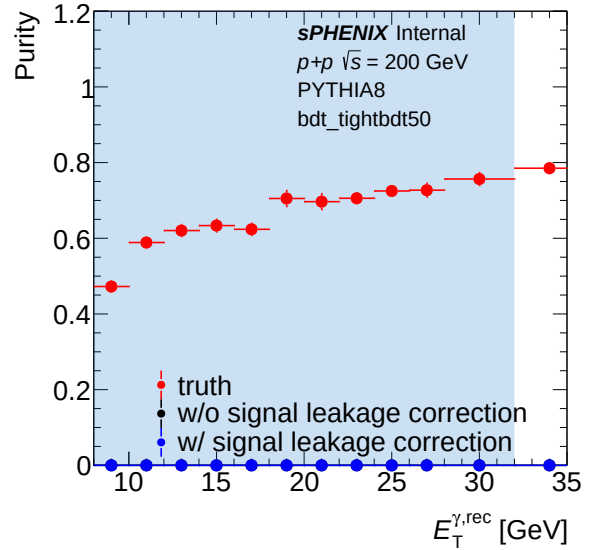
(a) Purity (data) — bdt_nom



(b) Purity (data) — bdt_tightbdt50

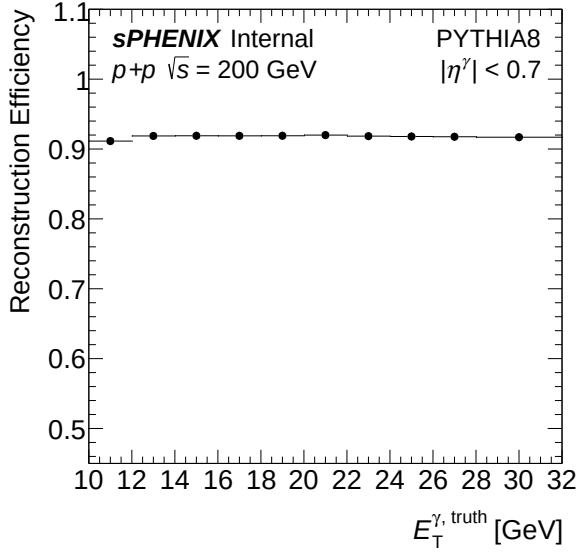


(a) Purity (MC) — bdt_nom

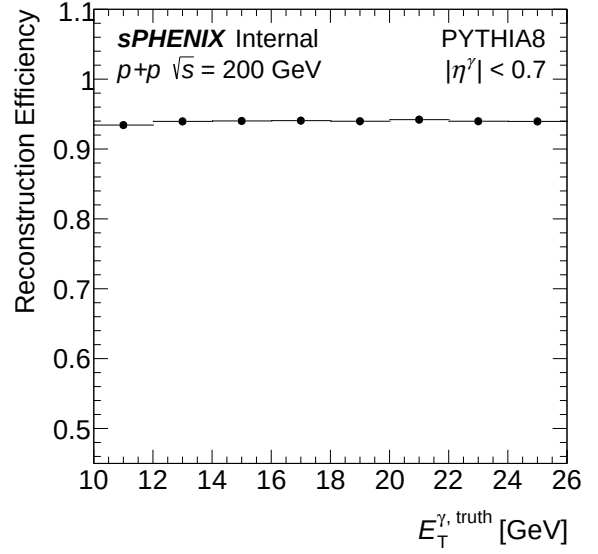


(b) Purity (MC) — bdt_tightbdt50

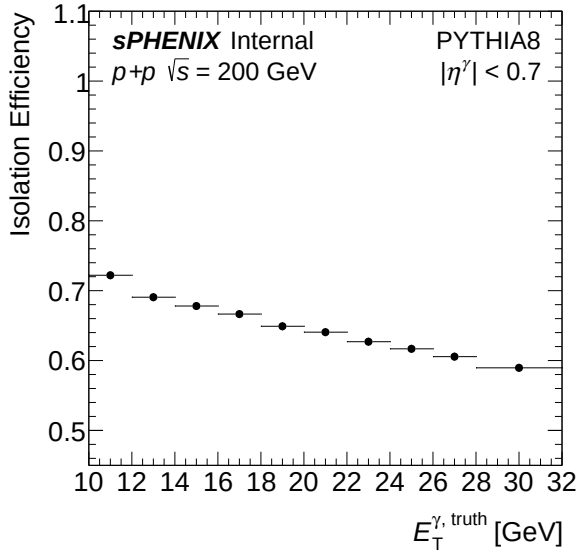
1.4 Efficiency



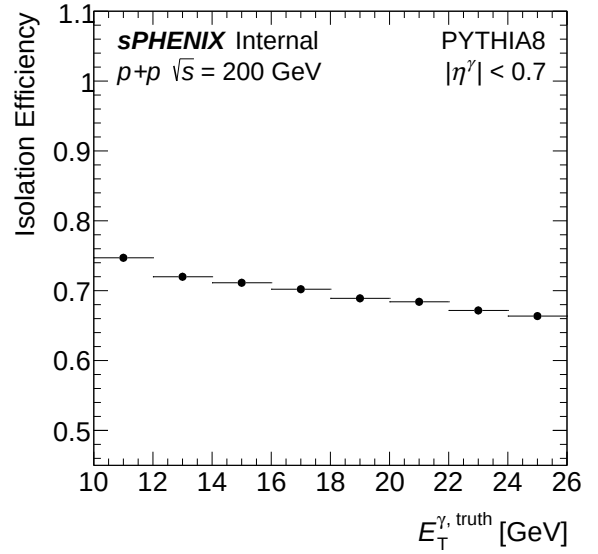
(a) Reconstruction efficiency — bdt_nom



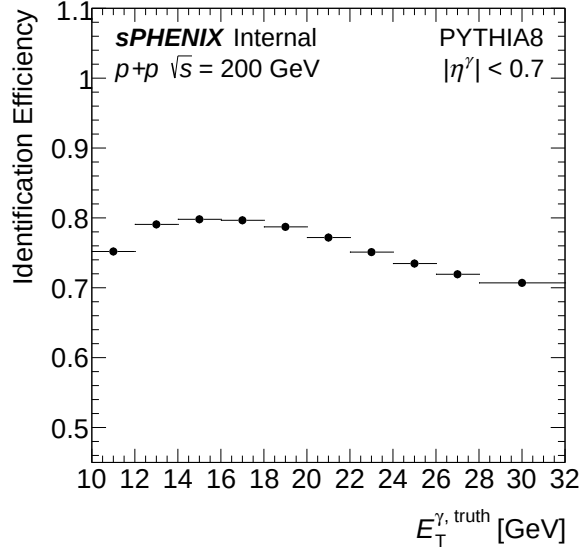
(b) Reconstruction efficiency — bdt_tightbdt50



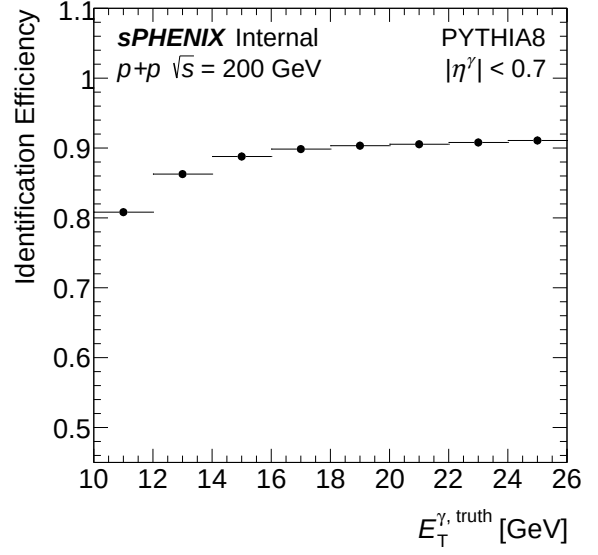
(a) Isolation efficiency — bdt_nom



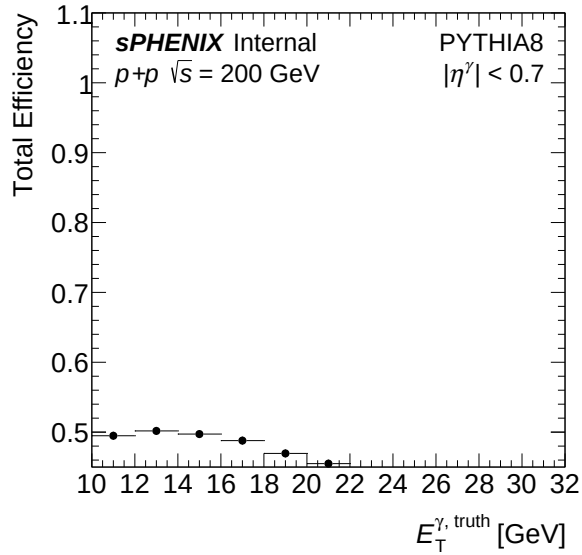
(b) Isolation efficiency — bdt_tightbdt50



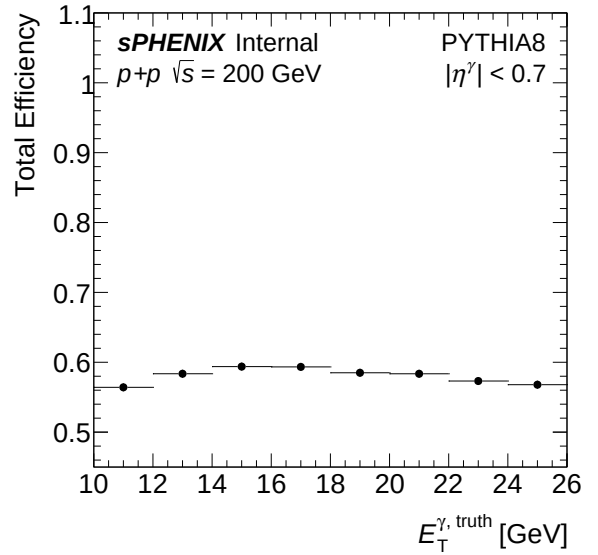
(a) Identification efficiency — bdt_nom



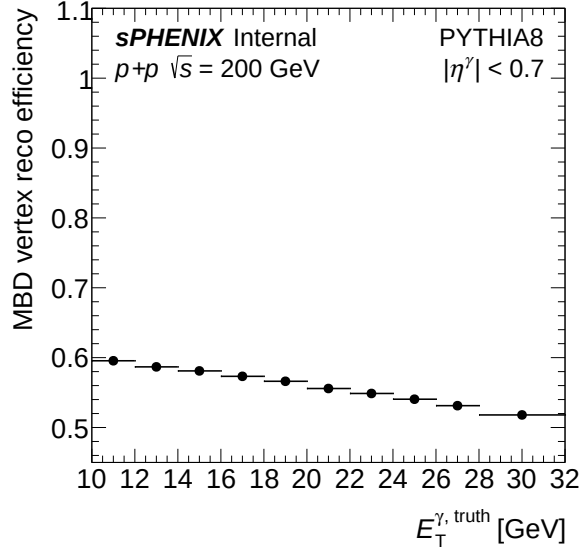
(b) Identification efficiency — bdt_tightbdt50



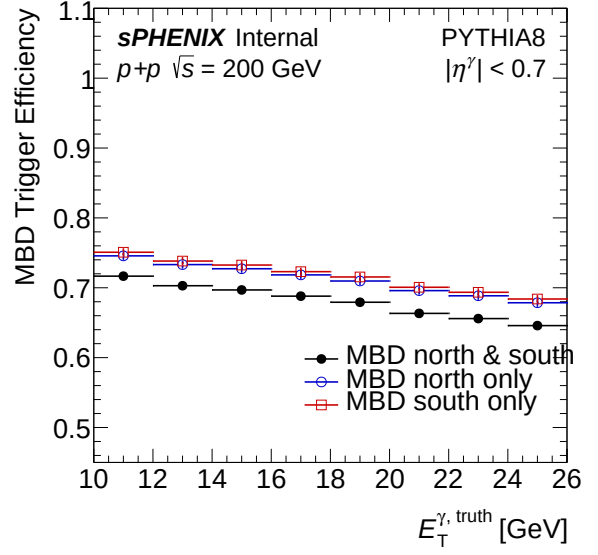
(a) Total efficiency — bdt_nom



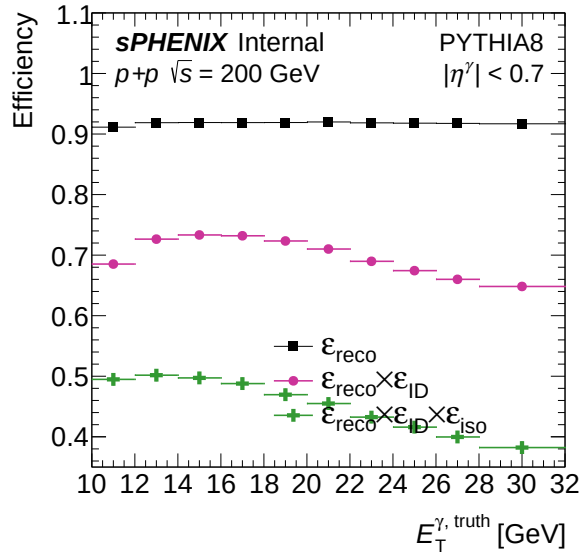
(b) Total efficiency — bdt_tightbdt50



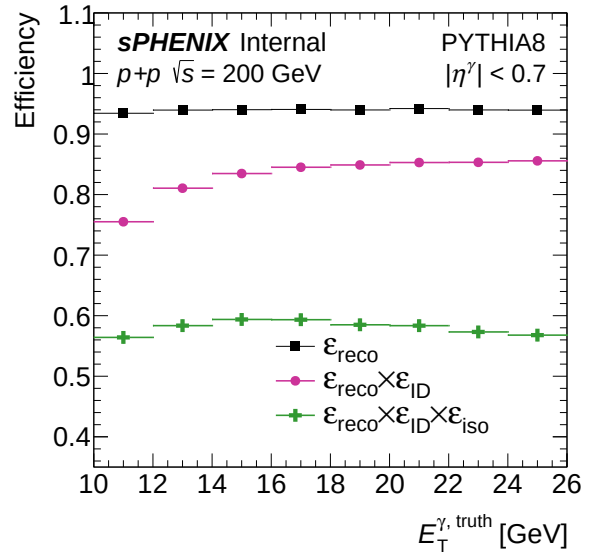
(a) MBD trigger efficiency — bdt_nom



(b) MBD trigger efficiency — bdt_tightbdt50

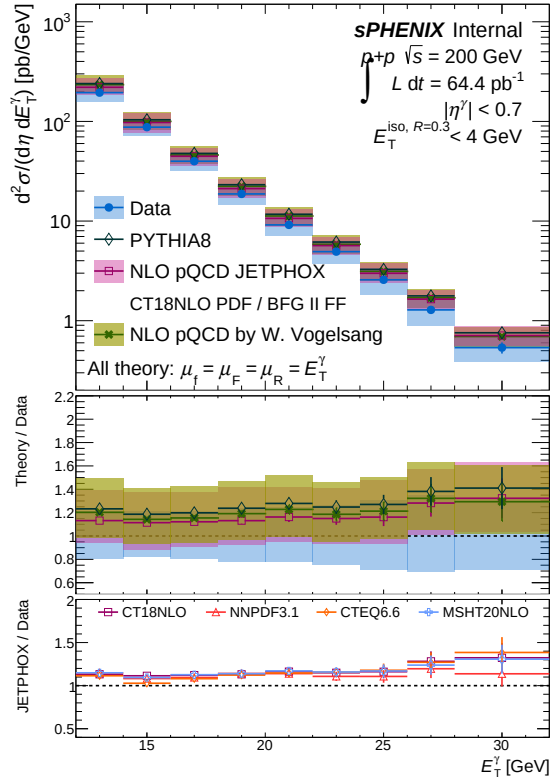
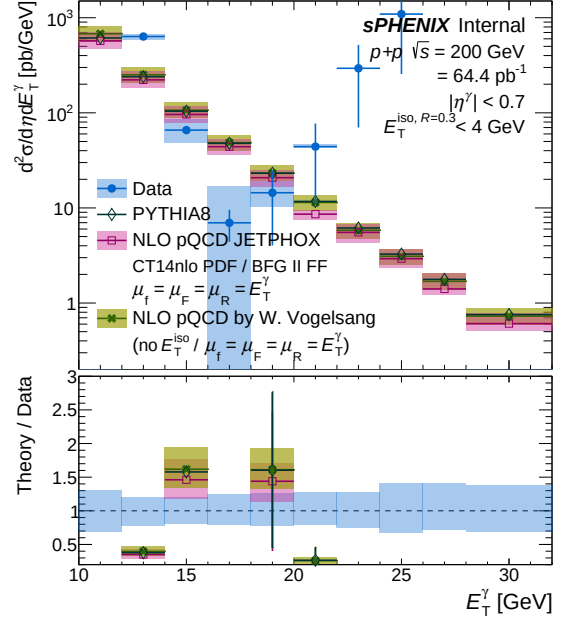


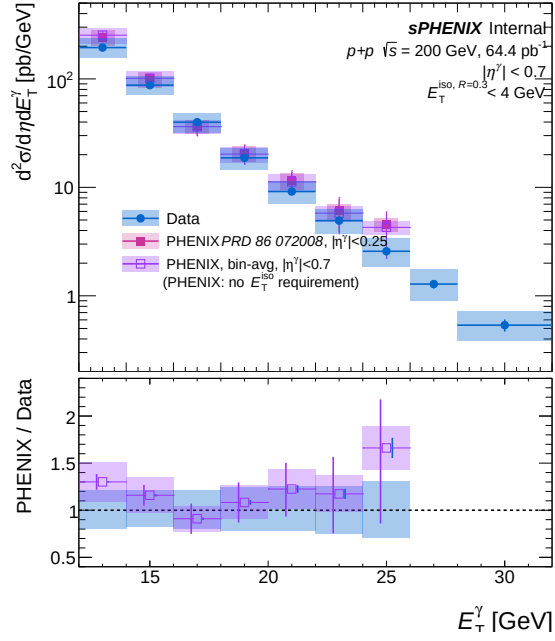
(a) Combined photon efficiency — bdt_nom



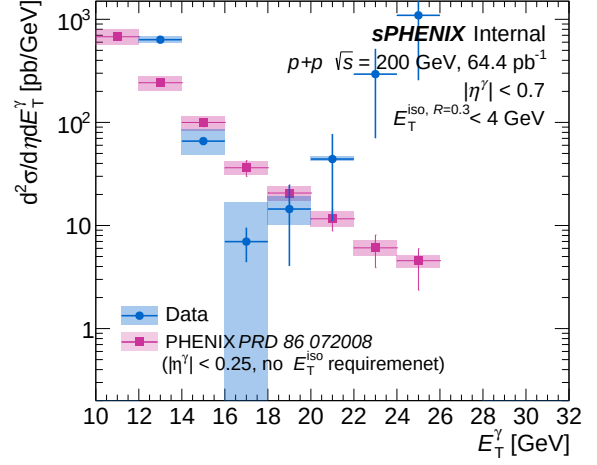
(b) Combined photon efficiency — bdt_tightbdt50

1.5 Final result

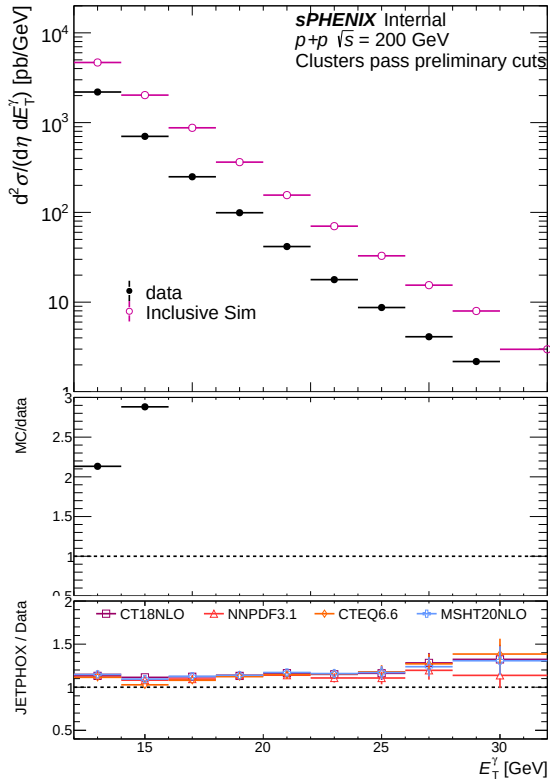
(a) Final cross section result — **bdt_nom**(b) Final cross section result — **bdt_tightbdt50**



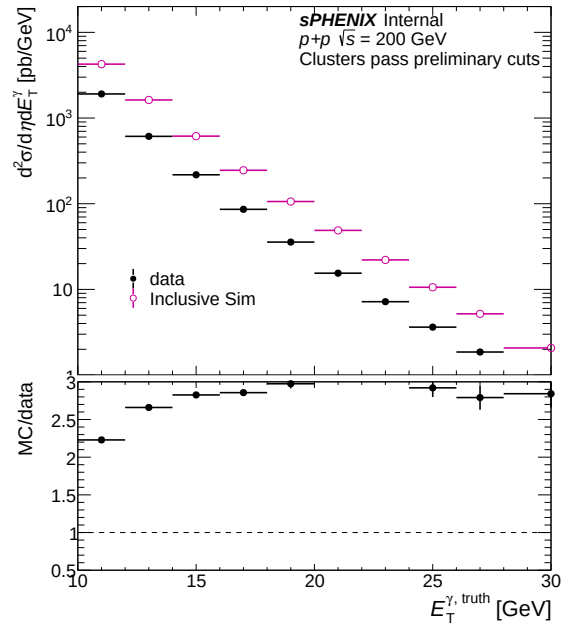
(a) Comparison with PHENIX — bdt_nom



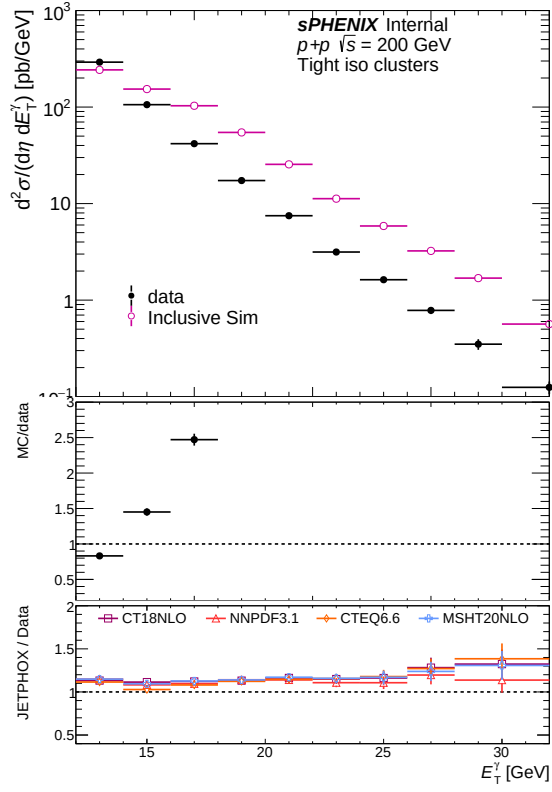
(b) Comparison with PHENIX — bdt_tightbdt50



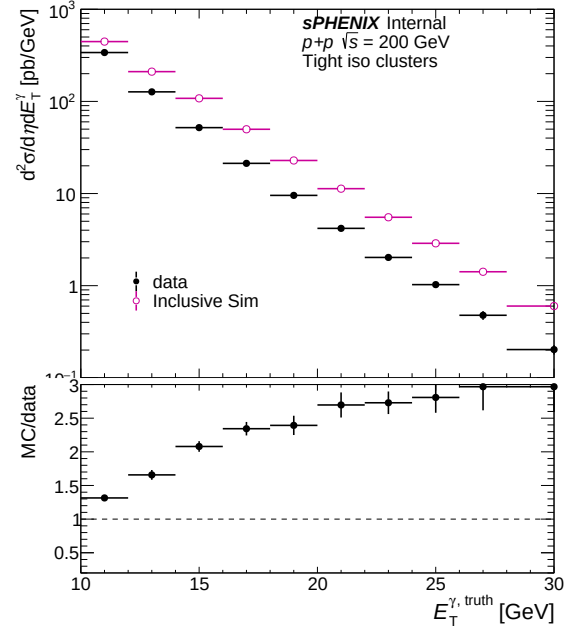
(a) Common cluster yield (data vs MC) — bdt_nom



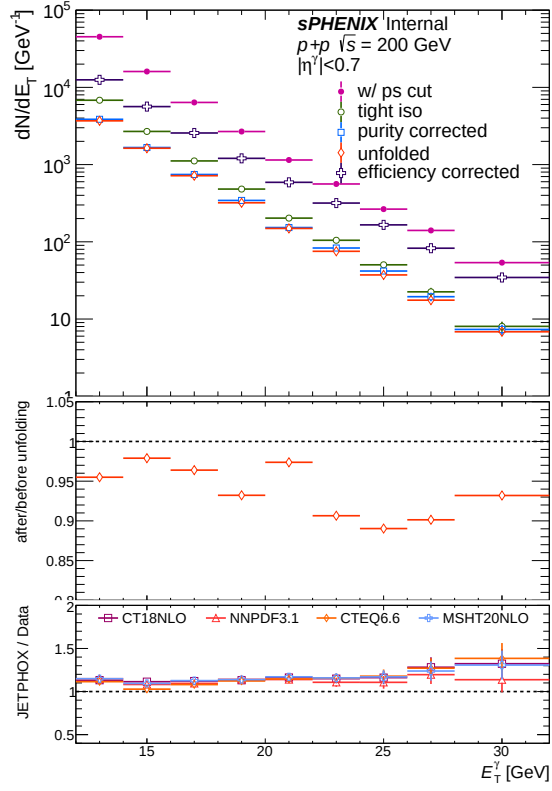
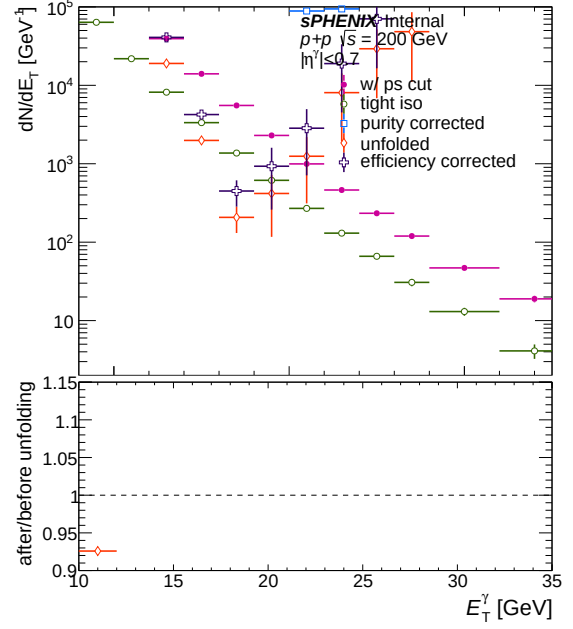
(b) Common cluster yield (data vs MC) — bdt_tightbdt50



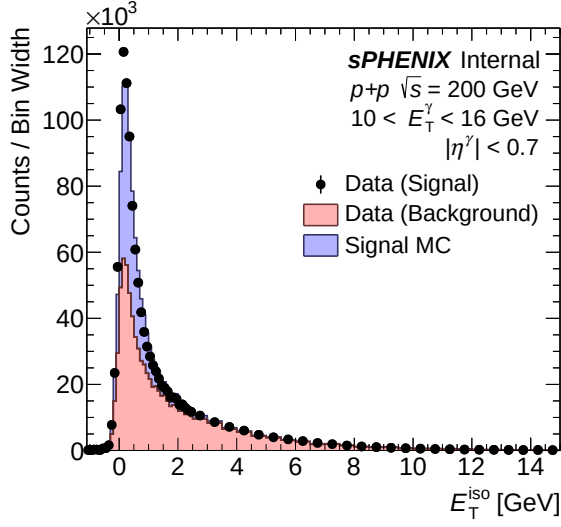
(a) Tight iso cluster yield (data vs MC) — bdt_nom



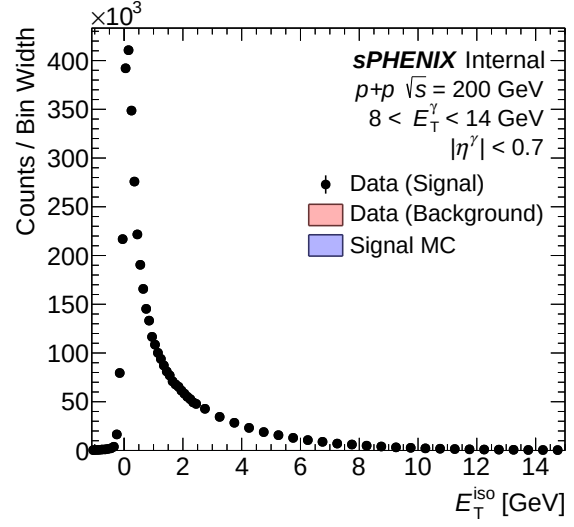
(b) Tight iso cluster yield (data vs MC) — bdt_tightbdt50

(a) All correction steps — **bdt_nom**(b) All correction steps — **bdt_tightbdt50**

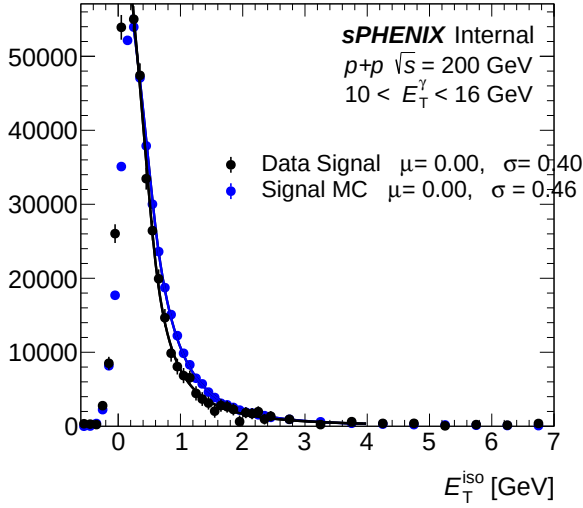
1.6 Isolation fit (CONF)



(a) IsoET template (pT bins 0-2) — bdt_nom

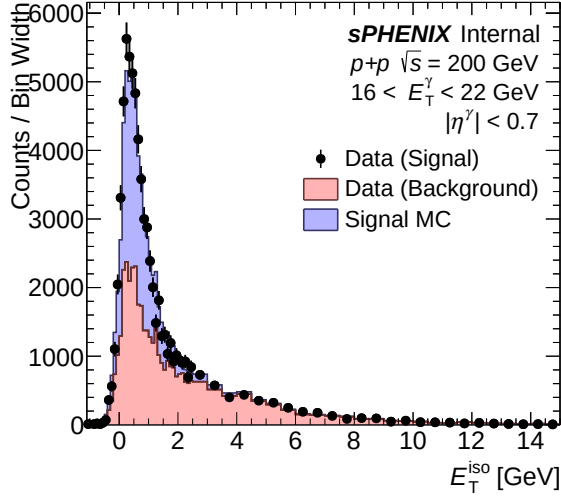


(b) IsoET template (pT bins 0-2) — bdt_tightbdt50

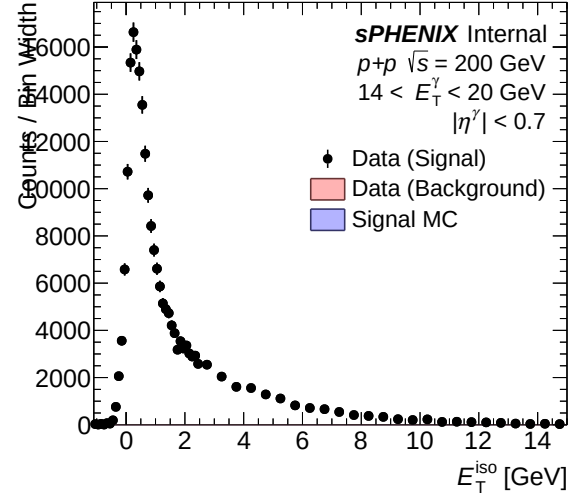


(a) IsoET fit template (pT bins 0-2) — bdt_nom

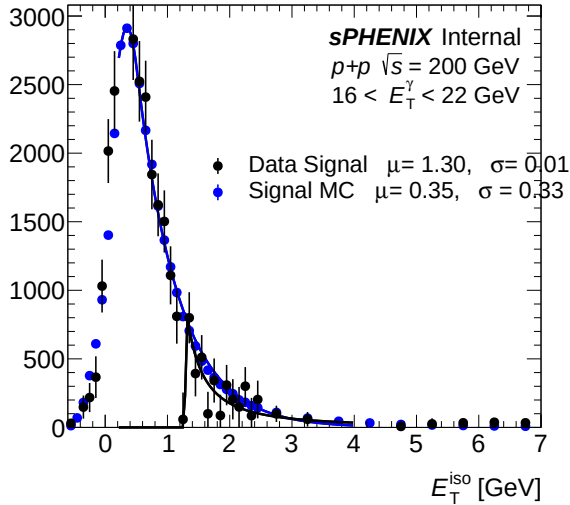
(b) IsoET fit template (pT bins 0-2) — bdt_tightbdt50



(a) IsoET template (pT bins 3-5) — bdt_nom

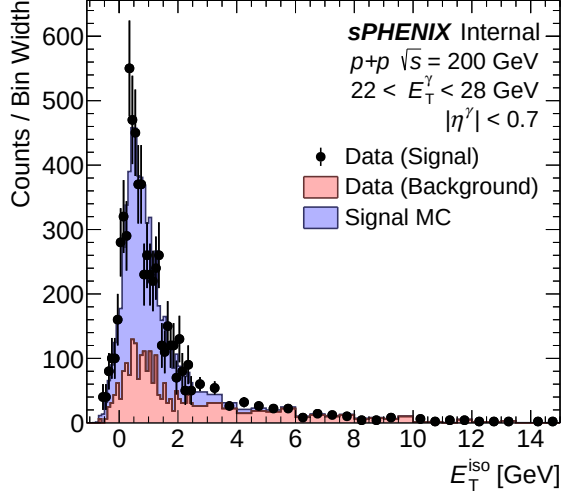


(b) IsoET template (pT bins 3-5) — bdt_tightbdt50

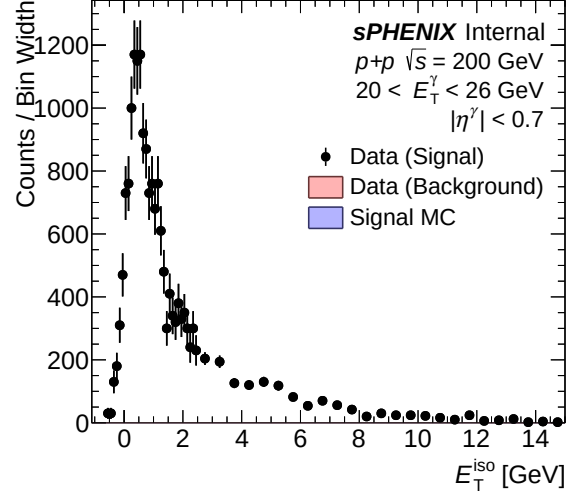


(a) IsoET fit template (pT bins 3-5) — bdt_nom

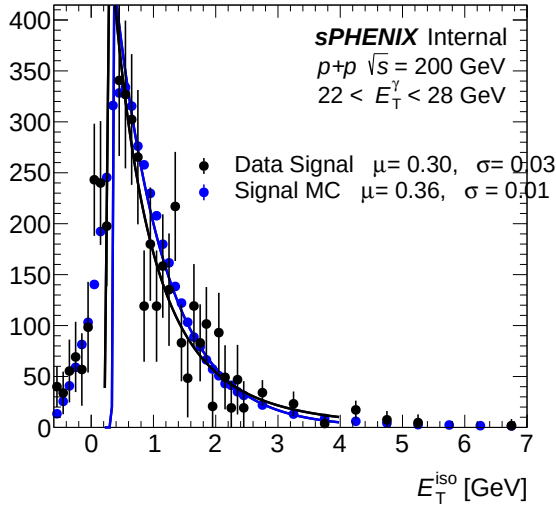
(b) IsoET fit template (pT bins 3-5) — bdt_tightbdt50



(a) IsoET template (pT bins 6–8) — bdt_nom

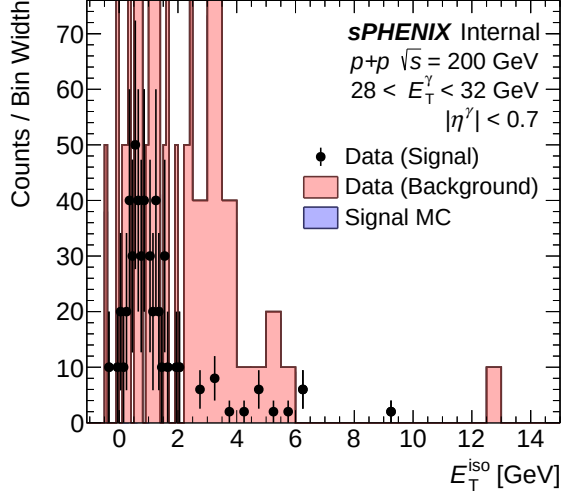
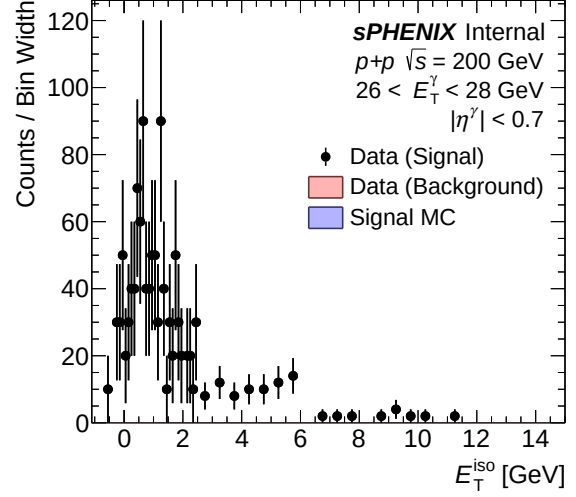
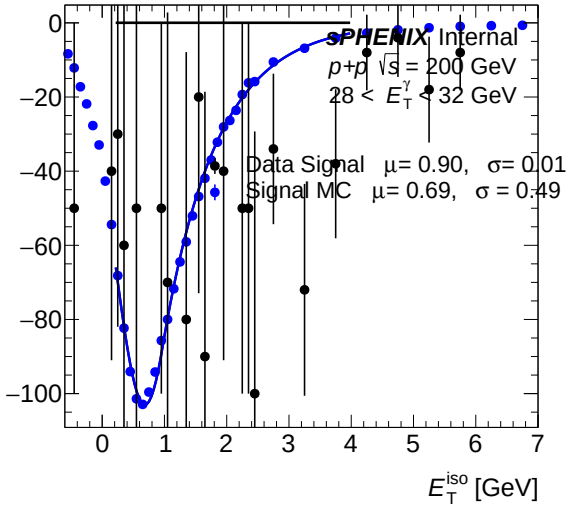


(b) IsoET template (pT bins 6–8) — bdt_tightbdt50

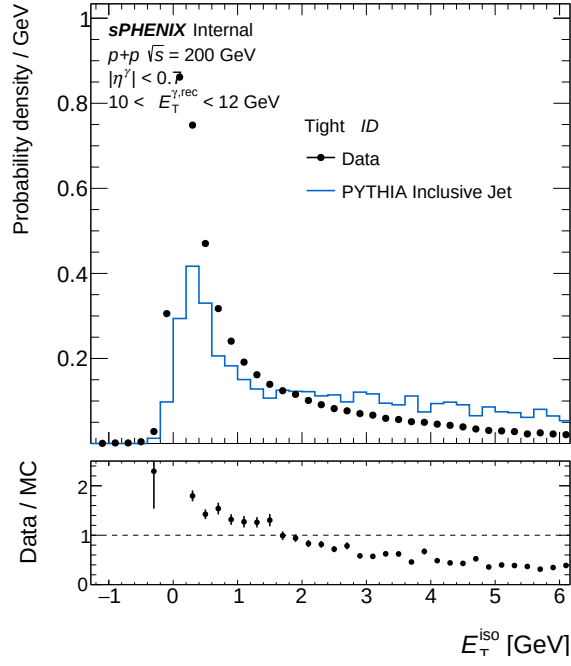


(a) IsoET fit template (pT bins 6–8) — bdt_nom

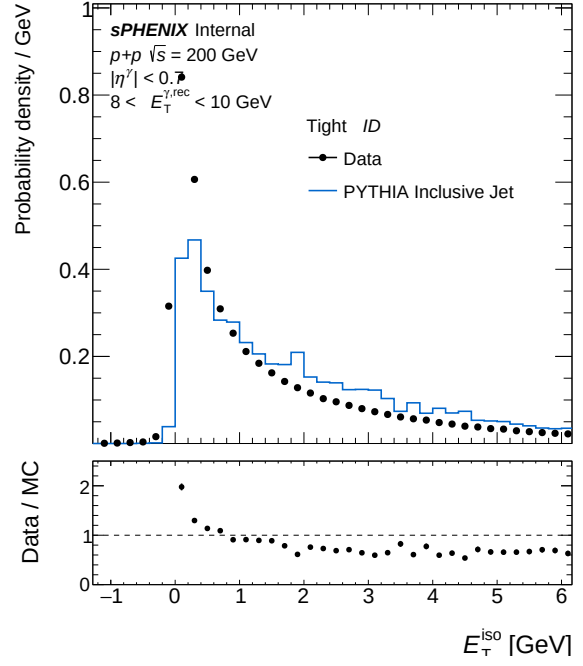
(b) IsoET fit template (pT bins 6–8) — bdt_tightbdt50

(a) IsoET template (pT bins 9-9) — `bdt_nom`(b) IsoET template (pT bins 9-9) — `bdt_tightbdt50`(a) IsoET fit template (pT bins 9-9) — `bdt_nom`(b) IsoET fit template (pT bins 9-9) — `bdt_tightbdt50`

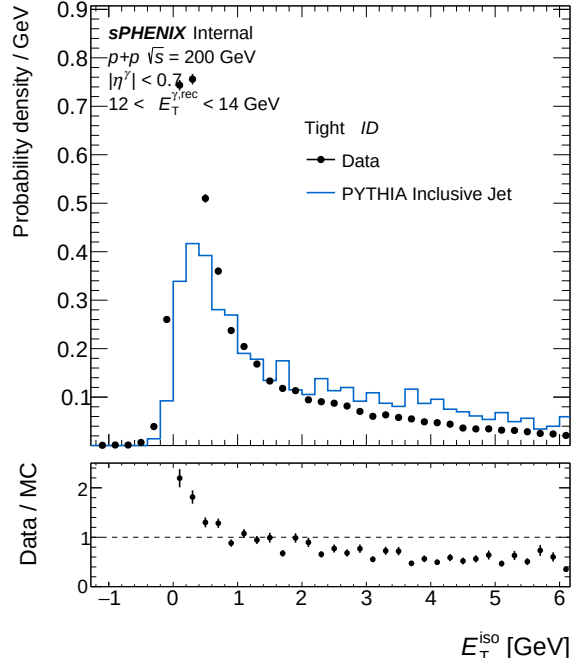
1.7 Isolation ET



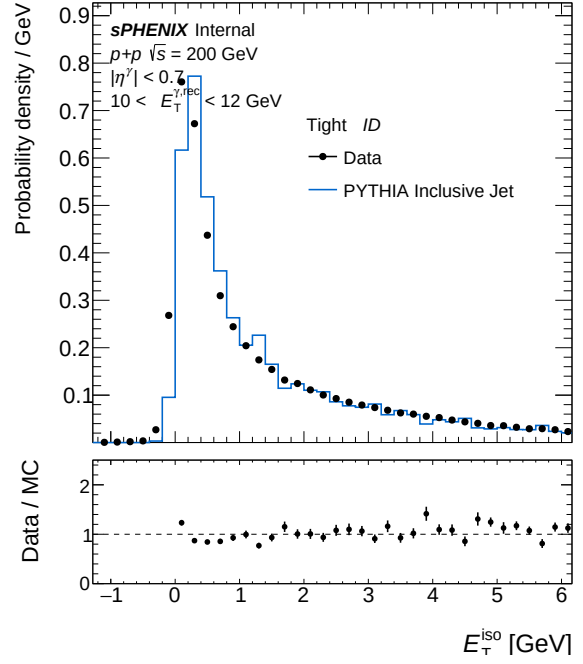
(a) IsoET Tight, pT bin 0 — bdt_nom



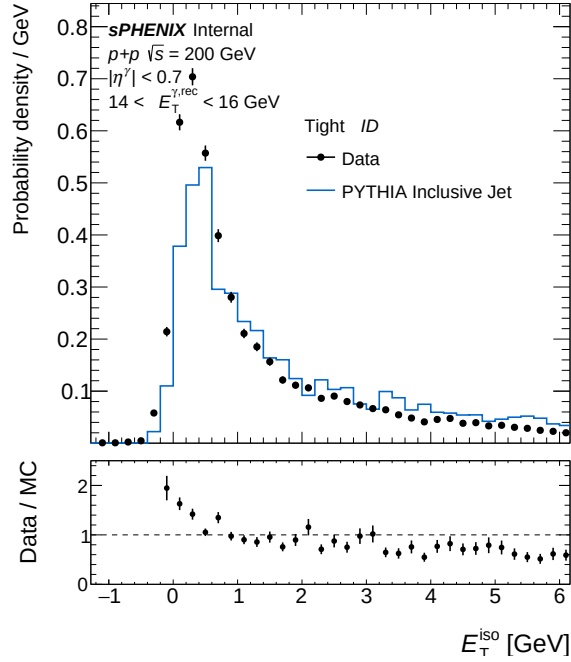
(b) IsoET Tight, pT bin 0 — bdt_tightbdt50



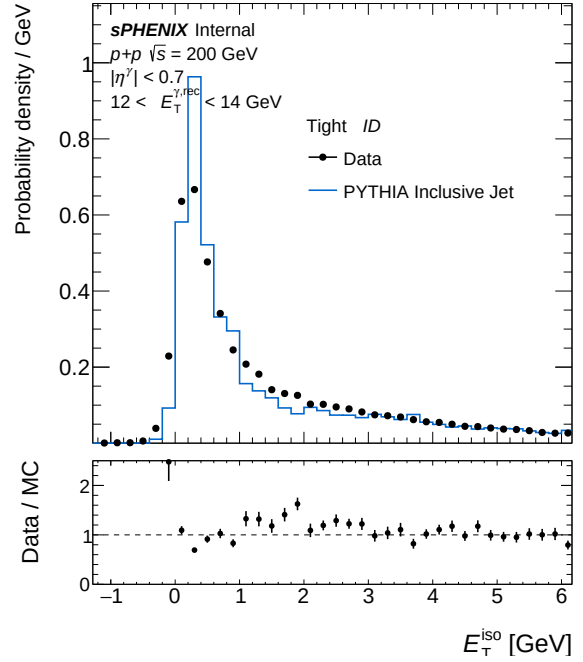
(a) IsoET Tight, pT bin 1 — bdt_nom



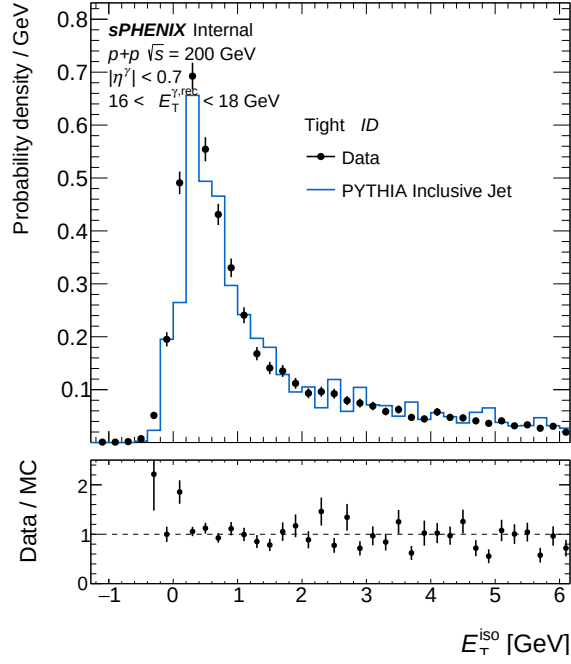
(b) IsoET Tight, pT bin 1 — bdt_tightbdt50



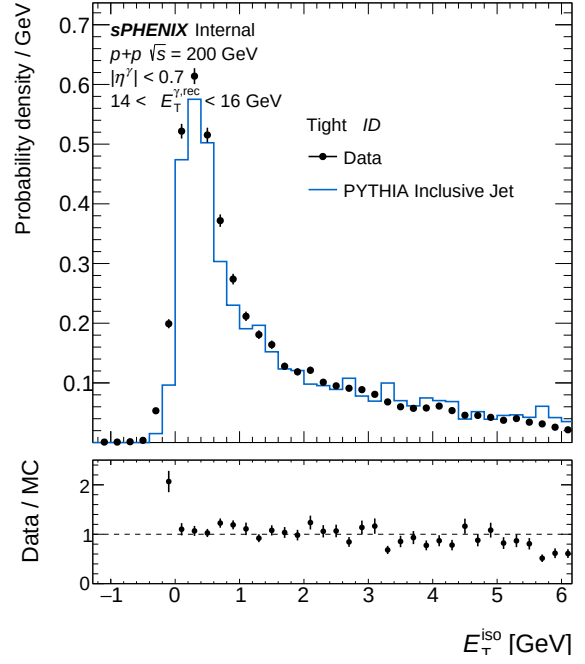
(a) IsoET Tight, pT bin 2 — bdt_nom



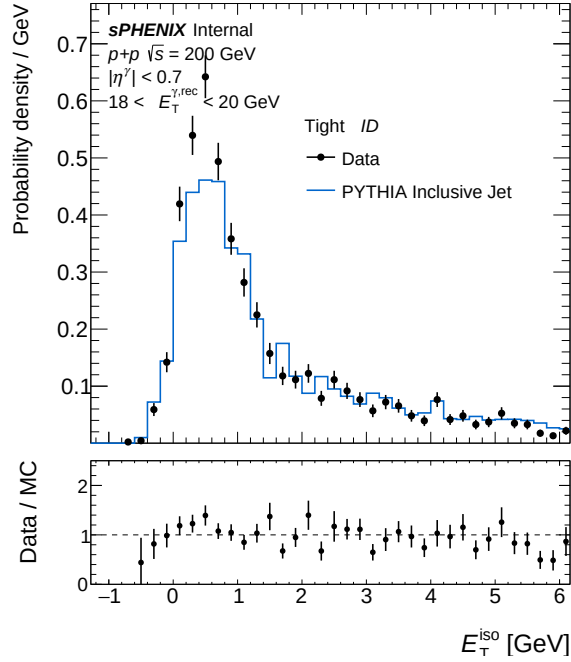
(b) IsoET Tight, pT bin 2 — bdt_tightbdt50



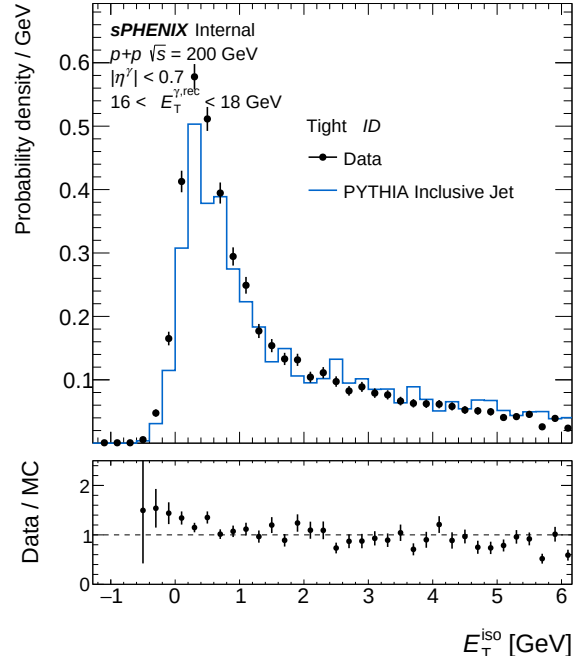
(a) IsoET Tight, pT bin 3 — bdt_nom



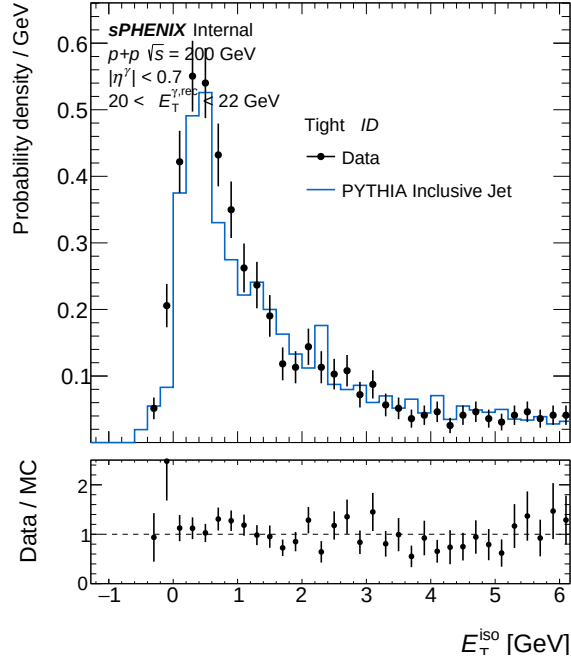
(b) IsoET Tight, pT bin 3 — bdt_tightbdt50



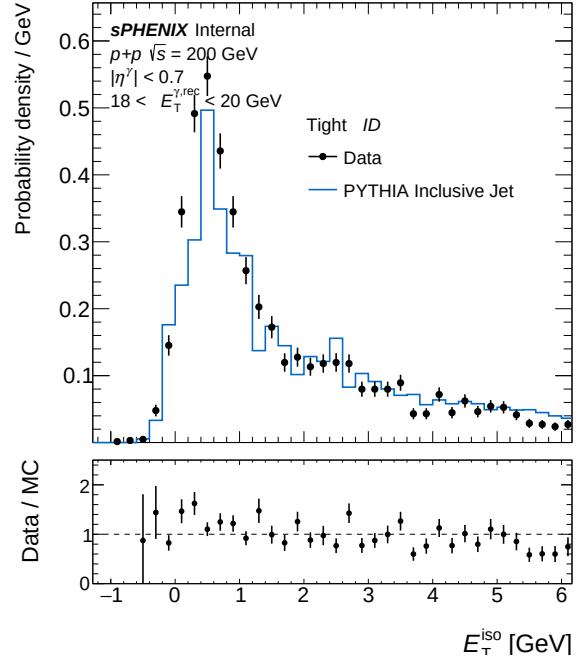
(a) IsoET Tight, pT bin 4 — bdt_nom



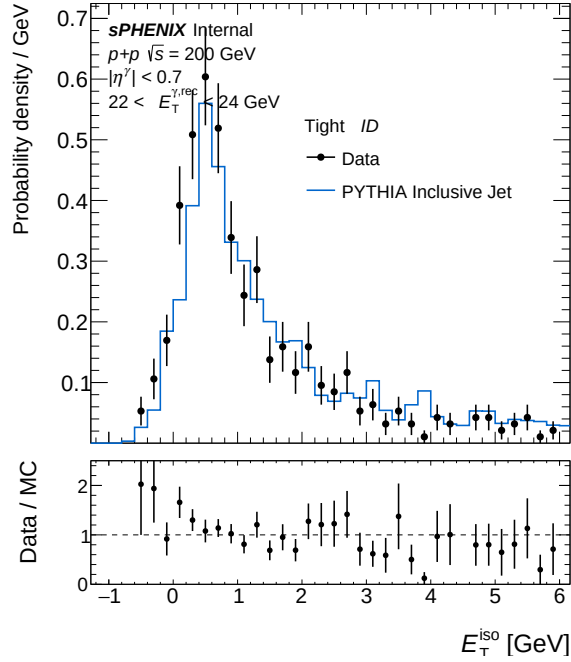
(b) IsoET Tight, pT bin 4 — bdt_tightbdt50



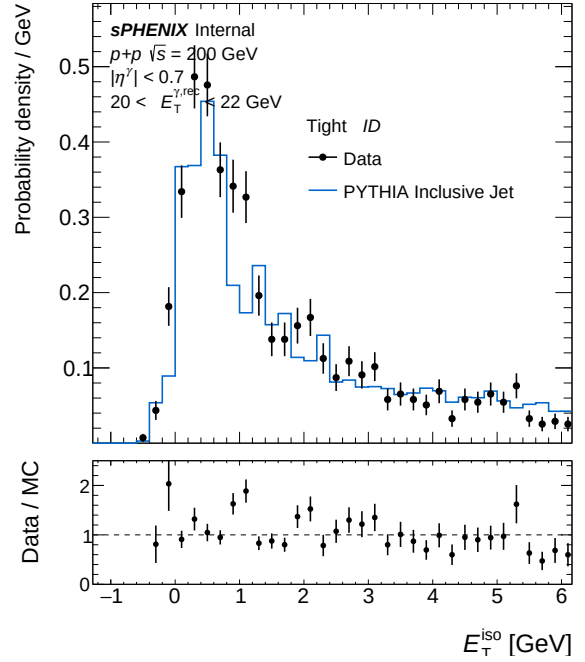
(a) IsoET Tight, pT bin 5 — bdt_nom



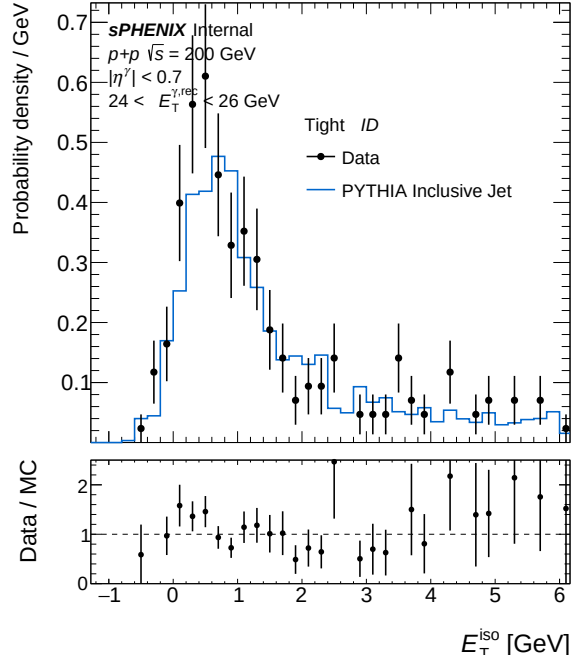
(b) IsoET Tight, pT bin 5 — bdt_tightbdt50



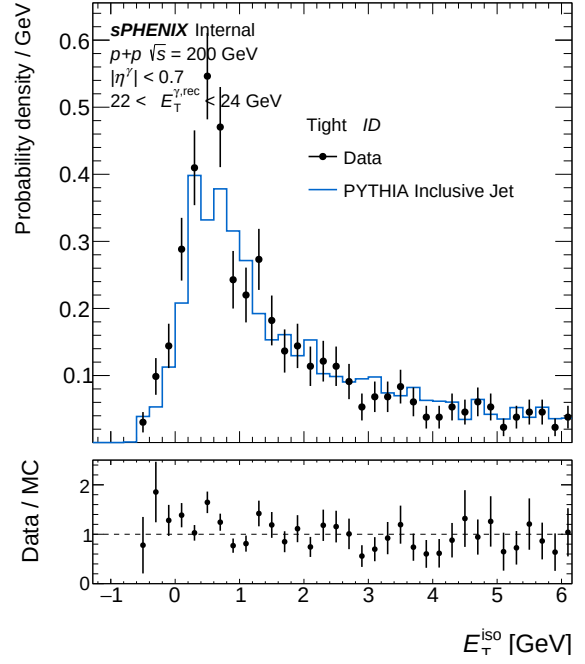
(a) IsoET Tight, pT bin 6 — bdt_nom



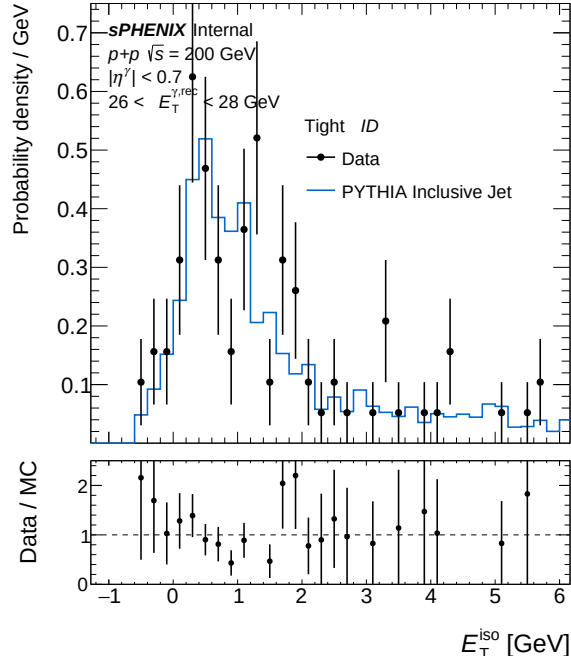
(b) IsoET Tight, pT bin 6 — bdt_tightbdt50



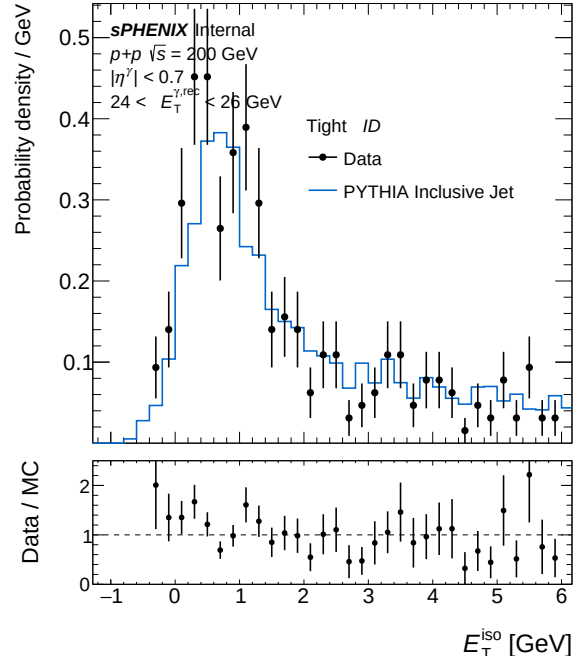
(a) IsoET Tight, pT bin 7 — bdt_nom



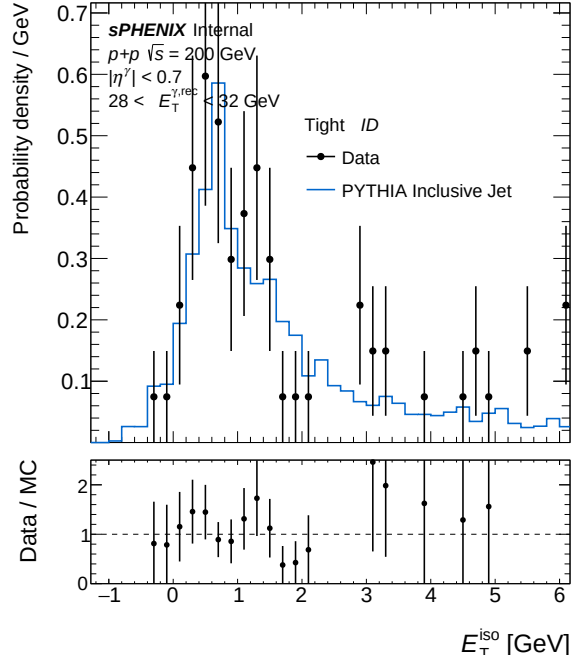
(b) IsoET Tight, pT bin 7 — bdt_tightbdt50



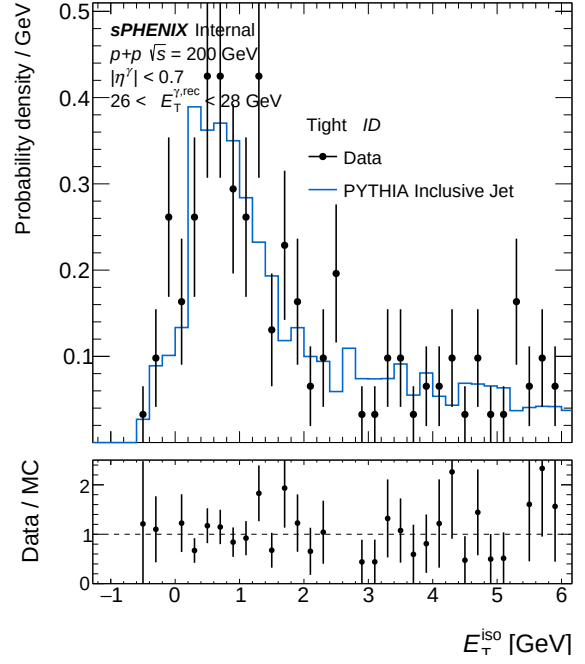
(a) IsoET Tight, pT bin 8 — bdt_nom



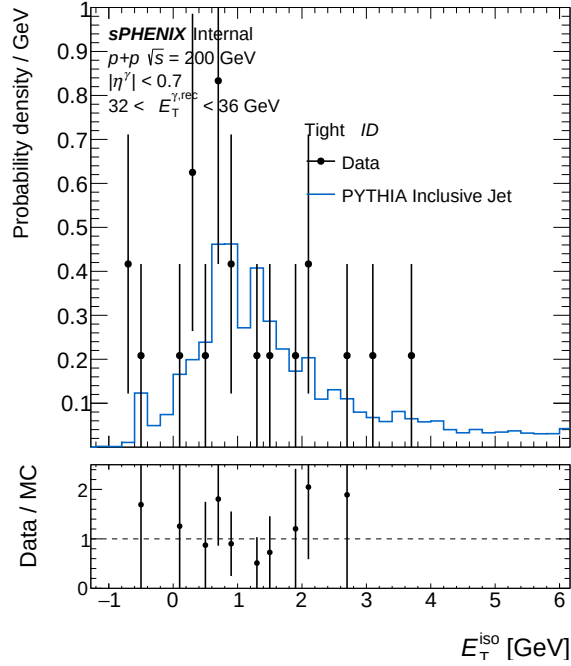
(b) IsoET Tight, pT bin 8 — bdt_tightbdt50



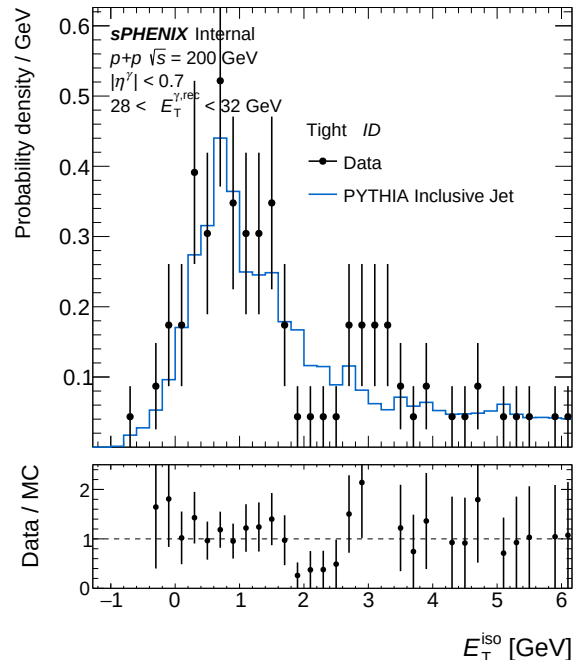
(a) IsoET Tight, pT bin 9 — bdt_nom



(b) IsoET Tight, pT bin 9 — bdt_tightbdt50



(a) IsoET Tight, pT bin 10 — bdt_nom



(b) IsoET Tight, pT bin 10 — bdt_tightbdt50

2 bdt_nom vs bdt_npb03

Config diff (A → B)

```

--- config_bdt_nom.yaml
+++ config_bdt_npb03.yaml
@@ -20,5 +20,5 @@
     response_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/MC_response"
     data_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/data_histo"
-   var_type: "bdt_nom"
+   var_type: "bdt_npb03"
     final_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/Photon.final"

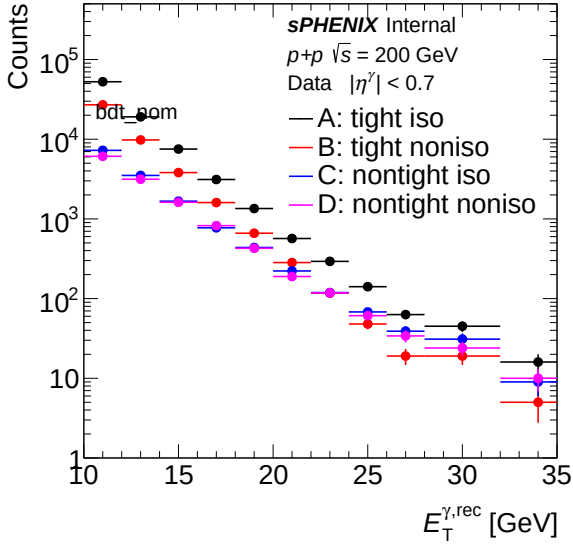
@@ -239,5 +239,5 @@

     npb_cut_on: 1
-   npb_score_cut: 0.5
+   npb_score_cut: 0.3

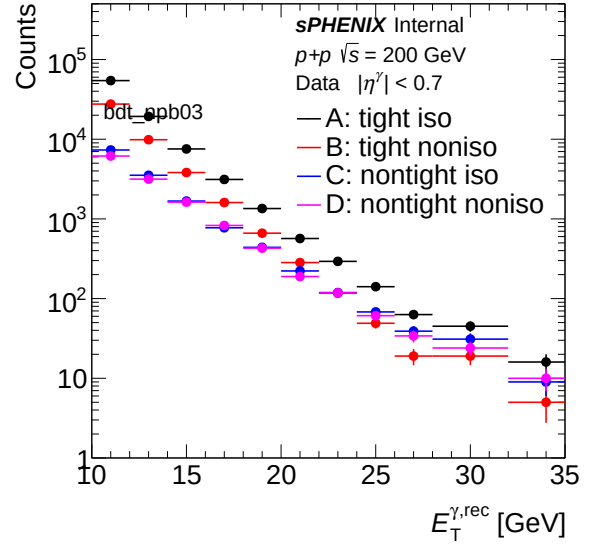
     prob_max: 1.0

```

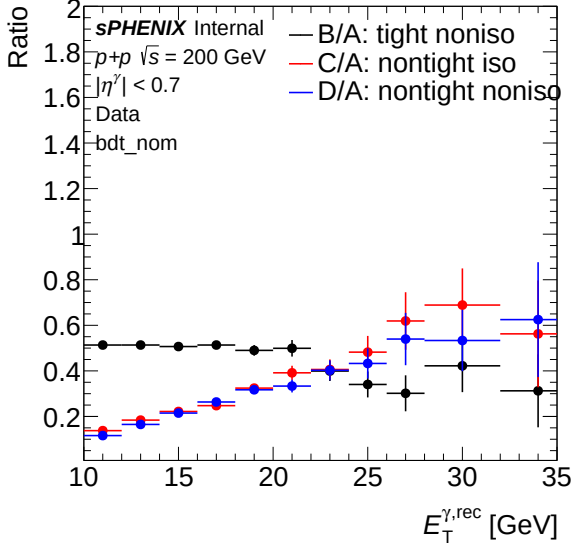
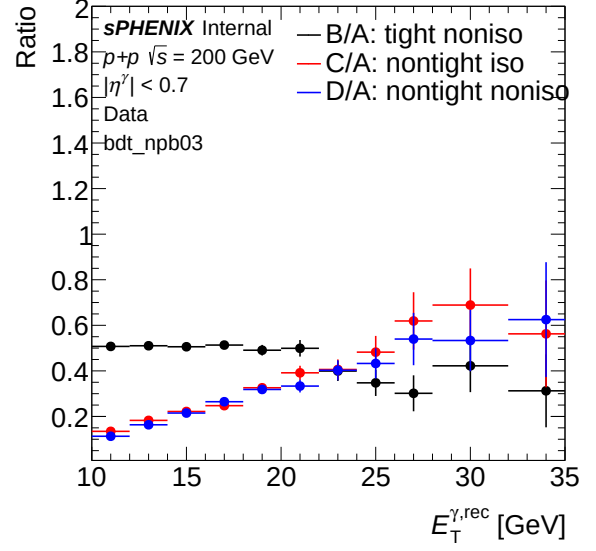
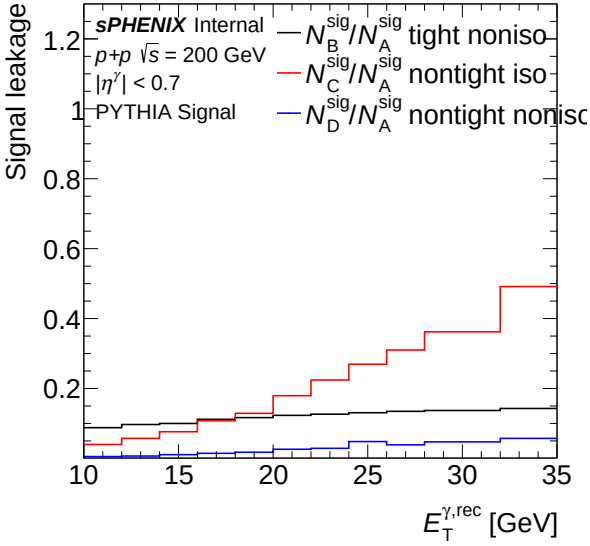
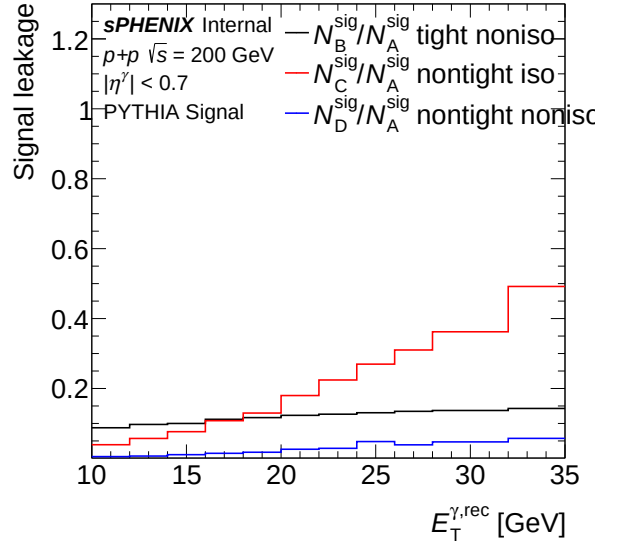
2.1 Sideband (Data)



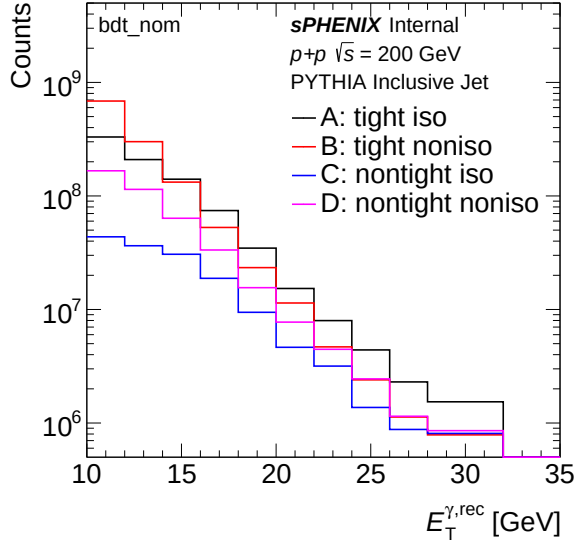
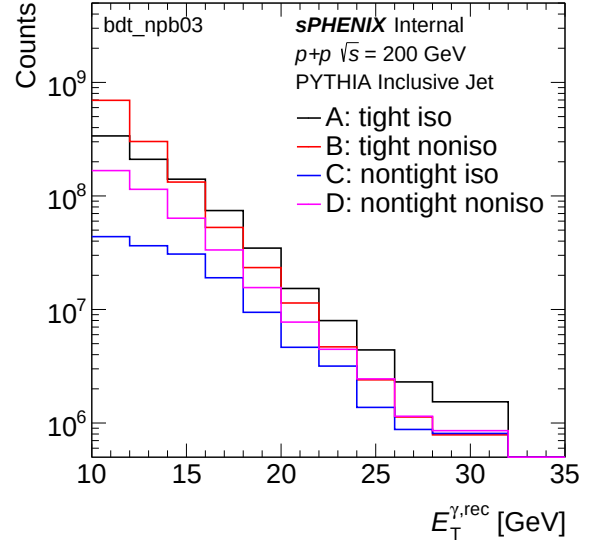
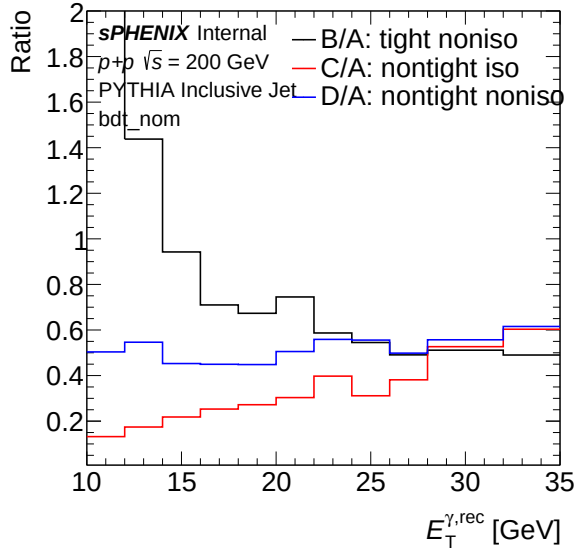
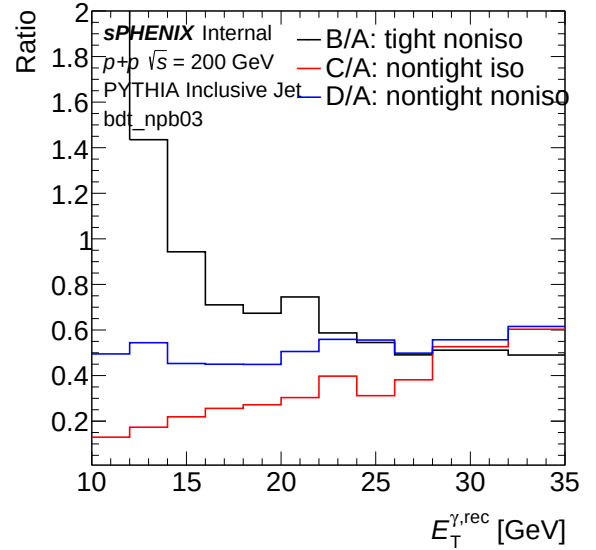
(a) Sideband ET spectrum (data) — bdt_nom

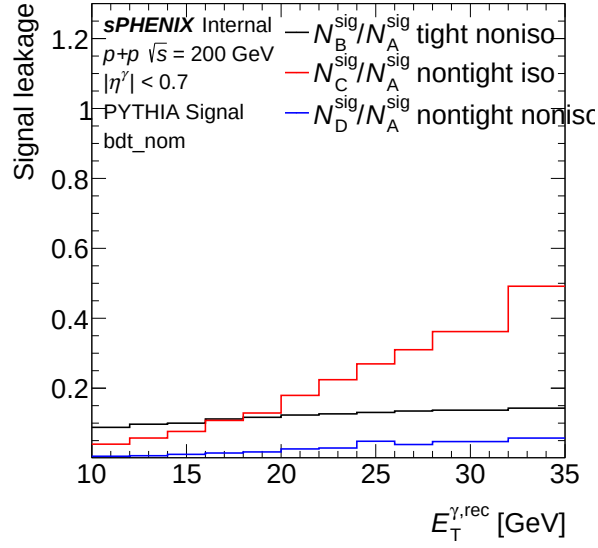


(b) Sideband ET spectrum (data) — bdt_npb03

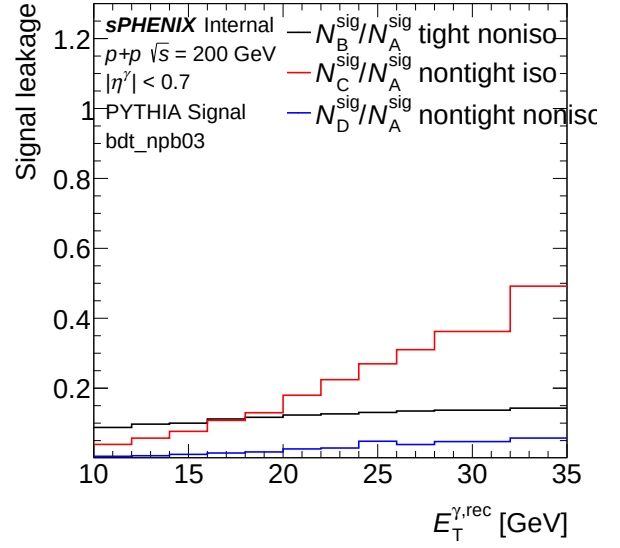
(a) Sideband ET ratio (data) — bdt_nom (b) Sideband ET ratio (data) — bdt_npb03 (a) Leakage fraction vs ET (data) — bdt_nom (b) Leakage fraction vs ET (data) — bdt_npb03

2.2 Sideband (MC)

(a) Sideband ET spectrum (MC) — `bdt_nom`(b) Sideband ET spectrum (MC) — `bdt_npb03`(a) Sideband ET ratio (MC) — `bdt_nom`(b) Sideband ET ratio (MC) — `bdt_npb03`

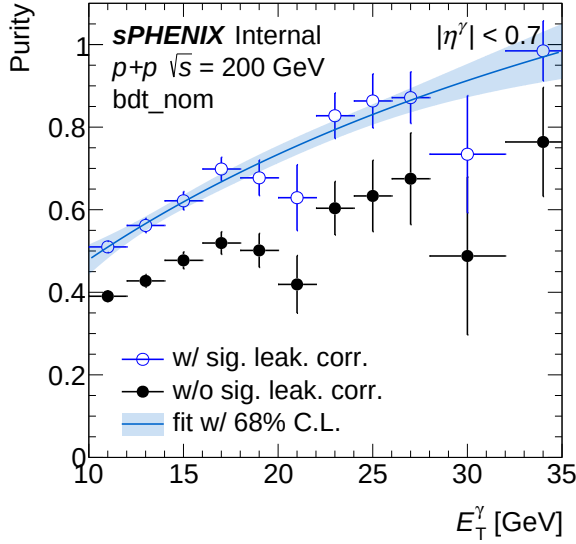


(a) Leakage fraction vs ET (MC) — bdt_nom

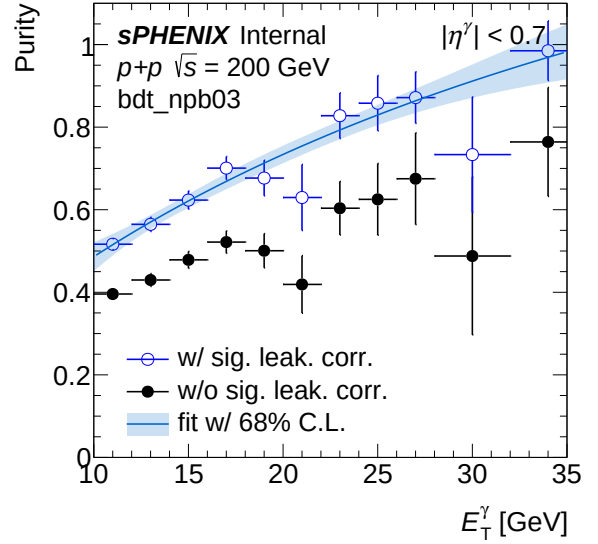


(b) Leakage fraction vs ET (MC) — bdt_npb03

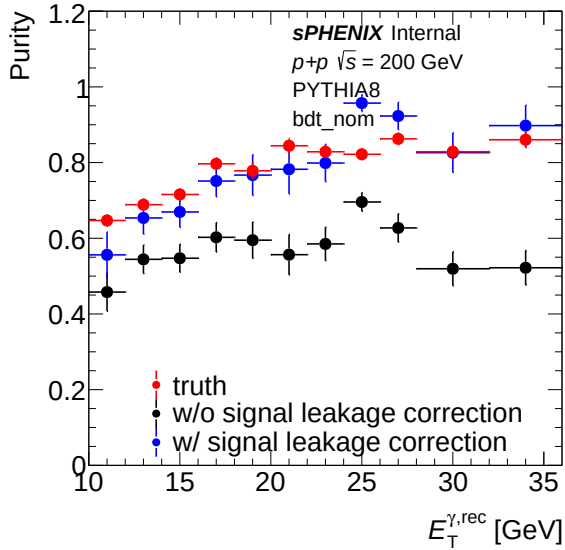
2.3 Purity



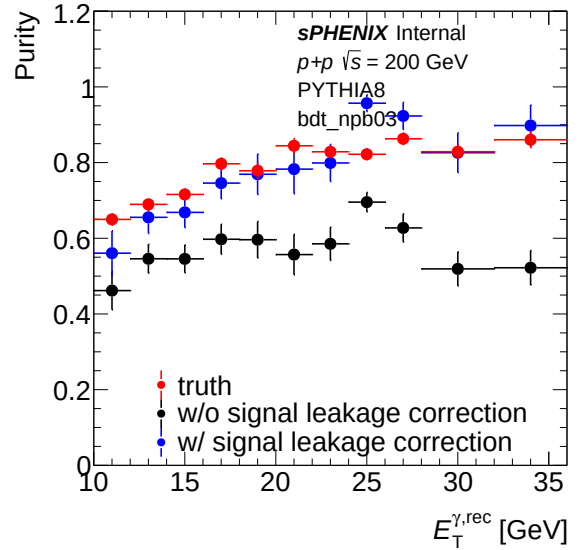
(a) Purity (data) — bdt_nom



(b) Purity (data) — bdt_npb03

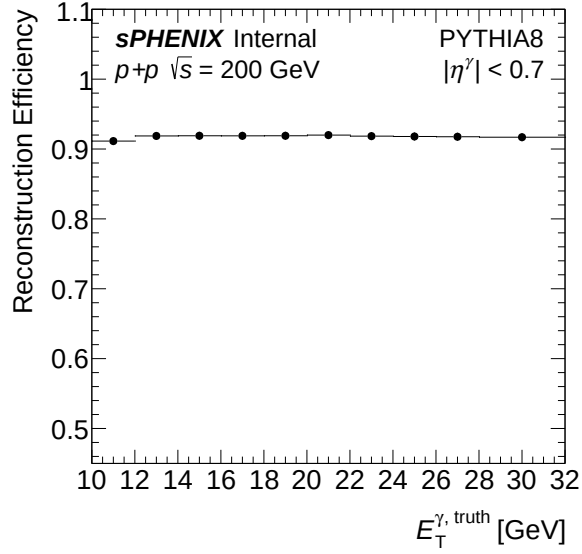


(a) Purity (MC) — bdt_nom

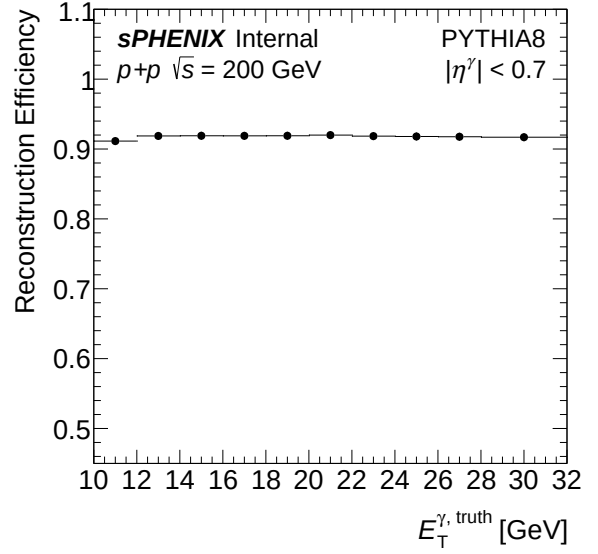


(b) Purity (MC) — bdt_npb03

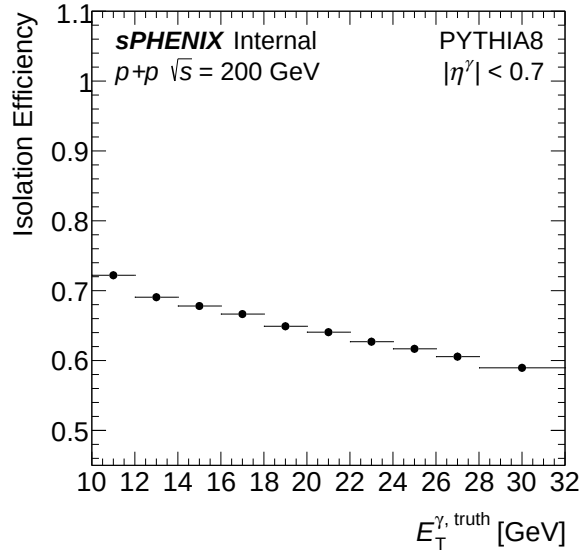
2.4 Efficiency



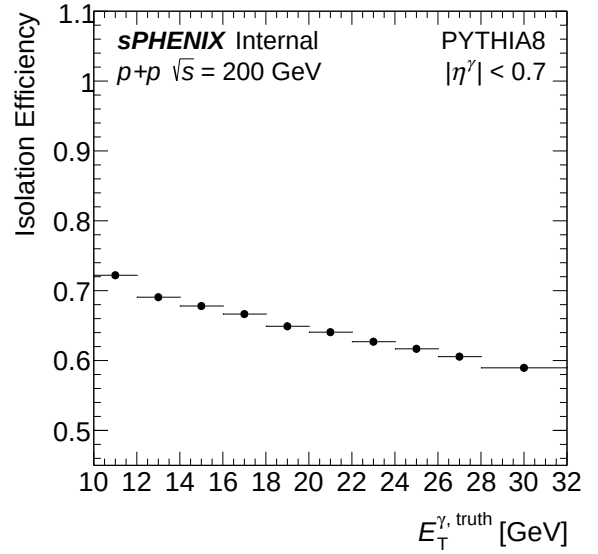
(a) Reconstruction efficiency — bdt_nom



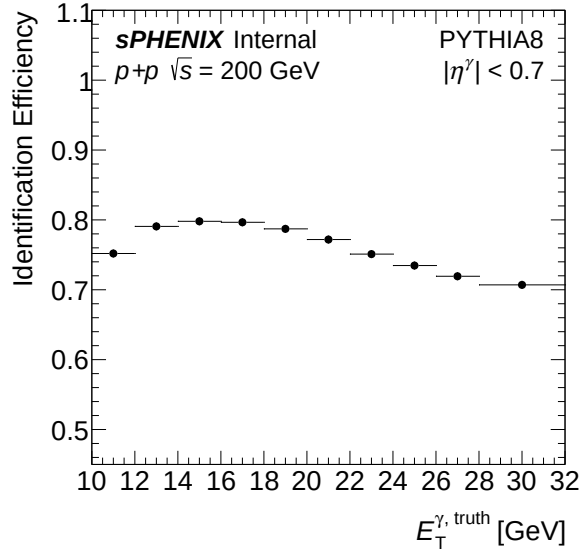
(b) Reconstruction efficiency — bdt_npb03



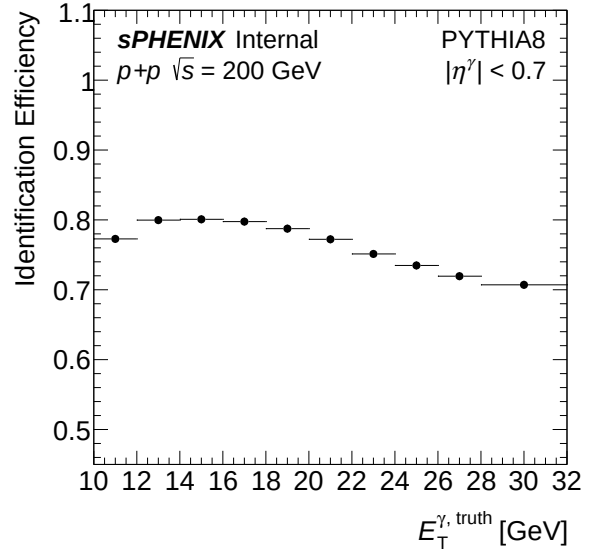
(a) Isolation efficiency — bdt_nom



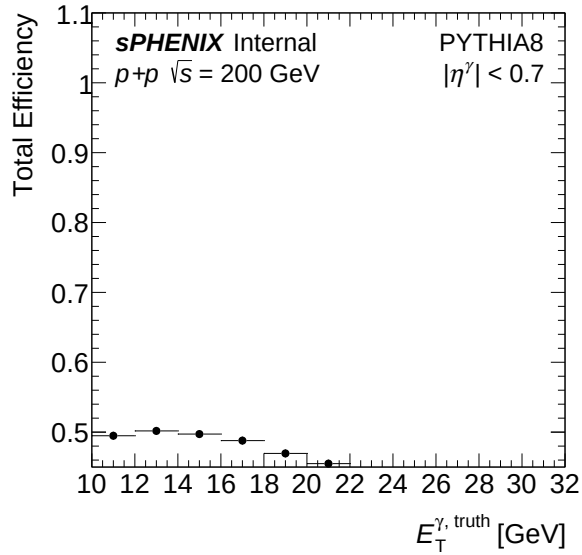
(b) Isolation efficiency — bdt_npb03



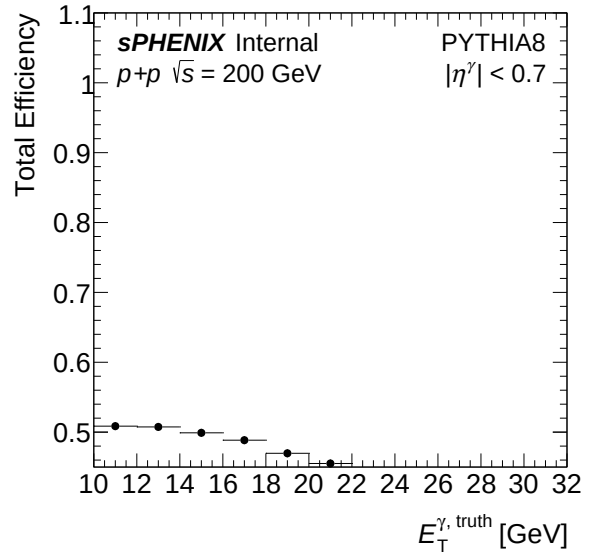
(a) Identification efficiency — bdt_nom



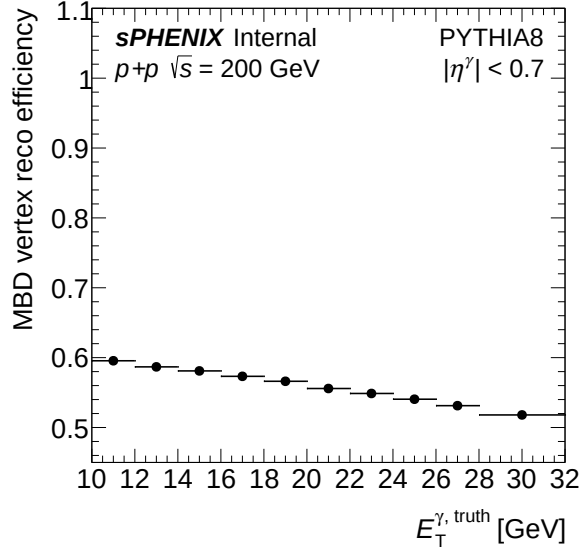
(b) Identification efficiency — bdt_npb03



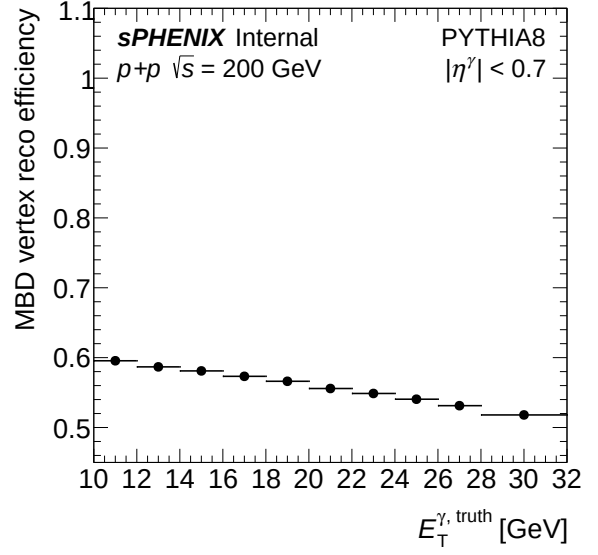
(a) Total efficiency — bdt_nom



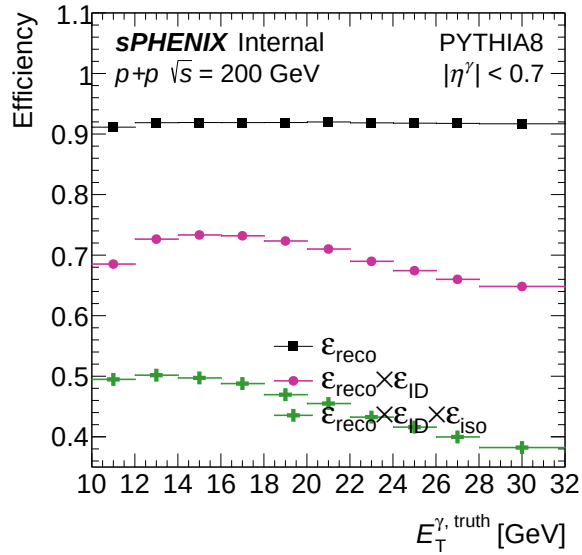
(b) Total efficiency — bdt_npb03



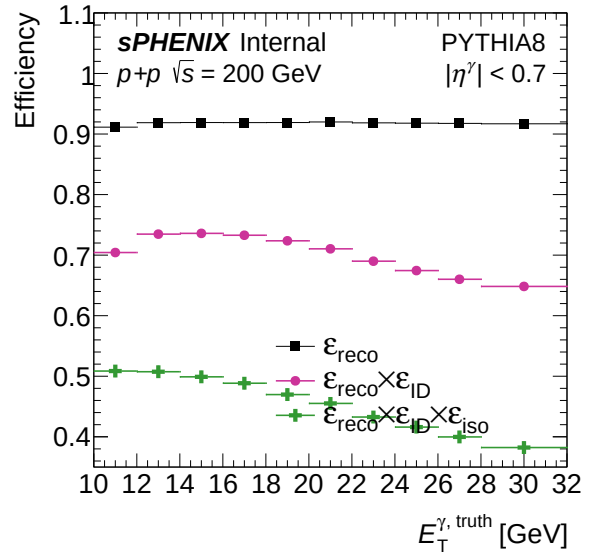
(a) MBD trigger efficiency — bdt_nom



(b) MBD trigger efficiency — bdt_npb03

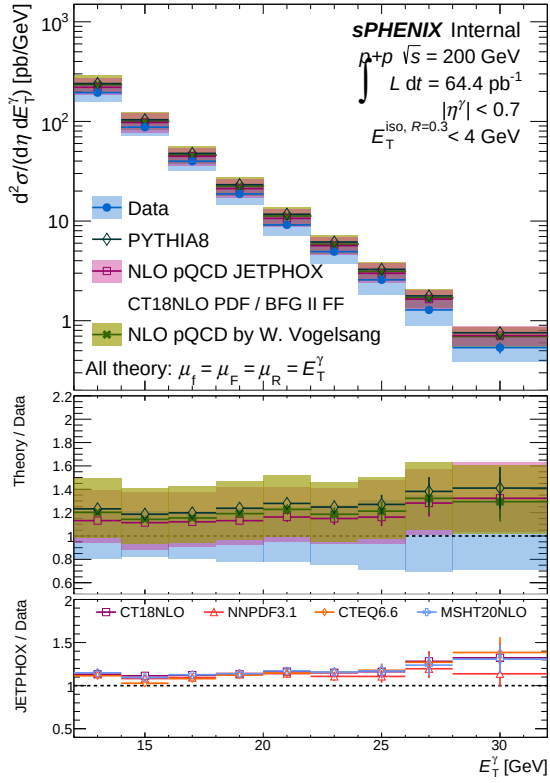
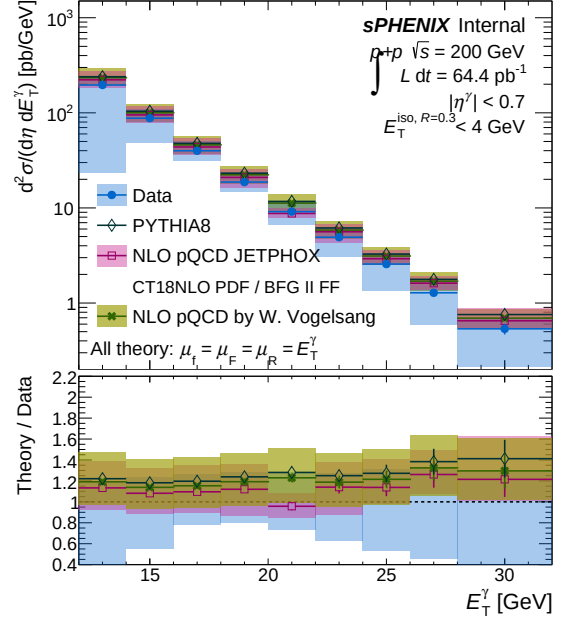


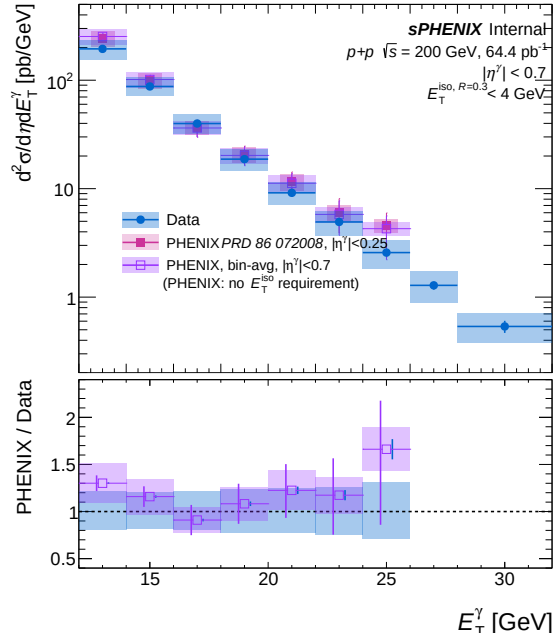
(a) Combined photon efficiency — bdt_nom



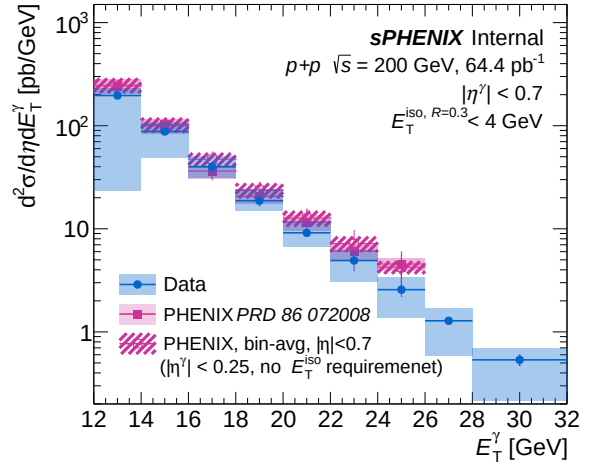
(b) Combined photon efficiency — bdt_npb03

2.5 Final result

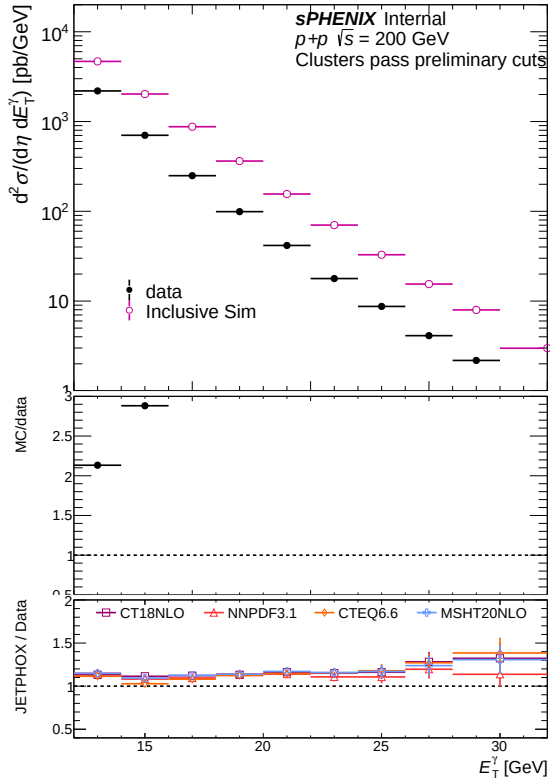
(a) Final cross section result — **bd_t_nom**(b) Final cross section result — **bd_t_npb03**



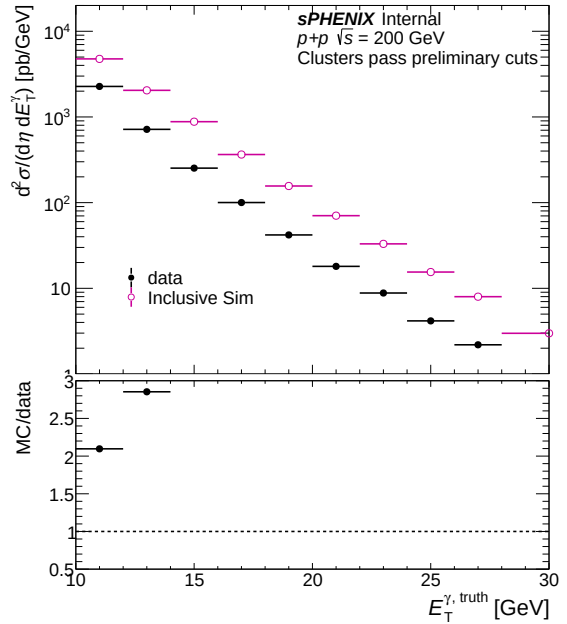
(a) Comparison with PHENIX — bdt_nom



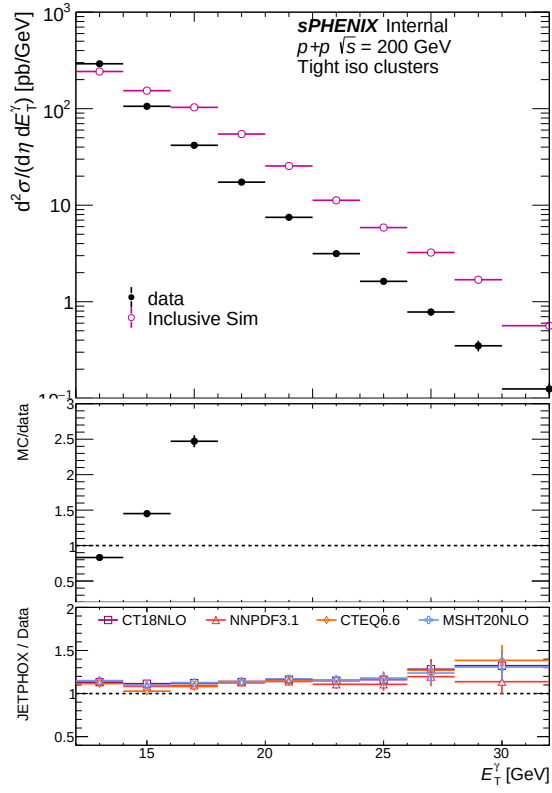
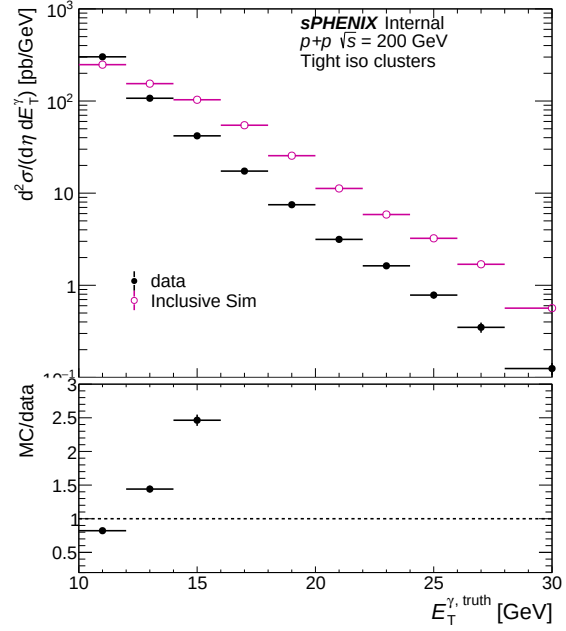
(b) Comparison with PHENIX — bdt_npb03

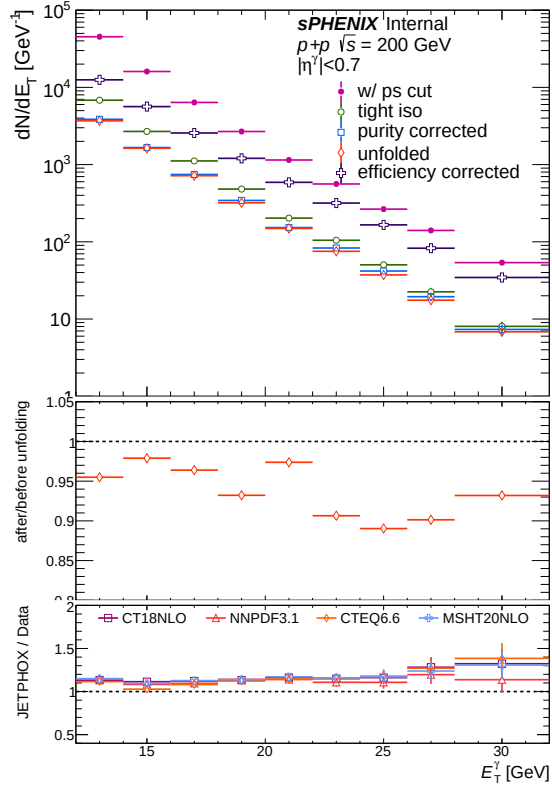
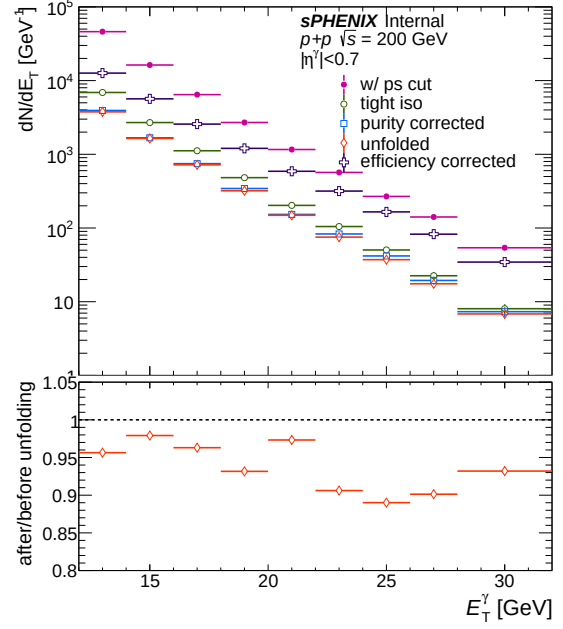


(a) Common cluster yield (data vs MC) — bdt_nom

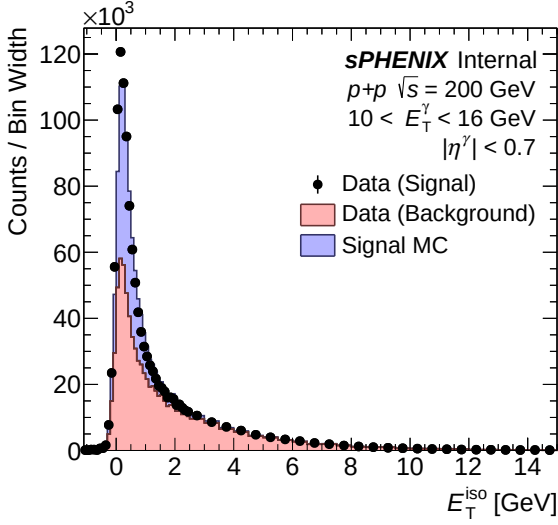


(b) Common cluster yield (data vs MC) — bdt_npb03

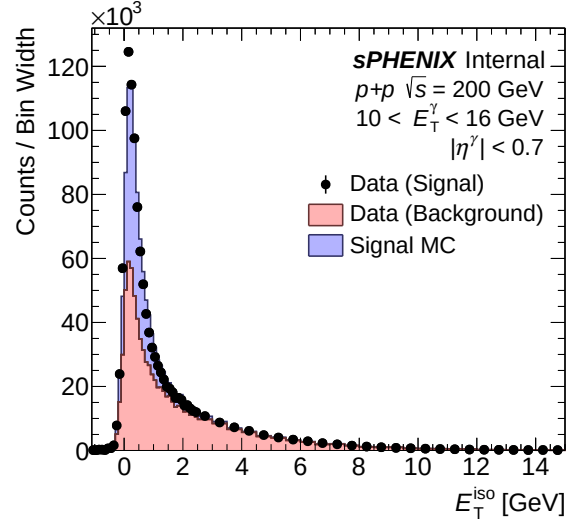
(a) Tight iso cluster yield (data vs MC) — **bdt_nom**(b) Tight iso cluster yield (data vs MC) — **bdt_npb03**

(a) All correction steps — **bdt_nom**(b) All correction steps — **bdt_npb03**

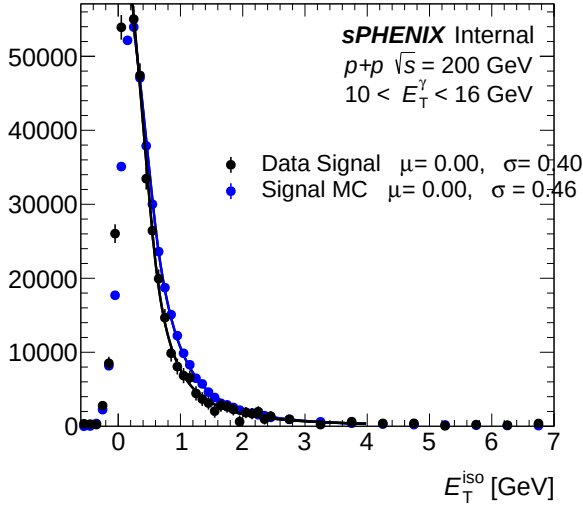
2.6 Isolation fit (CONF)



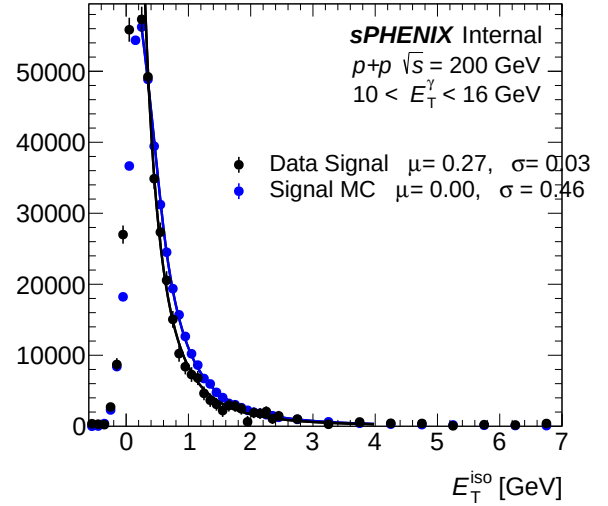
(a) IsoET template (pT bins 0-2) — bdt_nom



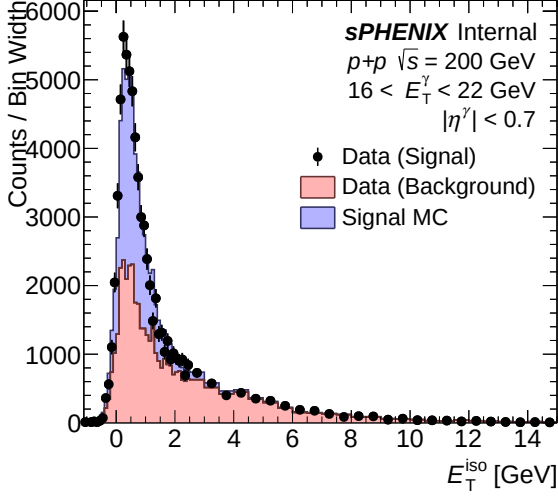
(b) IsoET template (pT bins 0-2) — bdt_npb03



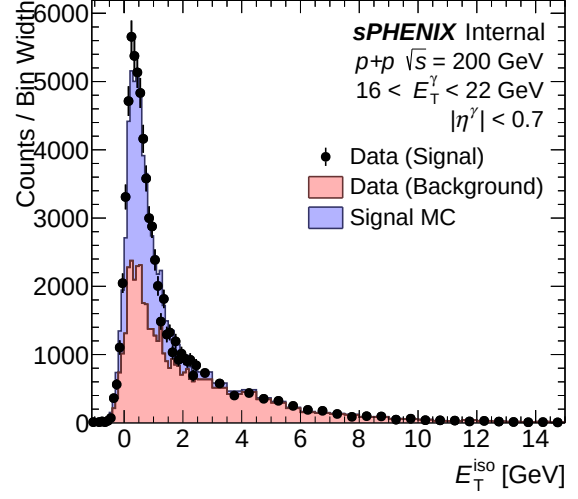
(a) IsoET fit template (pT bins 0-2) — bdt_nom



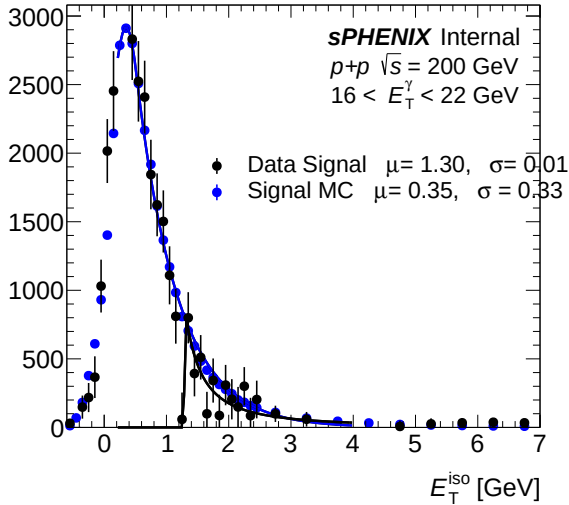
(b) IsoET fit template (pT bins 0-2) — bdt_npb03



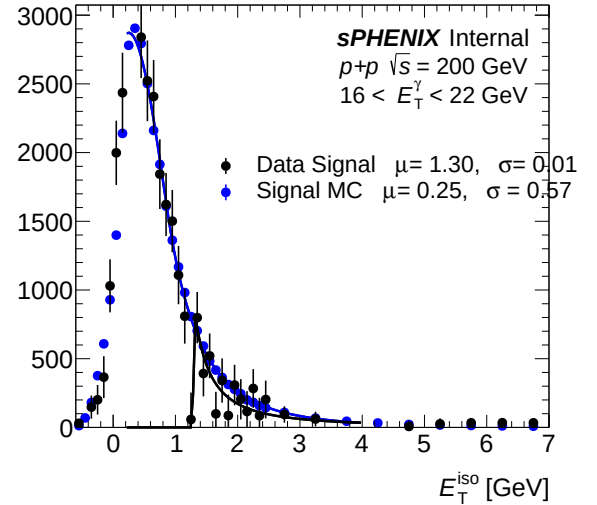
(a) IsoET template (pT bins 3-5) — bdt_nom



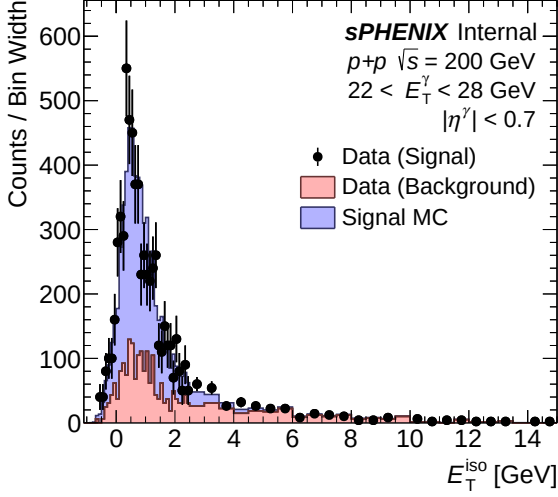
(b) IsoET template (pT bins 3-5) — bdt_npb03



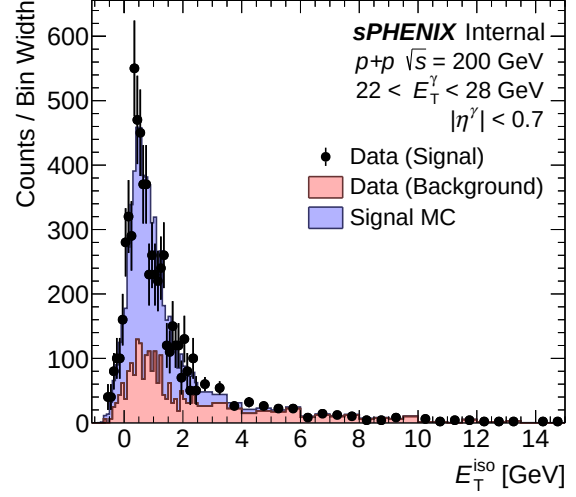
(a) IsoET fit template (pT bins 3-5) — bdt_nom



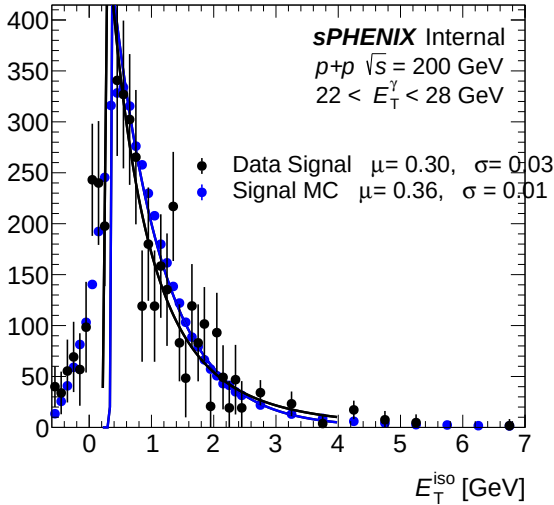
(b) IsoET fit template (pT bins 3-5) — bdt_npb03



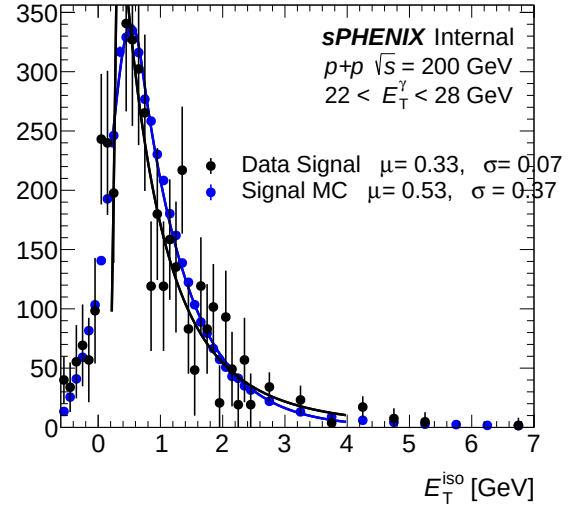
(a) IsoET template (pT bins 6–8) — bdt_nom



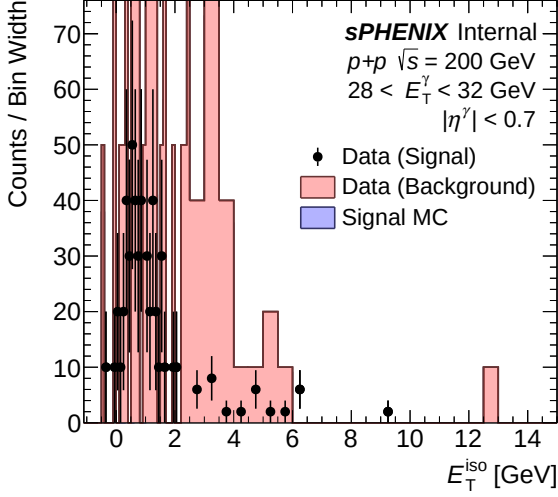
(b) IsoET template (pT bins 6–8) — bdt_npb03



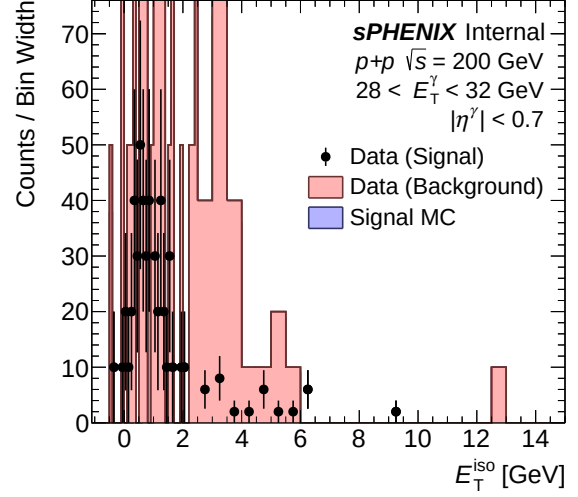
(a) IsoET fit template (pT bins 6–8) — bdt_nom



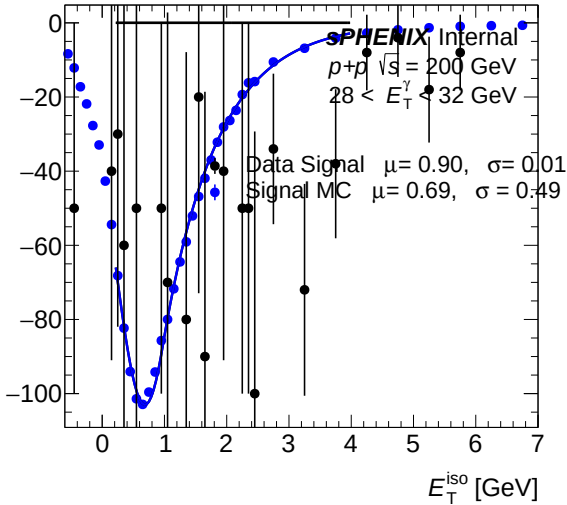
(b) IsoET fit template (pT bins 6–8) — bdt_npb03



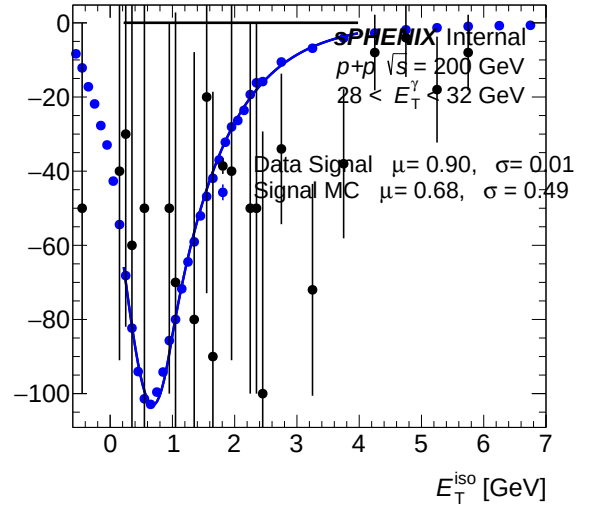
(a) IsoET template (pT bins 9–9) — bdt_nom



(b) IsoET template (pT bins 9–9) — bdt_npb03

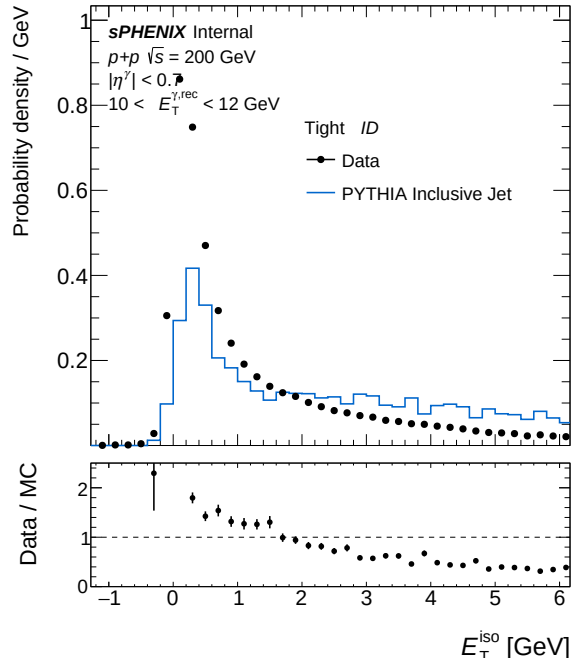
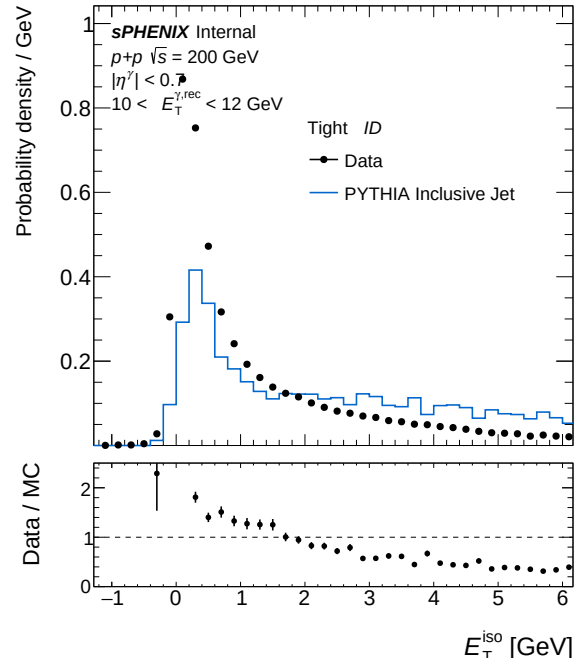


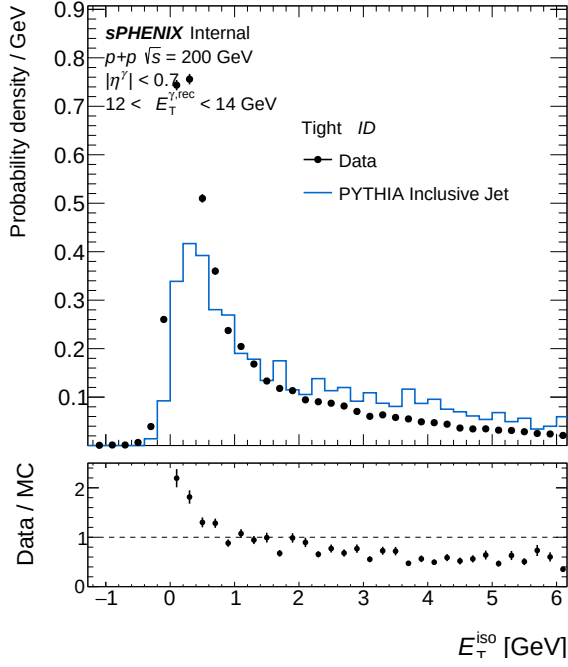
(a) IsoET fit template (pT bins 9–9) — bdt_nom



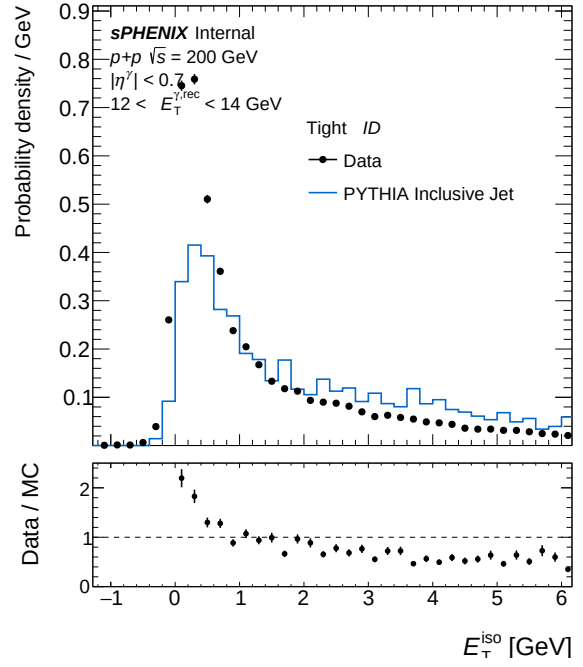
(b) IsoET fit template (pT bins 9–9) — bdt_npb03

2.7 Isolation ET

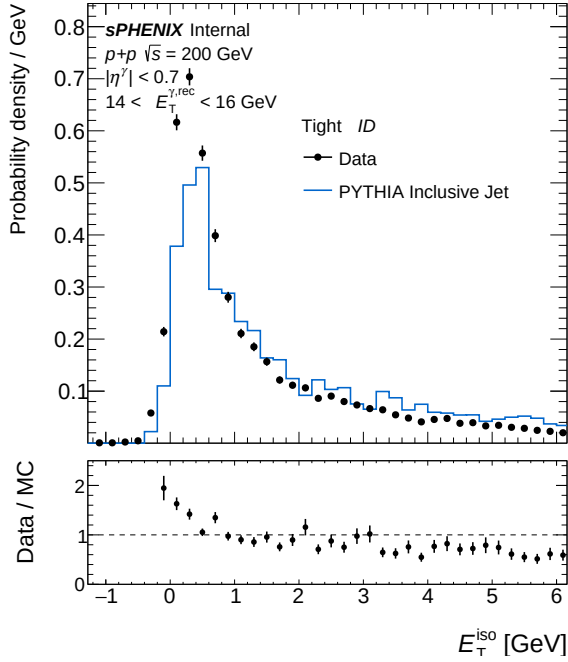
(a) IsoET Tight, pT bin 0 — `bdt_nom`(b) IsoET Tight, pT bin 0 — `bdt_npb03`



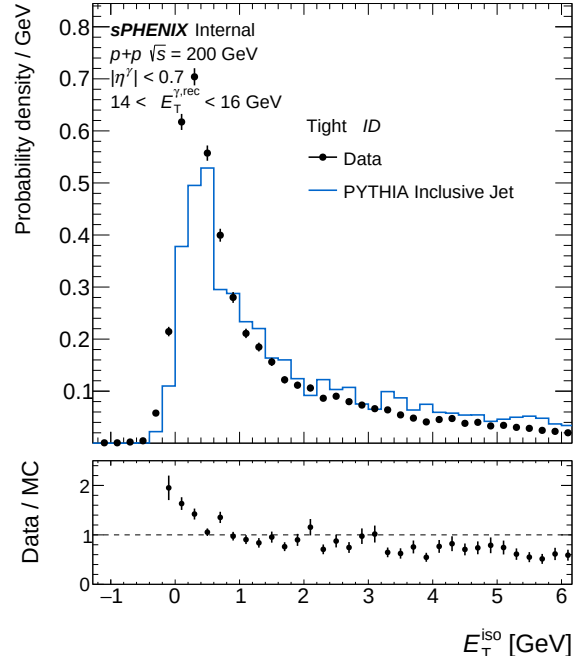
(a) IsoET Tight, pT bin 1 — bdt_nom



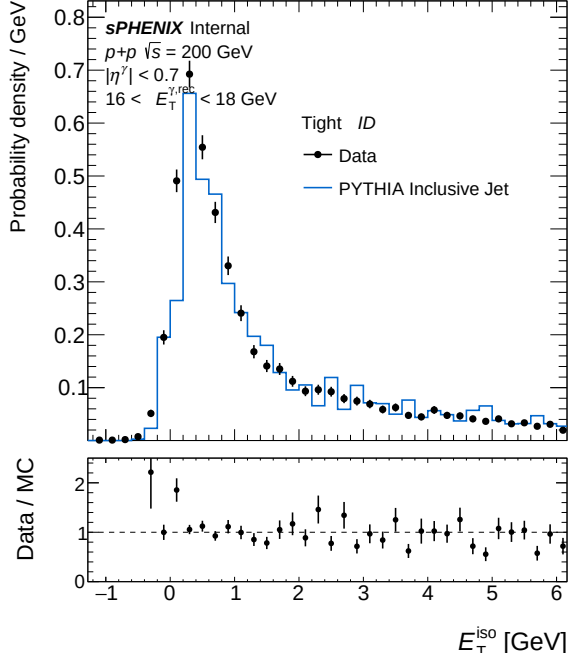
(b) IsoET Tight, pT bin 1 — bdt_npb03



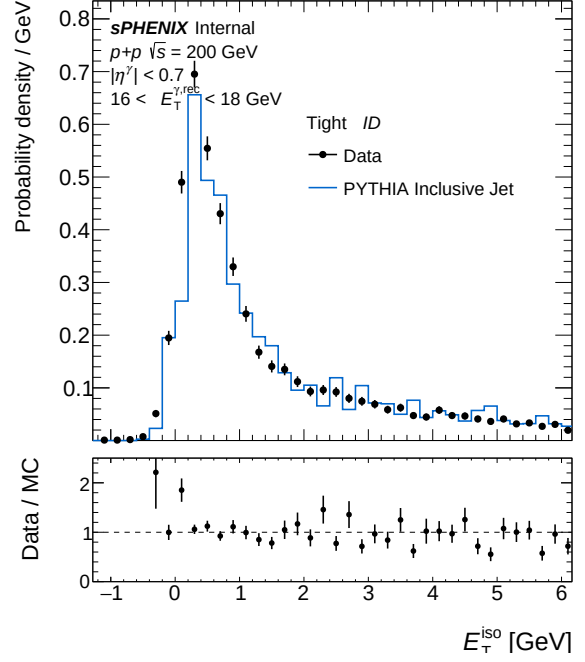
(a) IsoET Tight, pT bin 2 — bdt_nom



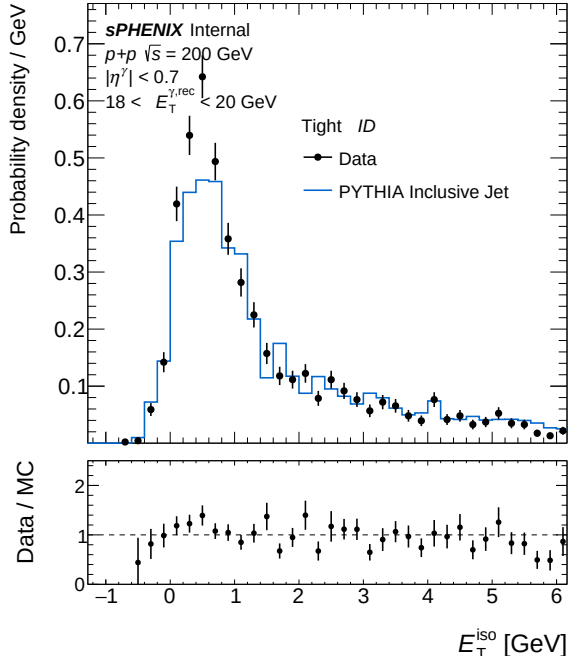
(b) IsoET Tight, pT bin 2 — bdt_npb03



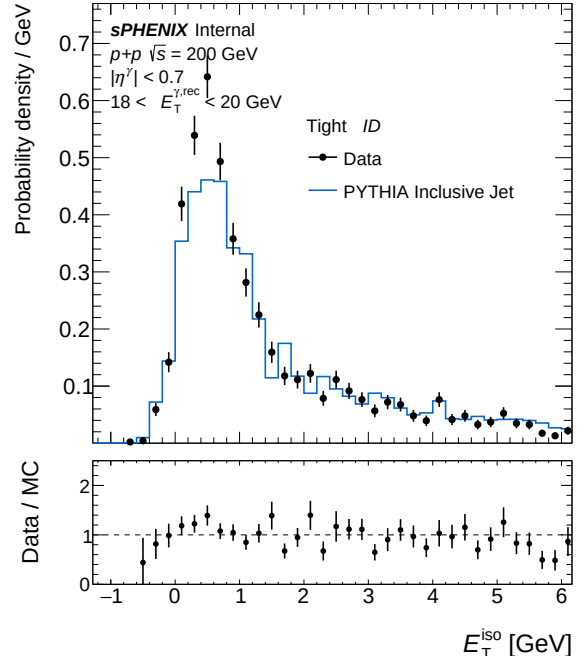
(a) IsoET Tight, pT bin 3 — bdt_nom



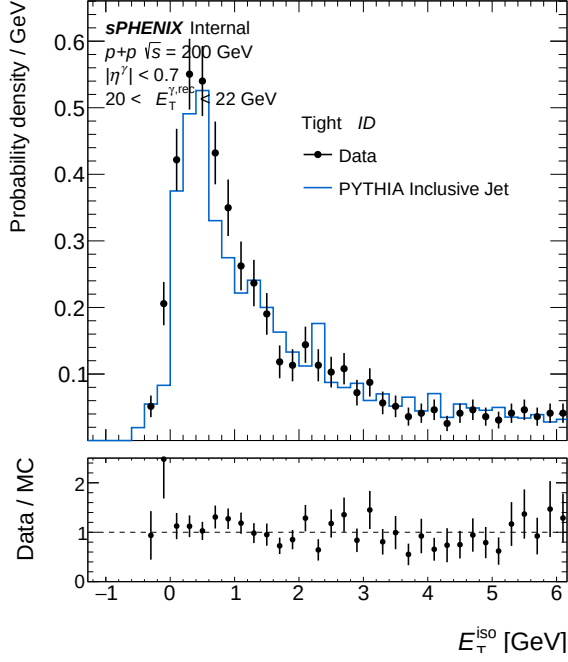
(b) IsoET Tight, pT bin 3 — bdt_npb03



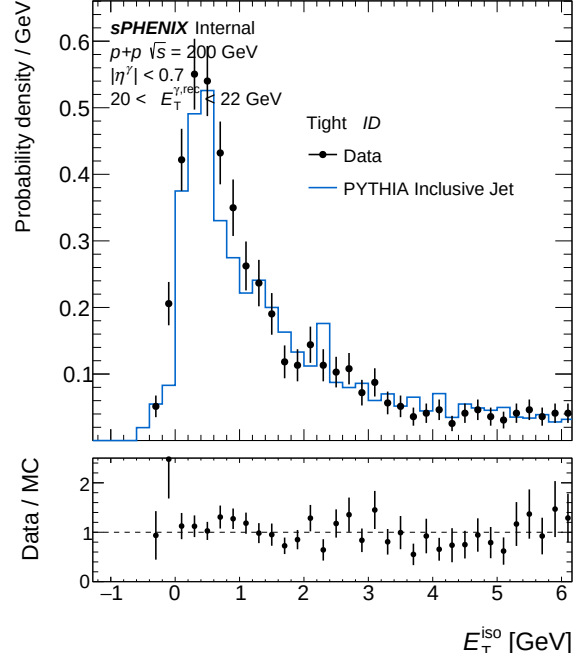
(a) IsoET Tight, pT bin 4 — bdt_nom



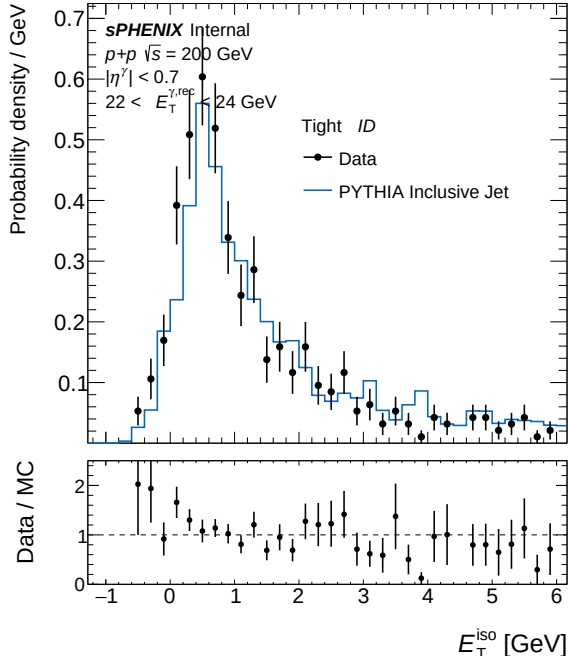
(b) IsoET Tight, pT bin 4 — bdt_npb03



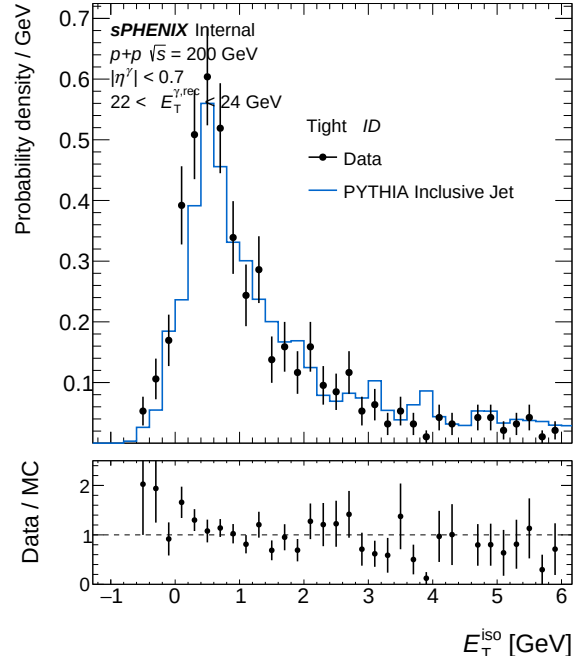
(a) IsoET Tight, pT bin 5 — bdt_nom



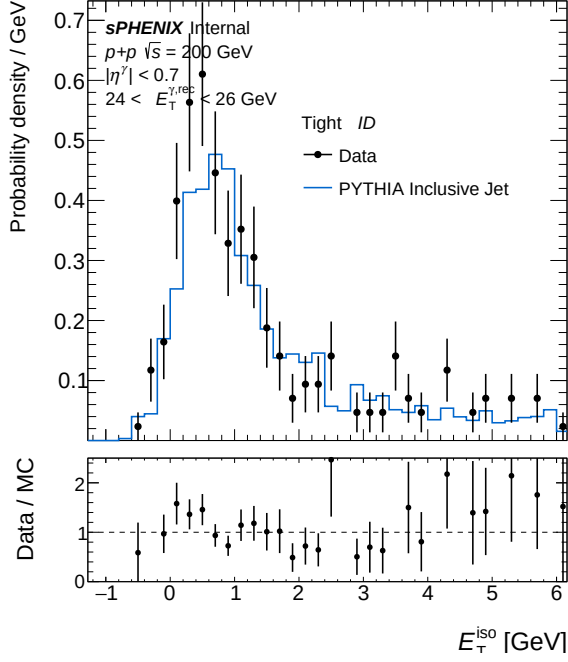
(b) IsoET Tight, pT bin 5 — bdt_npb03



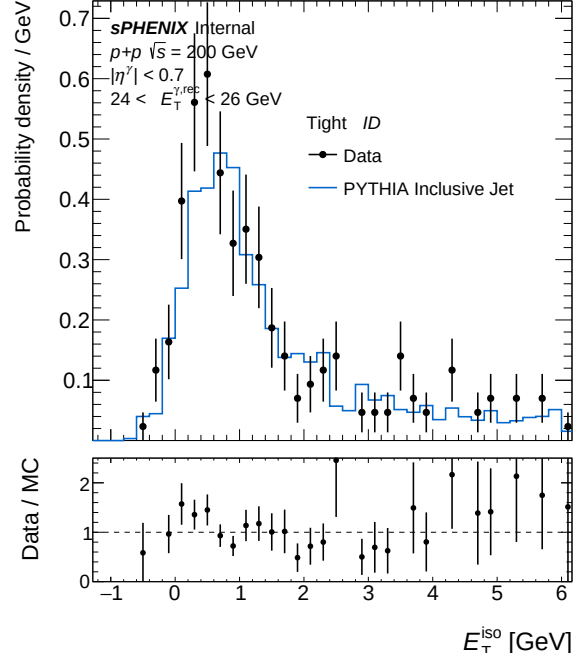
(a) IsoET Tight, pT bin 6 — bdt_nom



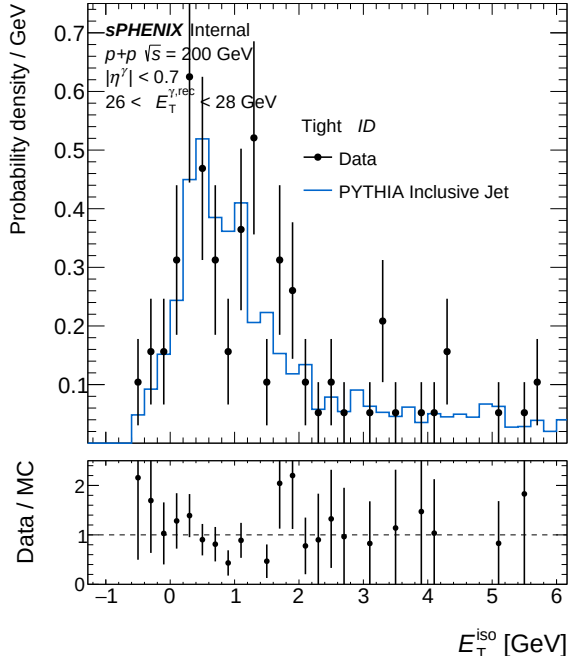
(b) IsoET Tight, pT bin 6 — bdt_npb03



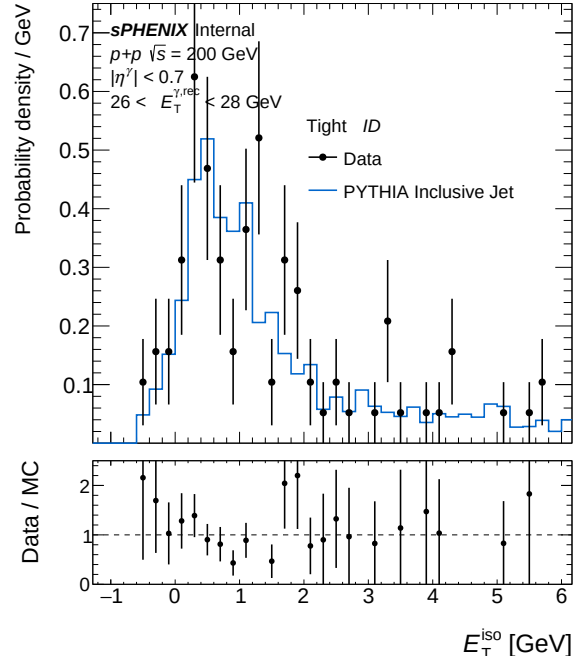
(a) IsoET Tight, pT bin 7 — bdt_nom



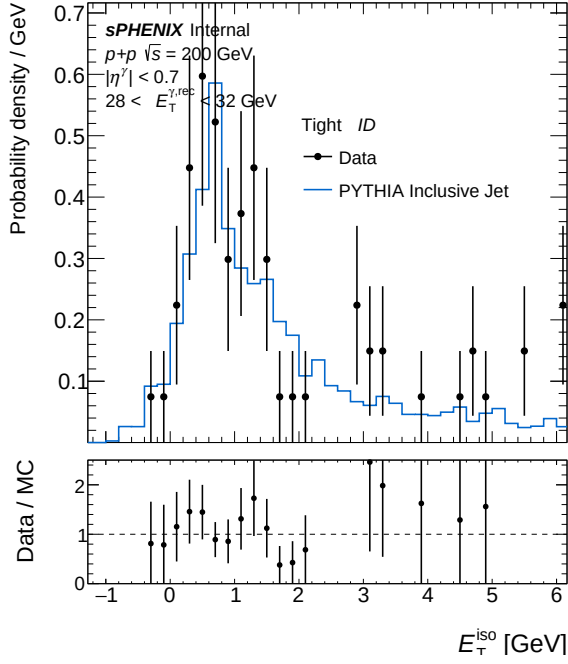
(b) IsoET Tight, pT bin 7 — bdt_npb03



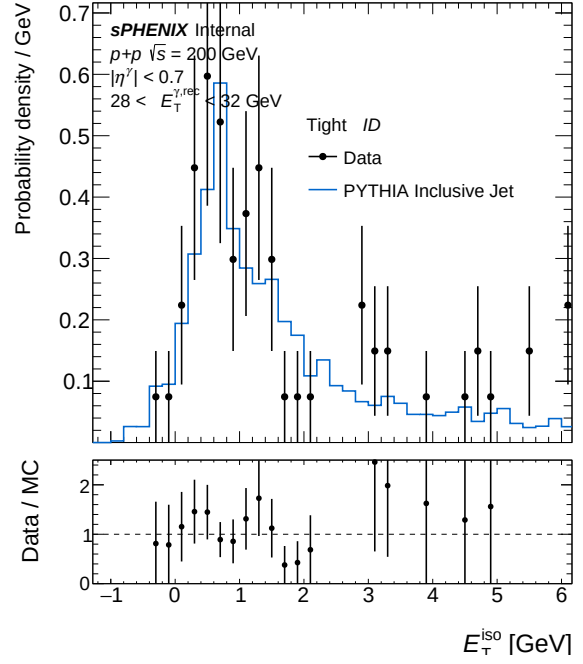
(a) IsoET Tight, pT bin 8 — bdt_nom



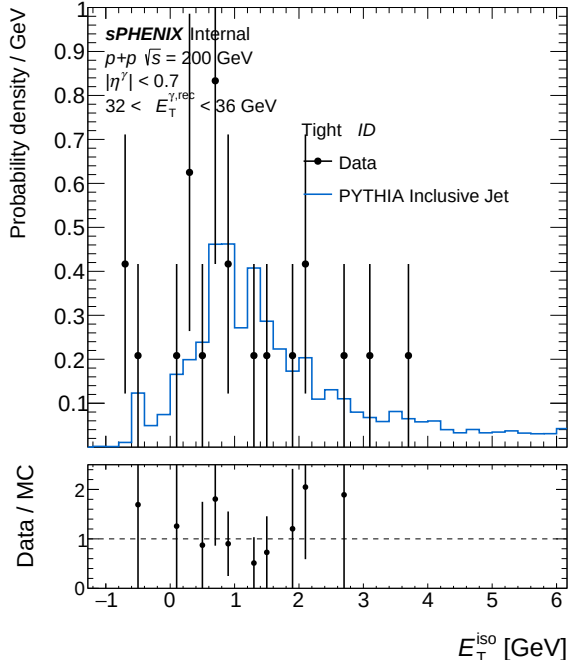
(b) IsoET Tight, pT bin 8 — bdt_npb03



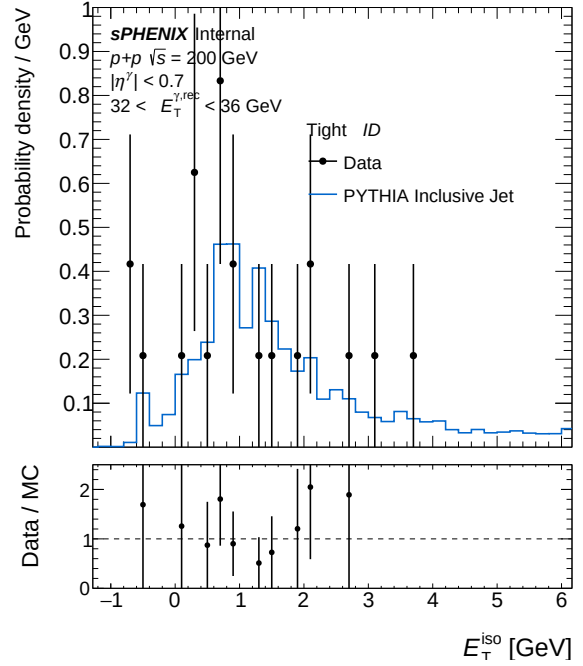
(a) IsoET Tight, pT bin 9 — bdt_nom



(b) IsoET Tight, pT bin 9 — bdt_npb03



(a) IsoET Tight, pT bin 10 — bdt_nom



(b) IsoET Tight, pT bin 10 — bdt_npb03

3 bdt_nom vs bdt_nosplit

Config diff (A → B)

```

--- config_bdt_nom.yaml
+++ config_bdt_nosplit.yaml
@@ -1,293 +0,0 @@
-#config.yaml
-#bdt nominal
-input:
-  photon_jet_file_root_dir: "/sphenix/user/shuhangli/ppg12/FunWithxgboost/"
-  photon_jet_file_branch_dir: "/bdt_split.root"
-  #photon_jet_file_branch_dir: "/bdt_tree.root"
-  data_file: "/sphenix/user/shuhangli/ppg12/anatreemaker/macro_maketree/data/ana521/condorout/part.*.root"
-  tree: "slimtree"
-  cluster_node_name: "CLUSTERINFO_CEMC"
-  bdt_model_name: "base_E"
-  # Optional: use different BDT models in different ET bins.
-  # bdt_et_bin_edges has N+1 values; bdt_et_bin_models has N values.
-  # Clusters outside all bins use bdt_model_name as fallback.
-  bdt_et_bin_edges: [8, 15, 35]
-  bdt_et_bin_models: ["base_v3E", "base_v3E"]
-  unfold_input: "results/MC_efficiency_photon5.root"
-
-output:
-  eff_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/MC_efficiency"
-  response_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/MC_response"
-  data_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/data_histo"
-  var_type: "bdt_nom"
-  final_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/Photon_final"
-
-analysis:
-  isbackground: 0
-  eta_bins: [-0.7, 0.7]
-  #eta_bins: [-0.7, 0.7]
-  vertex_cut: 60.0
-  #vertex_cut: 60.0
-  truth_iso_max: 4.0
-  reco_iso_min: -20.0
-  reco_iso_max_b: 0.490
-  reco_iso_max_s: 0.037
-  #reco_iso_max_b: 0
-  #reco_iso_max_s: 0.0293956
-  #reco_iso_max_b: 0.330117
-  #reco_iso_max_s: 0.025
-  #reco_iso_max_b: 1.08128
-  #reco_iso_max_s: 0.0299107
-  #reco_iso_max_b: -0.109478
-  #reco_iso_max_s: 0.0369506
-  #reco_iso_max_b: 3.0
-  #reco_iso_max_s: 0.0
-  reco_noniso_min_shift: 0.8
-  reco_noniso_max: 20.0
-

```

```

- # L1 trigger-efficiency correction on data clusters: weight *= 1/eps_L1(ET^reco)
- # Gumbel-CDF parametrization eps_L1(ET) = p0 * exp(-exp(-(ET - mu)/beta))
- # measured in plot_trigger_bit30.C with scaled[10]/live[30].
- apply_trigger_eff_correction: 1
- trigger_eff_p0: 1.000
- trigger_eff_mu: -2.04
- trigger_eff_beta: 3.33
- reco_min_ET: 5.0
- cone_size: 3
- eff_dR: 0.1
- trigger_used: 30
- pT_bins: [10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 32, 36]
- pT_bins_truth: [8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 32, 36, 45]
- use_topo_iso: 2 # 0: standard cone (size set by cone_size), 1: cluster_iso_topo_03, 2: cluster_iso_topo_04
- iso_threshold: 1
- iso_hcalonly: 0
- iso_emcalinnerr: 0.0
- n_nt_fail: 1
- weta_fail: 0
- wphi_fail: 0
- et1_fail: 0
- e11_to_e33_fail: 0
- e32_to_e35_fail: 0
- bdt_fail: 0
- weta_on: 1
- wphi_on: 0
- et1_on: 0
- e11_to_e33_on: 0
- e32_to_e35_on: 0
- et2_on: 0
- et3_on: 0
- et4_on: 0
- bdt_on: 1
- mc_iso_shift: 0.1
- mc_iso_scale: 1.2
- vertex_reweight_on: 1
- # Path to data vertex scan output file (h_vertexz). Used for on-the-fly vertex reweighting.
- # Leave empty ("") to use the pre-made vertex_reweight_file instead.
- vertex_scan_data_file: ""
- # Data-driven iterative truth-vertex reweight (standalone tool in
- # efficiencytool/truth_vertex_reweight/). When on, forces vertex_reweight_on
- # off and applies w(z_h) (single MC) or w(z_h)*w(z_mb) (double MC) per event.
- # Flip truth_vertex_reweight_on to 1 to switch the nominal 1.5 mrad
- # analysis from the f(z_r) reco-level reweight to the truth-level one.
- truth_vertex_reweight_on: 0
- truth_vertex_reweight_file: "/sphenix/user/shuhangli/ppg12/efficiencytool/truth_vertex_reweight/output"
- fit_option: 1
- cluster_escale: 1.0
- cluster_eres: 0.04
- mc_purity_correction: 0
- mc_purity_correction_file: ""
-
- # Data-driven tower mask (phi-symmetry z<-2 at tight-selection level).
- # Nominal production applies this mask symmetrically to data and MC to

```

```

- # correct for MC not emulating all Run-24 data dead towers. Switched on
- # as nominal on 2026-04-23. Other single-level masks (preselect, common,
- # OR) are available as systematic cross-checks via VARIANTS in
- # make_bdt_variations.py.
- tower_mask_on: 1
- tower_mask_file: "/sphenix/user/shuhangli/ppg12/efficiencytool/tower_masks_bdt_nom.root"
- tower_mask_name: "mask_phisymm_tight"
-
- tight:
-   reta77_min: 0.0
-   reta77_max: 1.0
-
-   rhad33_max: 1.0
-   rhad33_min: 0.00
-
-   w72_max: 1.0
-   w72_min: 0.0
-
-   re11_E_max: 1.0
-   re11_E_min: 0.0
-
-   CNN_max: 1.0
-   CNN_min: 0.0
-   #CNN_min: 0.8
-
-   #correlation
-   et1_max: 1.0
-   et1_min: 0.5
-   et1_min_b: 0.5
-   et1_min_s: 0.0
-   #correlation
-   e32_over_e35_max: 1.0
-   e32_over_e35_min: 0.8
-
-   #correlation
-   weta_cogx_max: 1.0
-   weta_cogx_min: 0.0
-   weta_cogx_max_b: 1.0
-   weta_cogx_max_s: 0.0
-
-   wphi_cogx_max: 1.0
-   wphi_cogx_min: 0.0
-   wphi_cogx_max_b: 1.0
-   wphi_cogx_max_s: 0.0
-
-   #correlation
-   prob_max: 1.0
-   prob_min: 0.0
-
-   #small correlation
-   e11_over_e33_max: 1.0
-   e11_over_e33_min: 0.0
-

```

```

- #small correlation
- et4_max: 1.0
- et4_min: 0.0
-
- et2_max: 1.0
- et2_min: 0.0
-
- et3_max: 1.0
- et3_min: 0.0
-
- #w32 small correlation
- w32_max: 1.0
- w32_min: 0.0
-
- #bdt
- bdt_max: 1.0
- bdt_min: 0.7
- bdt_min_slope: -0.003333333333333336
- bdt_min_intercept: 0.8333333333333334
-
-
- non_tight:
-   reta77_min: 0.0
-   reta77_max: 1.0
-
-   rhad33_max: 1.0
-   rhad33_min: 0.00
-
-   w72_max: 1.0
-   w72_min: 0.0
-
-   re11_E_max: 1.0
-   re11_E_min: 0.0
-
-   CNN_max: 1.0
-   CNN_min: 0.0
-   #correlation
-   et1_max: 1.0
-   et1_min: 0.6
-   #correlation
-   e32_over_e35_max: 1.0
-   e32_over_e35_min: 0.8
-
-   #correlation
-   weta_cogx_max: 1.0
-   weta_cogx_min: 0.0
-   weta_cogx_max_b: 1.0
-   weta_cogx_max_s: 0.0
-
-
-   wphi_cogx_max: 1.0
-   wphi_cogx_min: 0.0
-   wphi_cogx_max_b: 1.0
-   wphi_cogx_max_s: 0.0

```



```

-
- #correlation
- prob_max: 1.0
- prob_min: 0.0
-
- #small correlation
- e11_over_e33_max: 1.0
- e11_over_e33_min: 0.0
-
- #small correlation
- et4_max: 1.0
- et4_min: 0.0
-
- #w32 small correlation
- w32_max: 1.0
- w32_min: 0.0
-
- #bdt
- bdt_max: 0.5
- bdt_max_slope: 0.003333333333333336
- bdt_max_intercept: 0.6666666666666666
- bdt_min: 0.02
-
- bdt_min_intercept: 0.7333333333333333
- bdt_min_slope: -0.01333333333333333
-
- common:
- wr_cogx_bound: 0.0
- cluster_weta_cogx_bound: 2.0
-
- npb_cut_on: 1
- npb_score_cut: 0.5
-
- prob_max: 1.0
- prob_min: 0.0
-
- e11_over_e33_max: 0.98
-
- e11_over_e33_min: 0.0
-
- e32_over_e35_max: 1.0
- e32_over_e35_min: 0.8
-
- et1_max: 1.0
- et1_min: 0.6
-
-
-
-
-
-
- # Run range for data processing (-1 = no bound)
- run_min: 47289
- run_max: 54000

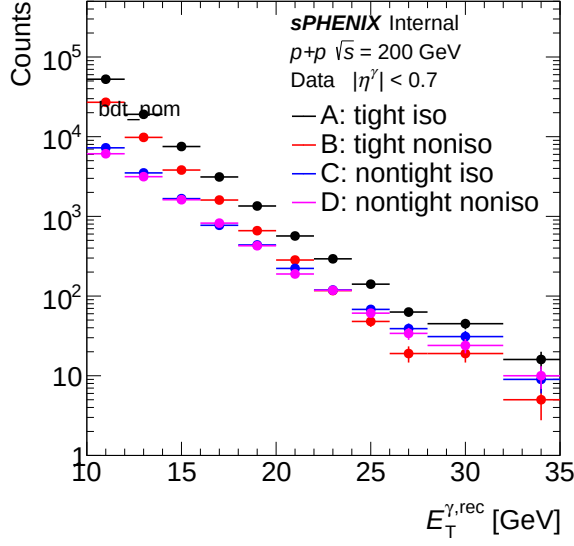
```

```

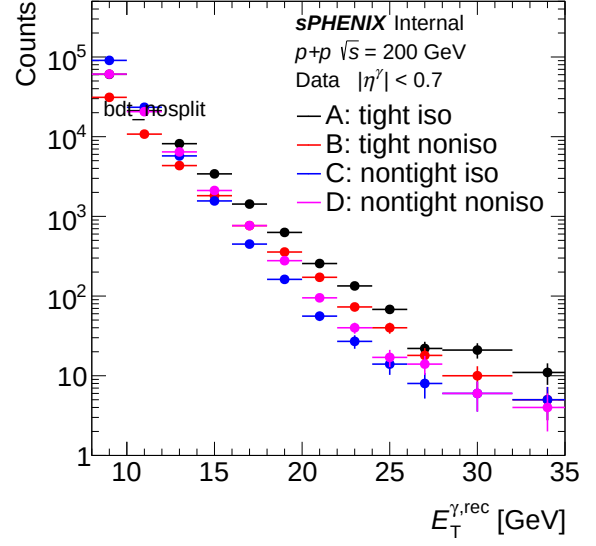
- # Optional run list file (one run number per line). If set, only listed runs are processed.
- run_list_file: ""
- # Total luminosity (pb^-1) for the selected run range.
- # Nominal switched to all-z (beam-delivered) 64.3718 pb^-1 on 2026-04-22;
- # paired with vertex_cut_truth=9999 to widen the MBD-eff truth denominator.
- lumi: 64.3718
-
- unfold:
-   resultit: 2
-   resultleak: 1
-   reweight: 1
-   # Truth-pT reweight for unfolding response prior. Re-derived 2026-04-27
-   # from R = (data unfolded with iter=1, no-truth-reweight variant) / mc_truth_prior
-   # (both unit-area normalized) on allz nominal with iso 0.490+0.037*ET and the
-   # L1 trigger-efficiency correction enabled. Fit range [8,35] GeV, rational(2,2)
-   # form (parameter-bounded). Iter=1 captures the prior-bias correction on the
-   # first round of iterative-prior reweighting. Loader: RecoEffCalculator_TTreeReader.C ~line 606.
-   truth_reweight:
-     formula: "([0] + [1]*x + [3]*x*x) / (1 + [2]*x + [4]*x*x)"
-     params: [4.05592, -0.984728, -0.478818, 0.0723232, 0.0522681]
-     xmin: 0
-     xmax: 100
-     clamp_pT_min: 10
-     clamp_pT_max: 30
-
-
- lumi_target: 64.3718
- vertex_cut_truth: 9999.0

```

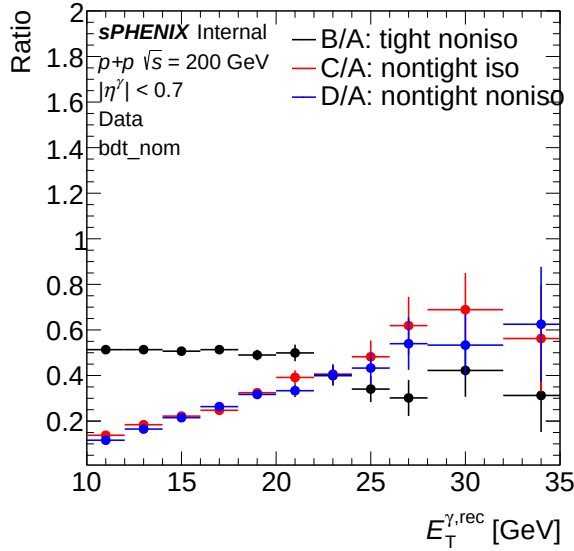
3.1 Sideband (Data)



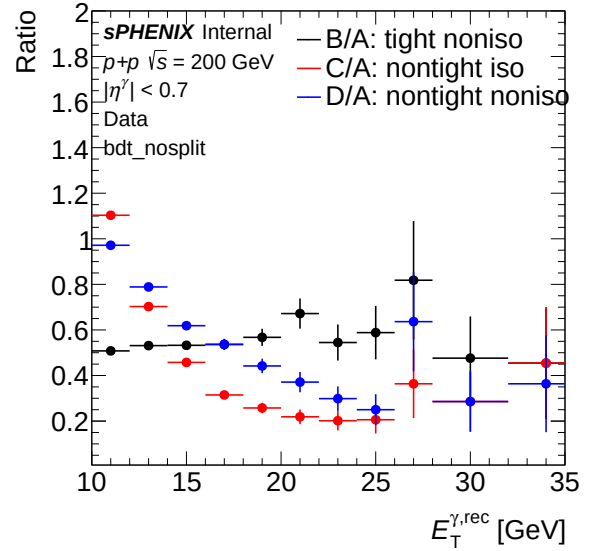
(a) Sideband ET spectrum (data) — bdt_nom



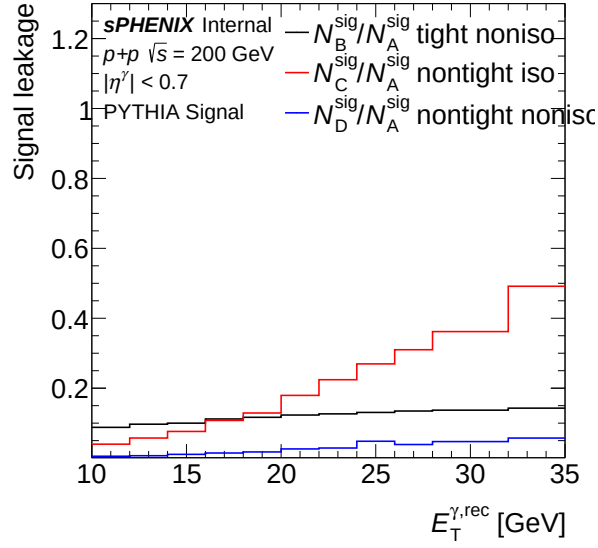
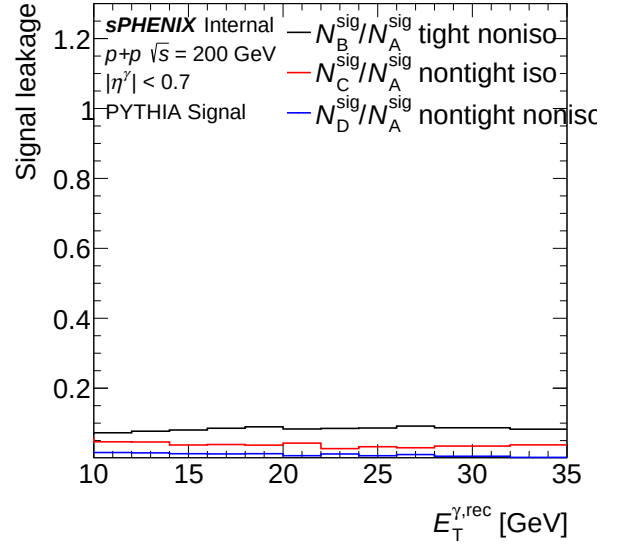
(b) Sideband ET spectrum (data) — bdt_nosplit



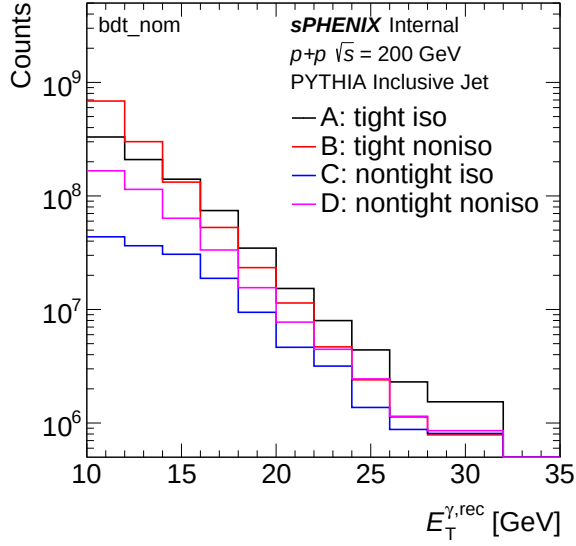
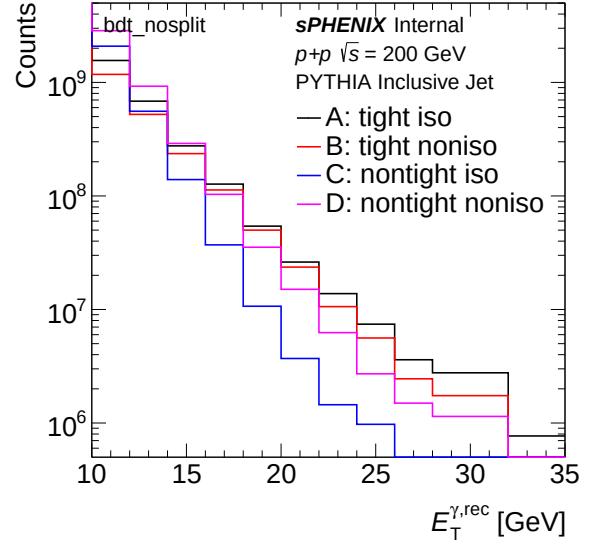
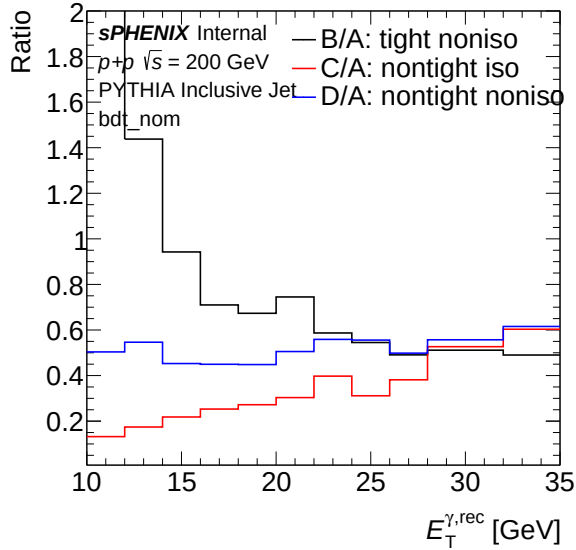
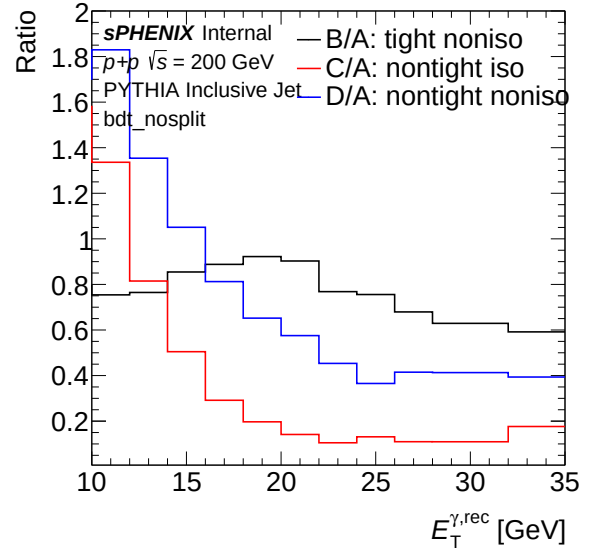
(a) Sideband ET ratio (data) — bdt_nom

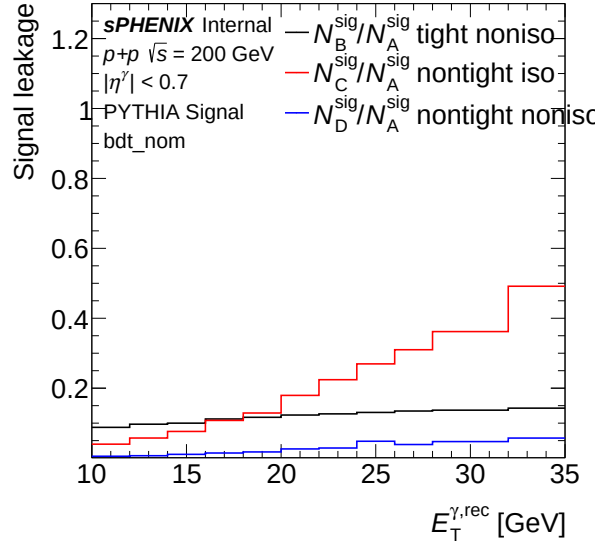
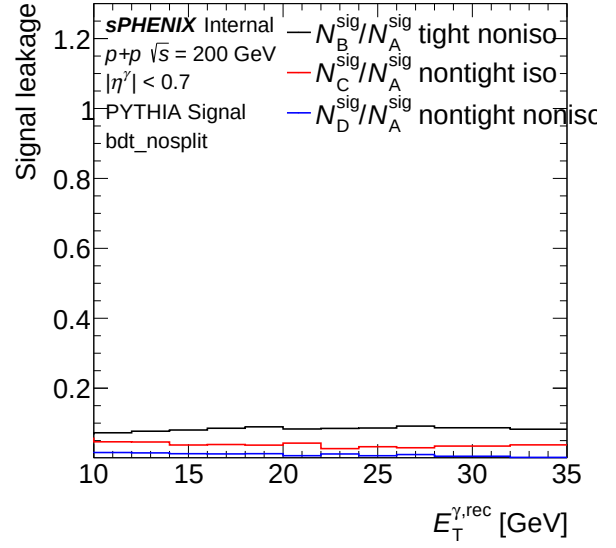


(b) Sideband ET ratio (data) — bdt_nosplit

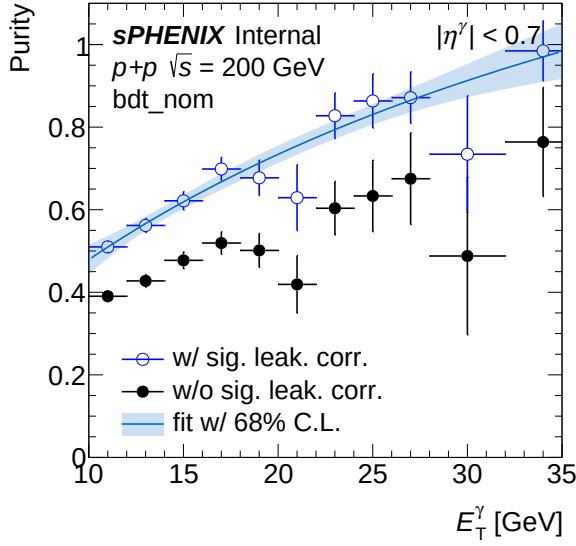
(a) Leakage fraction vs ET (data) — `bdt_nom`(b) Leakage fraction vs ET (data) — `bdt_nosplit`

3.2 Sideband (MC)

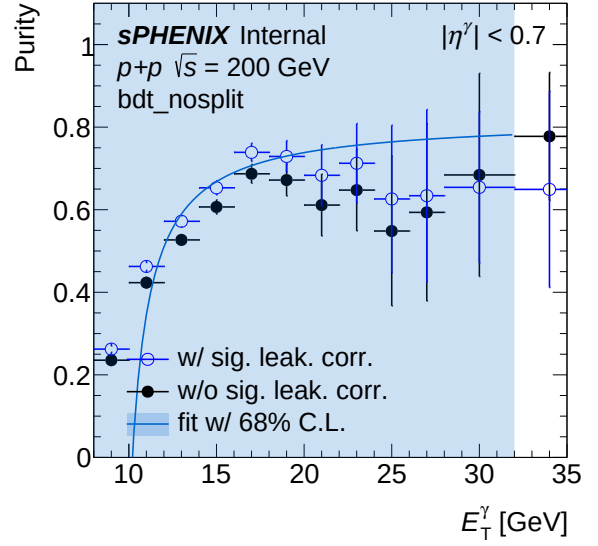
(a) Sideband ET spectrum (MC) — `bdt_nom`(b) Sideband ET spectrum (MC) — `bdt_nosplit`(a) Sideband ET ratio (MC) — `bdt_nom`(b) Sideband ET ratio (MC) — `bdt_nosplit`

(a) Leakage fraction vs ET (MC) — `bdt_nom`(b) Leakage fraction vs ET (MC) — `bdt_nosplit`

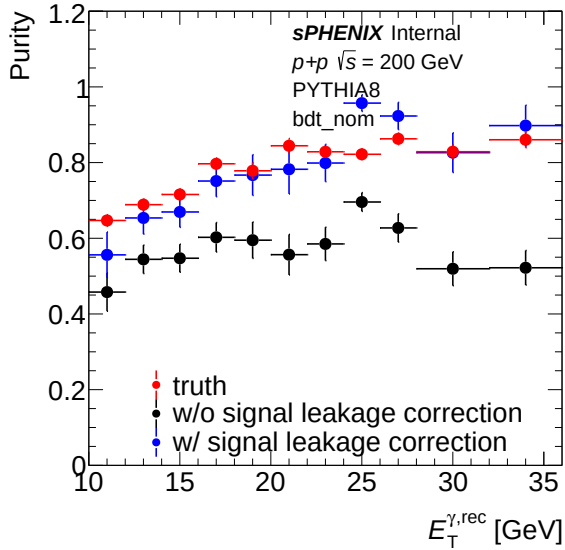
3.3 Purity



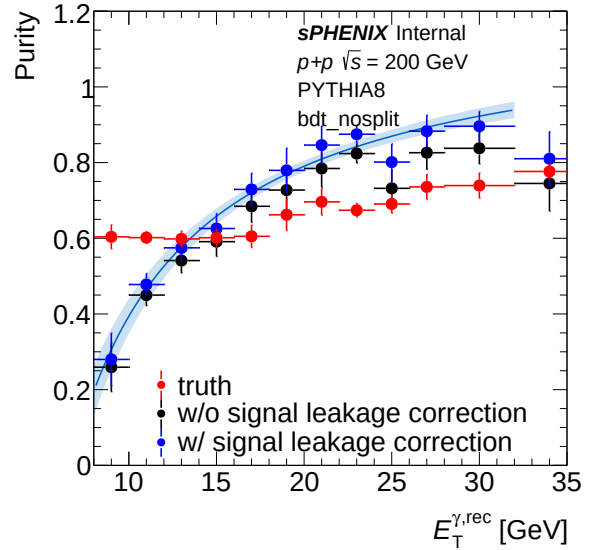
(a) Purity (data) — bdt_nom



(b) Purity (data) — bdt_nosplit

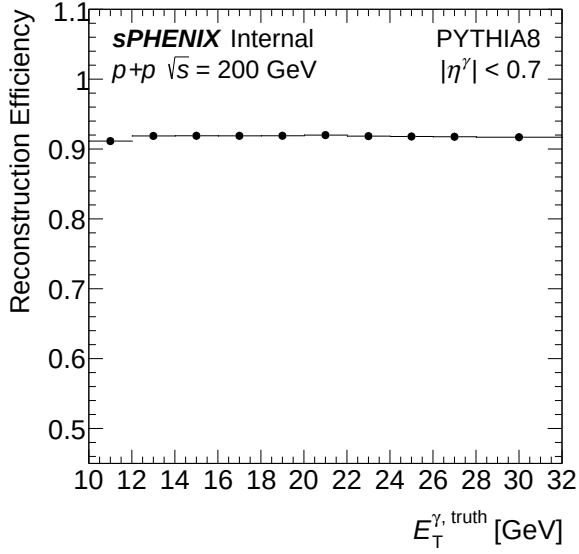


(a) Purity (MC) — bdt_nom

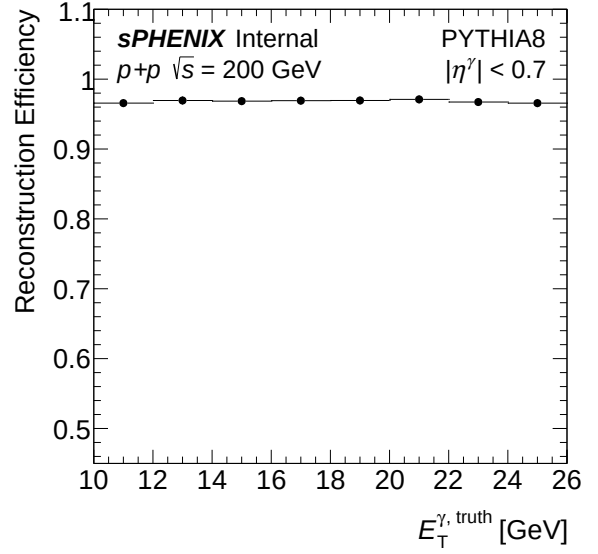


(b) Purity (MC) — bdt_nosplit

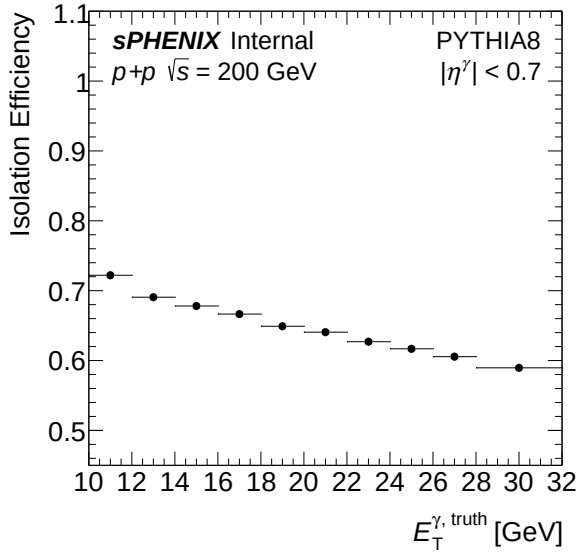
3.4 Efficiency



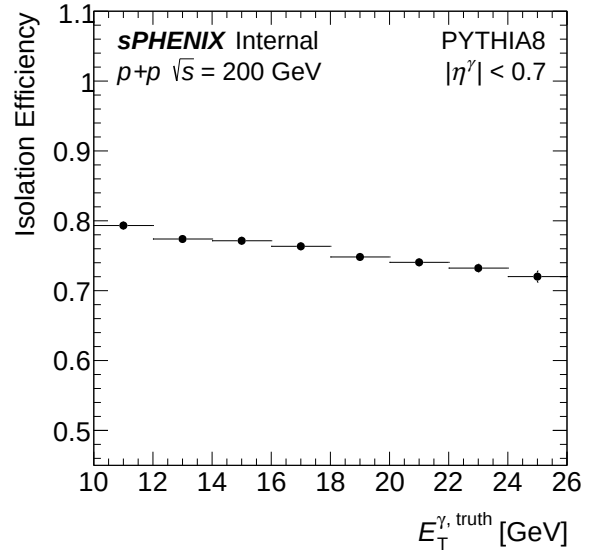
(a) Reconstruction efficiency — bdt_nom



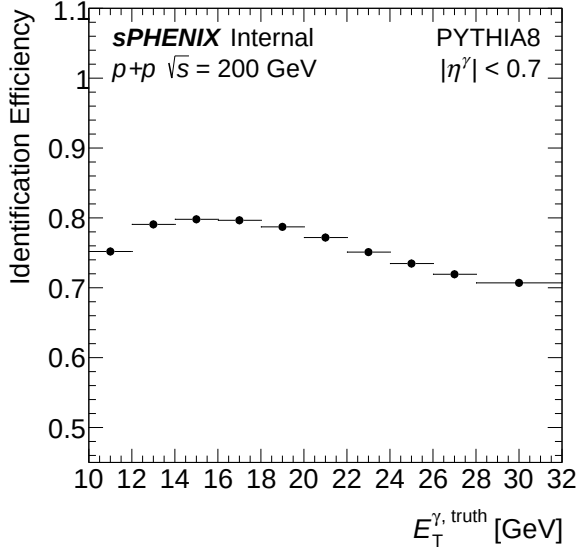
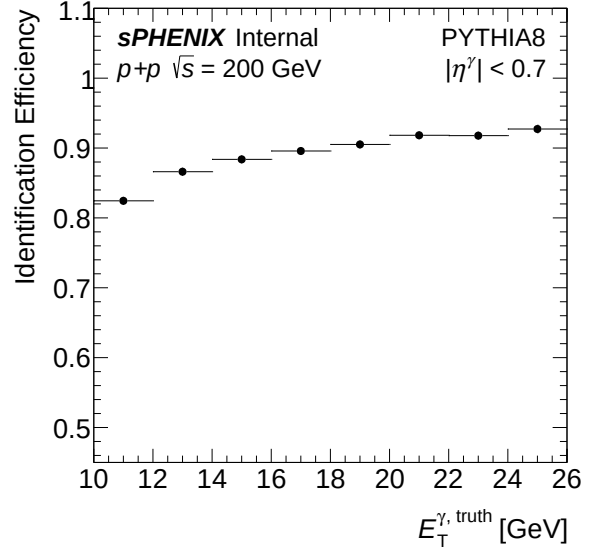
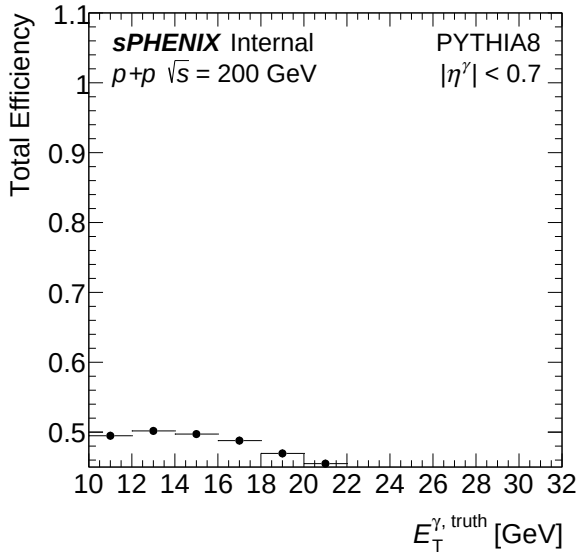
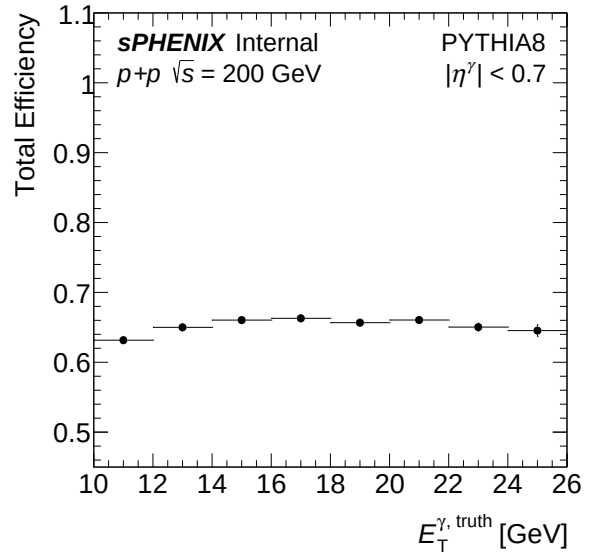
(b) Reconstruction efficiency — bdt_nosplit

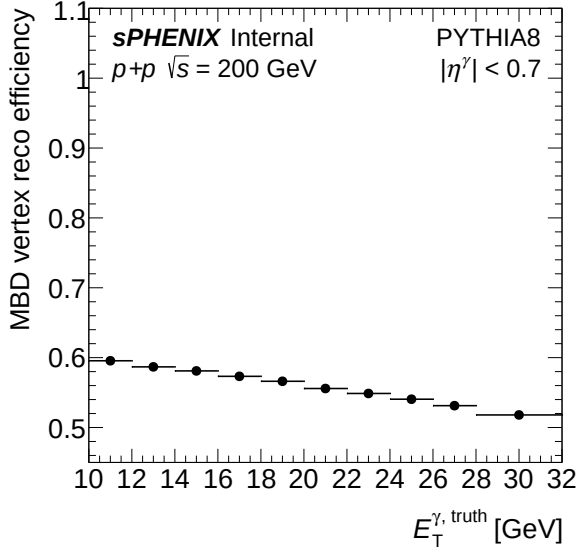


(a) Isolation efficiency — bdt_nom

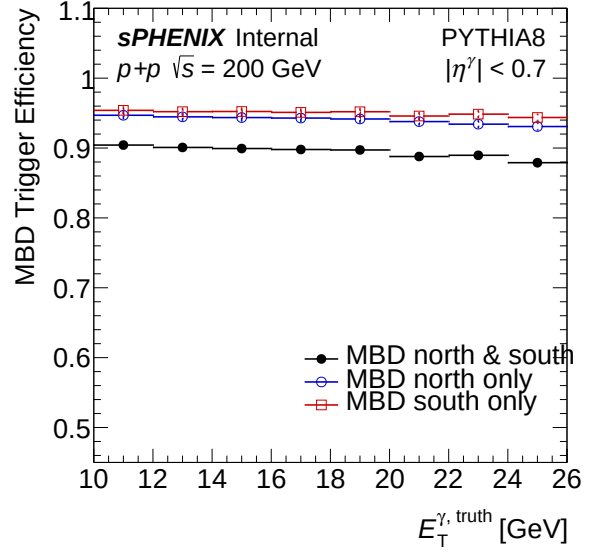


(b) Isolation efficiency — bdt_nosplit

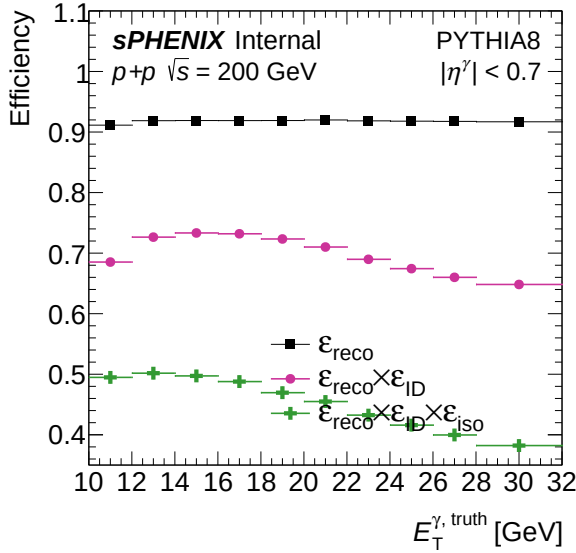
(a) Identification efficiency — `bdt_nom`(b) Identification efficiency — `bdt_nosplit`(a) Total efficiency — `bdt_nom`(b) Total efficiency — `bdt_nosplit`



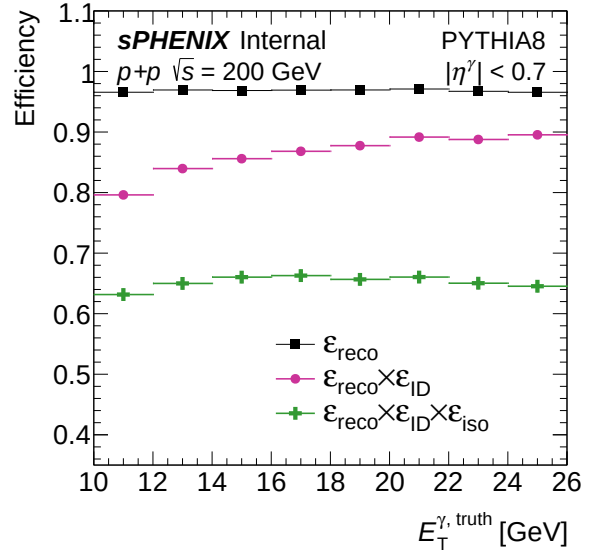
(a) MBD trigger efficiency — bdt_nom



(b) MBD trigger efficiency — bdt_nosplit

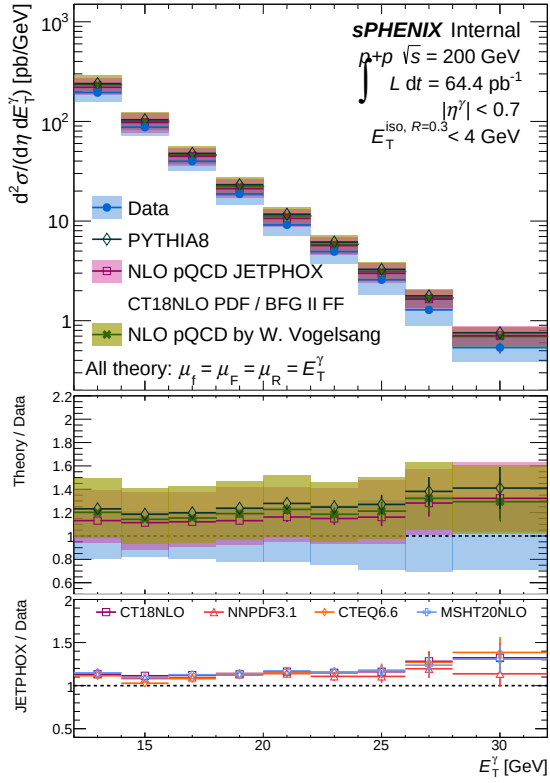
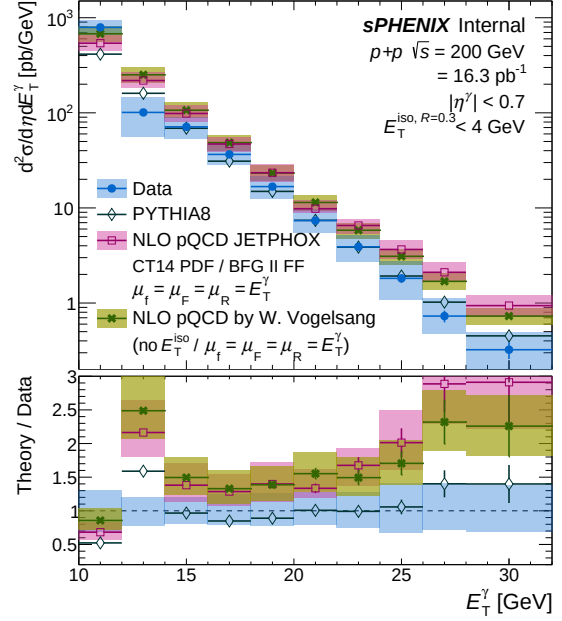


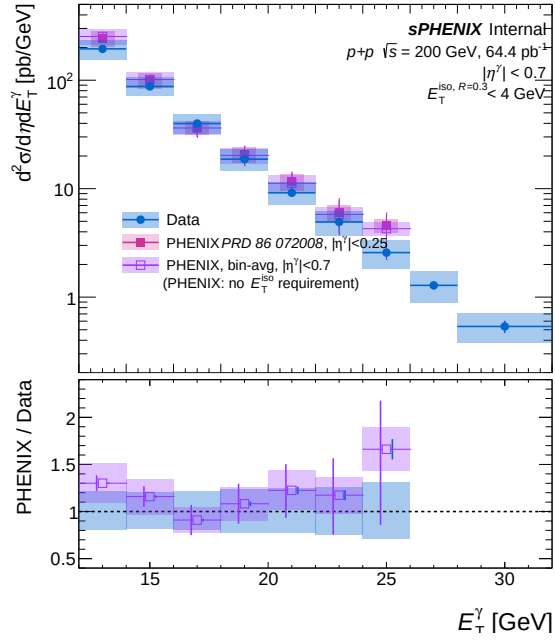
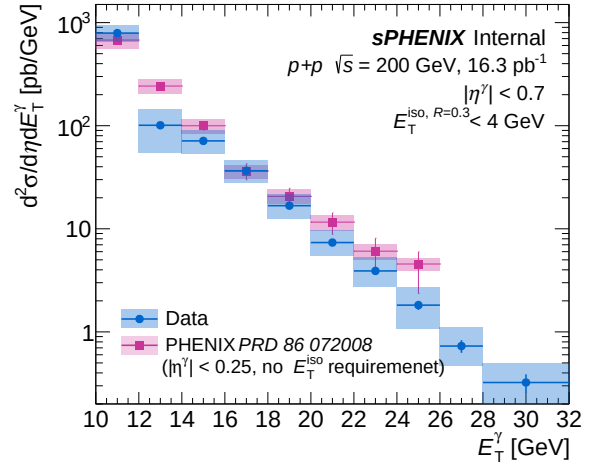
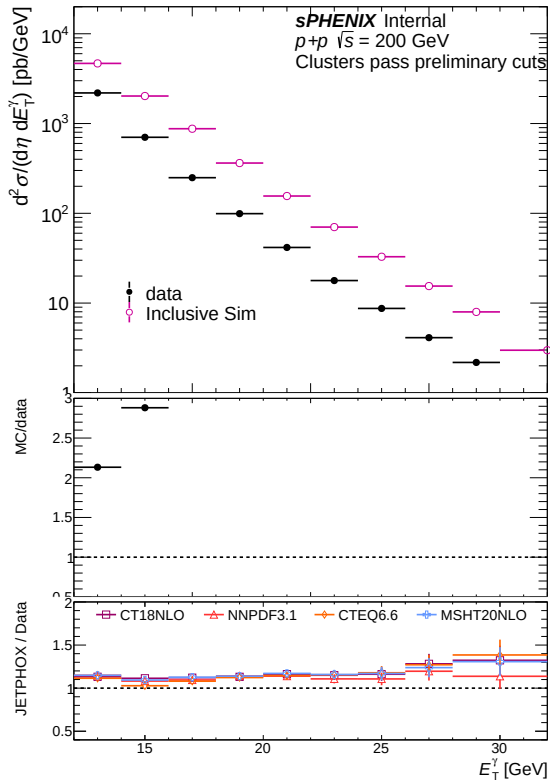
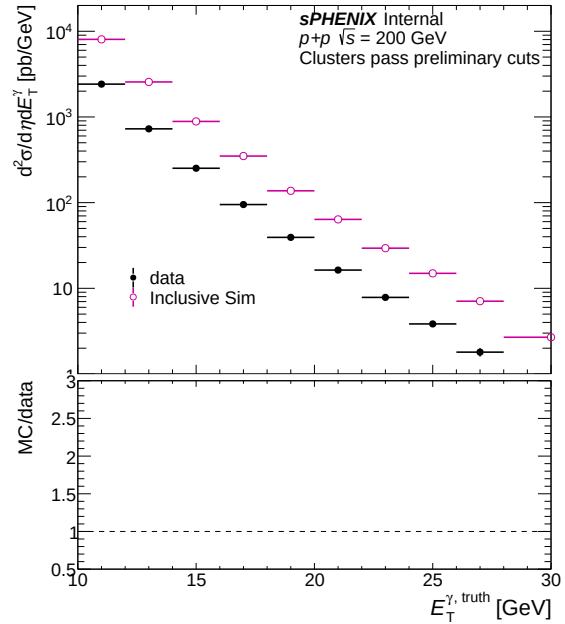
(a) Combined photon efficiency — bdt_nom

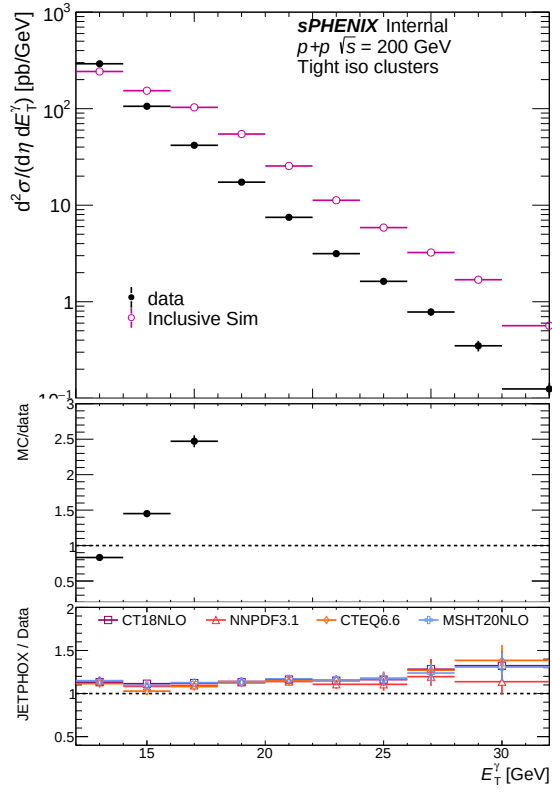
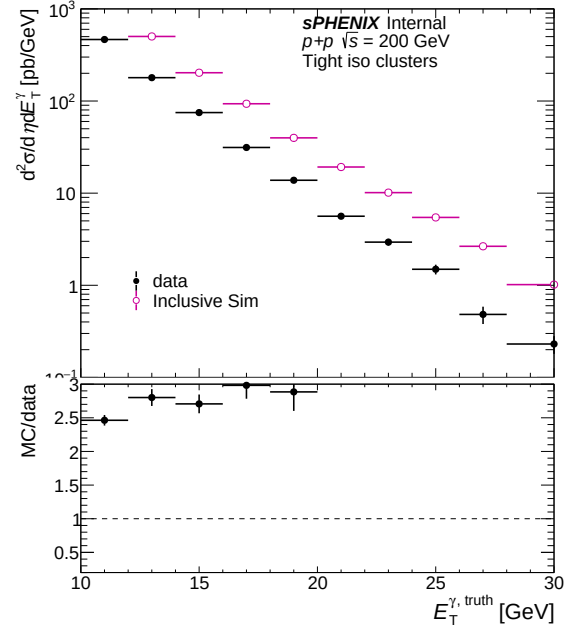


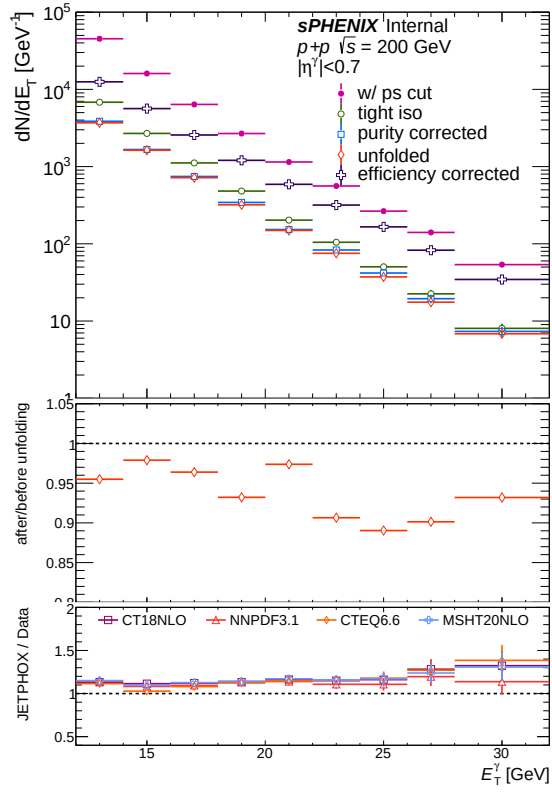
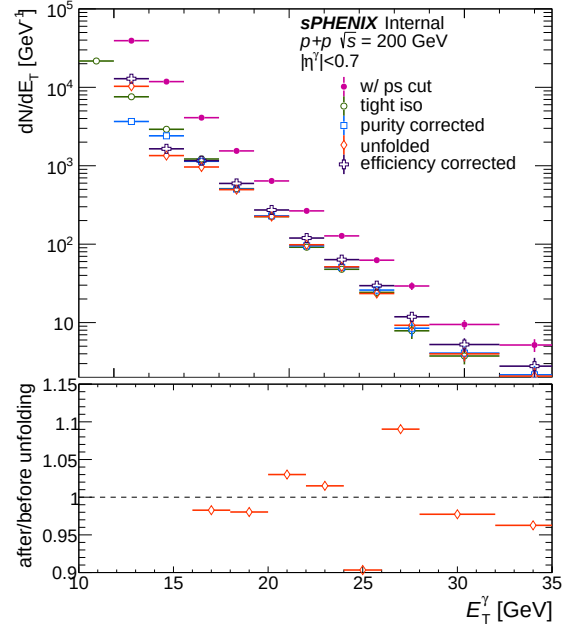
(b) Combined photon efficiency — bdt_nosplit

3.5 Final result

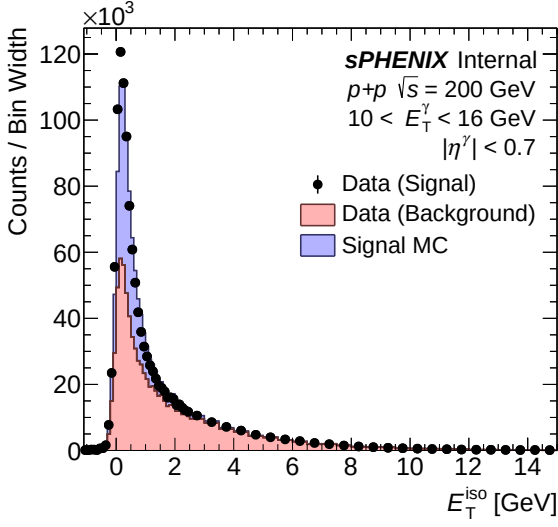
(a) Final cross section result — `bdt_nom`(b) Final cross section result — `bdt_nosplit`

(a) Comparison with PHENIX — `bdt_nom`(b) Comparison with PHENIX — `bdt_nosplit`(a) Common cluster yield (data vs MC) — `bdt_nom`(b) Common cluster yield (data vs MC) — `bdt_nosplit`

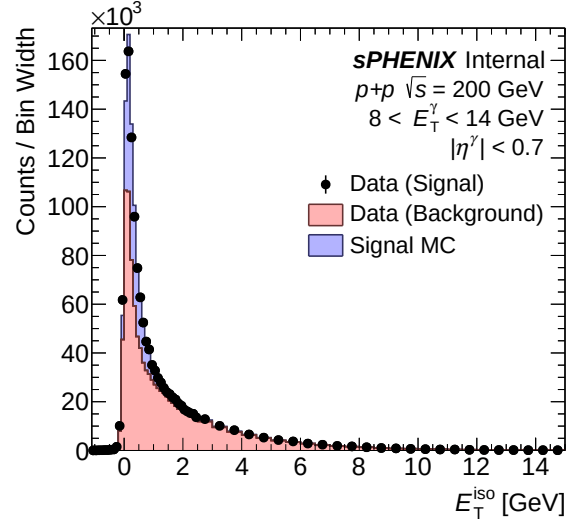
(a) Tight iso cluster yield (data vs MC) — `bdt_nom`(b) Tight iso cluster yield (data vs MC) — `bdt_nosplit`

(a) All correction steps — **bdt_nom**(b) All correction steps — **bdt_nosplit**

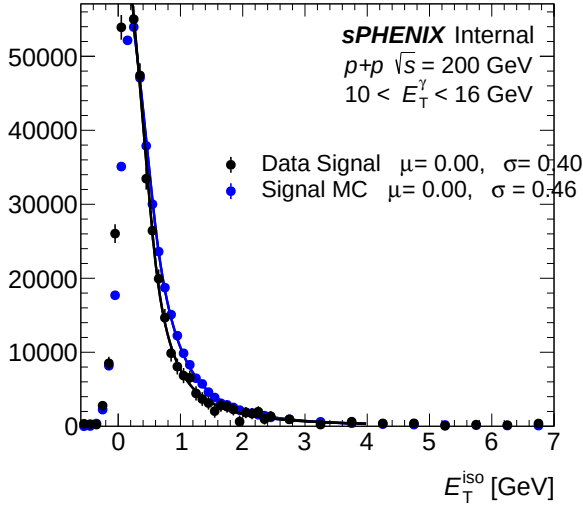
3.6 Isolation fit (CONF)



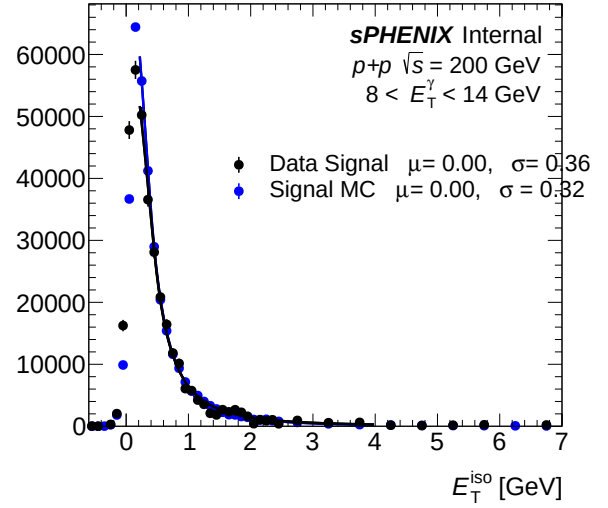
(a) IsoET template (pT bins 0-2) — bdt_nom



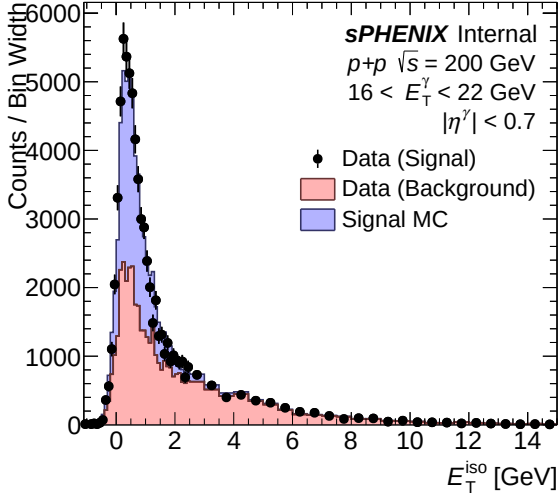
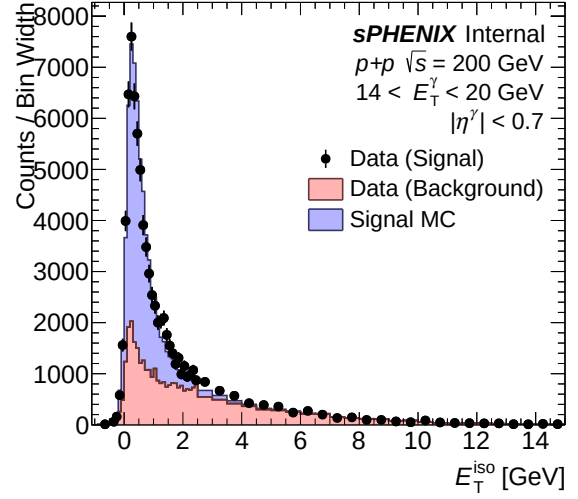
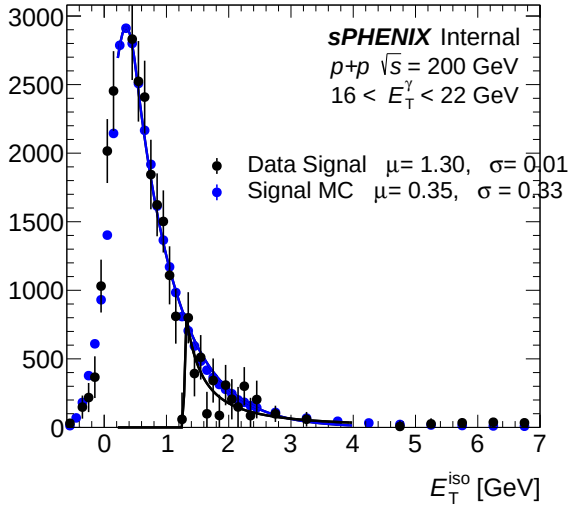
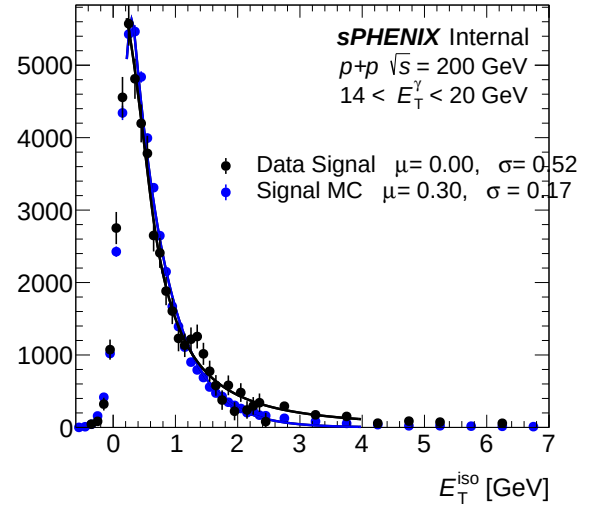
(b) IsoET template (pT bins 0-2) — bdt_nosplit

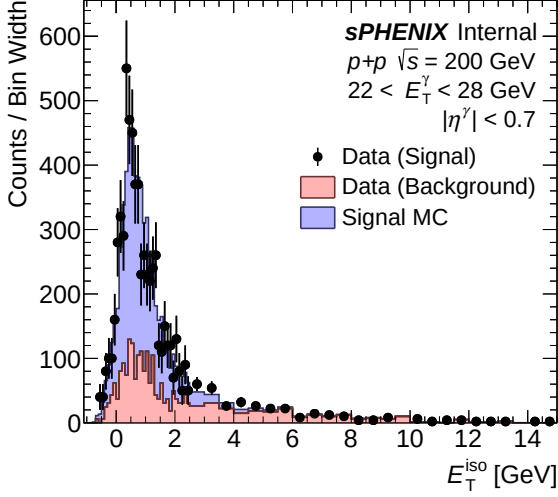


(a) IsoET fit template (pT bins 0-2) — bdt_nom

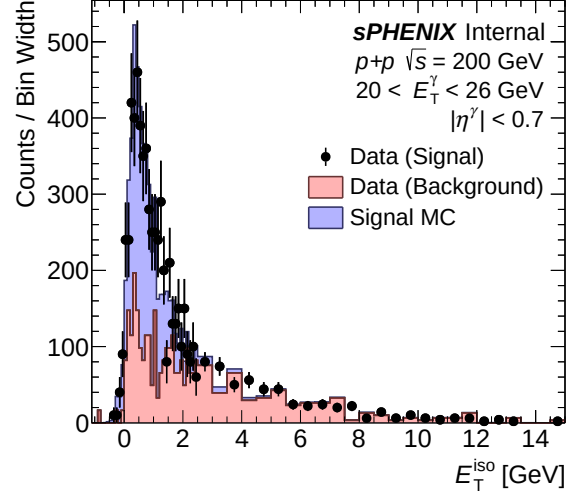


(b) IsoET fit template (pT bins 0-2) — bdt_nosplit

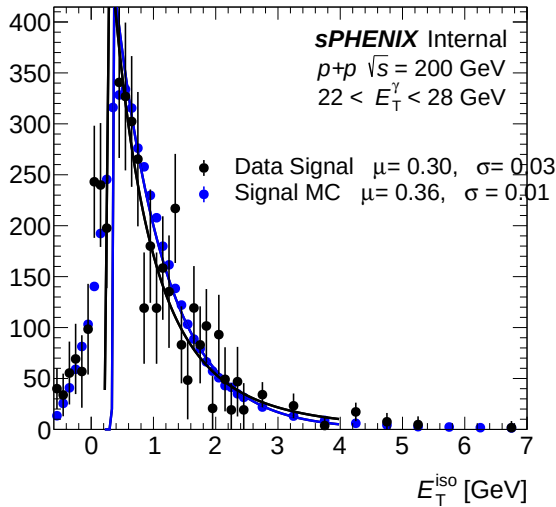
(a) IsoET template (pT bins 3–5) — `bdt_nom`(b) IsoET template (pT bins 3–5) — `bdt_nosplit`(a) IsoET fit template (pT bins 3–5) — `bdt_nom`(b) IsoET fit template (pT bins 3–5) — `bdt_nosplit`



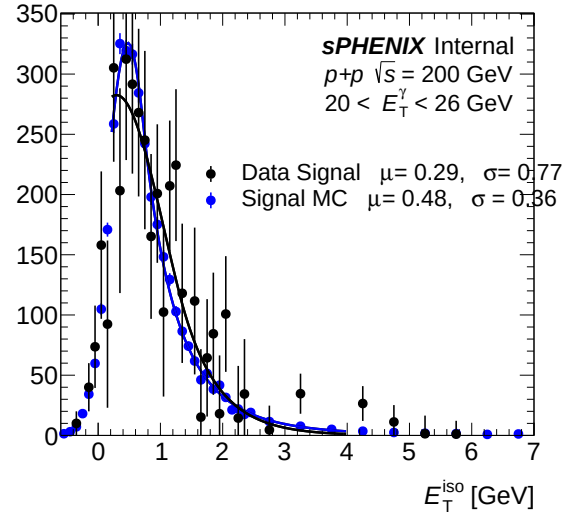
(a) IsoET template (pT bins 6–8) — bdt_nom



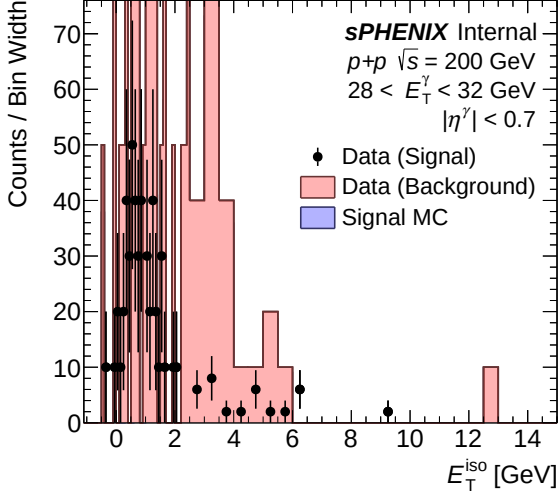
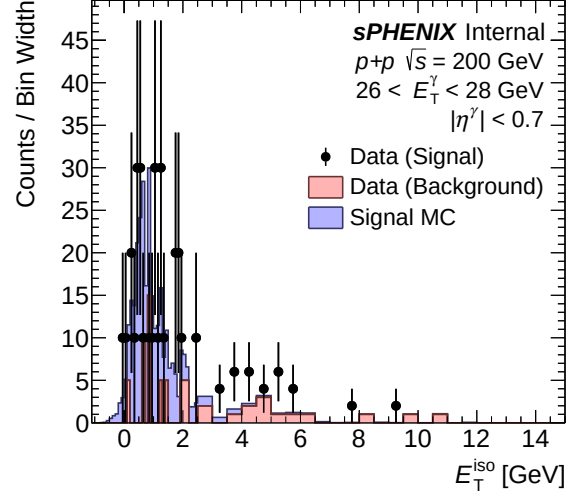
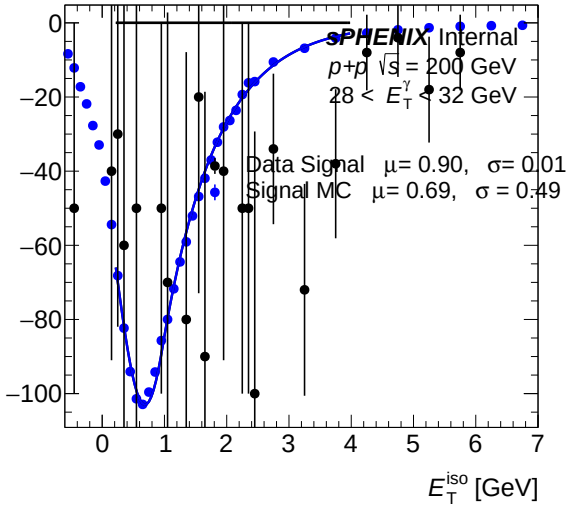
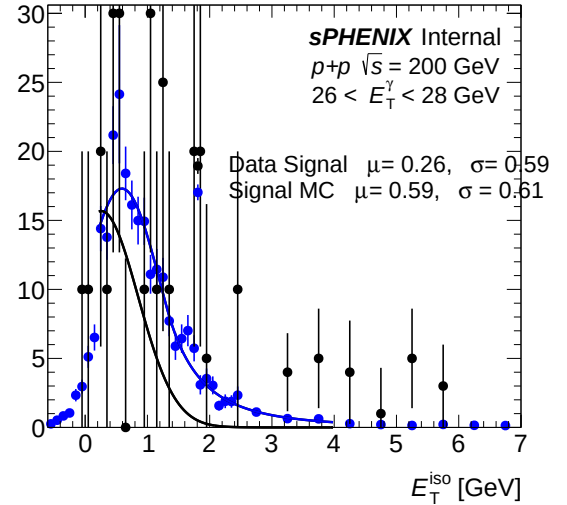
(b) IsoET template (pT bins 6–8) — bdt_nosplit



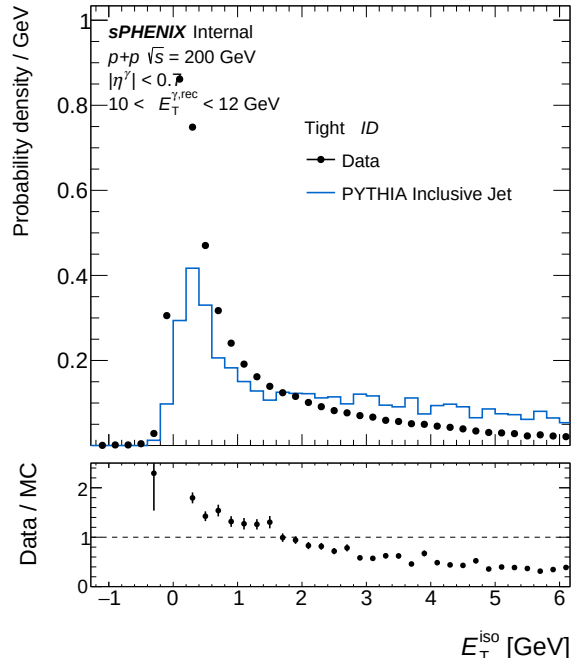
(a) IsoET fit template (pT bins 6–8) — bdt_nom



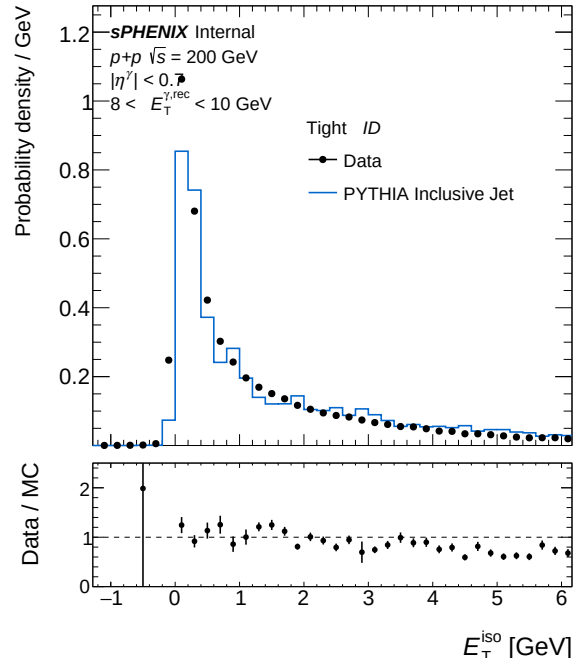
(b) IsoET fit template (pT bins 6–8) — bdt_nosplit

(a) IsoET template (pT bins 9–9) — `bdt_nom`(b) IsoET template (pT bins 9–9) — `bdt_nosplit`(a) IsoET fit template (pT bins 9–9) — `bdt_nom`(b) IsoET fit template (pT bins 9–9) — `bdt_nosplit`

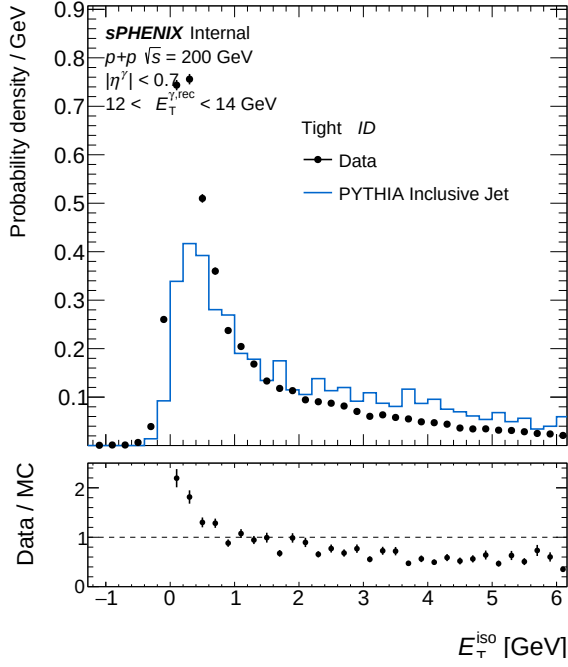
3.7 Isolation ET



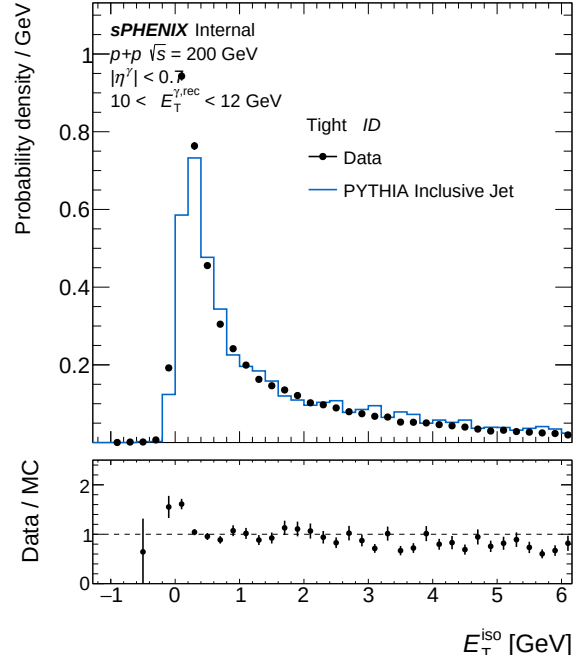
(a) IsoET Tight, pT bin 0 — bdt_nom



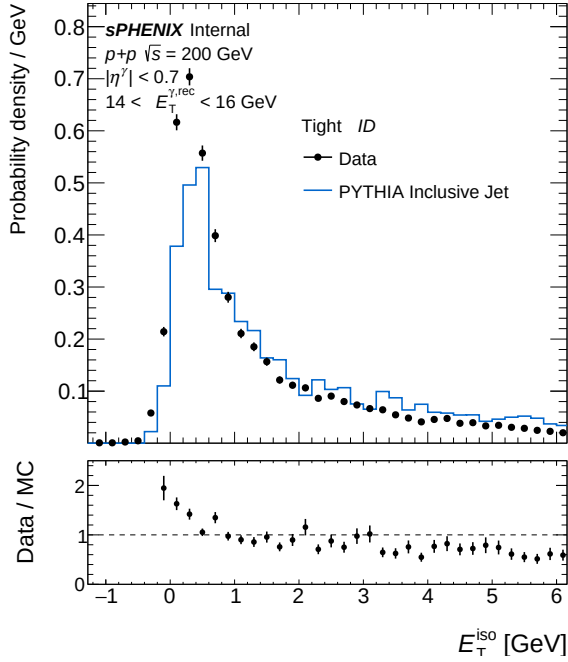
(b) IsoET Tight, pT bin 0 — bdt_nosplit



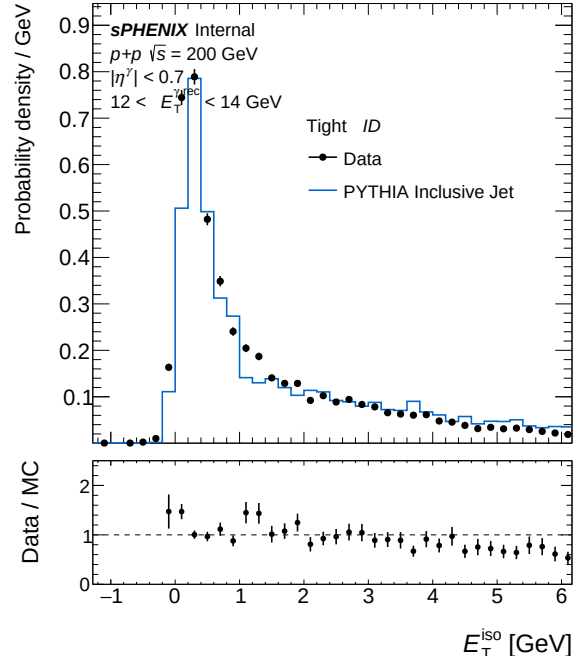
(a) IsoET Tight, pT bin 1 — bdt_nom



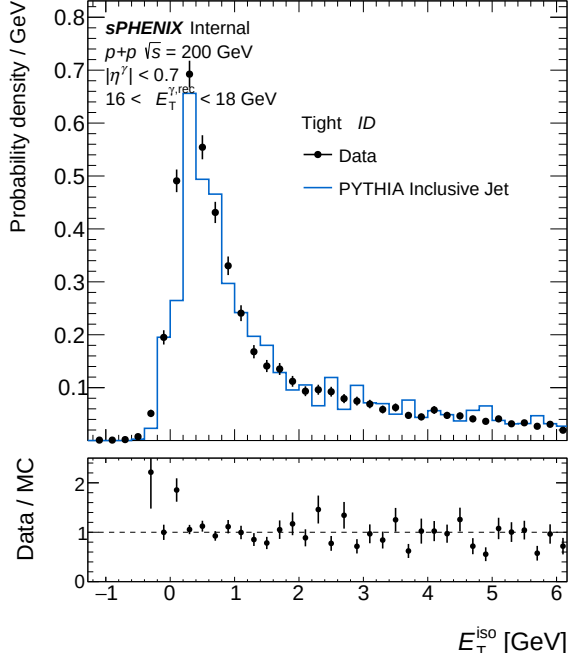
(b) IsoET Tight, pT bin 1 — bdt_nosplit



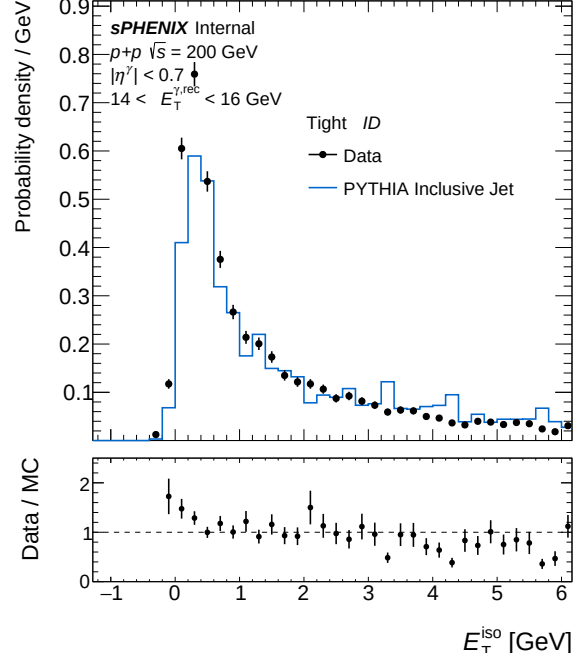
(a) IsoET Tight, pT bin 2 — bdt_nom



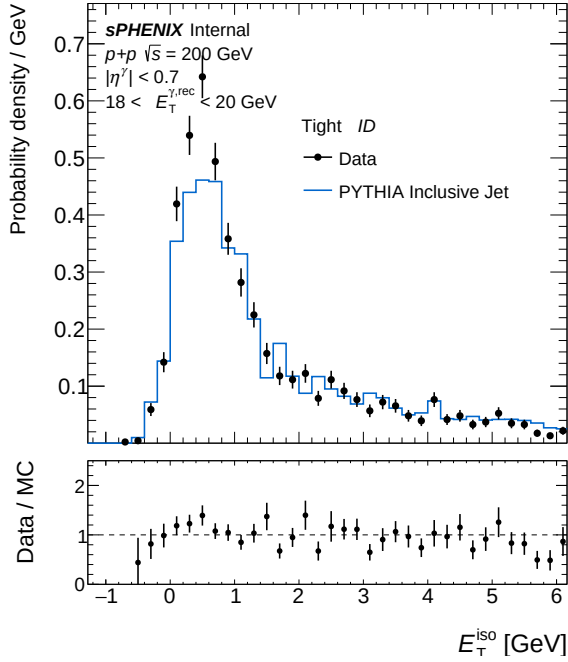
(b) IsoET Tight, pT bin 2 — bdt_nosplit



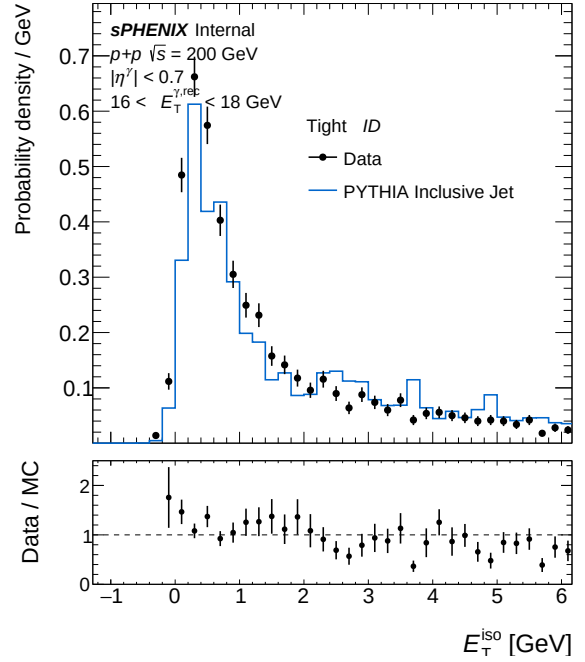
(a) IsoET Tight, pT bin 3 — bdt_nom



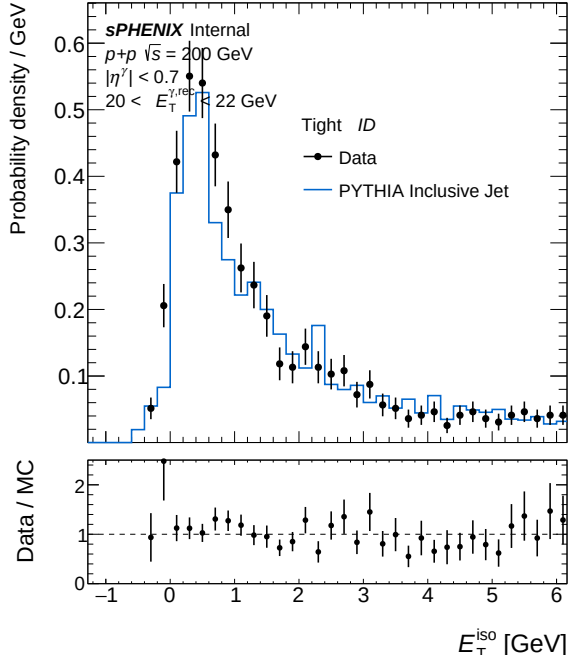
(b) IsoET Tight, pT bin 3 — bdt_nosplit



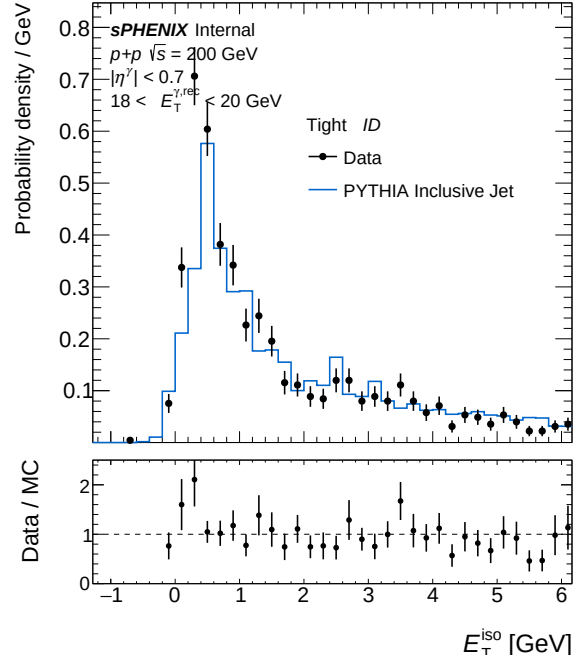
(a) IsoET Tight, pT bin 4 — bdt_nom



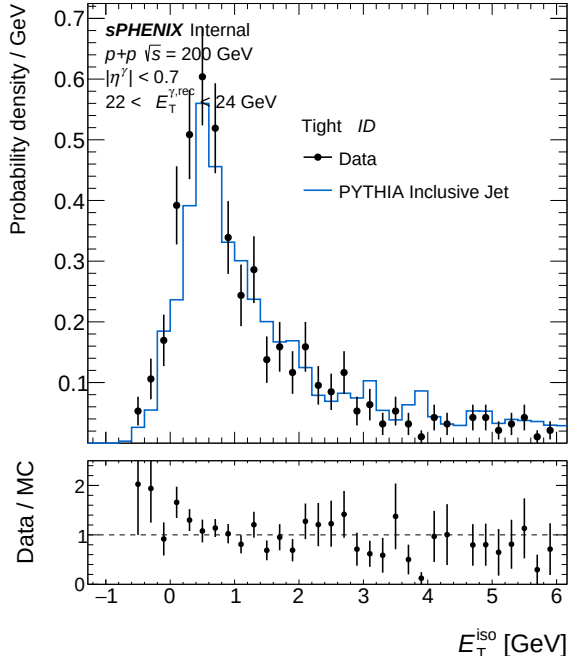
(b) IsoET Tight, pT bin 4 — bdt_nosplit



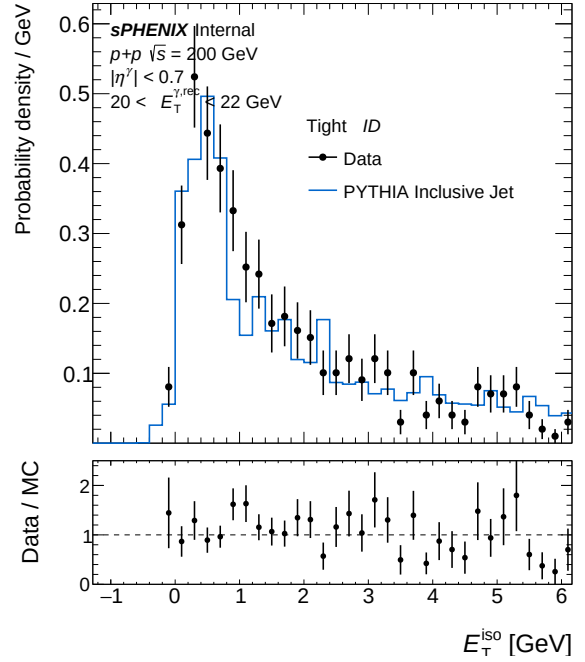
(a) IsoET Tight, pT bin 5 — bdt_nom



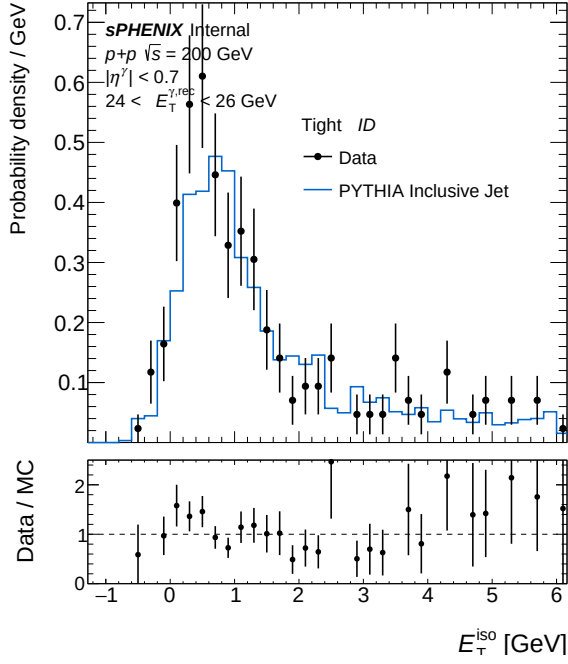
(b) IsoET Tight, pT bin 5 — bdt_nosplit



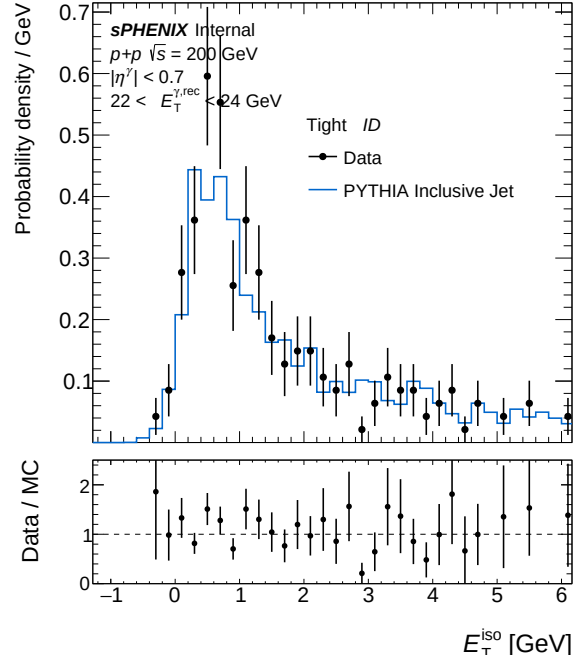
(a) IsoET Tight, pT bin 6 — bdt_nom



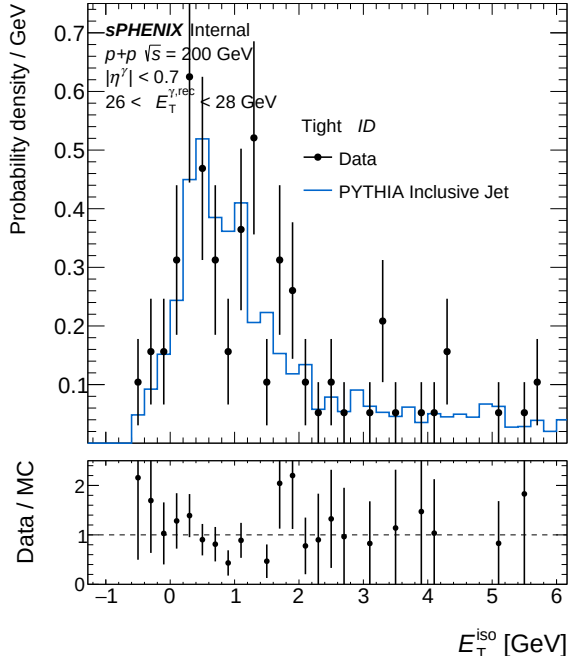
(b) IsoET Tight, pT bin 6 — bdt_nosplit



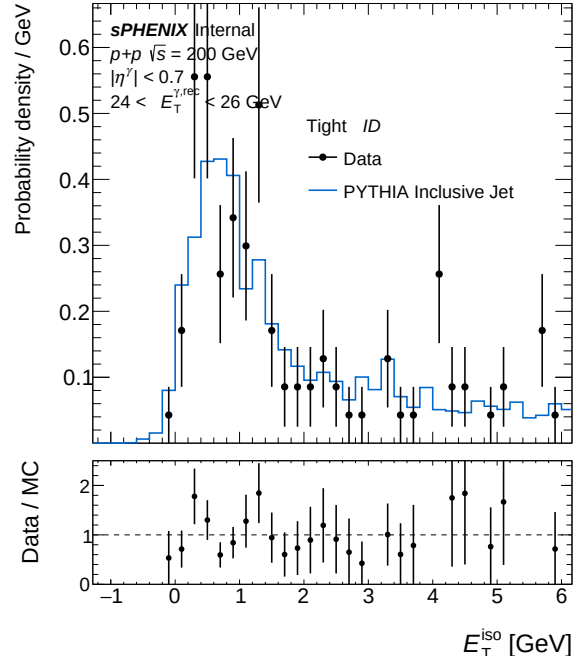
(a) IsoET Tight, pT bin 7 — bdt_nom



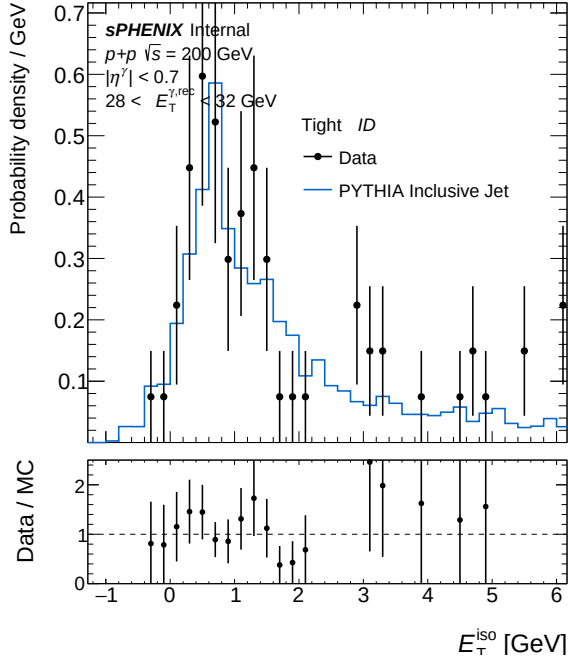
(b) IsoET Tight, pT bin 7 — bdt_nosplit



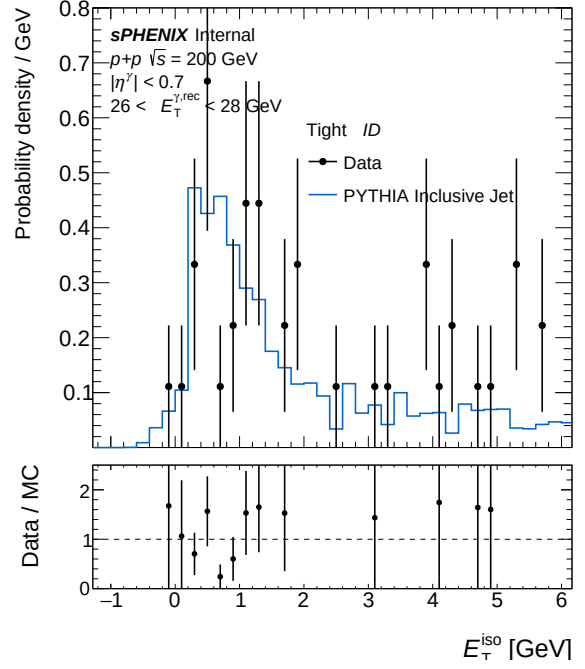
(a) IsoET Tight, pT bin 8 — bdt_nom



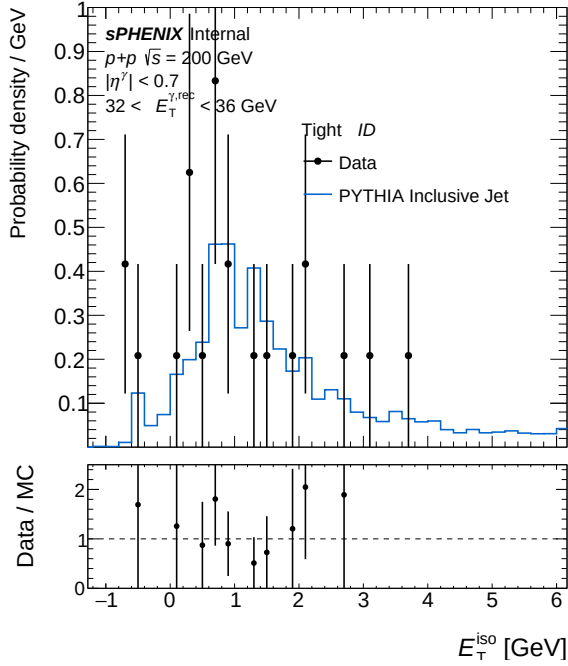
(b) IsoET Tight, pT bin 8 — bdt_nosplit



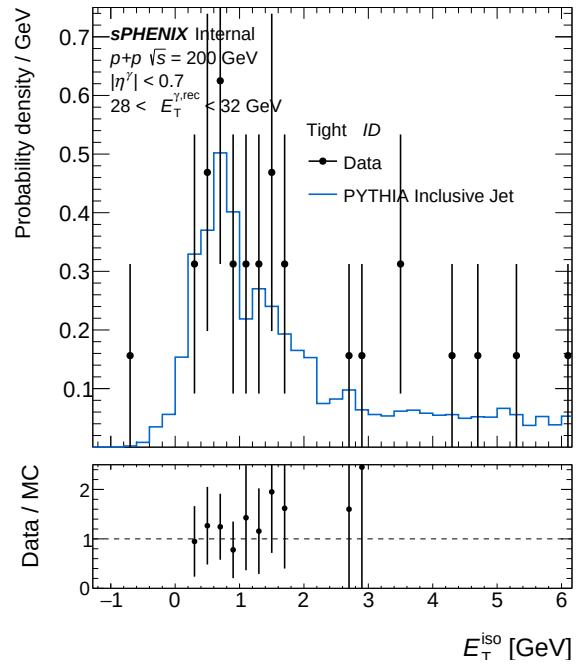
(a) IsoET Tight, pT bin 9 — bdt_nom



(b) IsoET Tight, pT bin 9 — bdt_nosplit



(a) IsoET Tight, pT bin 10 — bdt_nom



(b) IsoET Tight, pT bin 10 — bdt_nosplit

4 bdt_nom vs bdt_vtxreweight0

Config diff (A $\rightarrow$ B)

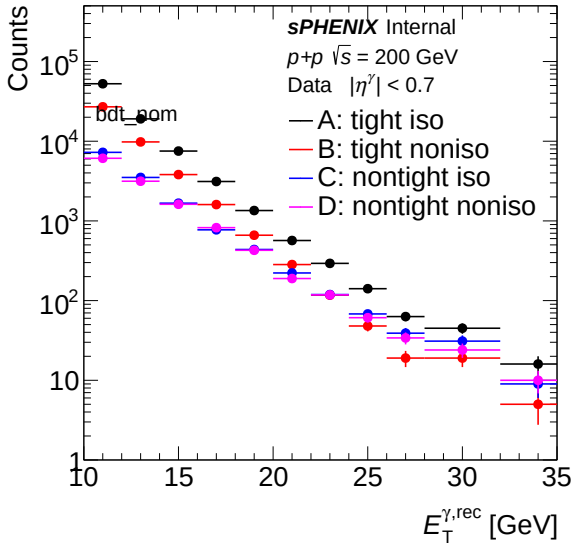
```

--- config_bdt_nom.yaml
+++ config_bdt_vtxreweight0.yaml
@@ -20,5 +20,5 @@
    response_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/MC_response"
    data_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/data_histo"
-   var_type: "bdt_nom"
+   var_type: "bdt_vtxreweight0"
    final_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/Photon.final"

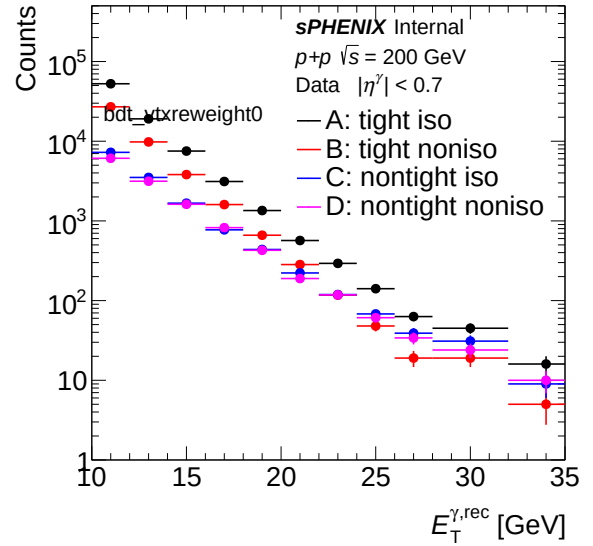
@@ -81,5 +81,5 @@
    mc_iso_shift: 0.1
    mc_iso_scale: 1.2
-   vertex_reweight_on: 1
+   vertex_reweight_on: 0
    # Path to data vertex scan output file (h_vertexz). Used for on-the-fly vertex reweighting.
    # Leave empty ("") to use the pre-made vertex_reweight_file instead.

```

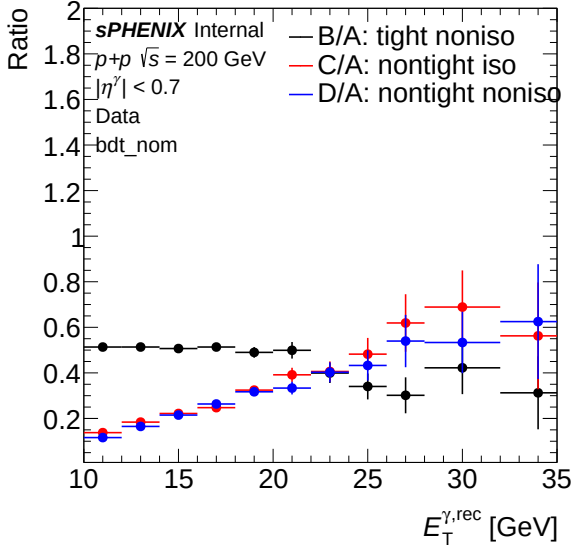
4.1 Sideband (Data)



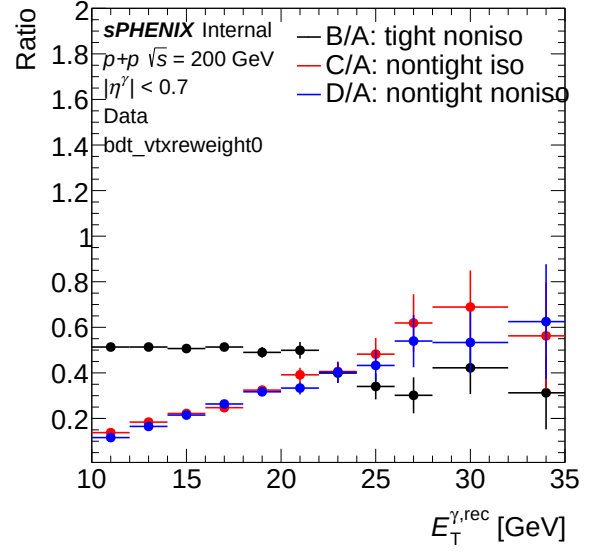
(a) Sideband ET spectrum (data) — bdt_nom



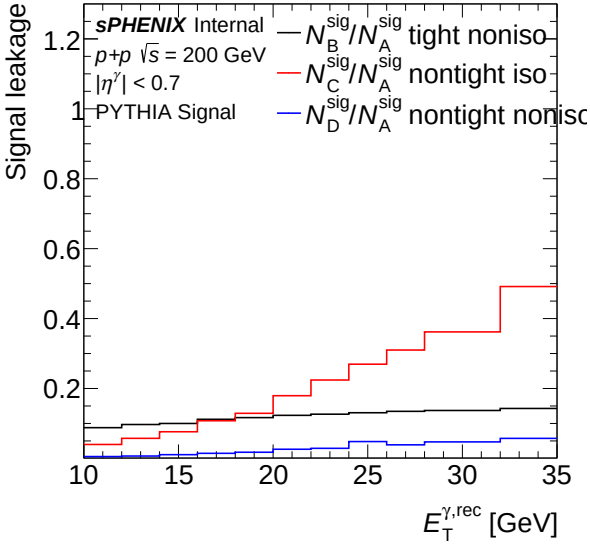
(b) Sideband ET spectrum (data) — bdt_vtxreweight0



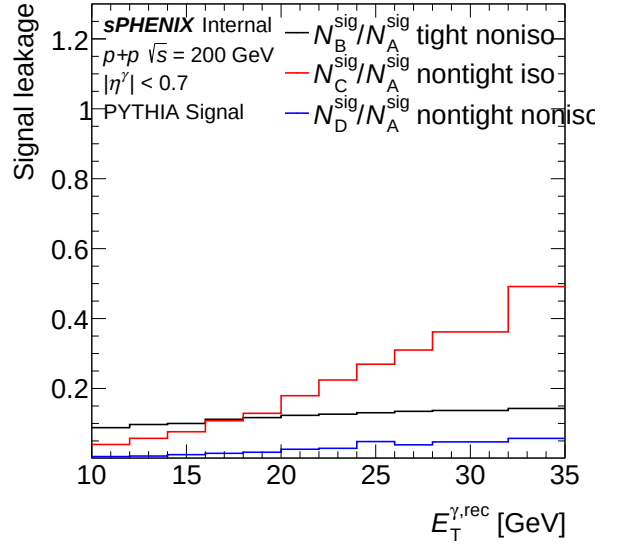
(a) Sideband ET ratio (data) — bdt_nom



(b) Sideband ET ratio (data) — bdt_vtxreweight0

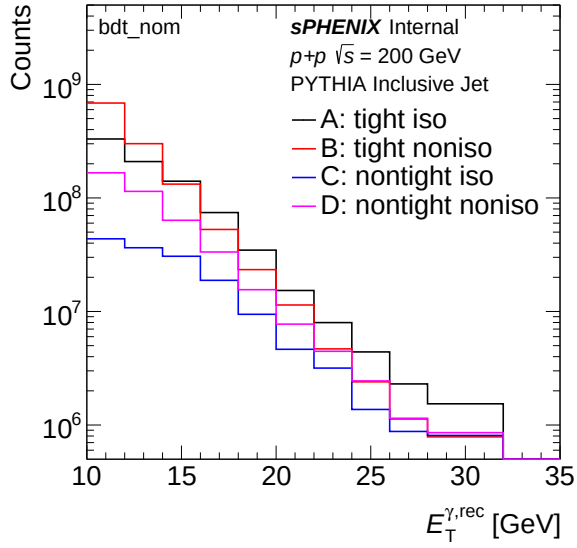
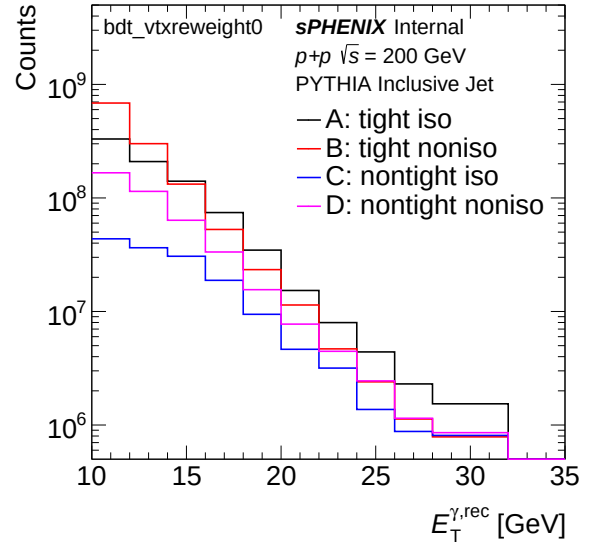
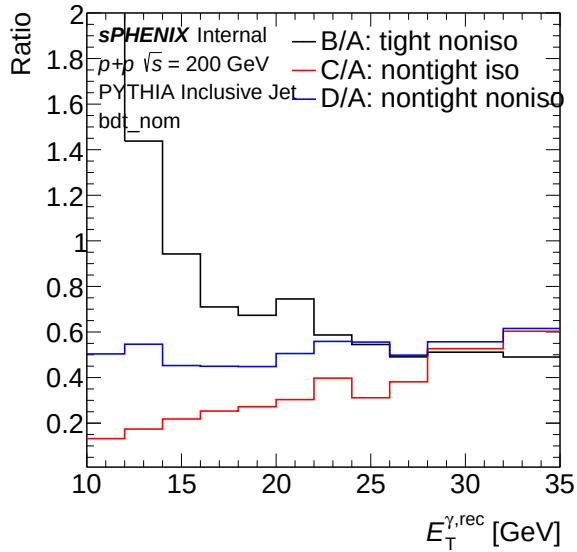
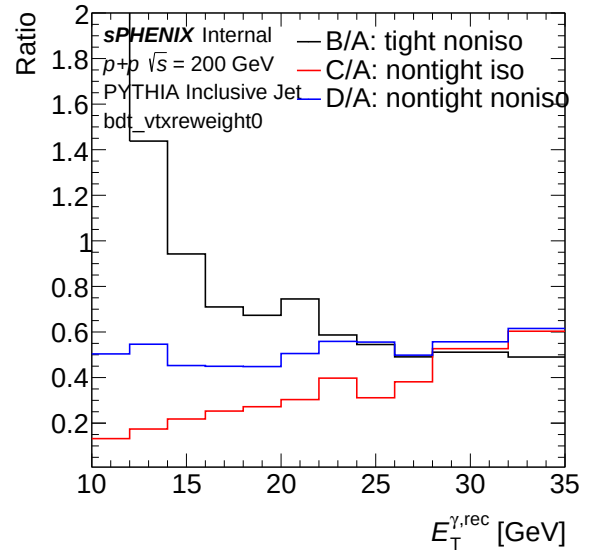


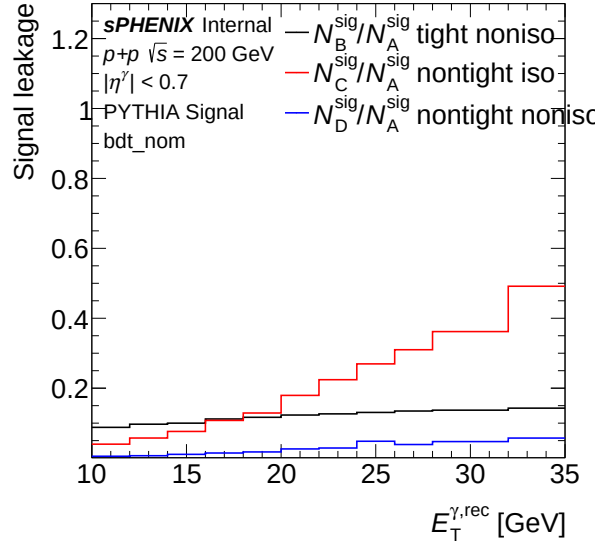
(a) Leakage fraction vs ET (data) — bdt_nom



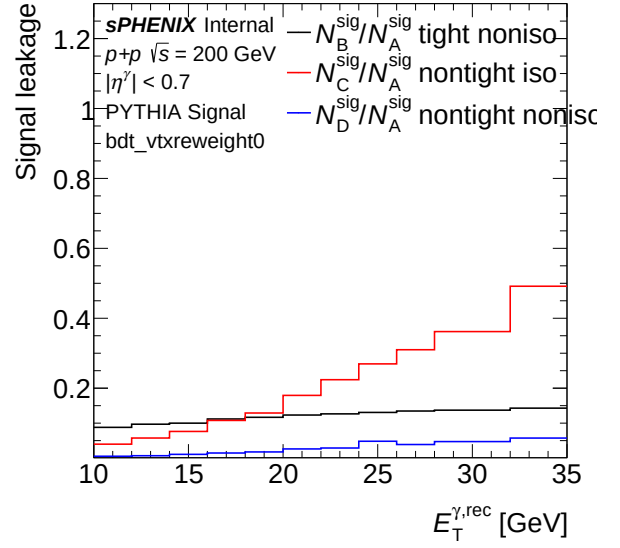
(b) Leakage fraction vs ET (data) — bdt_vtxreweight0

4.2 Sideband (MC)

(a) Sideband ET spectrum (MC) — `bdt_nom`(b) Sideband ET spectrum (MC) — `bdt_vtxreweight0`(a) Sideband ET ratio (MC) — `bdt_nom`(b) Sideband ET ratio (MC) — `bdt_vtxreweight0`

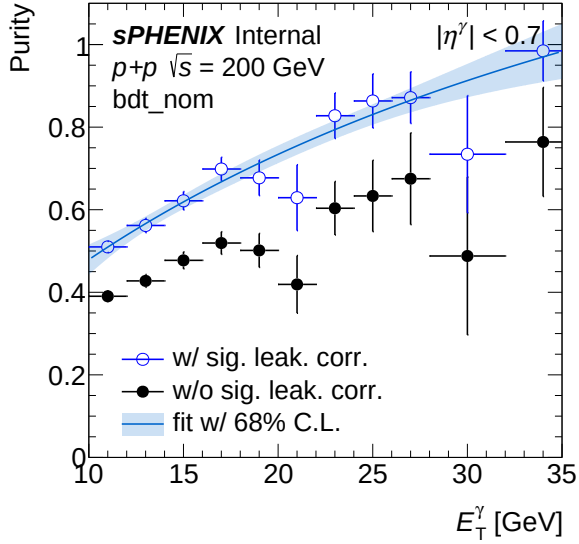


(a) Leakage fraction vs ET (MC) — bdt_nom

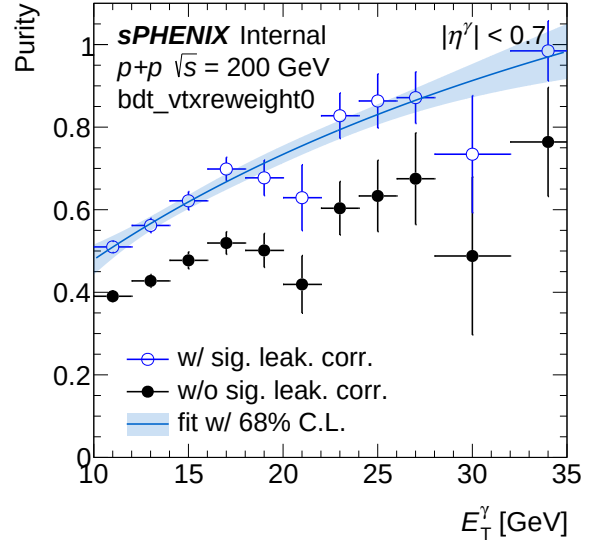


(b) Leakage fraction vs ET (MC) — bdt_vtxreweight0

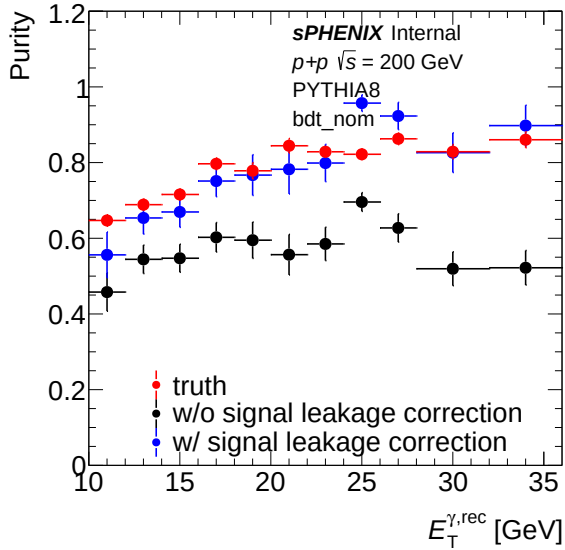
4.3 Purity



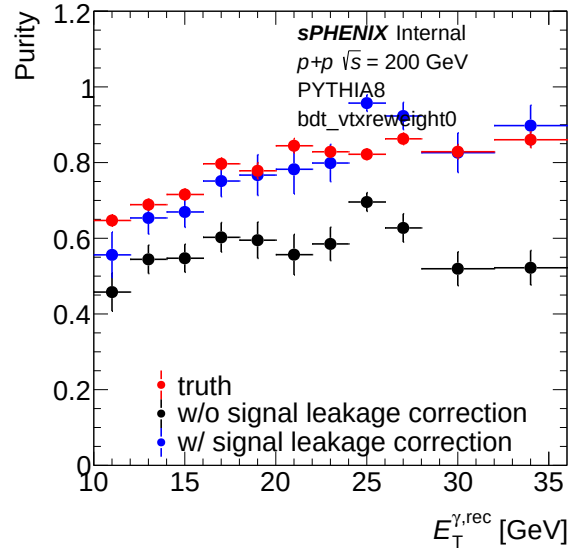
(a) Purity (data) — bdt_nom



(b) Purity (data) — bdt_vtxreweight0

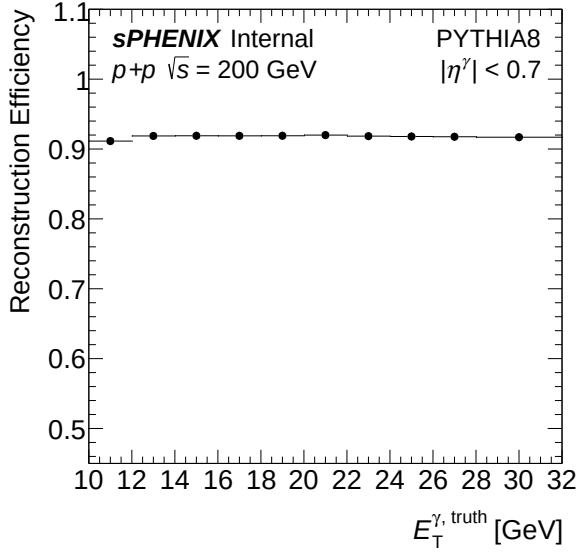


(a) Purity (MC) — bdt_nom

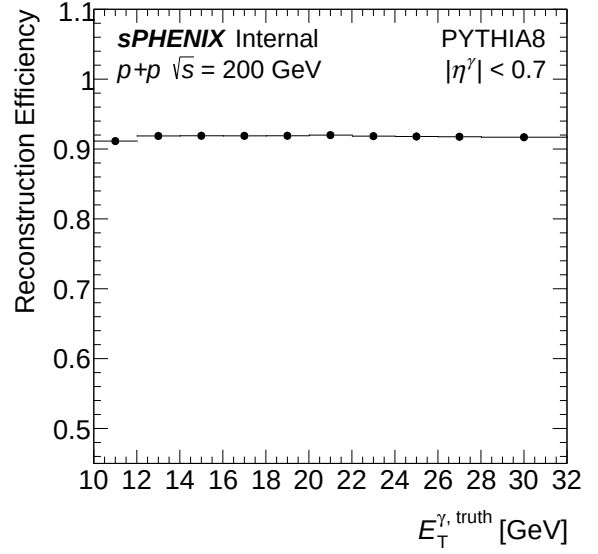


(b) Purity (MC) — bdt_vtxreweight0

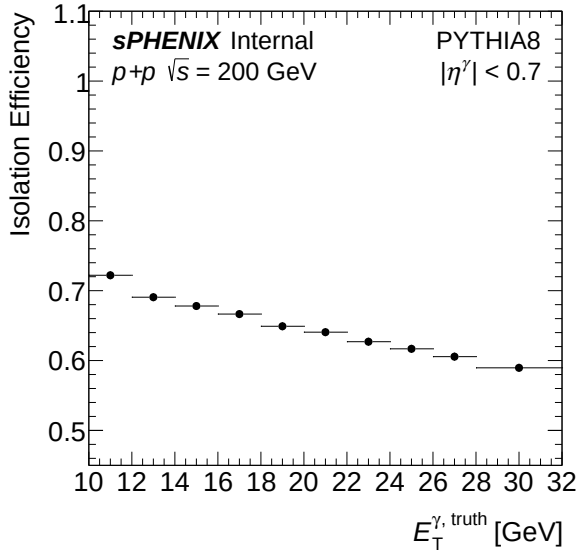
4.4 Efficiency



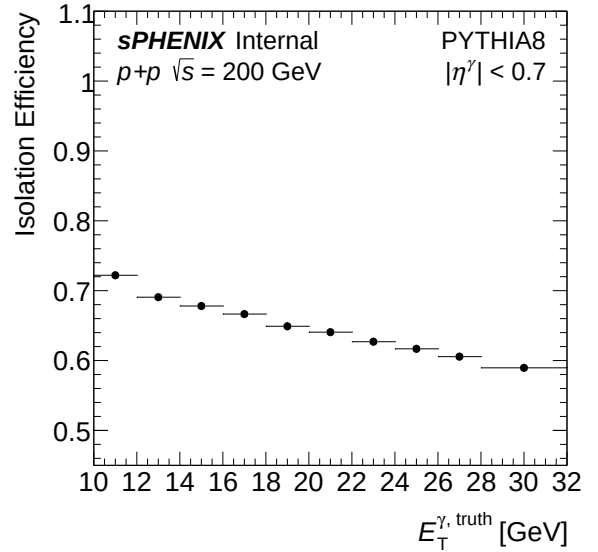
(a) Reconstruction efficiency — bdt_nom



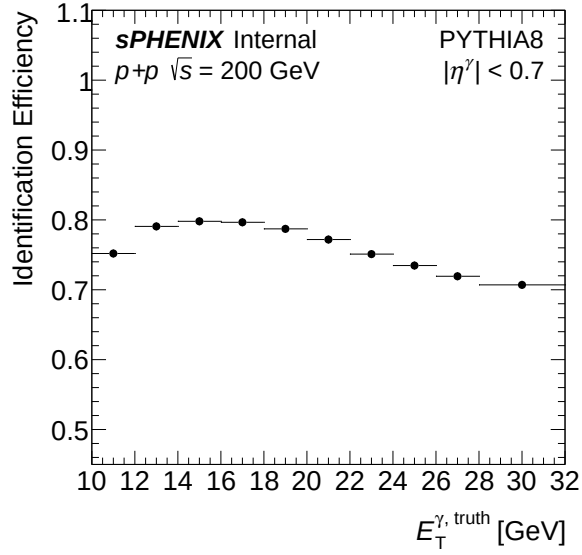
(b) Reconstruction efficiency — bdt_vtxreweight0



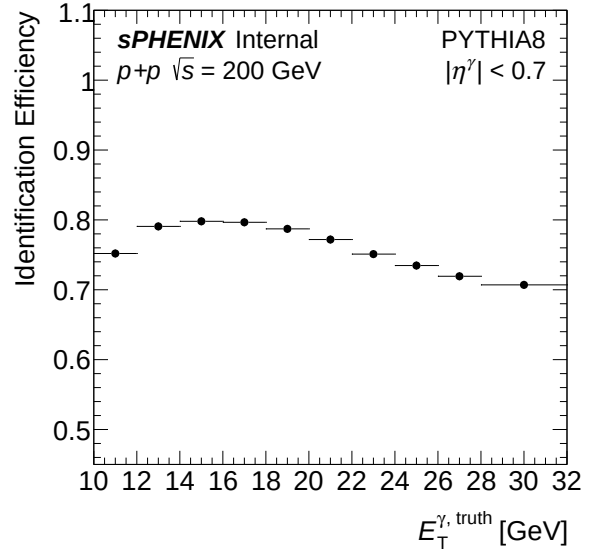
(a) Isolation efficiency — bdt_nom



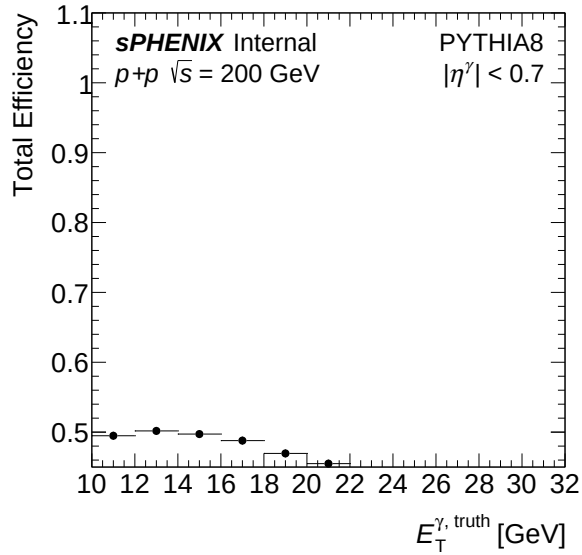
(b) Isolation efficiency — bdt_vtxreweight0



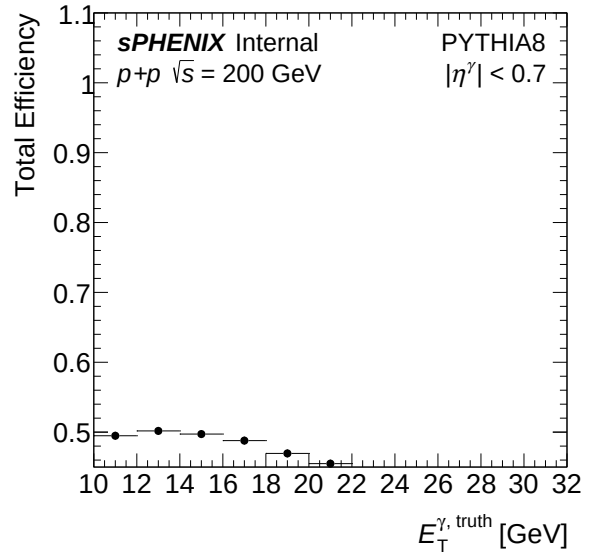
(a) Identification efficiency — bdt_nom



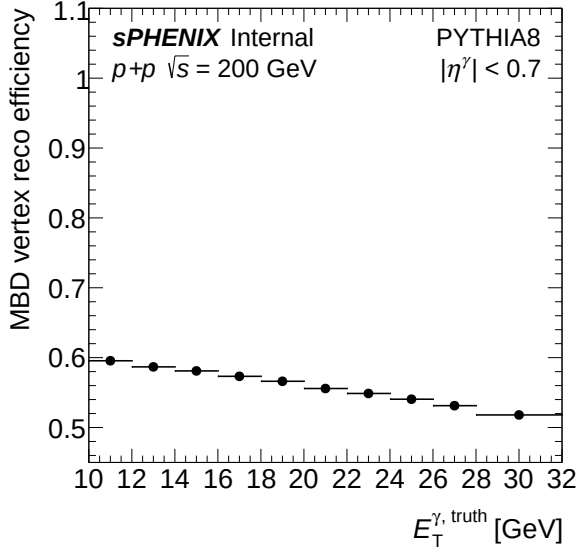
(b) Identification efficiency — bdt_vtxreweight0



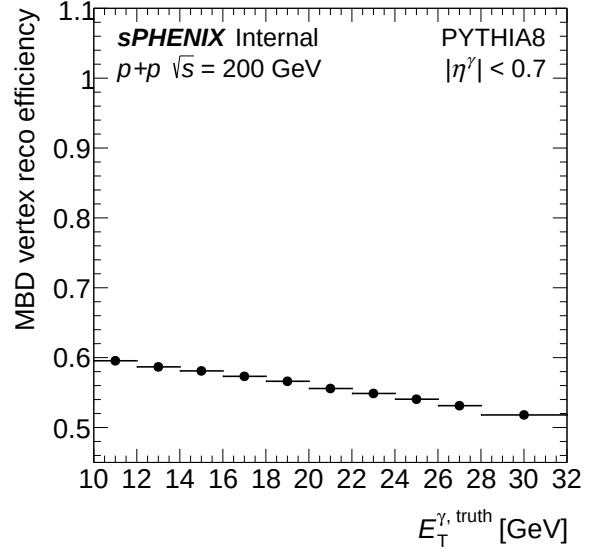
(a) Total efficiency — bdt_nom



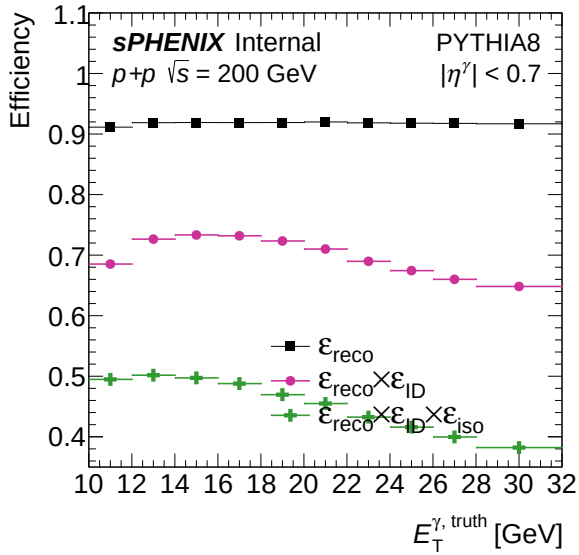
(b) Total efficiency — bdt_vtxreweight0



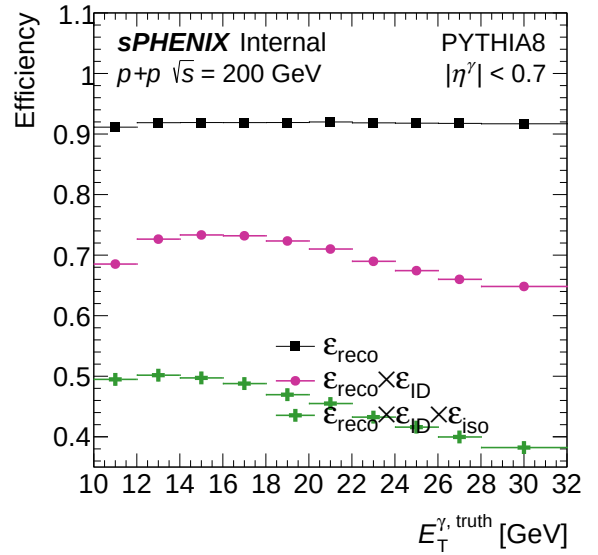
(a) MBD trigger efficiency — bdt_nom



(b) MBD trigger efficiency — bdt_vtxreweight0

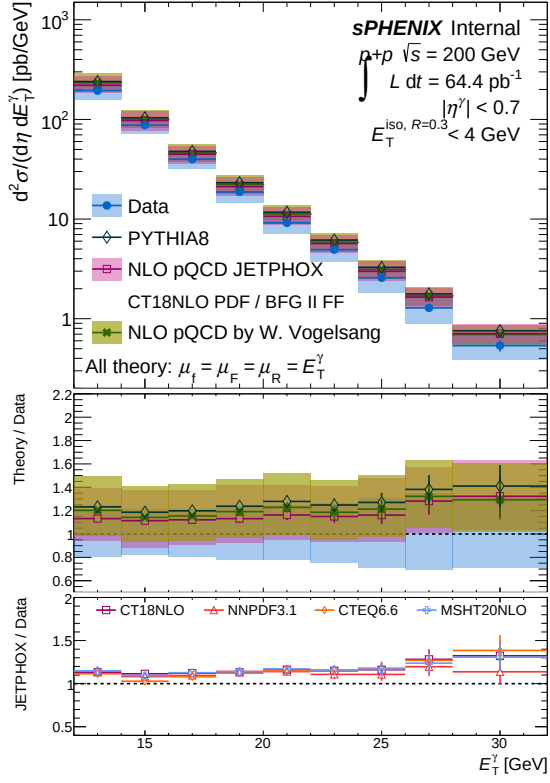
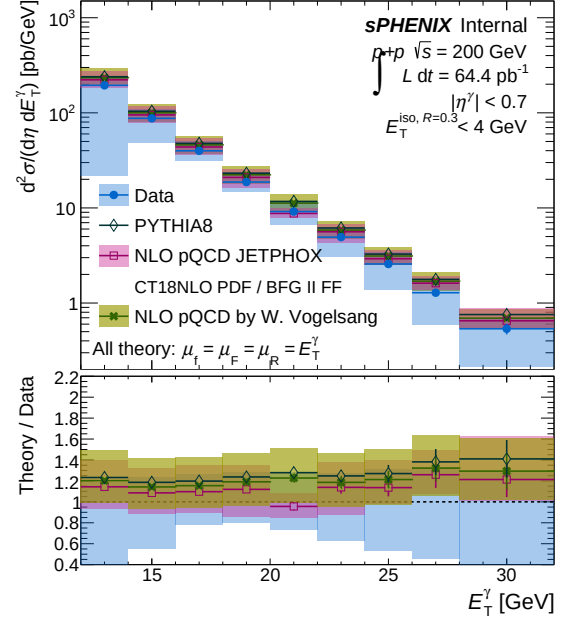


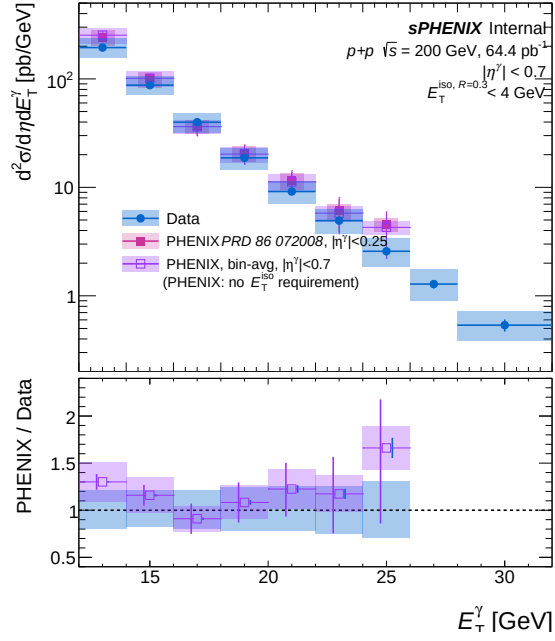
(a) Combined photon efficiency — bdt_nom



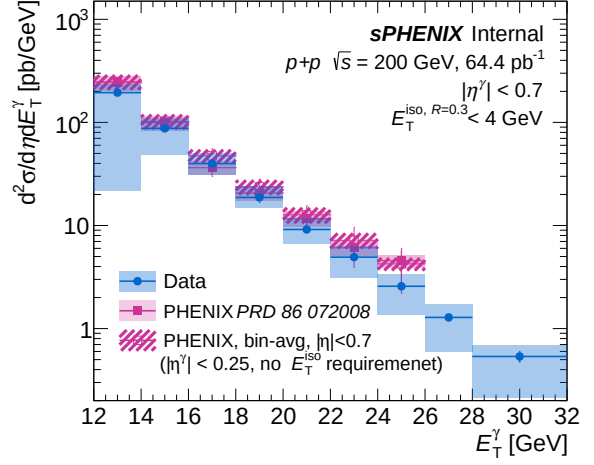
(b) Combined photon efficiency — bdt_vtxreweight0

4.5 Final result

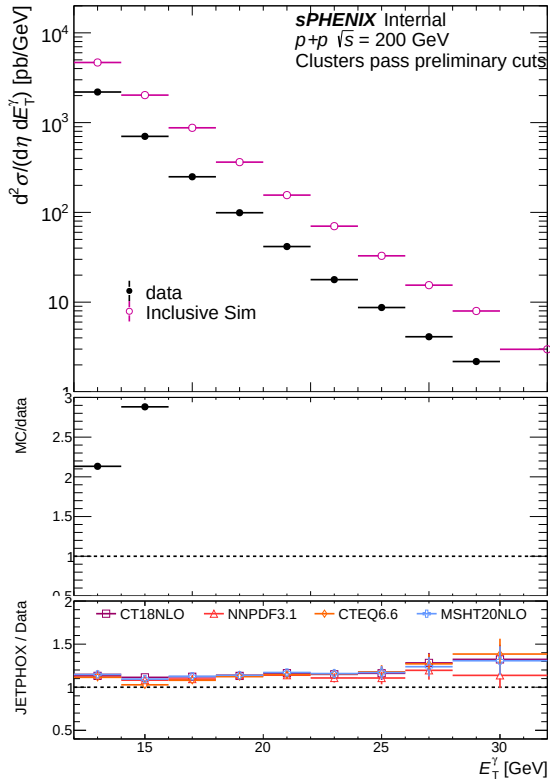
(a) Final cross section result — `bdt_nom`(b) Final cross section result — `bdt_vtxreweight0`



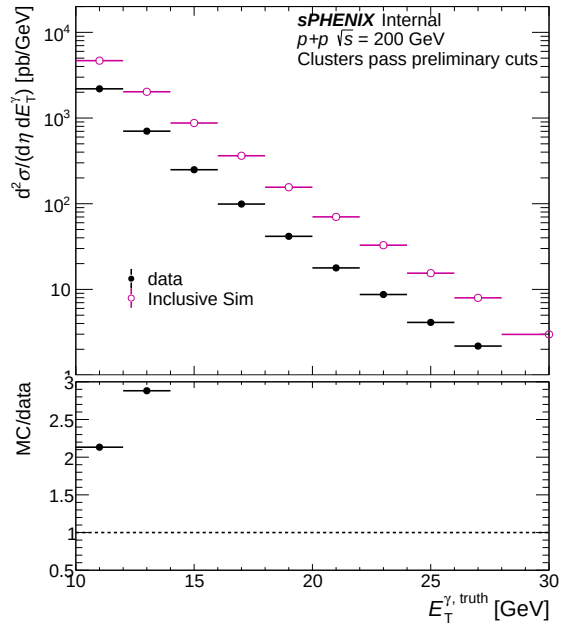
(a) Comparison with PHENIX — bdt_nom



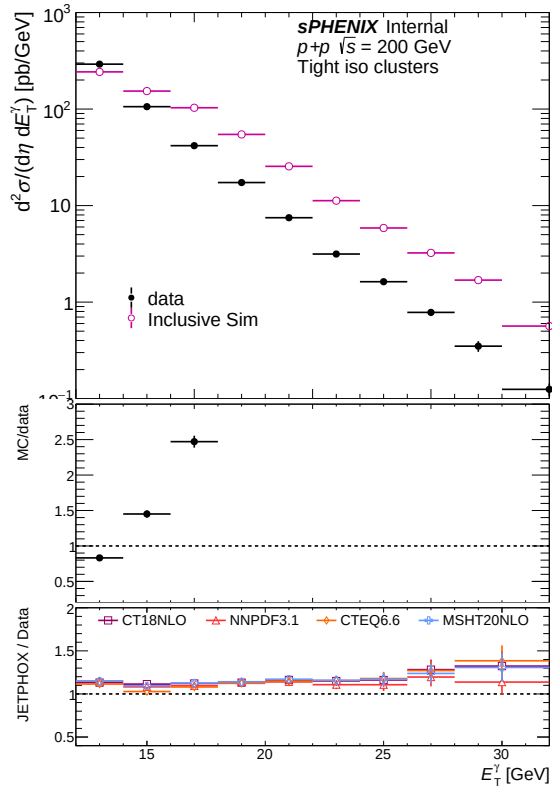
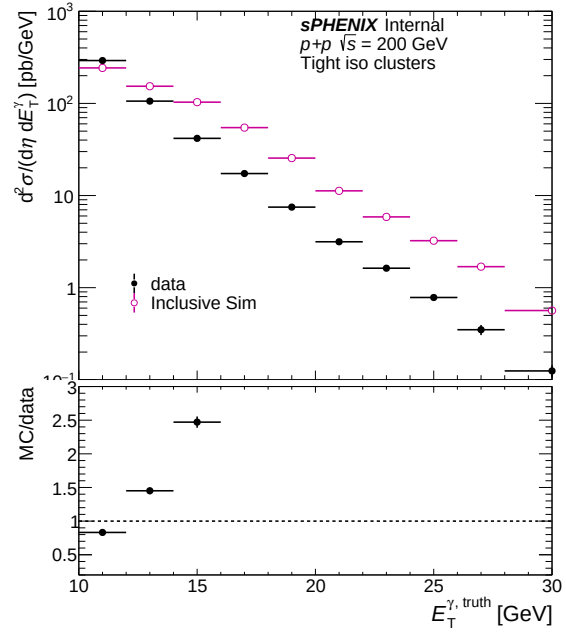
(b) Comparison with PHENIX — bdt_vtxreweight0

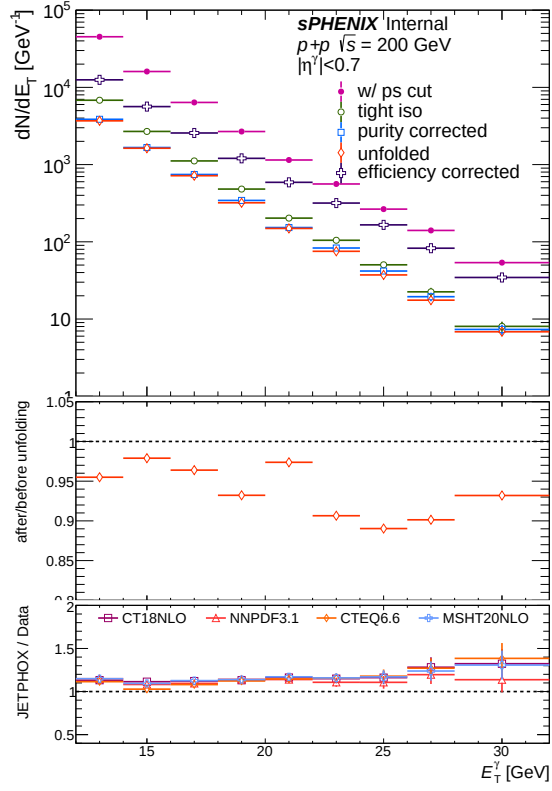


(a) Common cluster yield (data vs MC) — bdt_nom

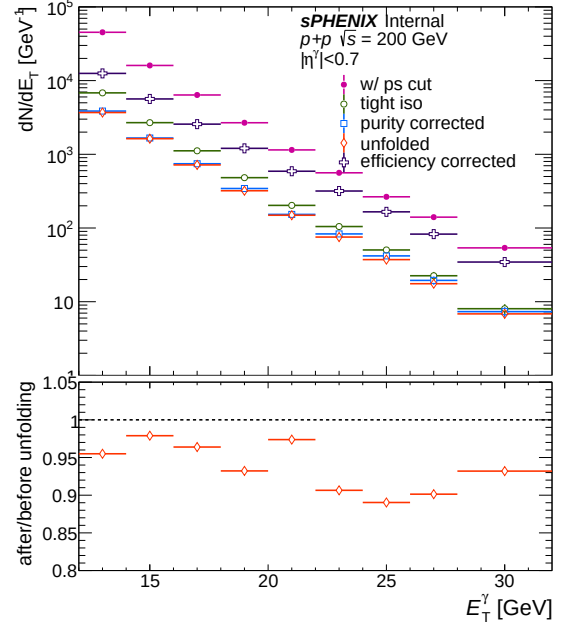


(b) Common cluster yield (data vs MC) — bdt_vtxreweight0

(a) Tight iso cluster yield (data vs MC) — `bdt_nom`(b) Tight iso cluster yield (data vs MC) — `bdt_vtxreweight0`

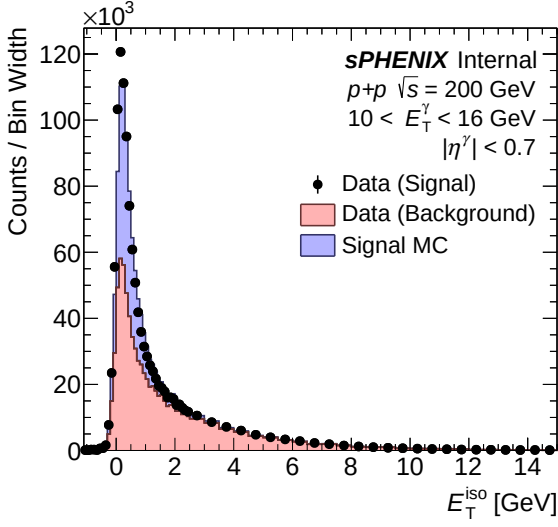


(a) All correction steps — bdt_nom

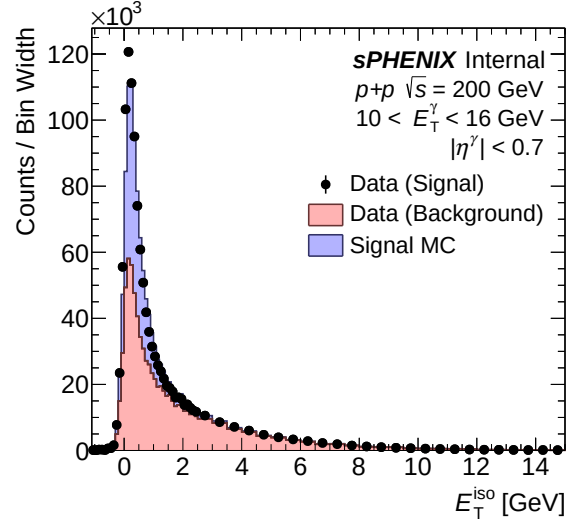


(b) All correction steps — bdt_vtxreweight0

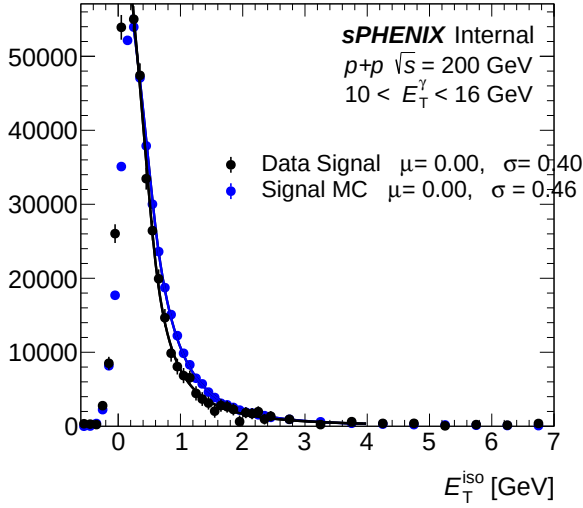
4.6 Isolation fit (CONF)



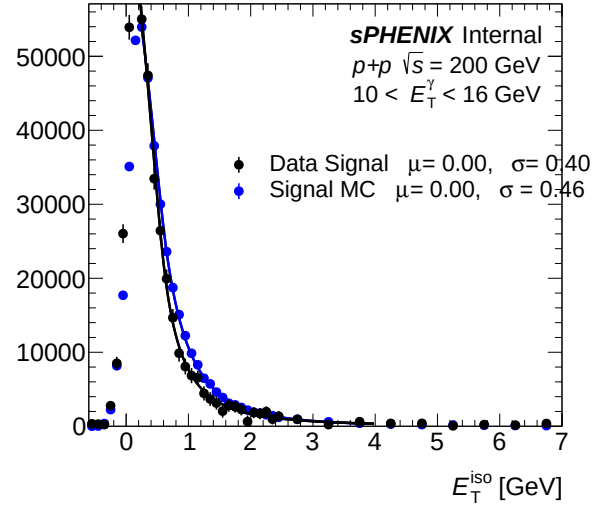
(a) IsoET template (pT bins 0-2) — bdt_nom



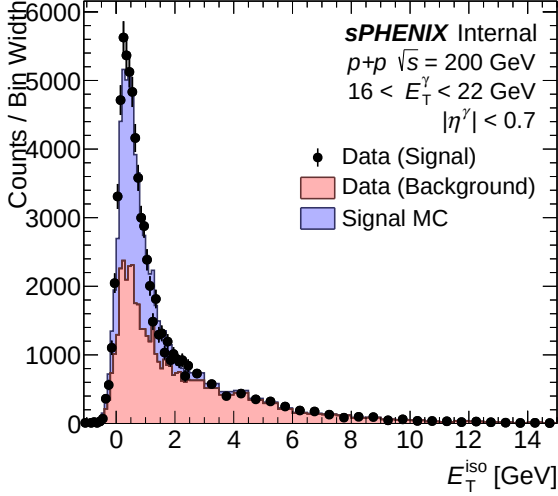
(b) IsoET template (pT bins 0-2) — bdt_vtxreweight0



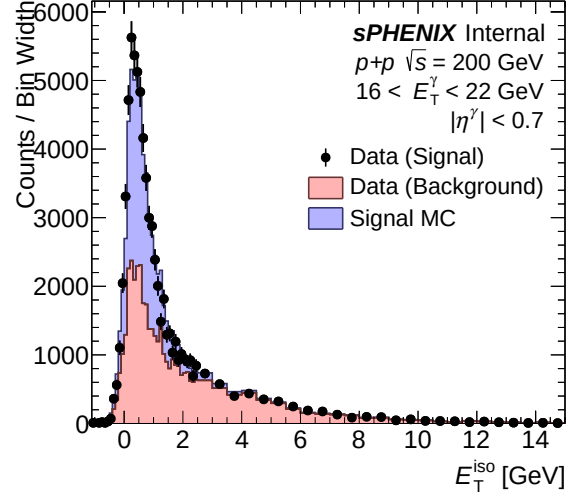
(a) IsoET fit template (pT bins 0-2) — bdt_nom



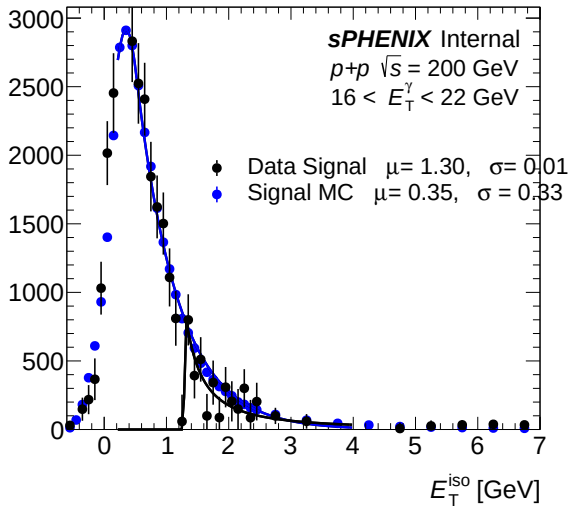
(b) IsoET fit template (pT bins 0-2) — bdt_vtxreweight0



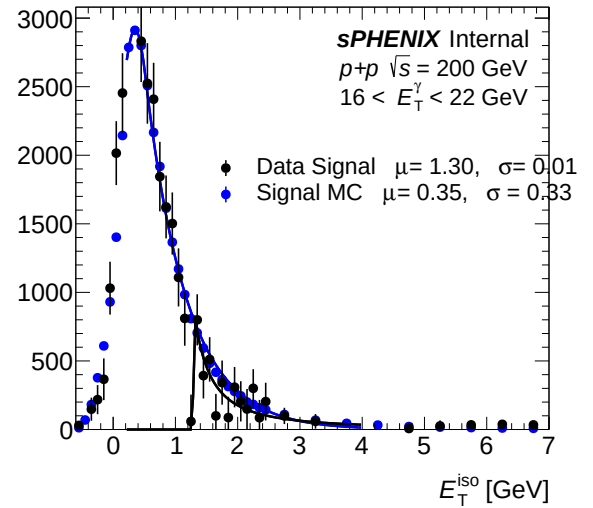
(a) IsoET template (pT bins 3–5) — bdt_nom



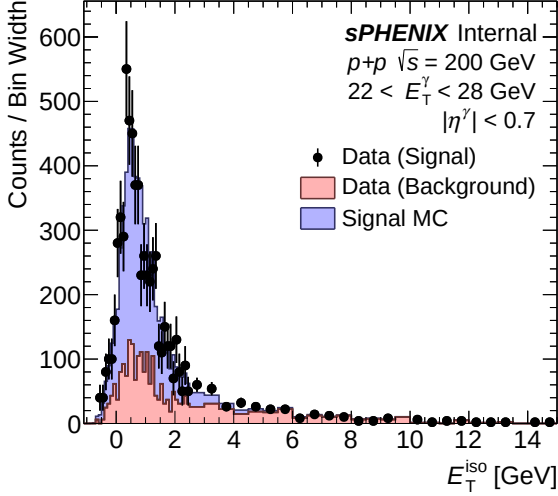
(b) IsoET template (pT bins 3–5) — bdt_vtxreweight0



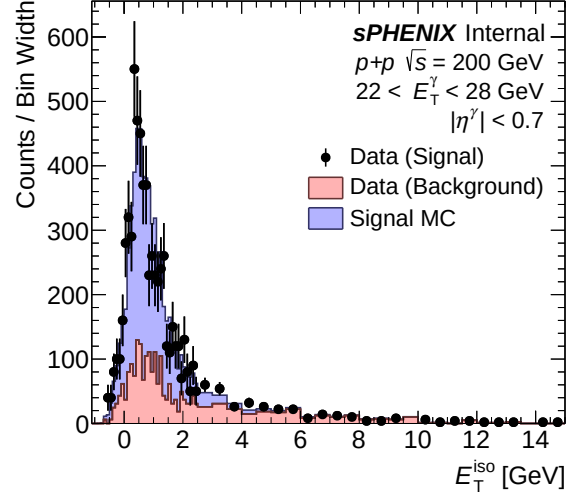
(a) IsoET fit template (pT bins 3–5) — bdt_nom



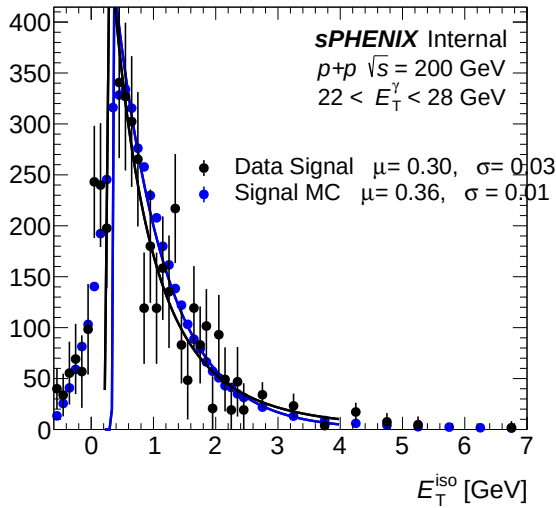
(b) IsoET fit template (pT bins 3–5) — bdt_vtxreweight0



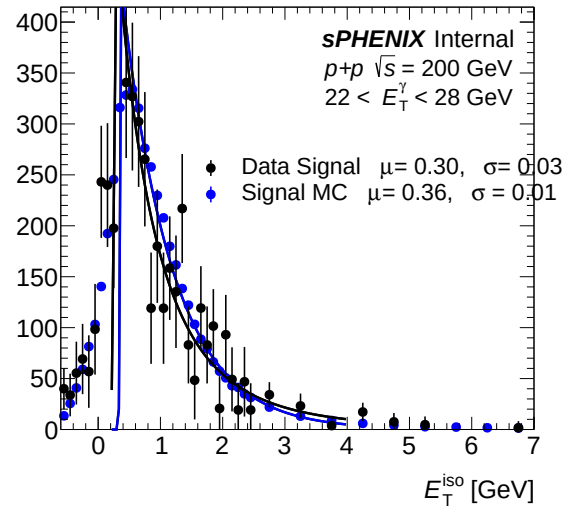
(a) IsoET template (pT bins 6–8) — bdt_nom



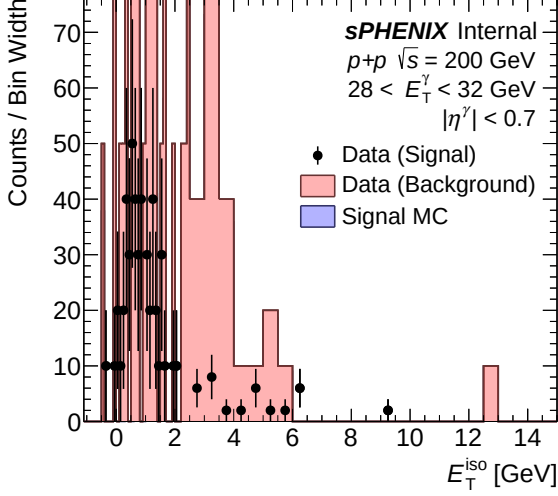
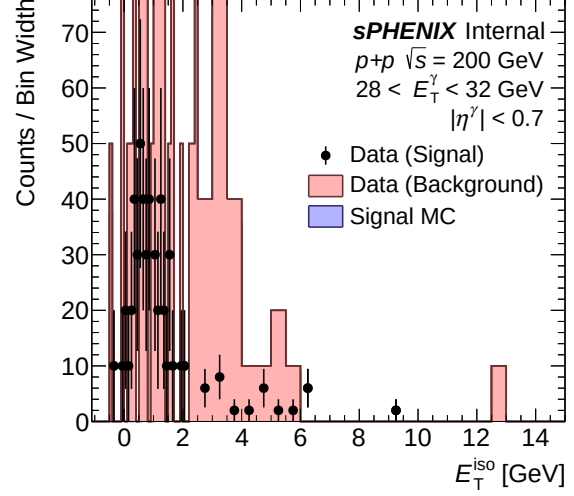
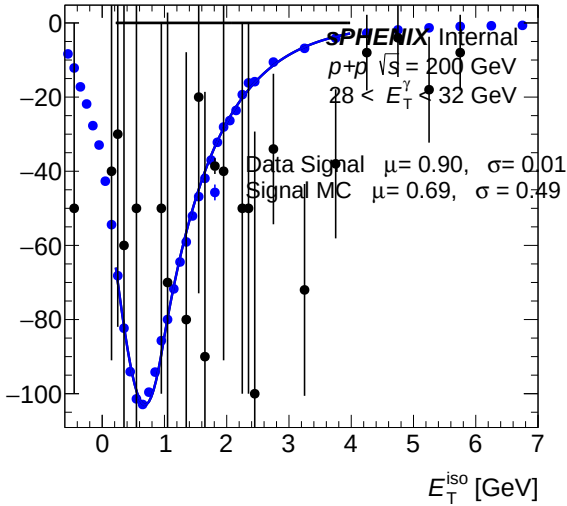
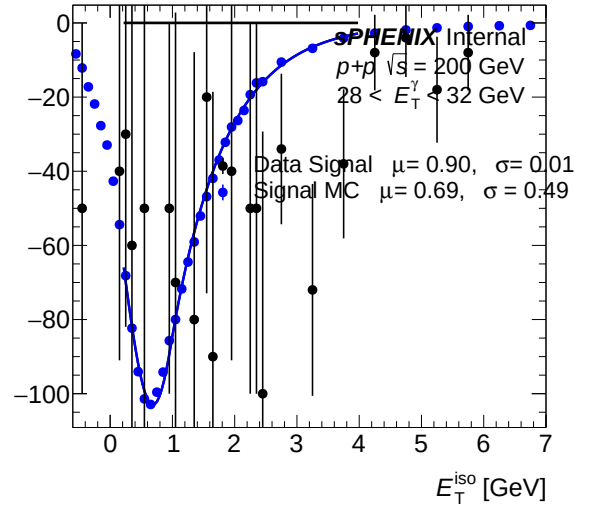
(b) IsoET template (pT bins 6–8) — bdt_vtxreweight0



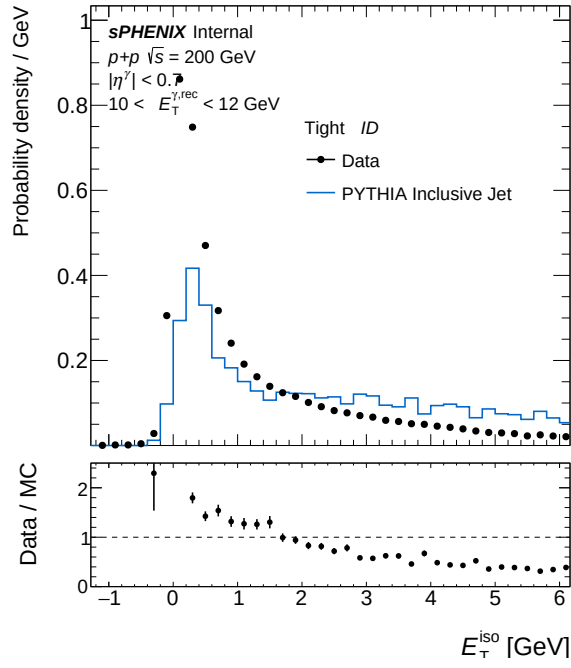
(a) IsoET fit template (pT bins 6–8) — bdt_nom



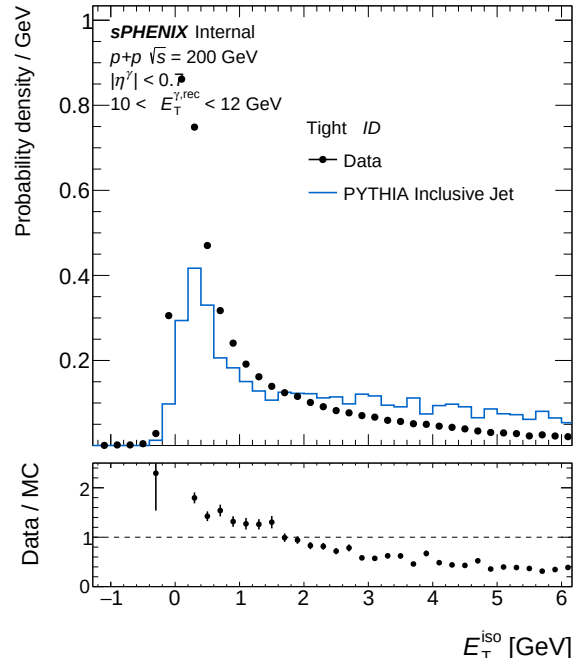
(b) IsoET fit template (pT bins 6–8) — bdt_vtxreweight0

(a) IsoET template (pT bins 9–9) — `bdt_nom`(b) IsoET template (pT bins 9–9) — `bdt_vtxreweight0`(a) IsoET fit template (pT bins 9–9) — `bdt_nom`(b) IsoET fit template (pT bins 9–9) — `bdt_vtxreweight0`

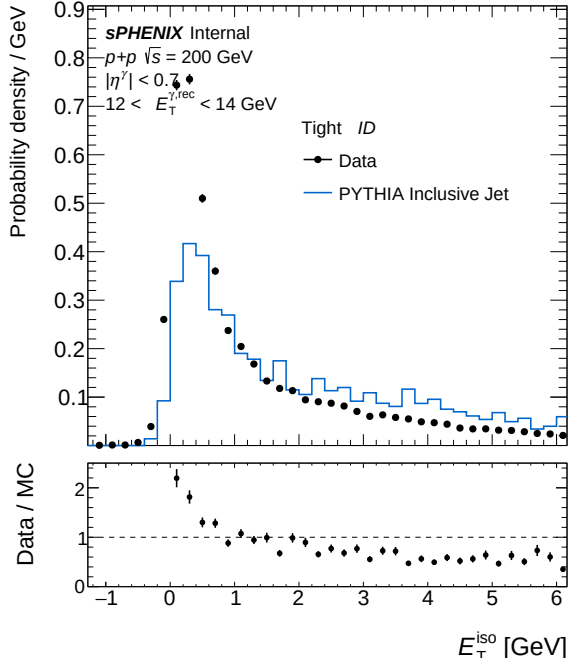
4.7 Isolation ET



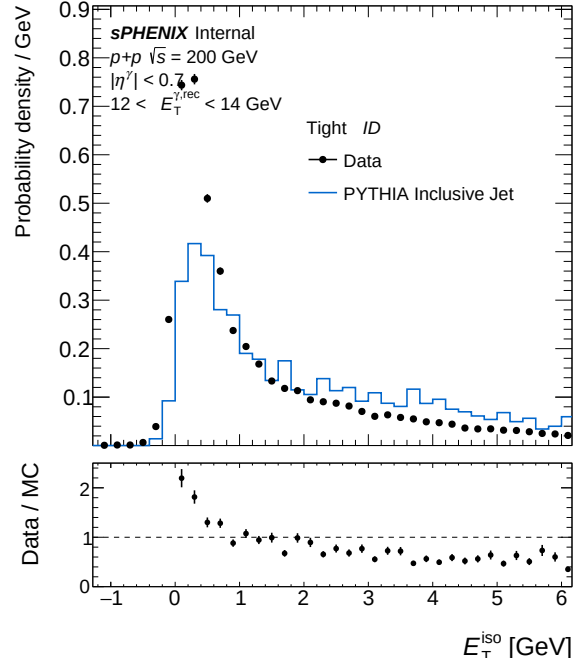
(a) IsoET Tight, pT bin 0 — bdt_nom



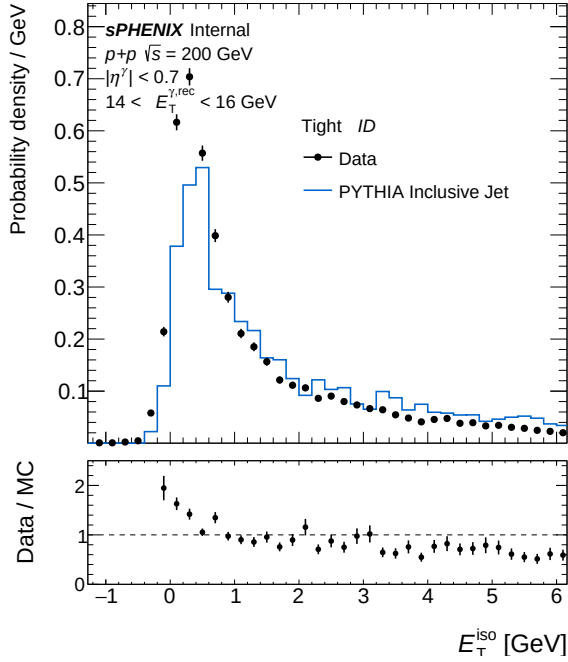
(b) IsoET Tight, pT bin 0 — bdt_vtxreweight0



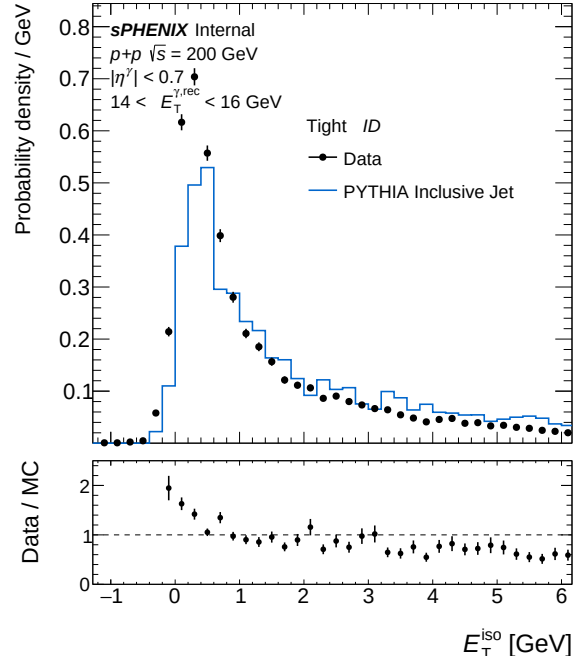
(a) IsoET Tight, pT bin 1 — bdt_nom



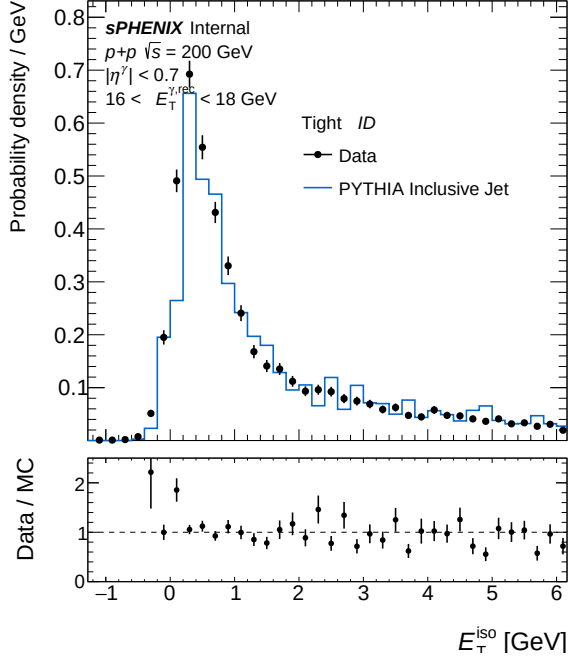
(b) IsoET Tight, pT bin 1 — bdt_vtxreweight0



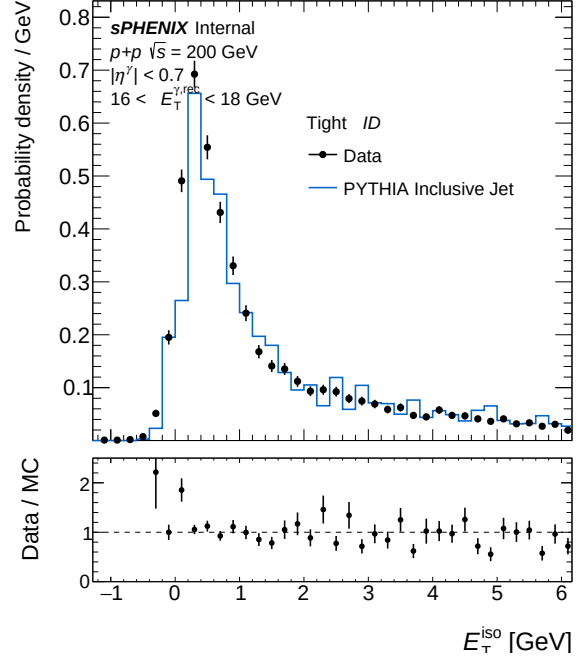
(a) IsoET Tight, pT bin 2 — bdt_nom



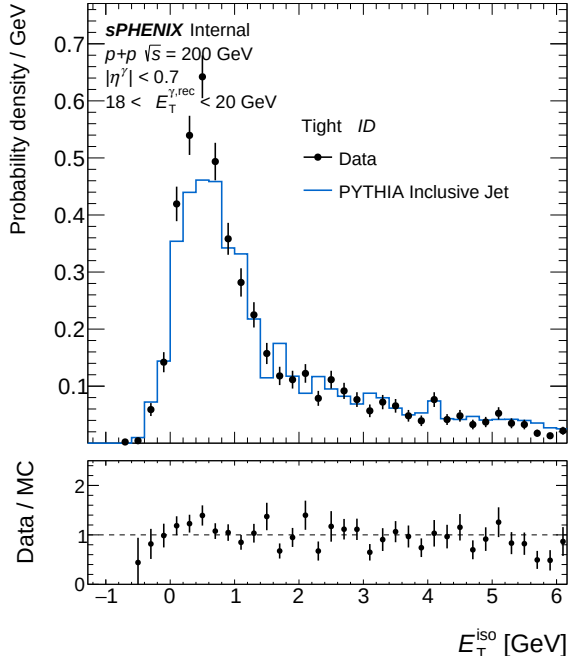
(b) IsoET Tight, pT bin 2 — bdt_vtxreweight0



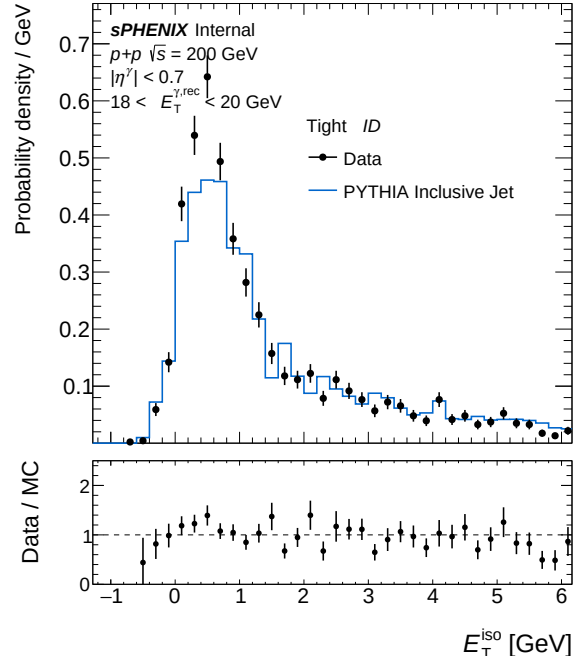
(a) IsoET Tight, pT bin 3 — bdt_nom



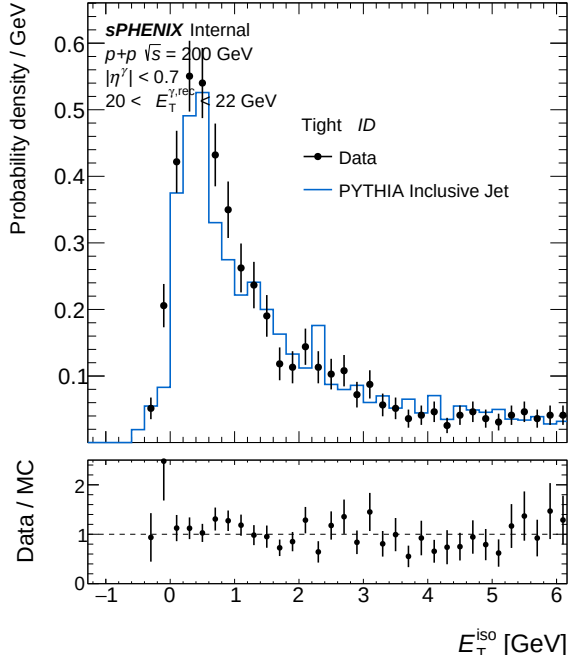
(b) IsoET Tight, pT bin 3 — bdt_vtxreweight0



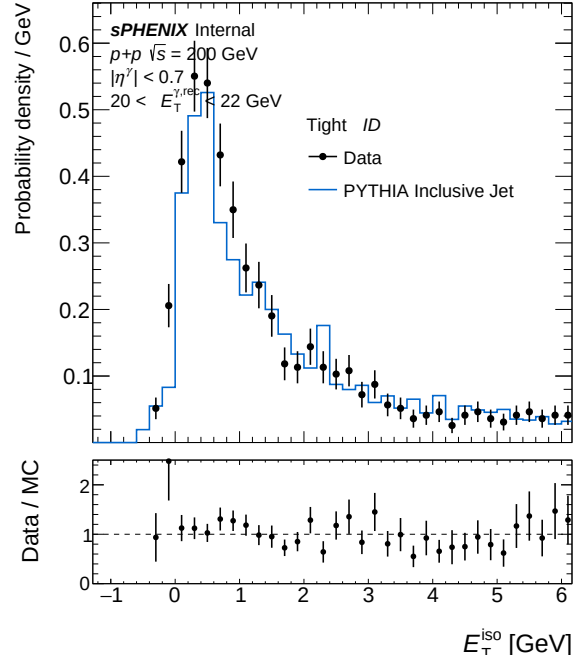
(a) IsoET Tight, pT bin 4 — bdt_nom



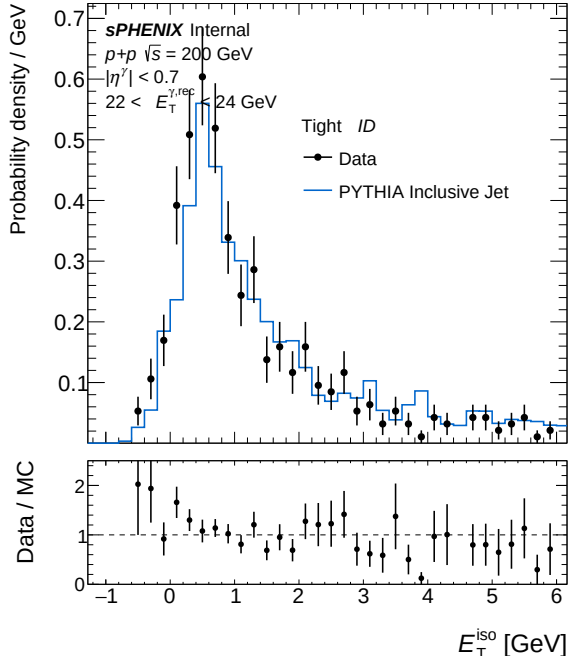
(b) IsoET Tight, pT bin 4 — bdt_vtxreweight0



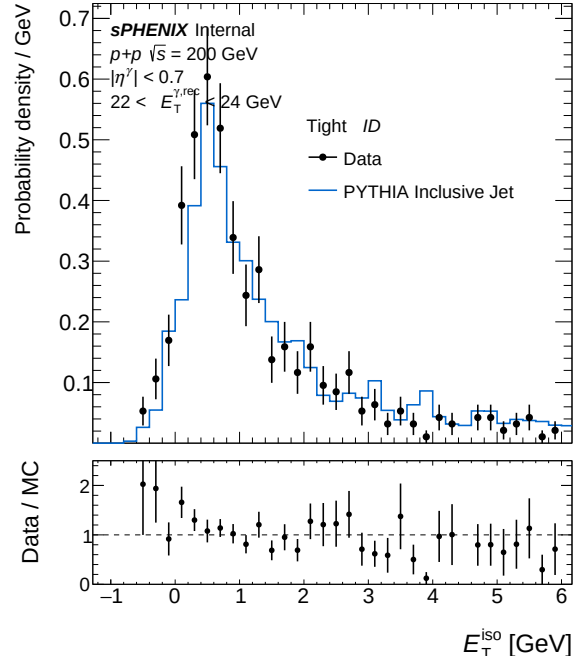
(a) IsoET Tight, pT bin 5 — bdt_nom



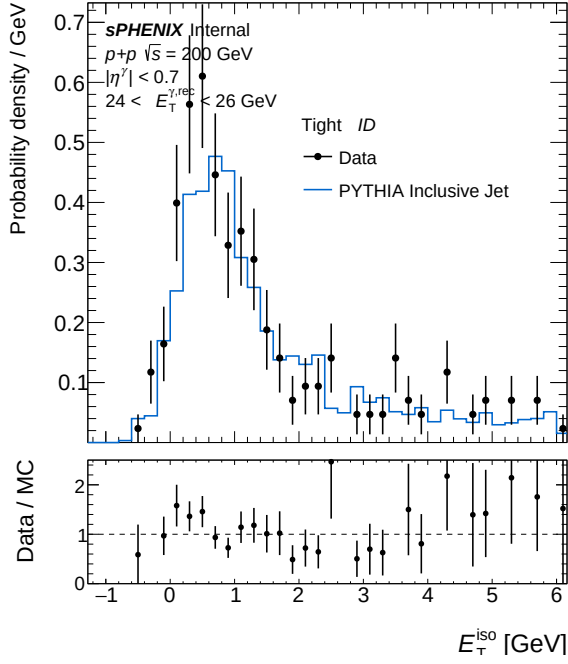
(b) IsoET Tight, pT bin 5 — bdt_vtxreweight0



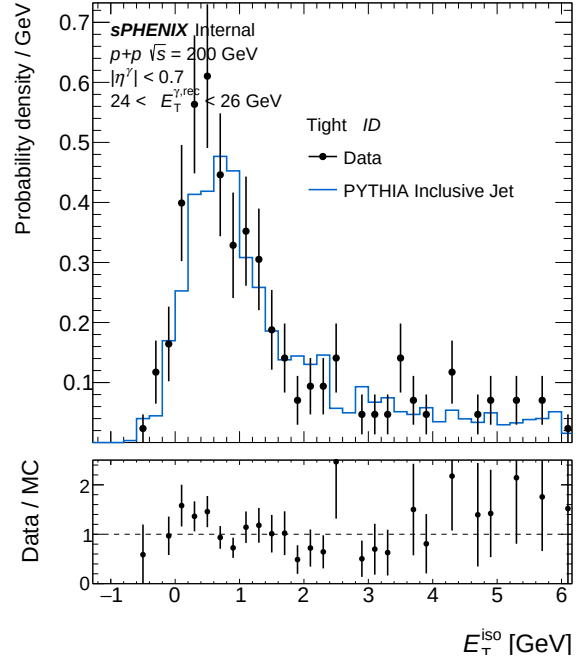
(a) IsoET Tight, pT bin 6 — bdt_nom



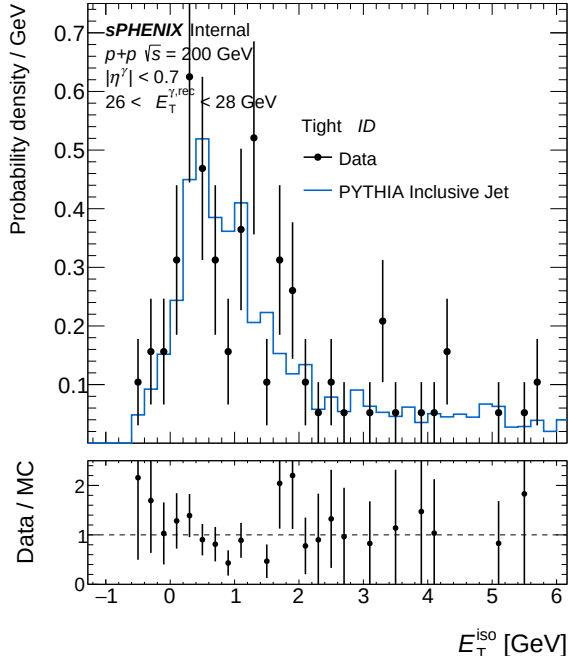
(b) IsoET Tight, pT bin 6 — bdt_vtxreweight0



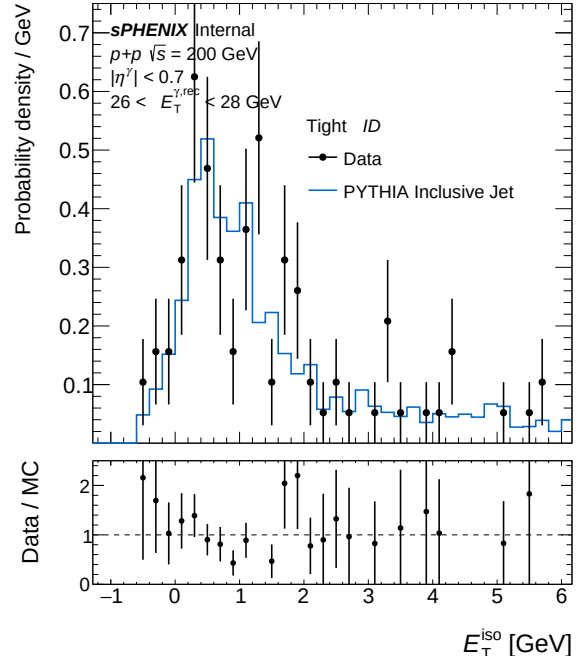
(a) IsoET Tight, pT bin 7 — bdt_nom



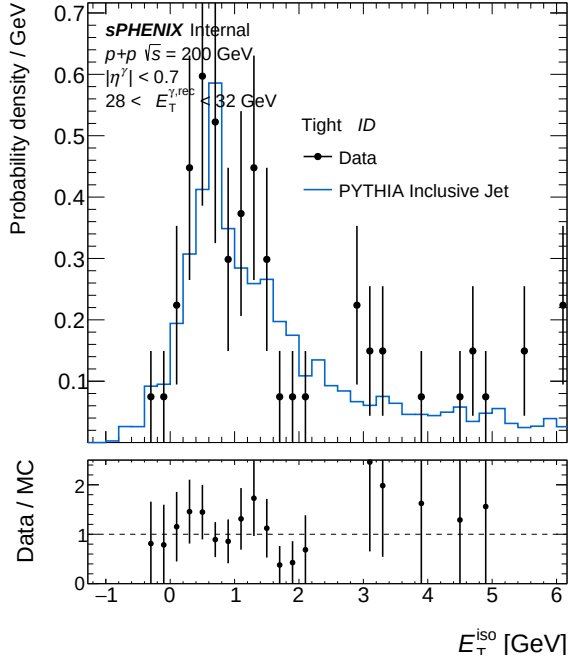
(b) IsoET Tight, pT bin 7 — bdt_vtxreweight0



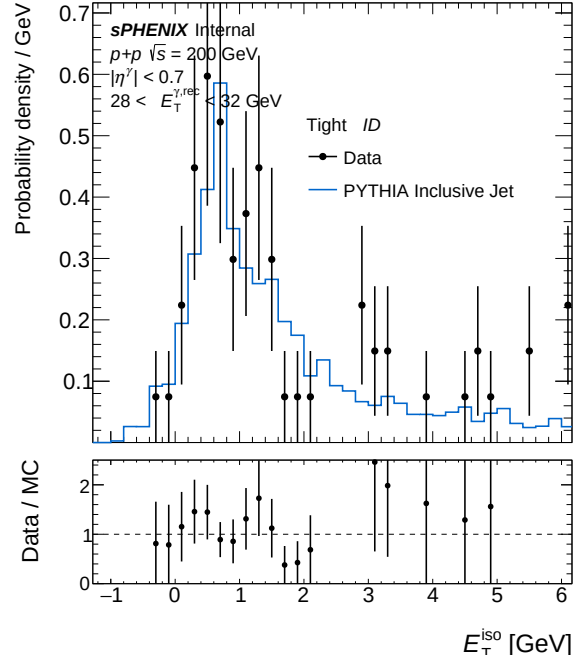
(a) IsoET Tight, pT bin 8 — bdt_nom



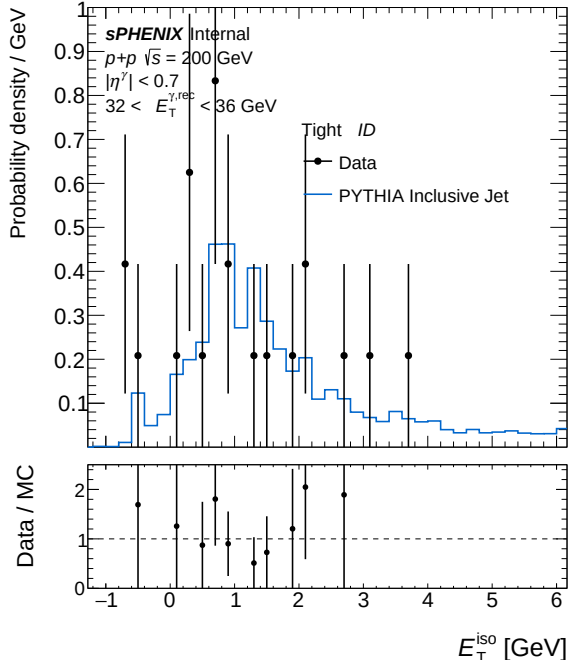
(b) IsoET Tight, pT bin 8 — bdt_vtxreweight0



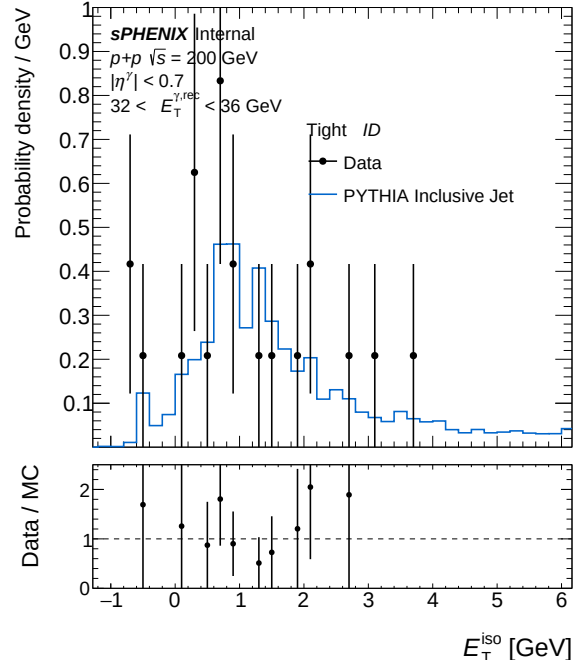
(a) IsoET Tight, pT bin 9 — bdt_nom



(b) IsoET Tight, pT bin 9 — bdt_vtxreweight0



(a) IsoET Tight, pT bin 10 — bdt_nom



(b) IsoET Tight, pT bin 10 — bdt_vtxreweight0

5 bdt_nom vs bdt_none

Config diff (A → B)

```

--- config_bdt_nom.yaml
+++ config_bdt_none.yaml
@@ -1,4 +1,6 @@
  #config.yaml
-#bdt nominal
+#bdt_none: legacy shower-shape photon ID (no BDT). Photon-ID cuts copied from
+#config_backup/config_bdt_none.yaml (pre-pipeline-refresh archive); all other
+#settings inherited from config_bdt_nom.yaml. All-range merge target.
  input:
    photon_jet_file_root_dir: "/sphenix/user/shuhangli/ppg12/FunWithxgboost/"
@@ -20,5 +22,5 @@
    response.outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/MC.response"
    data_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/data_histo"
-  var_type: "bdt_nom"
+  var_type: "bdt_none"
    final.outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/Photon.final"

@@ -63,4 +65,7 @@
  iso_hcalonly: 0
  iso_emcalinnerr: 0.0
+  # Legacy shower-shape selection: tight = pass all 5 cuts (weta, wphi, et1,
+  # e11/e33, e32/e35); non-tight = >=1 of those 5 fail at the looser non_tight
+  # cut values.
  nnt_fail: 1
  weta_fail: 0
@@ -71,12 +76,12 @@
  bdt_fail: 0
  weta_on: 1
-  wphi_on: 0
-  et1_on: 0
-  e11_to_e33_on: 0
-  e32_to_e35_on: 0
+  wphi_on: 1
+  et1_on: 1
+  e11_to_e33_on: 1
+  e32_to_e35_on: 1
  et2_on: 0
  et3_on: 0
  et4_on: 0
-  bdt_on: 1
+  bdt_on: 0
  mc_iso_shift: 0.1
  mc_iso_scale: 1.2
@@ -128,21 +133,21 @@
  et1_max: 1.0
  et1_min: 0.5
-  et1_min_b: 0.5
+  et1_min_b: 0.9
  et1_min_s: 0.0
  #correlation

```

```

    e32_over_e35_max: 1.0
-   e32_over_e35_min: 0.8
-
-   #correlation
-   weta_cogx_max: 1.0
+   e32_over_e35_min: 0.92
+
+   #correlation
+   weta_cogx_max: 0.20
    weta_cogx_min: 0.0
-   weta_cogx_max_b: 1.0
-   weta_cogx_max_s: 0.0
-
-
-   wphi_cogx_max: 1.0
+   weta_cogx_max_b: 0.15
+   weta_cogx_max_s: 0.006
+
+
+   wphi_cogx_max: 2.0
    wphi_cogx_min: 0.0
-   wphi_cogx_max_b: 1.0
-   wphi_cogx_max_s: 0.0
+   wphi_cogx_max_b: 0.15
+   wphi_cogx_max_s: 0.006

    #correlation
@@ -152,5 +157,5 @@
    #small correlation
    e11_over_e33_max: 1.0
-   e11_over_e33_min: 0.0
+   e11_over_e33_min: 0.4

    #small correlation
@@ -168,5 +173,5 @@
    w32_min: 0.0

-   #bdt
+   #bdt (inert: bdt_on=0)
    bdt_max: 1.0
    bdt_min: 0.7
@@ -198,13 +203,13 @@

    #correlation
-   weta_cogx_max: 1.0
+   weta_cogx_max: 0.6
    weta_cogx_min: 0.0
-   weta_cogx_max_b: 1.0
+   weta_cogx_max_b: 0.6
    weta_cogx_max_s: 0.0

-   wphi_cogx_max: 1.0
+   wphi_cogx_max: 0.8

```



```

wphi_cogx_min: 0.0
- wphi_cogx_max_b: 1.0
+ wphi_cogx_max_b: 0.8
wphi_cogx_max_s: 0.0

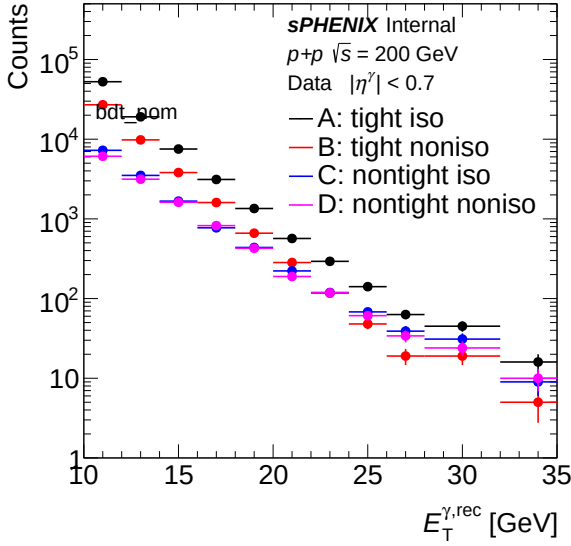
@@ -225,5 +230,5 @@
w32_min: 0.0

- #bdt
+ #bdt (inert: bdt_on=0)
bdt_max: 0.5
bdt_max_slope: 0.003333333333333336
@@ -238,5 +243,5 @@
cluster_weta_cogx_bound: 2.0

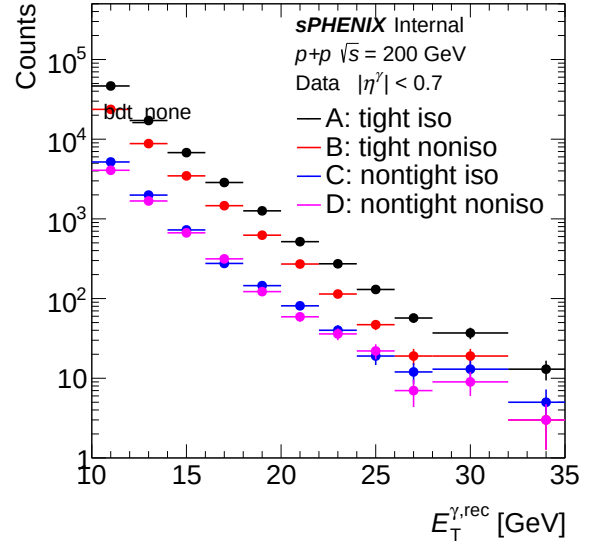
- npb_cut_on: 1
+ npb_cut_on: 0
npb_score_cut: 0.5

```

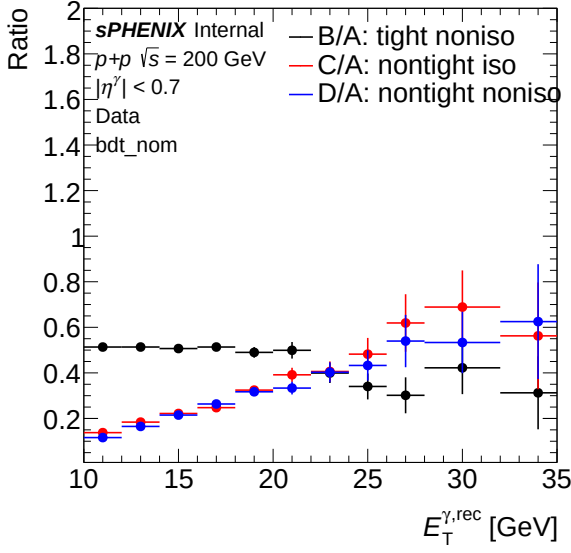
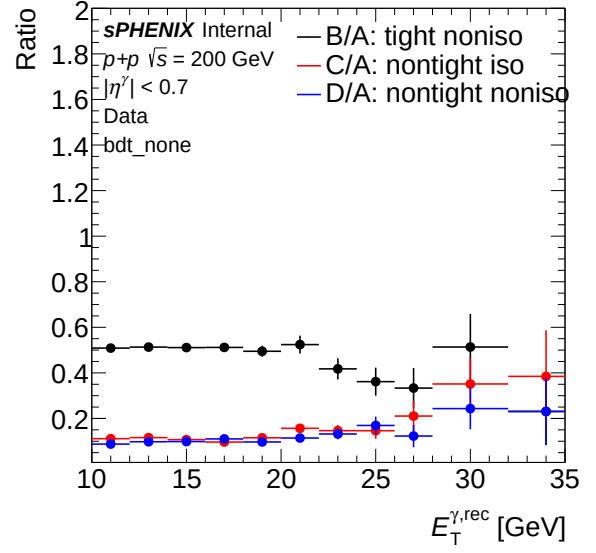
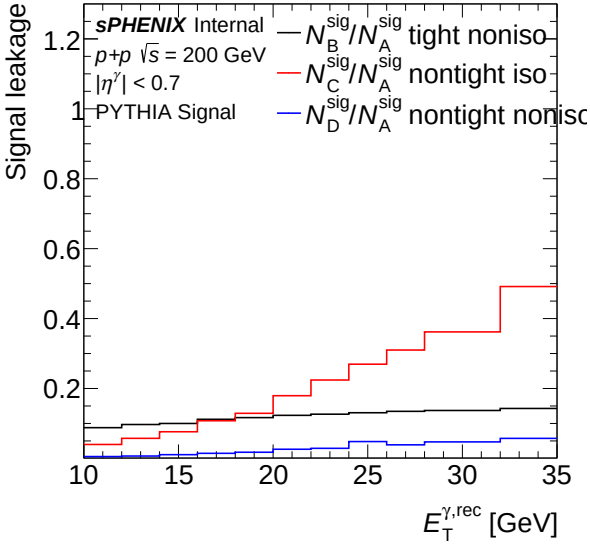
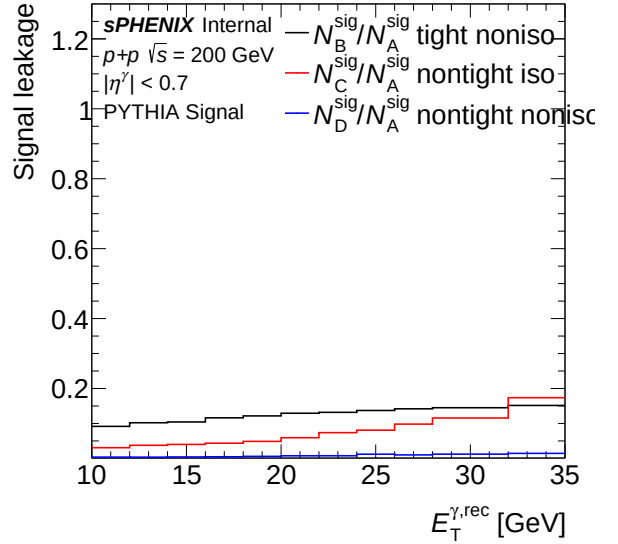
5.1 Sideband (Data)



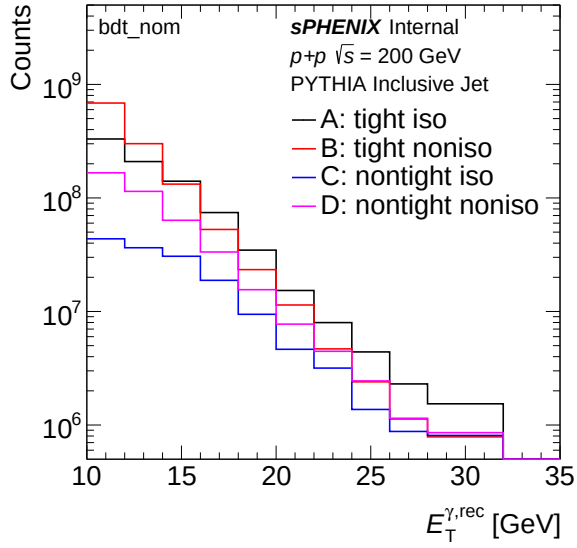
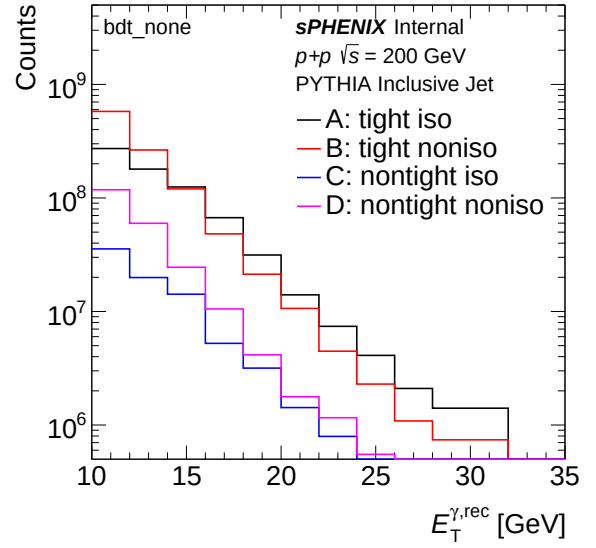
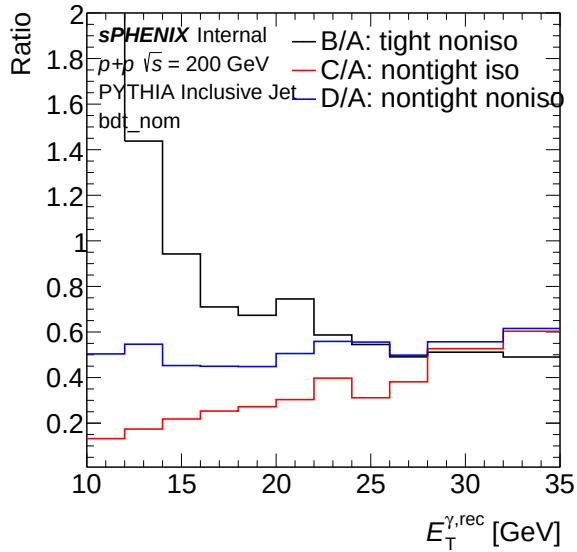
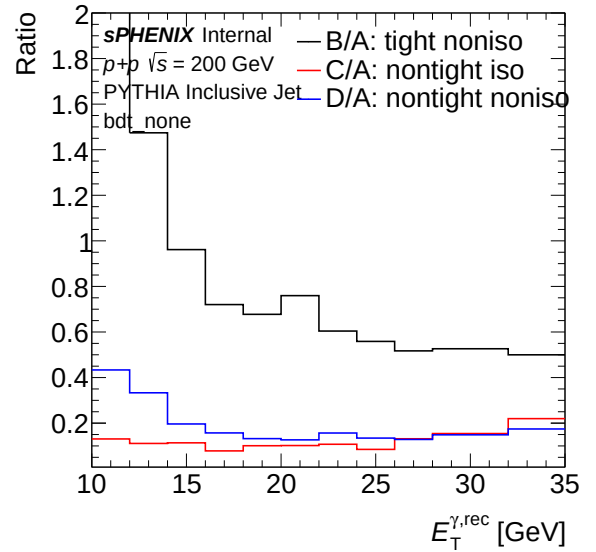
(a) Sideband ET spectrum (data) — **bdt_nom**

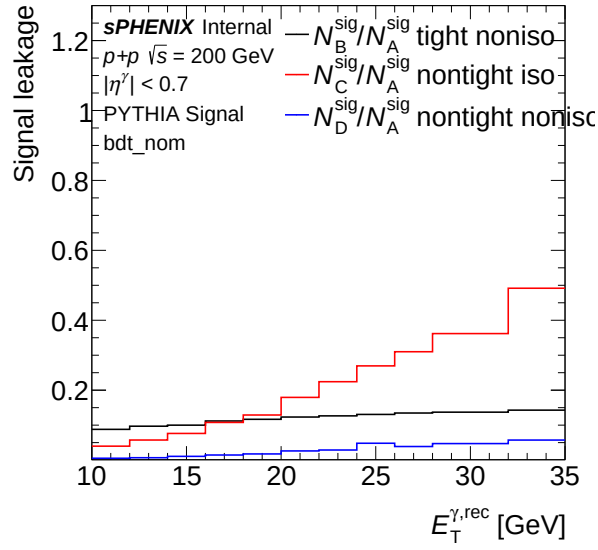
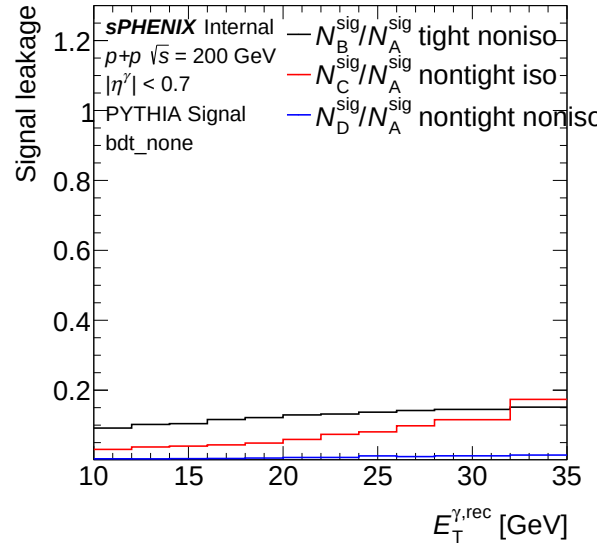


(b) Sideband ET spectrum (data) — **bdt_none**

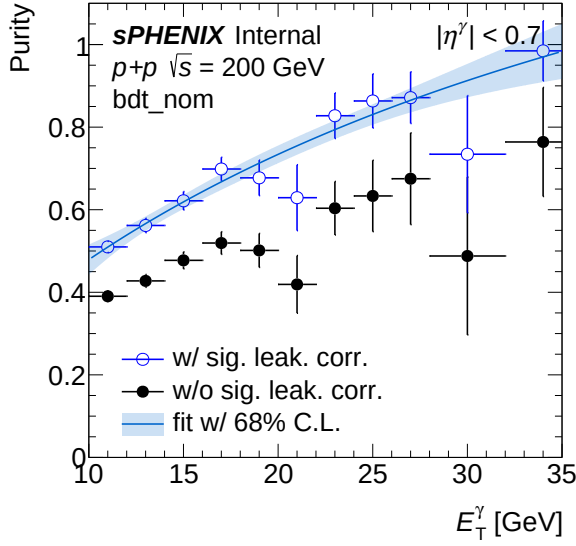
(a) Sideband ET ratio (data) — **bdt_nom**(b) Sideband ET ratio (data) — **bdt_none**(a) Leakage fraction vs ET (data) — **bdt_nom**(b) Leakage fraction vs ET (data) — **bdt_none**

5.2 Sideband (MC)

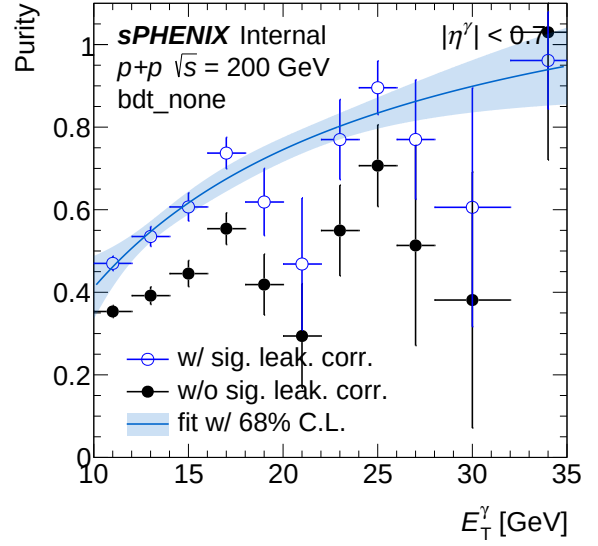
(a) Sideband ET spectrum (MC) — `bdt_nom`(b) Sideband ET spectrum (MC) — `bdt_none`(a) Sideband ET ratio (MC) — `bdt_nom`(b) Sideband ET ratio (MC) — `bdt_none`

(a) Leakage fraction vs ET (MC) — `bdt_nom`(b) Leakage fraction vs ET (MC) — `bdt_none`

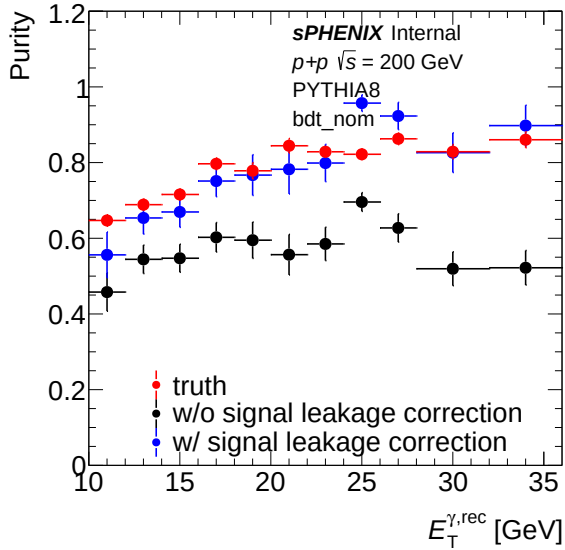
5.3 Purity



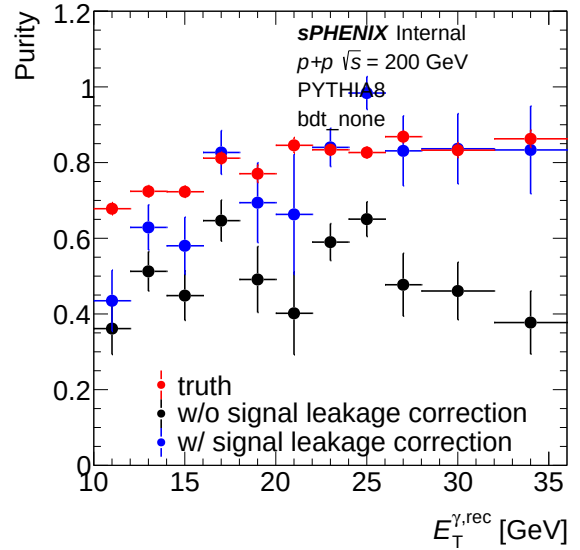
(a) Purity (data) — bdt_nom



(b) Purity (data) — bdt_none

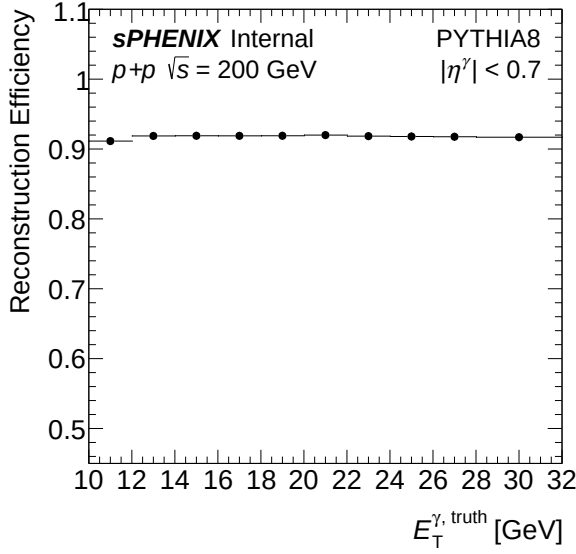


(a) Purity (MC) — bdt_nom

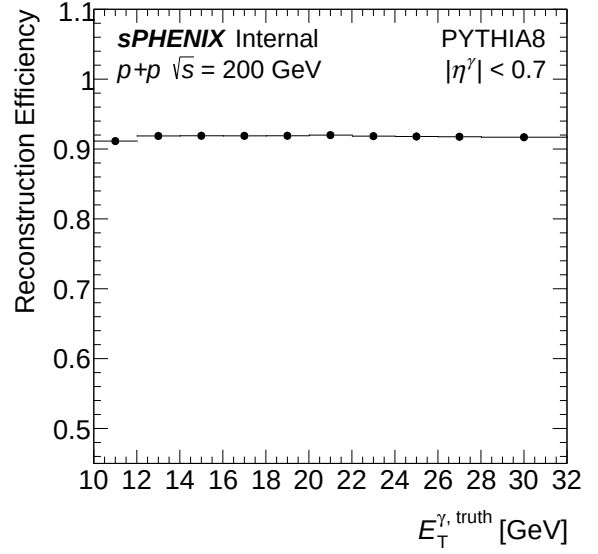


(b) Purity (MC) — bdt_none

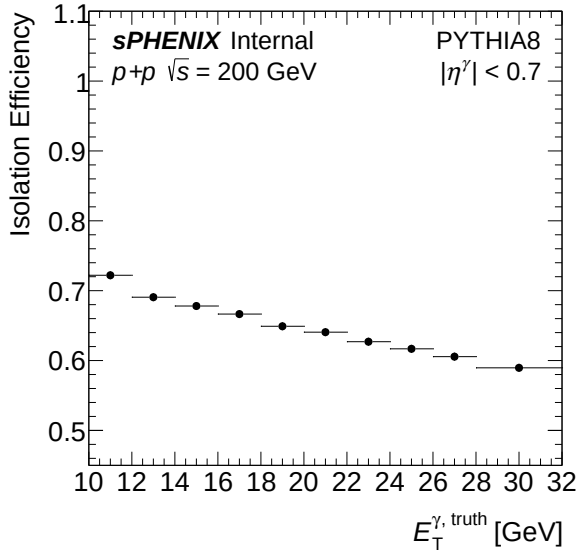
5.4 Efficiency



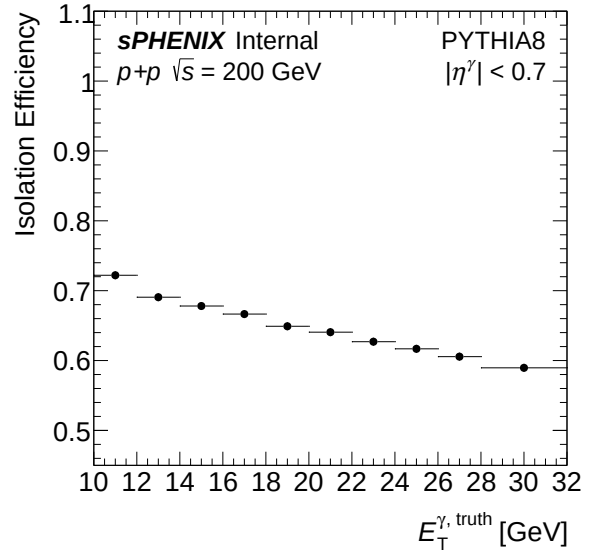
(a) Reconstruction efficiency — bdt_nom



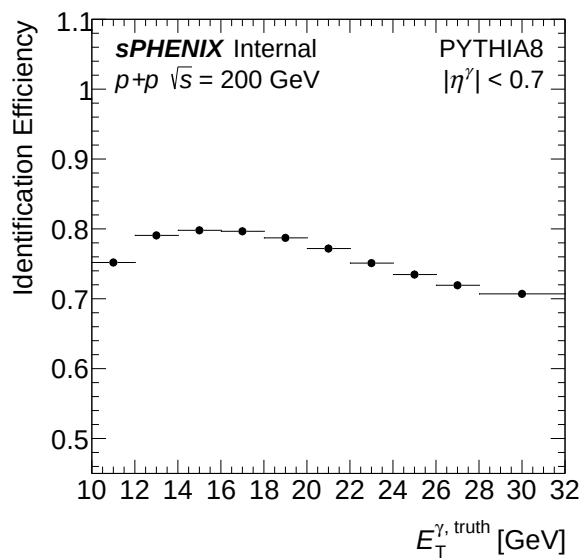
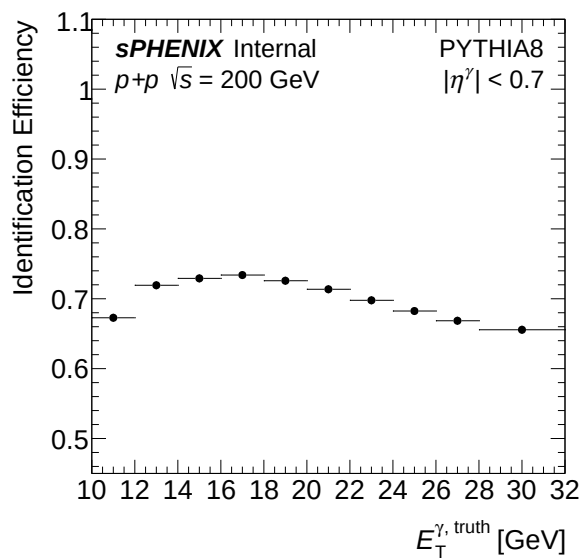
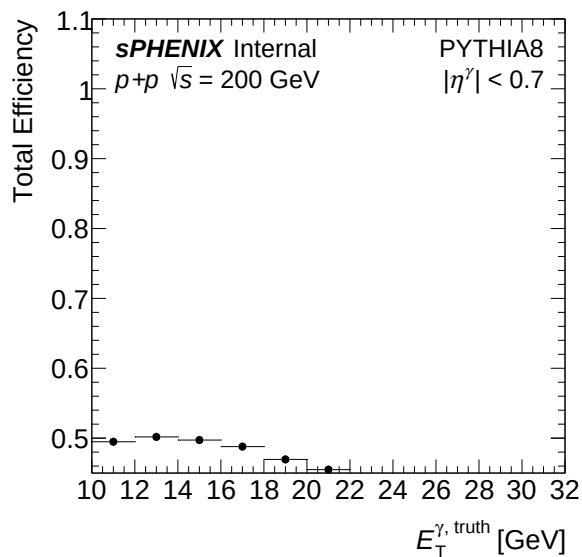
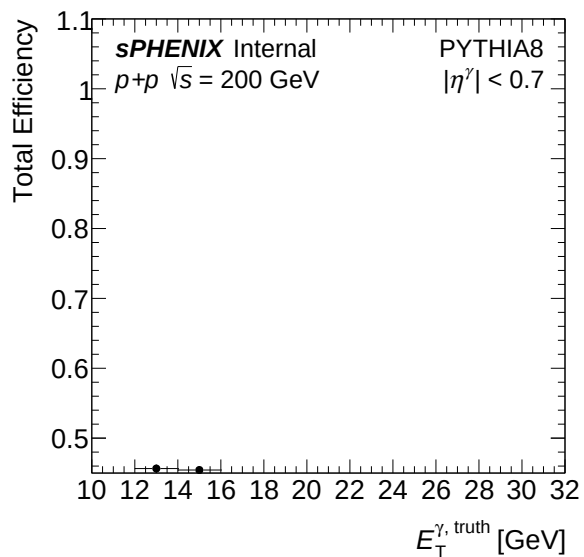
(b) Reconstruction efficiency — bdt_none

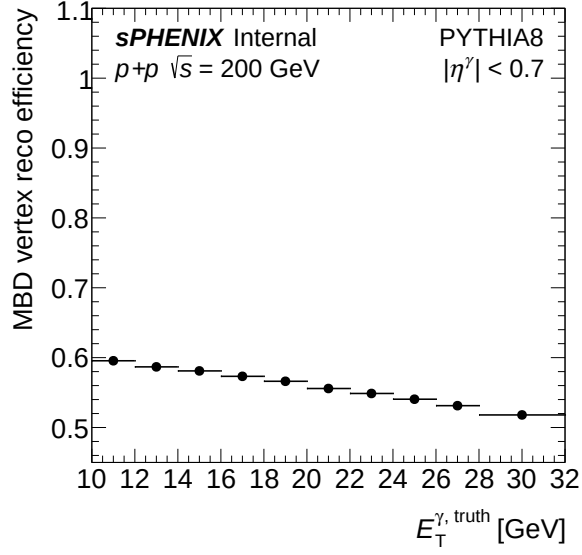


(a) Isolation efficiency — bdt_nom

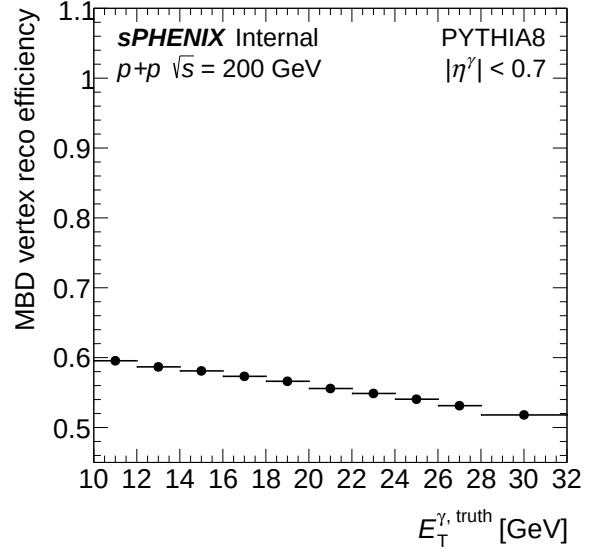


(b) Isolation efficiency — bdt_none

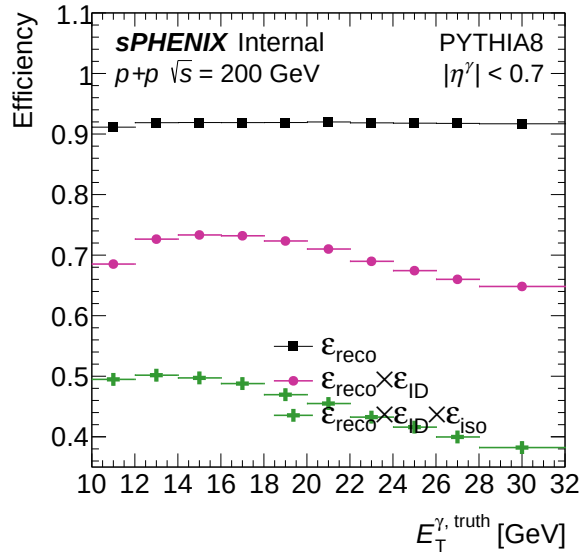
(a) Identification efficiency — `bdt_nom`(b) Identification efficiency — `bdt_none`(a) Total efficiency — `bdt_nom`(b) Total efficiency — `bdt_none`



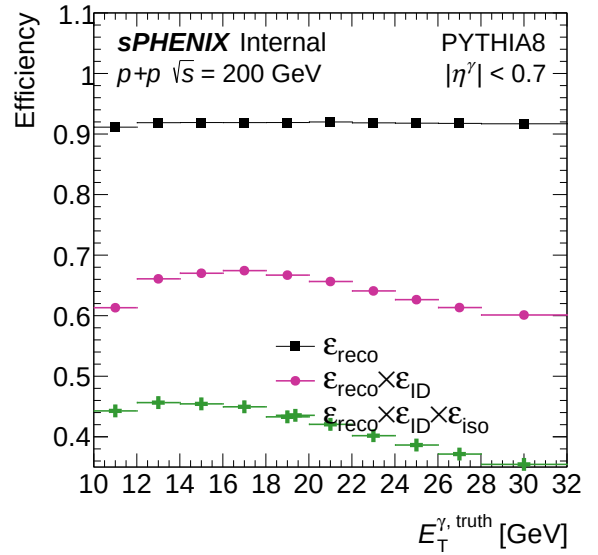
(a) MBD trigger efficiency — bdt_nom



(b) MBD trigger efficiency — bdt_none

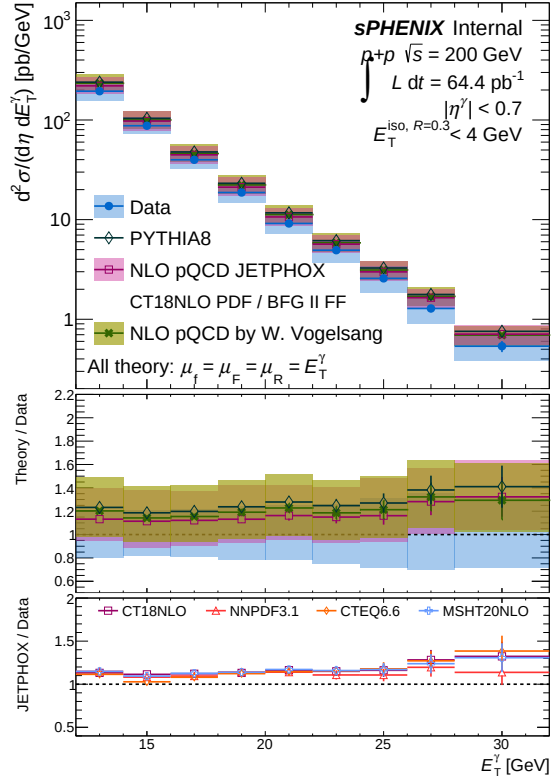
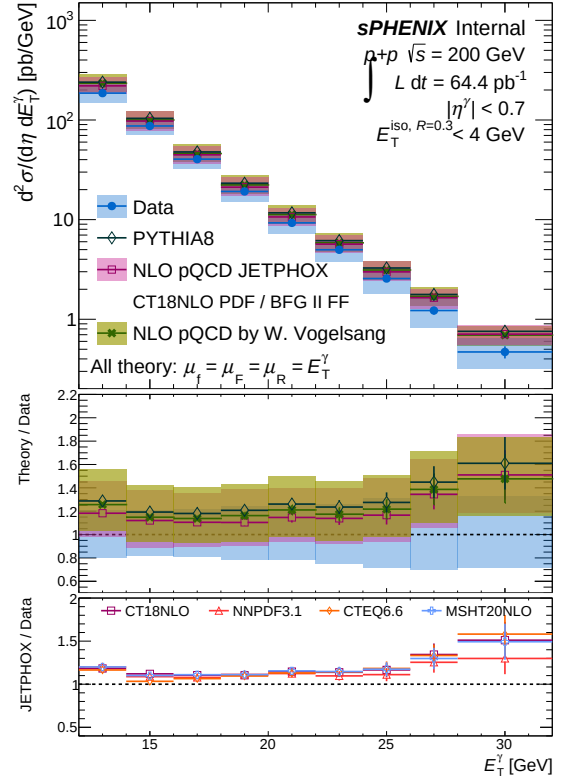


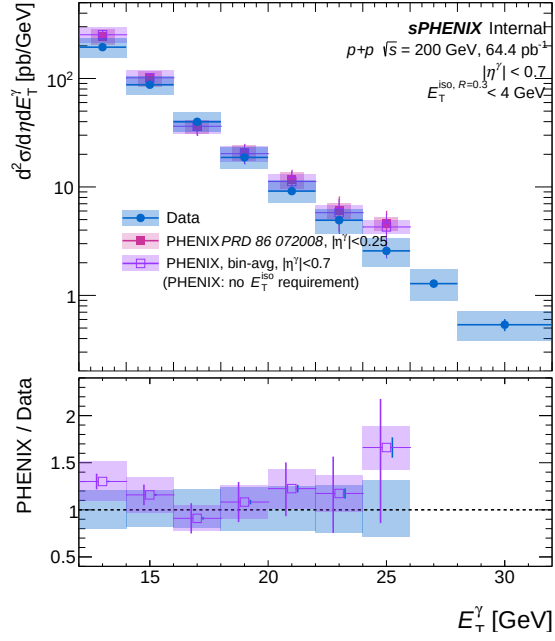
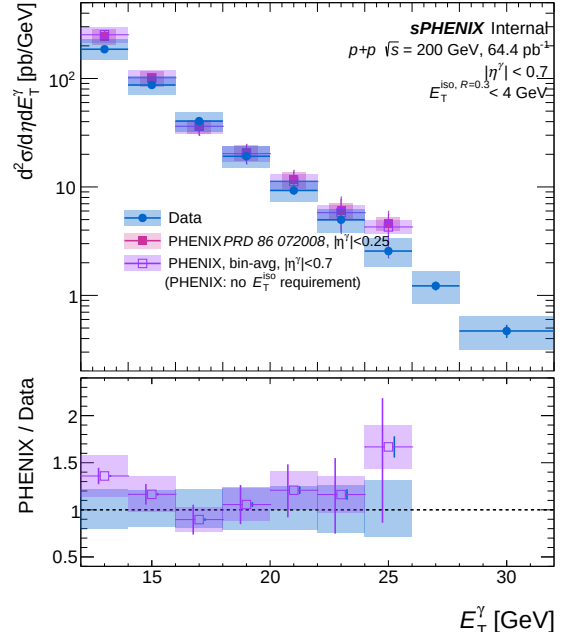
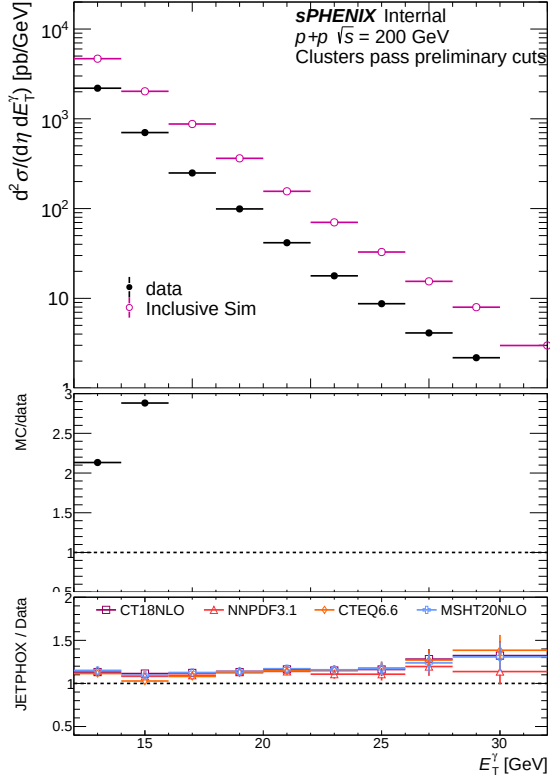
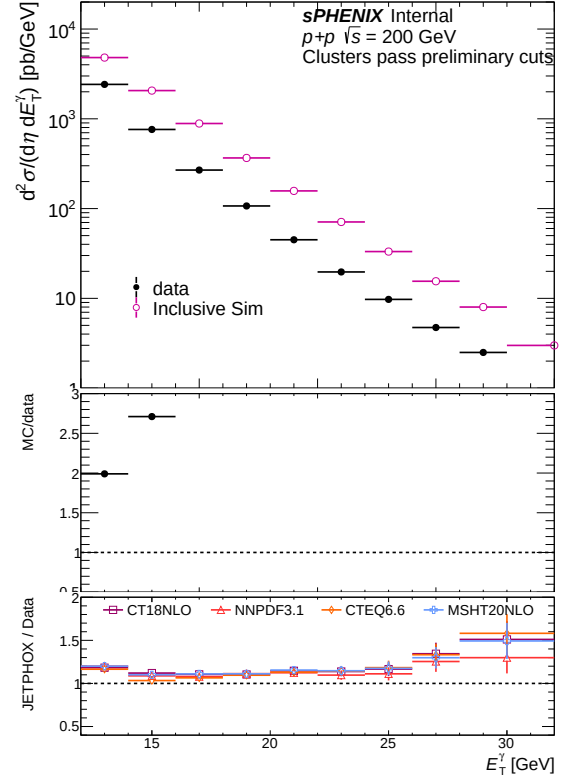
(a) Combined photon efficiency — bdt_nom

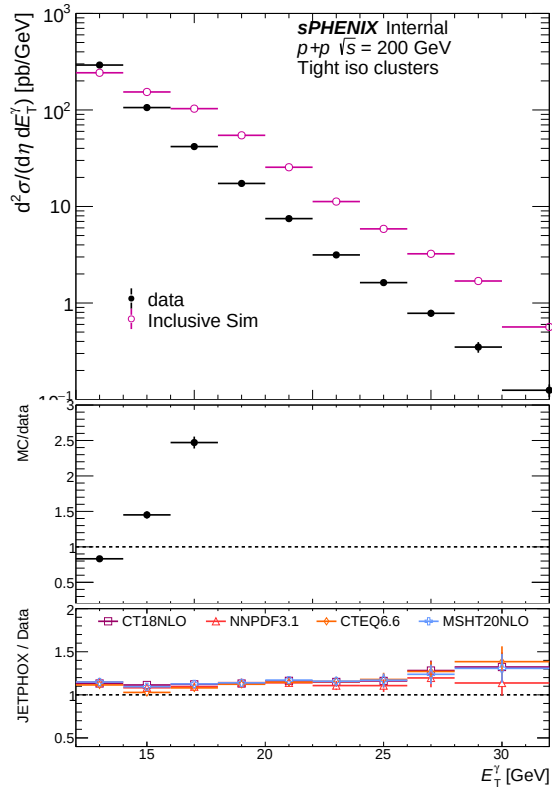
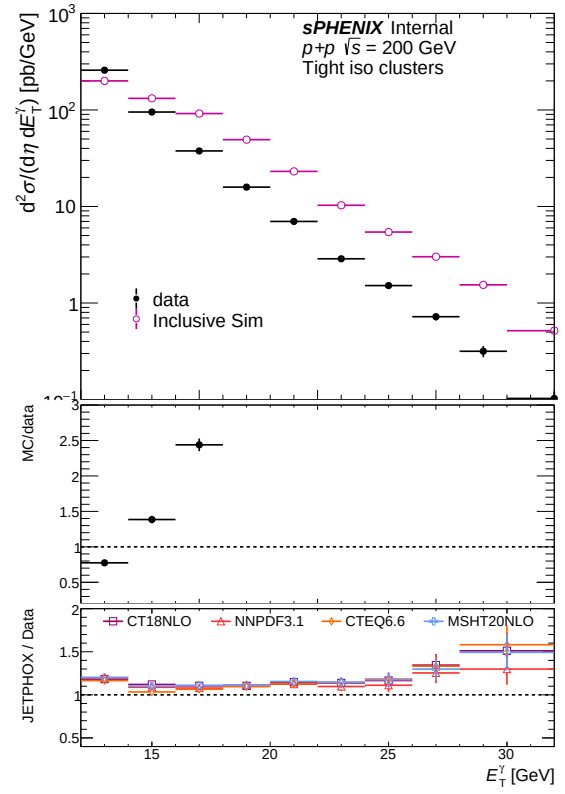


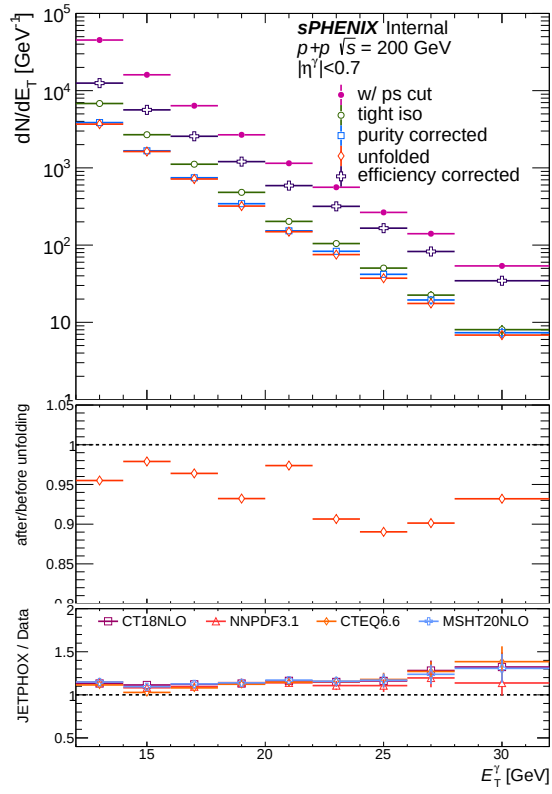
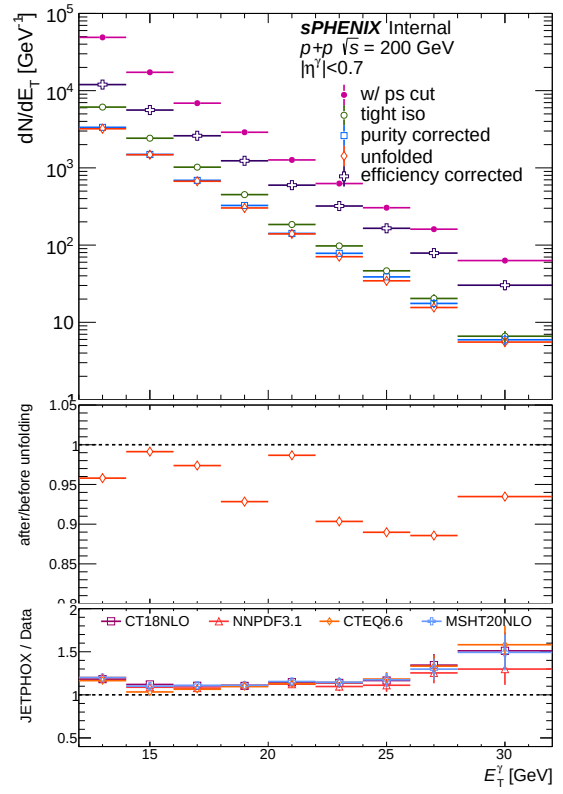
(b) Combined photon efficiency — bdt_none

5.5 Final result

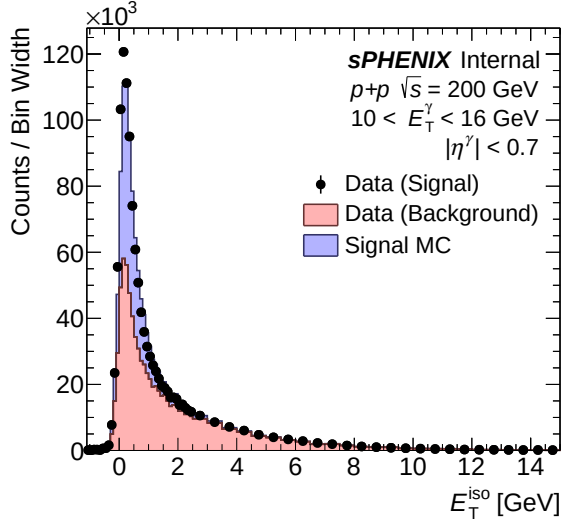
(a) Final cross section result — `bdt_nom`(b) Final cross section result — `bdt_none`

(a) Comparison with PHENIX — `bdt_nom`(b) Comparison with PHENIX — `bdt_none`(a) Common cluster yield (data vs MC) — `bdt_nom`(b) Common cluster yield (data vs MC) — `bdt_none`

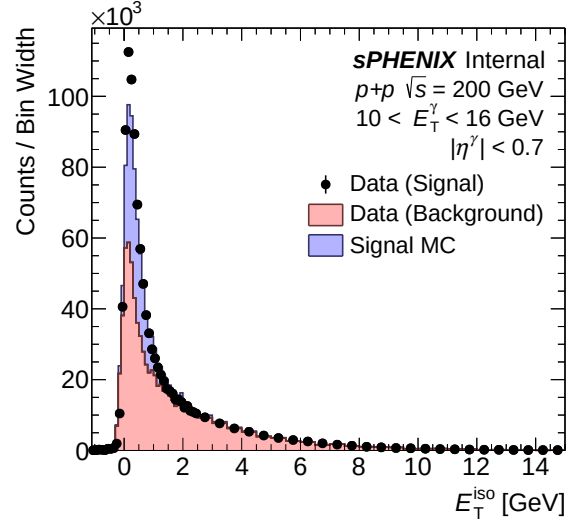
(a) Tight iso cluster yield (data vs MC) — `bdt_nom`(b) Tight iso cluster yield (data vs MC) — `bdt_none`

(a) All correction steps — `bdt_nom`(b) All correction steps — `bdt_none`

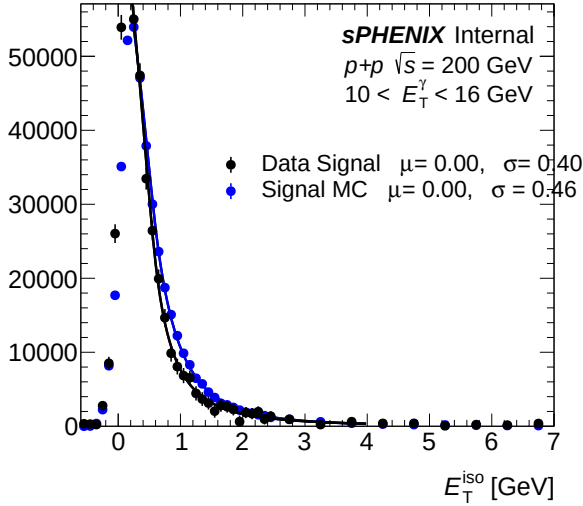
5.6 Isolation fit (CONF)



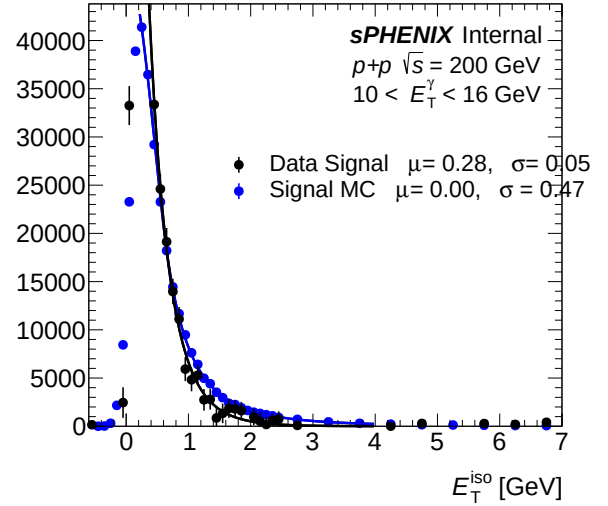
(a) IsoET template (pT bins 0-2) — bdt_nom



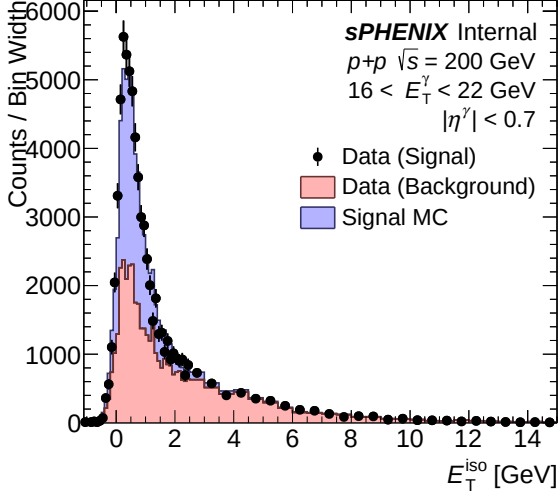
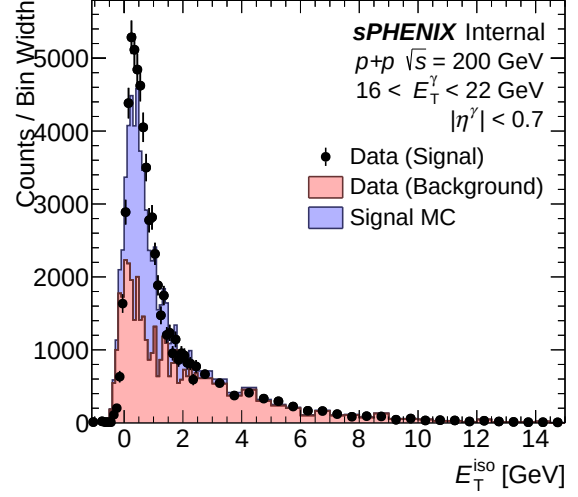
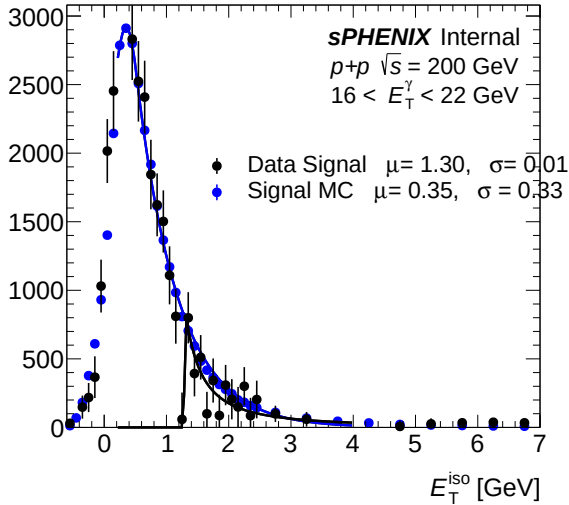
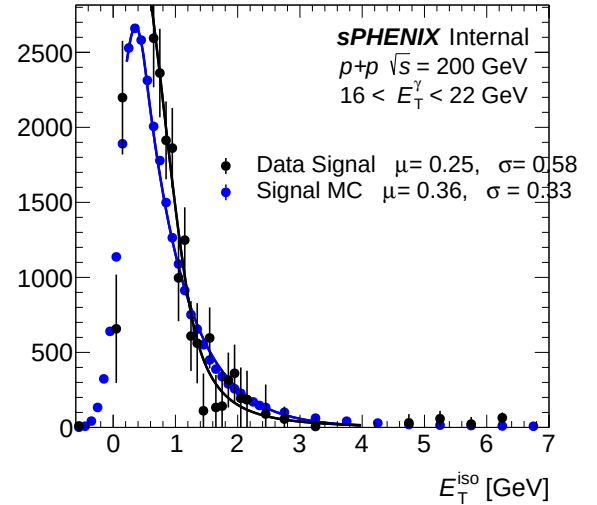
(b) IsoET template (pT bins 0-2) — bdt_none

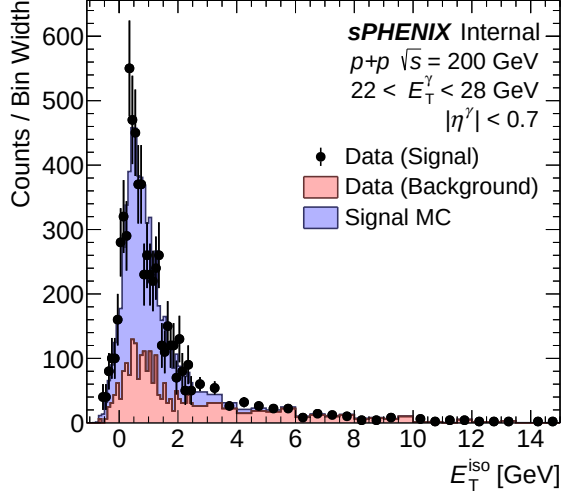
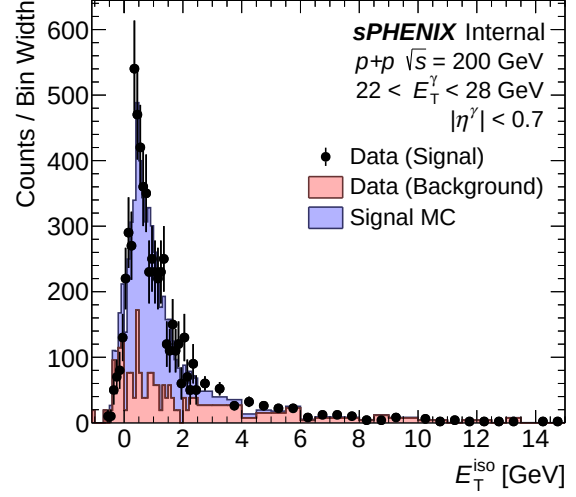
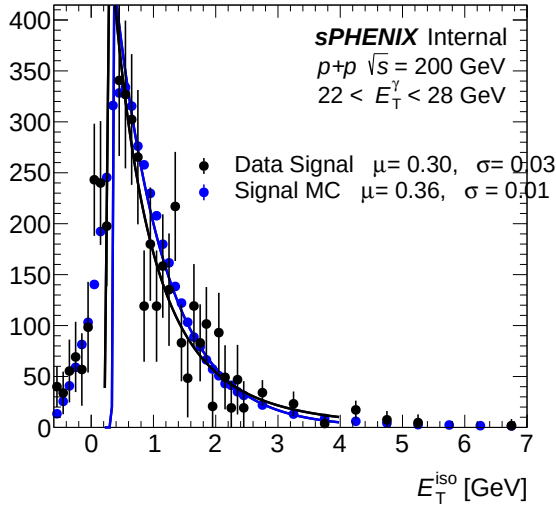
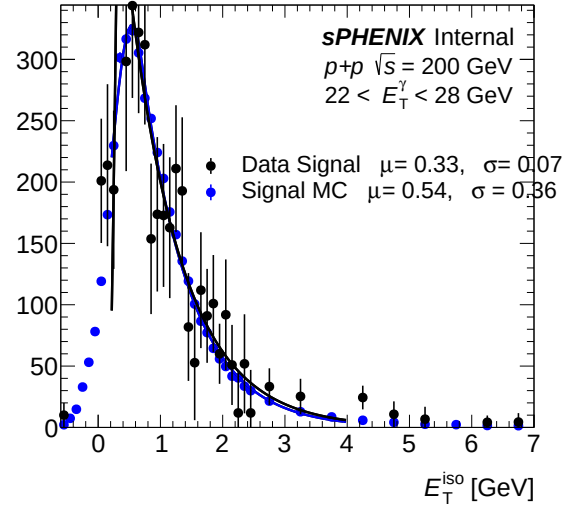


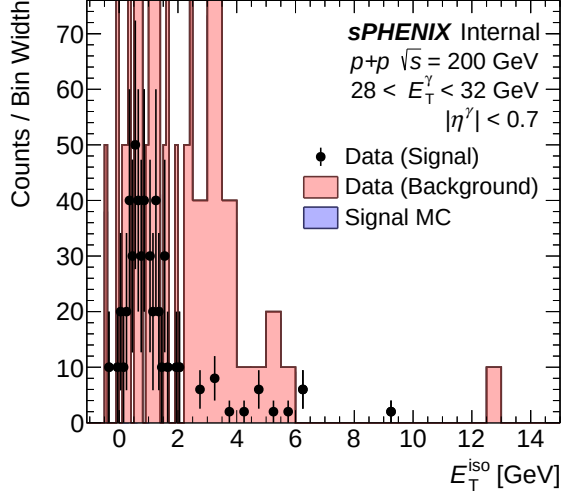
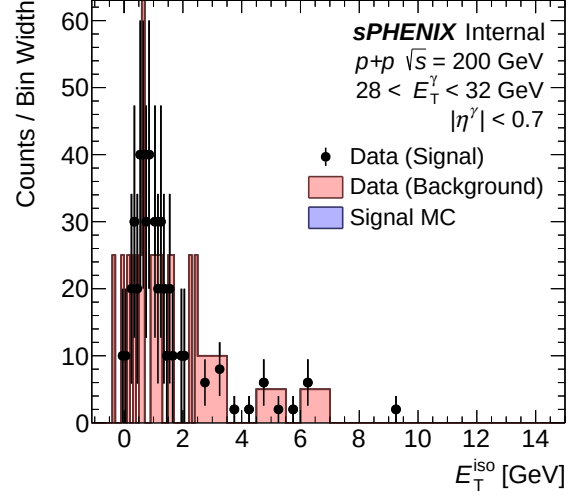
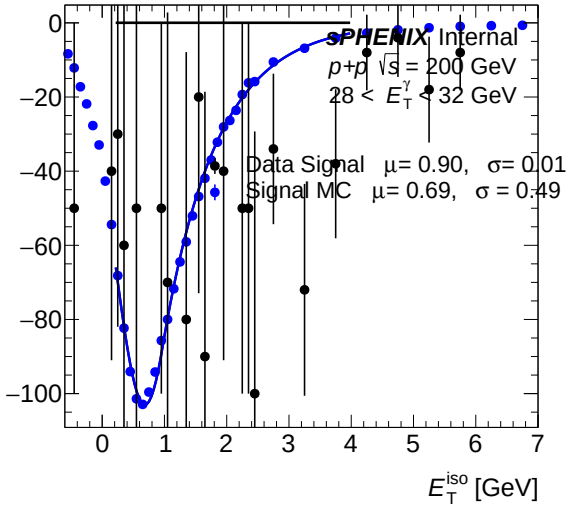
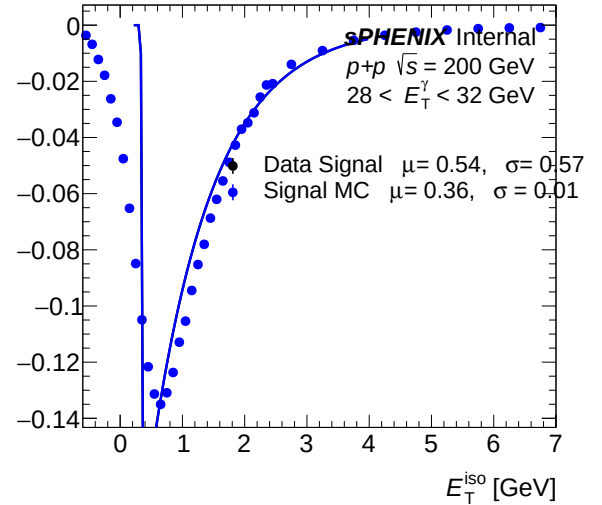
(a) IsoET fit template (pT bins 0-2) — bdt_nom



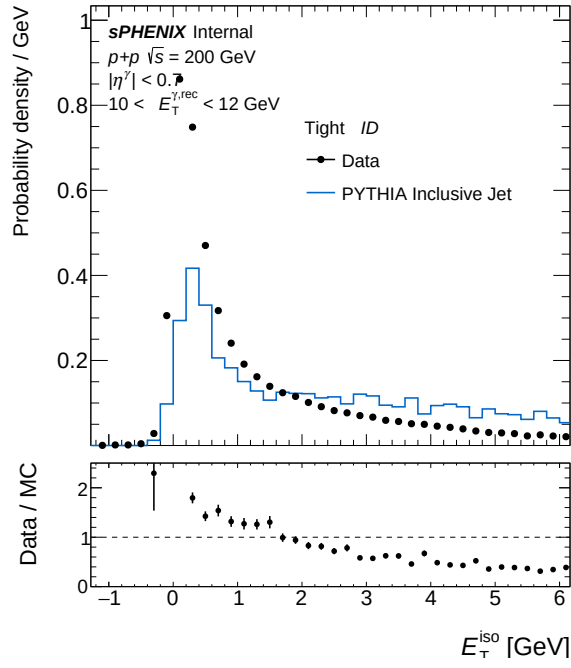
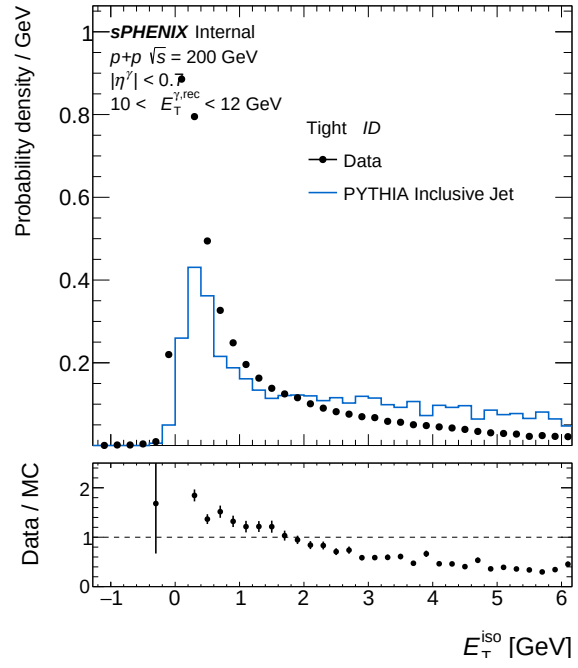
(b) IsoET fit template (pT bins 0-2) — bdt_none

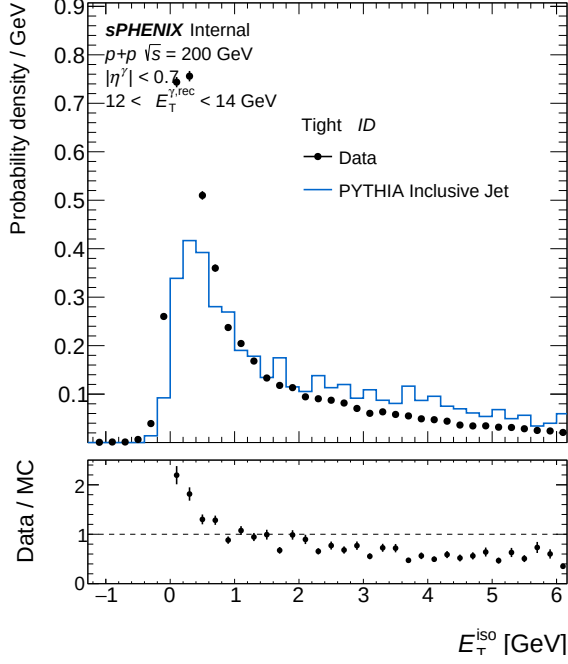
(a) IsoET template (pT bins 3–5) — `bdt_nom`(b) IsoET template (pT bins 3–5) — `bdt_none`(a) IsoET fit template (pT bins 3–5) — `bdt_nom`(b) IsoET fit template (pT bins 3–5) — `bdt_none`

(a) IsoET template (pT bins 6–8) — `bdt_nom`(b) IsoET template (pT bins 6–8) — `bdt_none`(a) IsoET fit template (pT bins 6–8) — `bdt_nom`(b) IsoET fit template (pT bins 6–8) — `bdt_none`

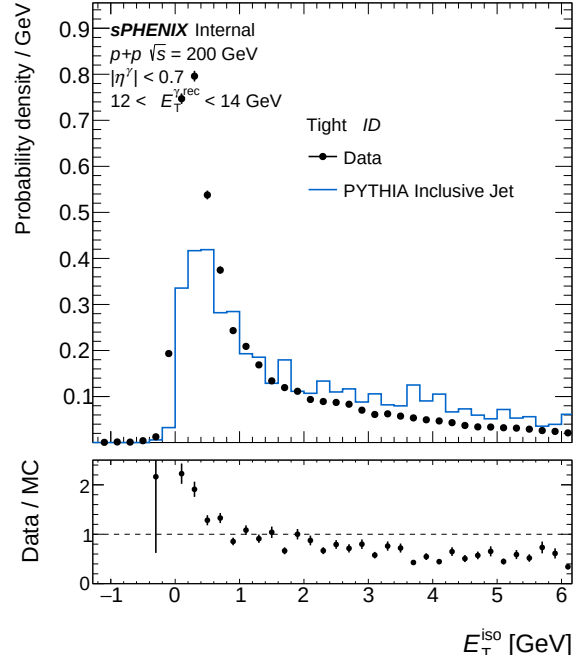
(a) IsoET template (pT bins 9–9) — `bdt_nom`(b) IsoET template (pT bins 9–9) — `bdt_none`(a) IsoET fit template (pT bins 9–9) — `bdt_nom`(b) IsoET fit template (pT bins 9–9) — `bdt_none`

5.7 Isolation ET

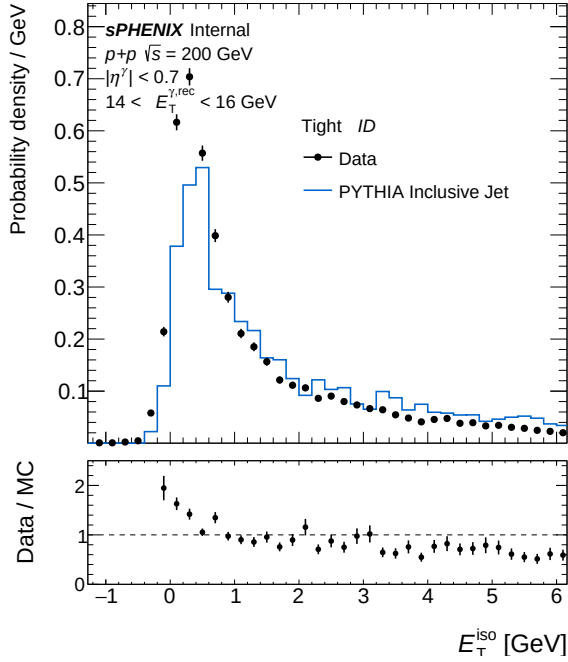
(a) IsoET Tight, pT bin 0 — `bdt_nom`(b) IsoET Tight, pT bin 0 — `bdt_none`



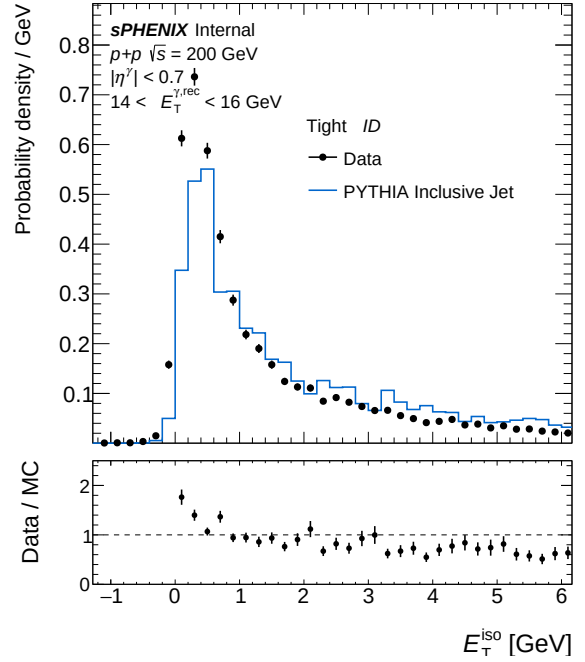
(a) IsoET Tight, pT bin 1 — bdt_nom



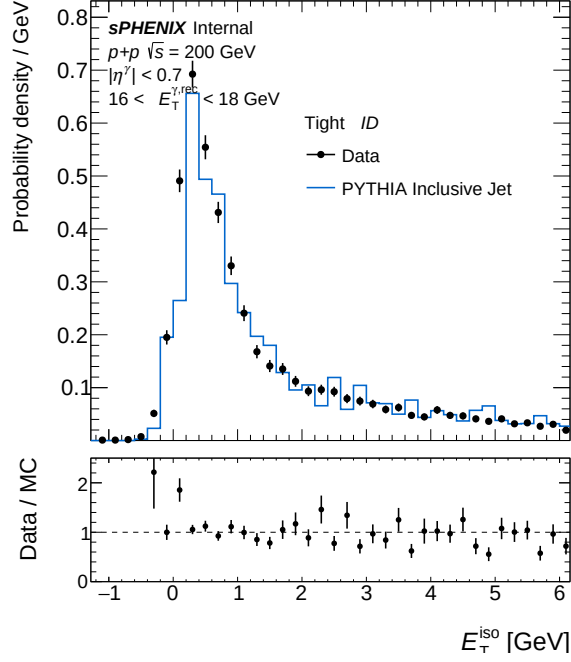
(b) IsoET Tight, pT bin 1 — bdt_none



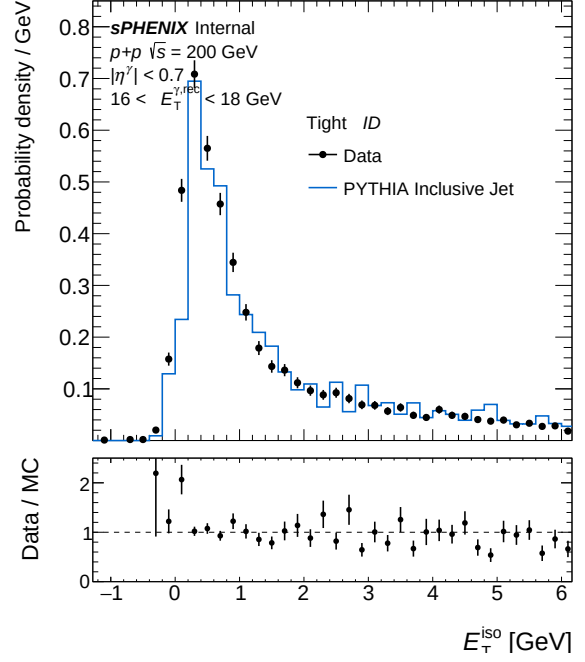
(a) IsoET Tight, pT bin 2 — bdt_nom



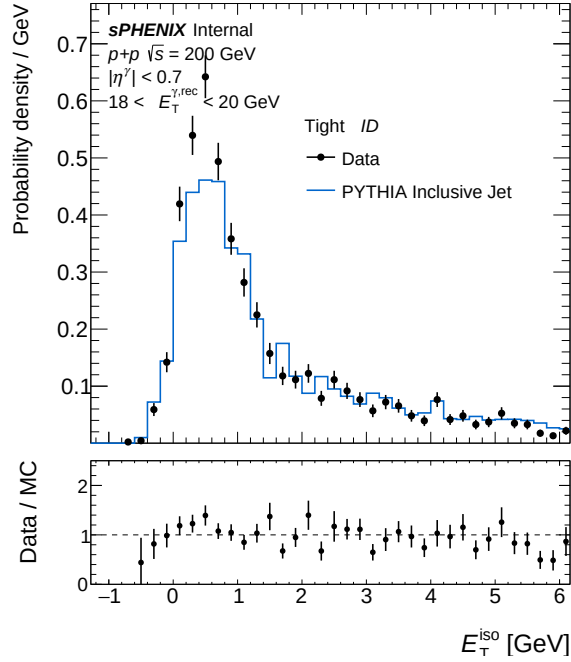
(b) IsoET Tight, pT bin 2 — bdt_none



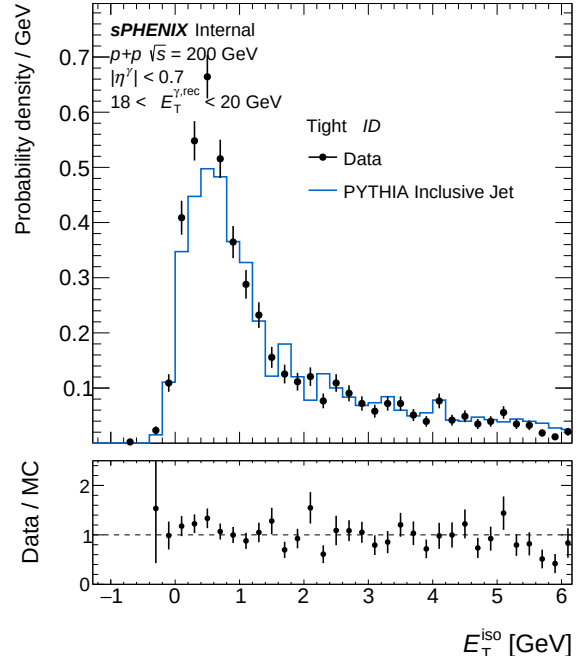
(a) IsoET Tight, pT bin 3 — bdt_nom



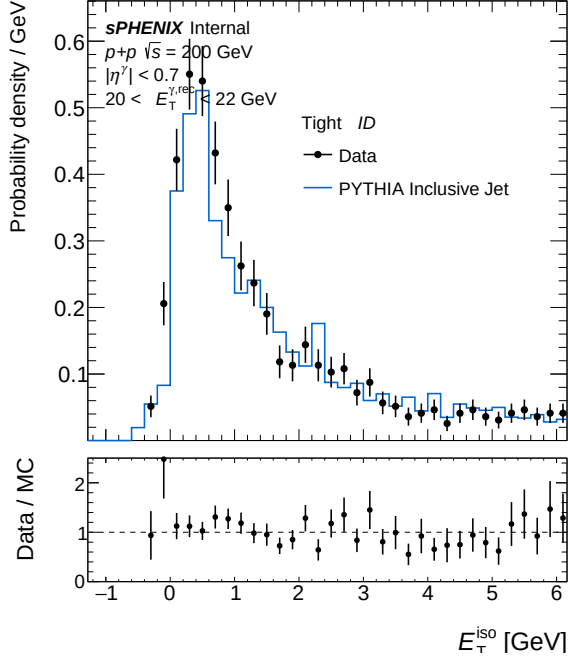
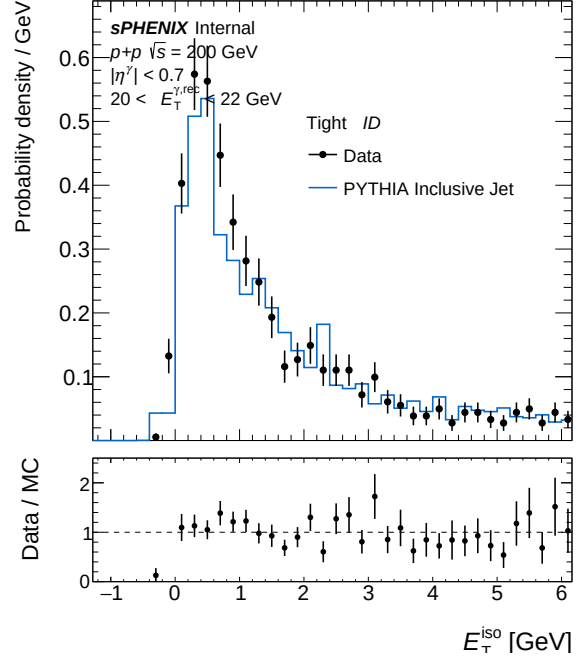
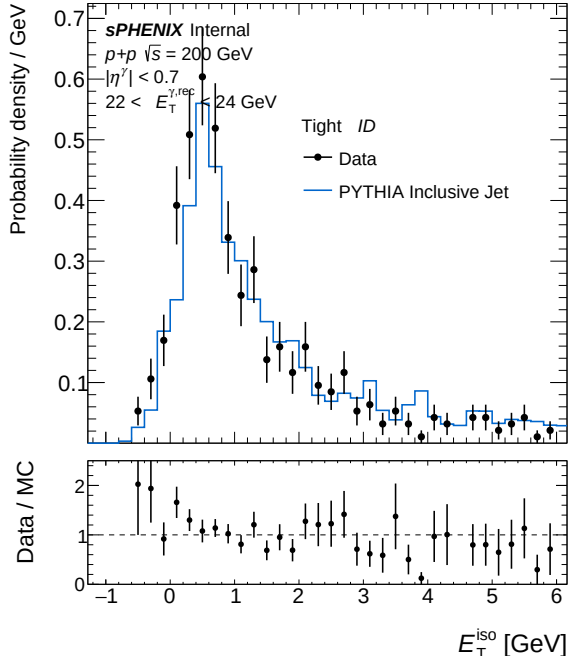
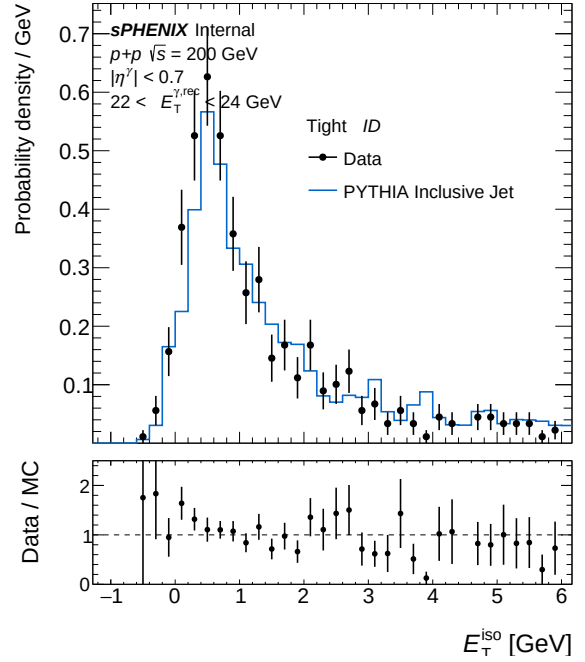
(b) IsoET Tight, pT bin 3 — bdt_none

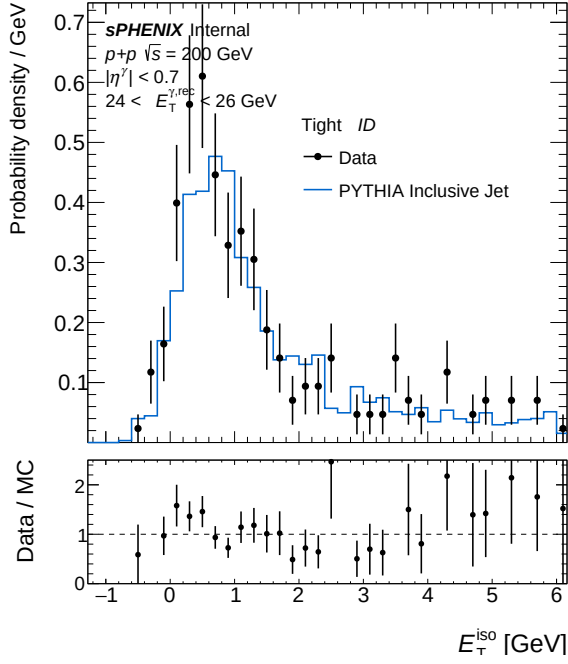
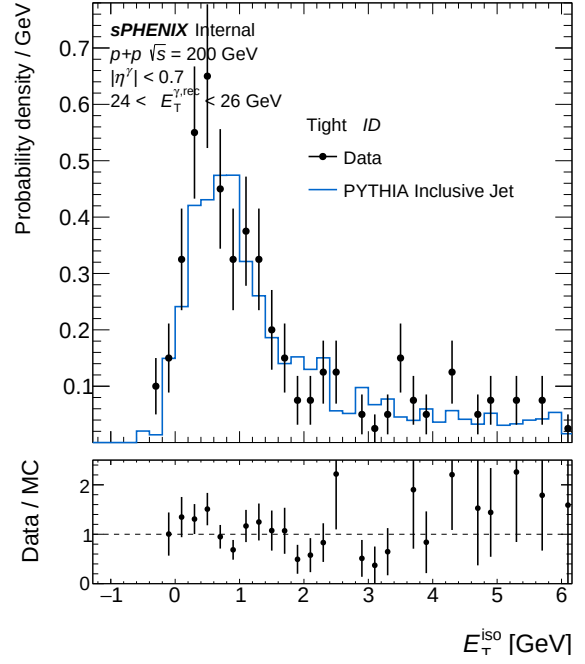
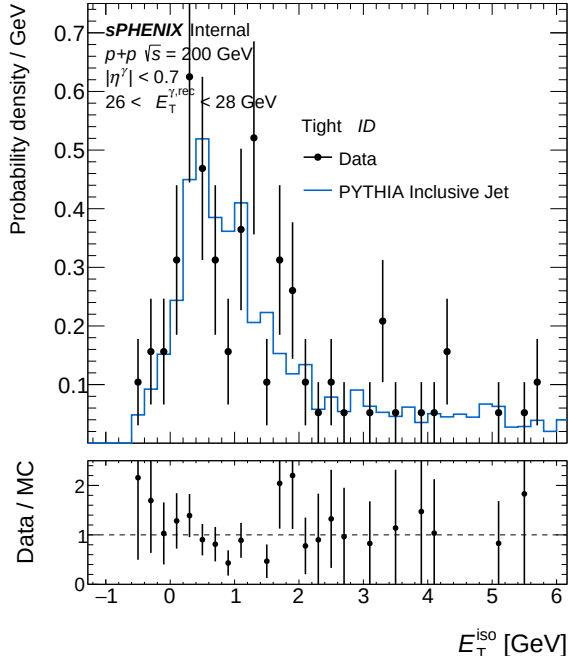
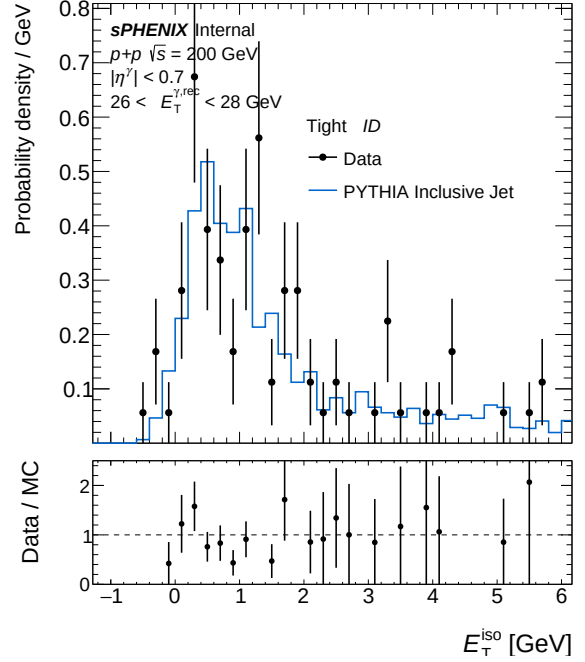


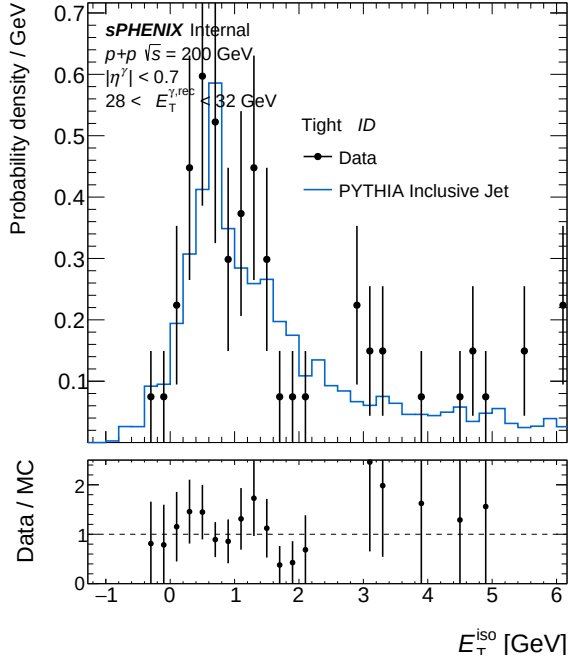
(a) IsoET Tight, pT bin 4 — bdt_nom



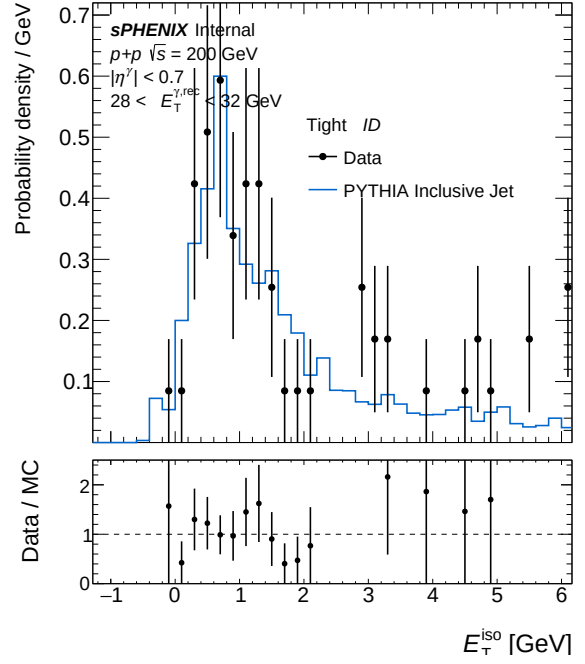
(b) IsoET Tight, pT bin 4 — bdt_none

(a) IsoET Tight, pT bin 5 — `bdt_nom`(b) IsoET Tight, pT bin 5 — `bdt_none`(a) IsoET Tight, pT bin 6 — `bdt_nom`(b) IsoET Tight, pT bin 6 — `bdt_none`

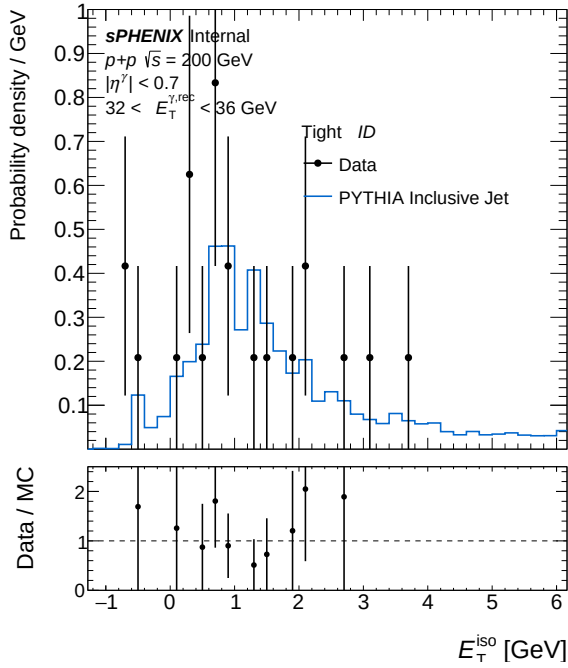
(a) IsoET Tight, pT bin 7 — `bdt_nom`(b) IsoET Tight, pT bin 7 — `bdt_none`(a) IsoET Tight, pT bin 8 — `bdt_nom`(b) IsoET Tight, pT bin 8 — `bdt_none`



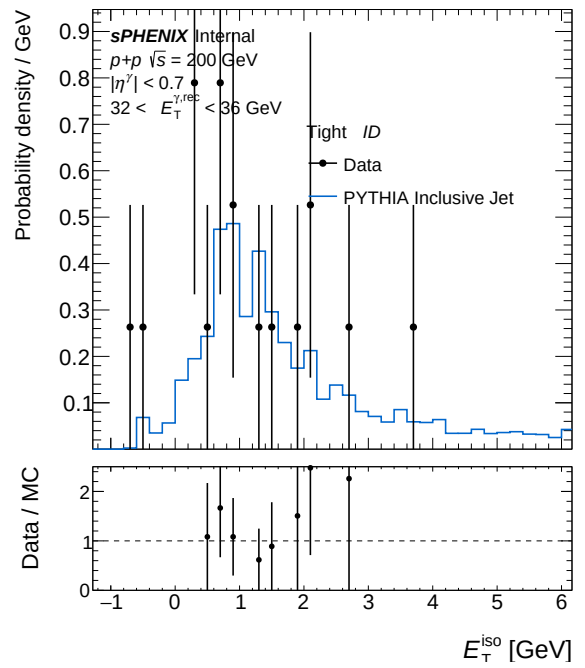
(a) IsoET Tight, pT bin 9 — bdt_nom



(b) IsoET Tight, pT bin 9 — bdt_none



(a) IsoET Tight, pT bin 10 — bdt_nom



(b) IsoET Tight, pT bin 10 — bdt_none